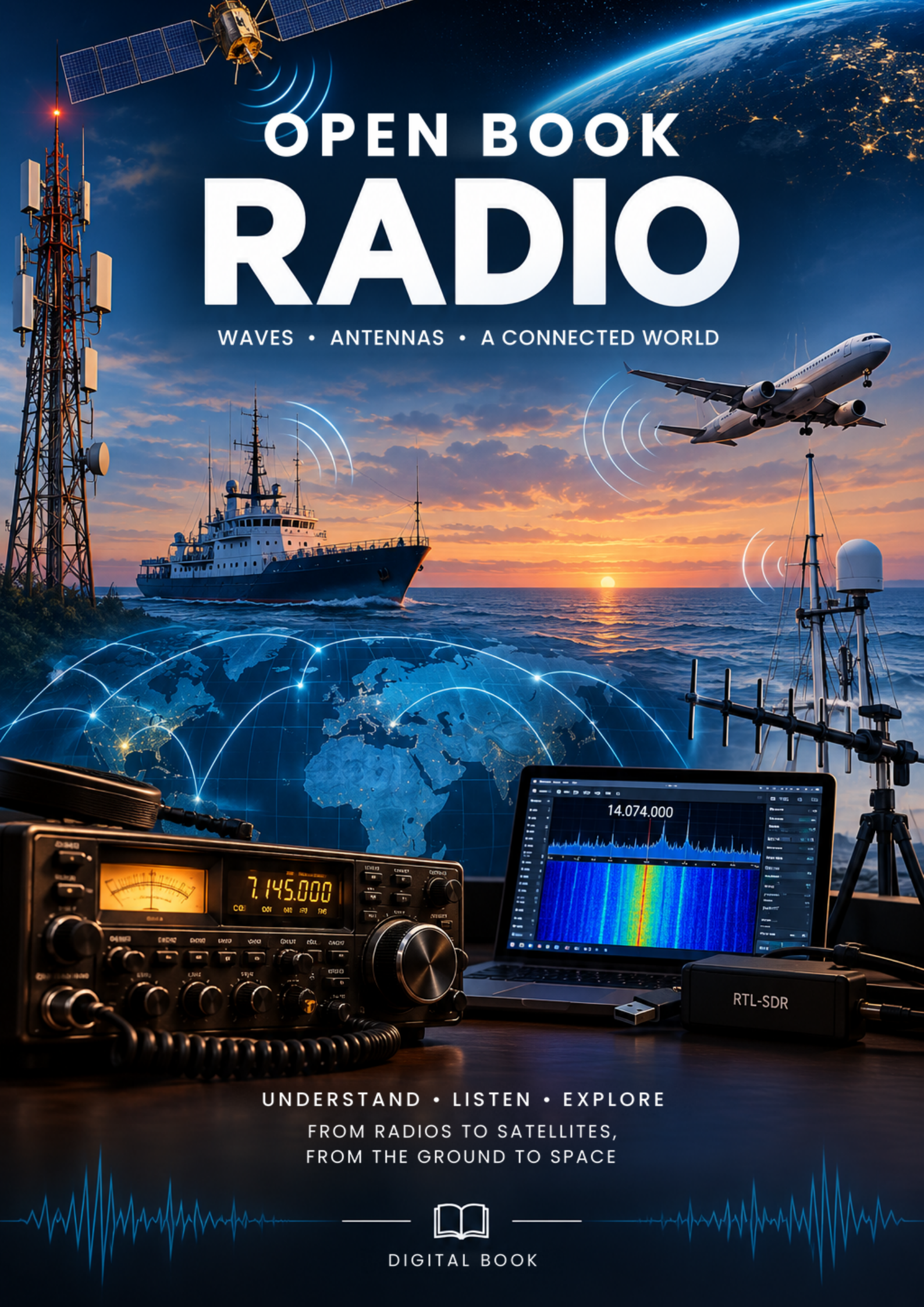

OPEN BOOK
RADIO
WAVES • ANTENNAS • A CONNECTED WORLD
7.145.000
14.074.000
RTL-SDR
UNDERSTAND • LISTEN • EXPLORE
FROM RADIOS TO SATELLITES,
FROM THE GROUND TO SPACE
DIGITAL BOOK

# Read this before you start

A free, open and shared guide

*This book is a gift: it is shared, not sold.*

## Free, and meant to stay free

This guide is **free** and **non-commercial**. It must not be sold, resold or offered for payment in any form. You can read it, copy it and pass it on freely, as long as you distribute it as is and free of charge.

## A collective work

The idea of this book came from a simple need: to gather in a few clear sheets what you need to understand, listen to and get by with radio. It was designed by its author and written with the help of Claude, Anthropic's assistant. It also owes a lot to the open source community and to everyone who shares their knowledge: firmware developers, free software authors, ham radio operators, documentation writers and enthusiasts who publish their plans, manuals and feedback. The projects mentioned (firmwares, applications, tools) remain the property of their authors and are covered by their licences: see their repositories to learn more about them.

## Mistakes are possible

This guide was prepared with care, but it may contain **errors, inaccuracies and omissions**. Values, frequencies, keys and firmware versions change quickly: some information is marked "from memory" or "to verify", and the antenna plans are theoretical. Always check official sources before relying on it, especially before flashing a radio or transmitting. It is provided as is, without warranty, and you use it at your own responsibility.

## Obey the law, everywhere and always

Rules on listening, transmitting, frequencies, equipment and the confidentiality of communications **vary from country to country**, and change over time. Everyone must know and obey the laws and regulations in force in their own country. Listening is generally allowed, but what you hear must not be disclosed. Transmitting most often requires a licence or approved equipment, and transmitting on distress frequencies is forbidden. This guide does not encourage any illegal use and declines all responsibility in case of non-compliance with regulations.

## Thank you

Thanks to everyone who, by publishing what they know, makes this kind of book possible. If you spot a mistake or have better information, share it in turn: that is how this guide will improve.

# Radio Listening Guide

70 sheets to understand, listen to and identify signals, with a UV-K1 or a UV-K5 V3, its open source firmwares and an SDR dongle.

Any radio with equivalent specifications can also be used for basic listening.

**Version V3 beta · November 2026**


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| 1 | Getting started |
| 2 | Where to listen: the bands |
| 3 | Aviation |
| 4 | Sea and inland waterways |
| 5 | Radiosondes and other frequencies |
| 6 | Emergencies and safety |
| 7 | DATA modes and images |
| 8 | Radio language: codes and conventions |
| A00 | Transmitting and communication (depends on the country) |
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| A2 | DIY: antennas, radios, energy |


Values "from memory" must be checked before use; the sheets indicate what was confirmed online.

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| | **Sheet A32** · ATS Mini (V4): Si4732 receiver and firmwares | p. 125 |
| | **Sheet A33** · Autonomous power: batteries and solar | p. 127 |
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## How to use this book

> **Listening is free, transmitting is not.** This guide is about reception. Transmitting requires, depending on the service, an amateur radio callsign, approved equipment and compliance with local regulations. Never transmit on a distress frequency (121.5; 156.8; 243; 406 MHz; 2182 kHz) and never replay a command you have picked up. Private exchanges you hear must not be disclosed.

## How to read a sheet

- Each chapter has its own colour, repeated in the strip at the top of the page: the current chapter is shown large.
- The badge at the top right shows reliability: "checked online", "from memory, to verify" or "partly checked".
- Cross-references ("sheet 14") avoid repeating information: antenna sizes are in sheet 1, tools in sheet 3, emergencies and DSC in sheets 14, 21 and 22.

## Chapters and appendices

- **Chapters 1 to 8**: listening (bands, aviation, sea, radiosondes, emergencies, DATA modes, radio language). They apply to all of Europe (ITU region 1).
- **Appendix A00: Transmitting and communication (sheets A00a and A00b)**. **This section depends on the country**: licences, licence-free bands, power limits and procedures change from one country to another; it describes France. Everyone is free to keep it in their book or not, or to replace it with the one for their own country. The other sheets do not depend on it; a few cross-references cite A00a or A00b and become pointless if it is removed.
- **Appendix A1: UV-K1 and overlay applications (sheets A01 to A18)** and **appendix A2: DIY, antennas, radios, energy (sheets A19 to A34)**: the hardware, its firmwares and what you can build.

## Two tools for the whole guide

- **UV-K1** and its open source firmwares (Fusion, Labs, Apex, Falcon, GoguFW): VHF/UHF listening, HF with the SI4732, embedded applications. See sheets 3 and A01 to A07.
- **RTL-SDR dongle**: everything that needs decoding (ADS-B, ACARS, AIS, radiosondes, rtl_433, satellites). See sheets 3 and A01 to A07.

## Where to start

| I want to... | Sheets |
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| Understand the basics and choose the antenna | 1, 2, 3 |
| Know what I am hearing | 5, then the chapter for the service |
| Program my radio's channels (CHIRP) | 4 |
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| Track ships | 13, 14 |
| Track a weather balloon or sensors | 15, 16, 17, 18 |
| Understand Bluetooth and Wi-Fi | 19 |
| Know what to do if I hear a distress call | 21, 22 |
| Decode digital signals or images | 23 to 31 (including 29) |
| Find the origin of a signal | 32, 33 |
| Know whether and how I can transmit | A00a |
| Prepare a crisis communication plan | A00b |
| Build an antenna, a radio or its power supply | A19 to A34 |

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| ACARS | sheet 11 | Firmwares (Fusion, Labs, Apex, Falcon, GoguFW) | sheet A03 |
| ADS-B, squawk 7700 | sheet 10, 21 | ILS | sheet 12 |
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| Bluetooth, BLE | sheet 19 | Images in the spectrum | sheet 30 |
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| Wi-Fi, 2.4 GHz, 5 GHz, 6 GHz | sheet 19 | MMSI, MID | sheet 14 |
| ATS Mini, ATS Mini firmwares | sheet A32 | NAVTEX | sheet 14 |
| Meshtastic | sheet 17 | Propagation | sheet 1 |
| MeshCore | sheet 18 | Direction finding | sheet 33 |
| LoRa | sheet 17, 18, 8 | Radiosondes | sheet 15 |
| UV-K1 keys by firmware | sheet A08, A10, A11, A13 | Settings (mode, bandwidth, squelch) | sheet 2 |
| Labs overlay applications | sheet A09, A14, A18 | RTTY | sheet 24 |
| KISS, TNC | sheet 27 | rtl_433 (433 and 868 MHz sensors) | sheet 16 |
| Antennas: sizes to use | sheet 1 | Satellites (NOAA, Meteor, Iridium, Inmarsat) | sheet 31 |
| DIY radio, survival radio, radio to build | sheet A29, A30, A31, A32 | Selcall | sheet 25 |
| NanoVNA, SWR, antenna measurement | sheet 1 | SigID | sheet 32 |
| Balun, unun, choke | sheet A20, A21, A24 | SSTV | sheet 28 |
| Antennas to build, plans (dipole, J-pole, EFHW) | sheet A19, A21, A22, A23, A24 | Emergencies: frequencies | sheet 21 |
| APRS | sheet 26, 27 | UV Studio, CHIRP | sheet A03 |
| Aeronautical ATIS | sheet 9 | Transmitting, licence, callsign, amateur radio exam | sheet A00a, 6, 8 |
| Inland waterways ATIS | sheet 13 | MAYDAY, PAN-PAN, distress procedure | sheet A00a, 21, 22 |
| 406 MHz beacons (EPIRB, PLB, ELT) | sheet 22 | Communication plan, PACE, watch, emergency network | sheet A00b, 21 |
| Amateur radio bands | sheet 6 | Power, battery, solar, autonomy | sheet A33, A34 |
| Broadcast (AM, FM, shortwave) | sheet 7 | 12 V battery, 18650 cells, BMS, motorcycle terminals, USB | sheet A34 |
| CB, PMR446, LPD433, LoRa | sheet 8 | DIY radio, antennas, power (appendix A2) | sheet A19, A33 |
| RTL-SDR dongle | sheet 3 | Multiboot, firmware slots | sheet A02 |
| Si4732 (HF receiver) | sheet 3, A06, A31, A32 | BK4829, BK1080, UV-K1 chips | sheet A04, A05 |

| | | | |
|---|---|---|---|
| Q code, RST, CB code (10-codes) | sheet 34 | UV-K1 external controls | sheet A01 |
| CW | sheet 5, 32, 34 | CHIRP, .img and .csv file, Python driver, developer mode | sheet 4 |
| DSC | sheet 14 | Programming the radio channels | sheet 4 |
| FT8, FT4, PSK31 | sheet 23 | | |

# Waves, propagation and antennas

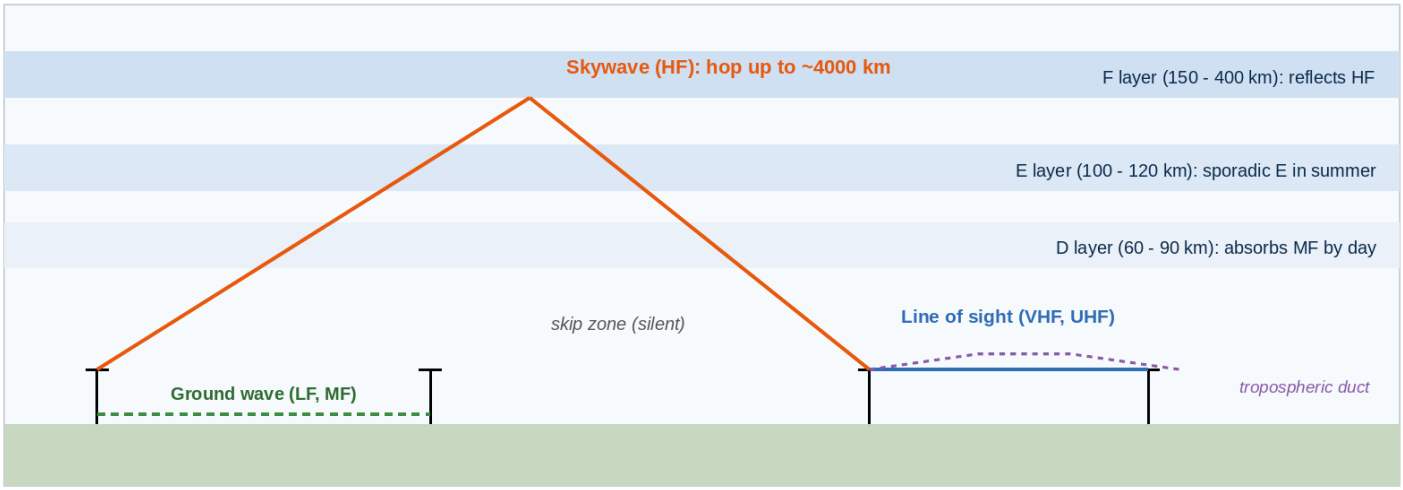
Principles, calculation formulas and sizes to use



## Radio waves in brief

- A radio wave is an electromagnetic wave: an electric field and a magnetic field travelling together at the speed of light, **c = 300,000 km/s** (299,792 km/s in vacuum).
- Frequency **f** (in Hz: oscillations per second) and wavelength **λ** (distance travelled during one oscillation) are linked by **λ = c / f**. The higher the frequency, the shorter the wavelength, so the smaller the antennas.
- **Polarisation**: direction of the electric field, vertical or horizontal depending on the antenna. Two cross-polarised antennas lose about 20 dB. In VHF/UHF FM, polarisation is vertical; in weak-signal SSB/CW, horizontal.
- An antenna transmits and receives in the same way (**reciprocity**): a good transmitting antenna is also a good listening antenna.

## How waves propagate




*Schematic: ground wave, sky wave reflected by the ionosphere, line of sight.*

- **Ground wave** (below ~3 MHz): follows the curvature of the Earth, better over the sea. By day, the D layer absorbs MF: only the ground wave carries (a few hundred km). At night the D layer disappears and the sky wave brings distant stations on MW.
- **Sky wave** (HF, 3 - 30 MHz): the ionosphere (E and F layers, ionised by the Sun) reflects the wave. One hop covers 2,000 to 4,000 km, several hops go around the world. Between the end of the ground wave and the point where the sky wave returns lies the **skip zone**.
- **MUF** (maximum usable frequency): above it, the wave goes through the ionosphere. It rises with solar activity (cycle of about 11 years) and by day. The high bands (15, 12, 10 m) open only when it is high; the low ones (80, 160 m) work mainly at night.
- **Line of sight** (VHF, UHF and above): the wave does not follow the Earth. Range is limited to the **radio horizon**, slightly beyond the optical horizon because of refraction.
- **Special propagation**: sporadic E (50 - 144 MHz, mostly in summer, 1,000 to 2,000 km), tropospheric ducting (temperature inversion: a few hundred km on VHF/UHF), meteor scatter (50 - 144 MHz), moonbounce (EME), satellites. Above 10 GHz, rain strongly attenuates the signal.

## Bands and dominant propagation

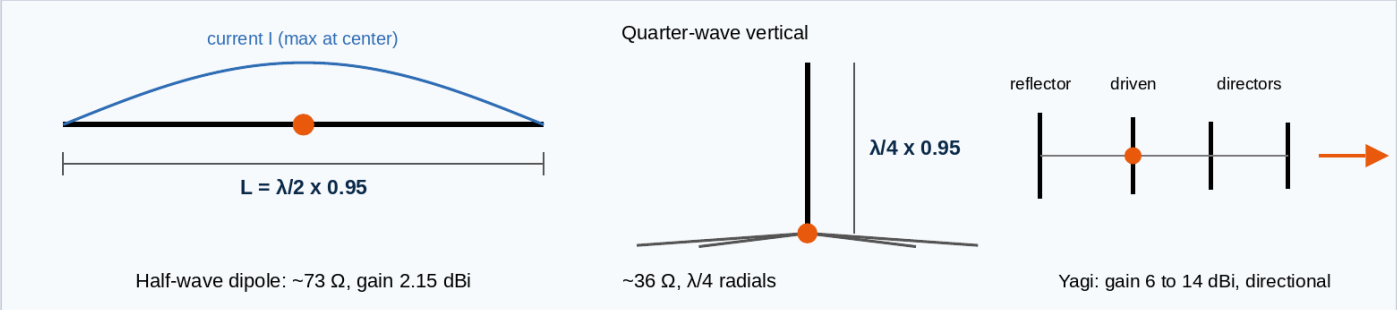
| Band (ITU) | Frequencies | Wavelength | Dominant propagation | Typical range |
|---|---|---|---|---|
| **LF** | 30 - 300 kHz | 10 km - 1 km | Ground wave | Several hundred to thousands of km |
| **MF** | 300 kHz - 3 MHz | 1 km - 100 m | Ground by day, sky at night | ~300 km by day, very far at night |
| **HF** | 3 - 30 MHz | 100 m - 10 m | Sky (ionosphere) | Regional to worldwide, depending on time and Sun |
| **VHF** | 30 - 300 MHz | 10 m - 1 m | Line of sight; Es, tropo | Radio horizon; exceptionally 1,000 km and more |
| **UHF** | 300 MHz - 3 GHz | 1 m - 10 cm | Line of sight; tropo | Radio horizon |
| **SHF** | 3 - 30 GHz | 10 cm - 1 cm | Strict line of sight | Point-to-point links, satellites |

## How an antenna works

# Waves, propagation and antennas

Principles, calculation formulas and sizes to use






*Half-wave dipole, quarter-wave vertical, Yagi.*

- An alternating current in a conductor radiates a wave. The antenna is efficient when it is **resonant**: its electrical length is a multiple of λ/4, the current is then maximum at the feed point and the impedance is purely resistive, so little energy is sent back to the transmitter.
- **Half-wave dipole**: two λ/4 elements, fed at the centre. Impedance of about 73 Ω in free space (close to 50 Ω as an "inverted V").
- **Quarter-wave vertical**: one λ/4 element above a ground plane (or λ/4 radials). The ground "reflects" the other half. Impedance of about 36 Ω. Radiates in all horizontal directions, at a low angle, useful for DX.
- **Gain**: an antenna does not add power, it concentrates the radiation. It is expressed in dBi (relative to an isotropic source) or dBd (relative to a dipole). +3 dB = power × 2; +6 dB = range × 2 in free space.
- **Height**: a horizontal antenna more than λ/2 above ground has a low take-off angle (DX). Lower (λ/8 to λ/4), it sends the energy upward: regional NVIS propagation on 80 and 40 m (0 - 500 km).
- **Balanced and unbalanced**: a dipole is balanced, a coax is unbalanced. A **balun** (1:1 or 4:1) stops current flowing back up the cable (stray radiation, noise). End-fed half-wave antennas (EFHW) use a **49:1 unun**.

## Calculation formulas

| Calculation | Formula | Example |
|---|---|---|
| **Wavelength** | λ (m) = 300 / f (MHz) | 145 MHz: 300 / 145 = 2.07 m |
| **Half-wave dipole** | L (m) = 142.5 / f (MHz) | 7.1 MHz: 142.5 / 7.1 = 20.1 m (2 × 10.0 m) |
| **Quarter-wave vertical** | L (m) = 71.25 / f (MHz) | 145 MHz: 71.25 / 145 = 49 cm |
| **5/8-wave vertical** | L (m) = 187.5 / f (MHz) | 145 MHz: 187.5 / 145 = 1.29 m |
| **Full-wave loop** | L (m) = 306 / f (MHz) | 14.2 MHz: 306 / 14.2 = 21.5 m perimeter |
| **Half-wave cable** | L (m) = 150 × VF / f (MHz) | 145 MHz, VF 0.66: 150 × 0.66 / 145 = 68 cm |
| **Yagi (orders of magnitude)** | Driven element 0.47 λ; reflector 0.49 λ; directors 0.43 to 0.45 λ | Spacing 0.15 to 0.25 λ |
| **Discone** | Cone side length (m) = 71.25 / fmin (MHz) | fmin 100 MHz: 71 cm |
| **Gain in dB** | G (dB) = 10 log (P2 / P1); dBi = dBd + 2.15 | × 10 in power: +10 dB |
| **Reflection coefficient** | Γ = (ZL - Z0) / (ZL + Z0) | 100 Ω on 50 Ω: Γ = 1/3 |
| **SWR** | SWR = (1 + \|Γ\|) / (1 - \|Γ\|) | Γ = 1/3: SWR = 2.0 |
| **Reflected power** | Preflected / P = $\Gamma^2$ | SWR 2: 11 %; SWR 3: 25 % |
| **EIRP / ERP** | EIRP (dBm) = P (dBm) - cable loss (dB) + G (dBi); ERP = EIRP - 2.15 dB | 5 W, 1 dB loss, 2.15 dBi: 5 W × 0.79 × 1.64 = 6.5 W EIRP |
| **Radio horizon** | d (km) = 4.12 × (√h1 + √h2), h in metres | Aircraft at 10,000 m, ground at 10 m: 4.12 × (100 + 3.2) = 425 km |
| **Free-space path loss** | A (dB) = 32.44 + 20 log f (MHz) + 20 log d (km) | 145 MHz, 100 km: 32.4 + 43.2 + 40 = 115.7 dB |
| **Link budget** | Preceived = Ptransmitted + Gt + Gr - A - losses | Every dB of loss costs 1 dB of signal |
| **Dish gain** | G = 10 log (η (π D / λ)$^2$), η ≈ 0.55 | D 0.6 m at 12 GHz: about 35 dBi |

**Velocity factor (VF)**: solid coax 0.66; foam-dielectric coax 0.80; twin lead 0.90 to 0.95. For a wire element in air, the 0.95 "shortening" factor is already included in the 142.5 and 71.25 lengths above.

# Waves, propagation and antennas

Principles, calculation formulas and sizes to use



## Sizes to use (theoretical values)

| Band / use | Frequency (MHz) | λ | λ/2 dipole | λ/4 vertical |
|---|---|---|---|---|
| **160 m** | 1.85 | 162.16 m | 77.03 m | 38.51 m |
| **80 m** | 3.65 | 82.19 m | 39.04 m | 19.52 m |
| **40 m** | 7.1 | 42.25 m | 20.07 m | 10.04 m |
| **30 m** | 10.12 | 29.64 m | 14.08 m | 7.04 m |
| **20 m** | 14.2 | 21.13 m | 10.04 m | 5.02 m |
| **15 m** | 21.2 | 14.15 m | 6.72 m | 3.36 m |
| **CB (channel 19)** | 27.185 | 11.04 m | 5.24 m | 2.62 m |
| **10 m** | 28.5 | 10.53 m | 5.00 m | 2.50 m |
| **6 m** | 50.15 | 5.98 m | 2.84 m | 1.42 m |
| **ILS: localizer** | 110 | 2.73 m | 1.30 m | 64.8 cm |
| **Aviation distress** | 121.5 | 2.47 m | 1.17 m | 58.6 cm |
| **Aviation (band centre)** | 125 | 2.40 m | 1.14 m | 57.0 cm |
| **ACARS** | 131.725 | 2.28 m | 1.08 m | 54.1 cm |
| **APT reception 137 MHz** | 137.5 | 2.18 m | 1.04 m | 51.8 cm |
| **2 m / APRS** | 145 | 2.07 m | 98.3 cm | 49.1 cm |
| **Marine VHF** | 156.8 | 1.91 m | 90.9 cm | 45.4 cm |
| **AIS** | 162 | 1.85 m | 88.0 cm | 44.0 cm |
| **ILS: glide (UHF)** | 332 | 90.4 cm | 42.9 cm | 21.5 cm |
| **Radiosondes** | 403 | 74.4 cm | 35.4 cm | 17.7 cm |
| **70 cm** | 433 | 69.3 cm | 32.9 cm | 16.5 cm |
| **PMR446** | 446 | 67.3 cm | 32.0 cm | 16.0 cm |
| **868 MHz sensors** | 868.3 | 34.6 cm | 16.4 cm | 8.2 cm |
| **ADS-B** | 1090 | 27.5 cm | 13.1 cm | 6.5 cm |
| **23 cm** | 1296 | 23.1 cm | 11.0 cm | 5.5 cm |

**Single reference table.** All the receiving sheets refer here for antenna size: do not recalculate elsewhere. Lengths are for wire elements in air (0.95 factor included); for the ILS, the dipole is horizontal, with two equal elements of half the length given. For 137 MHz APT, use a V dipole: two elements of about 52 cm forming a 120° angle, horizontal. Always measure: the wall, the mast and the height shift the resonance. Cut 3 to 5 % long, then shorten with an analyser.

## Which antenna for what?

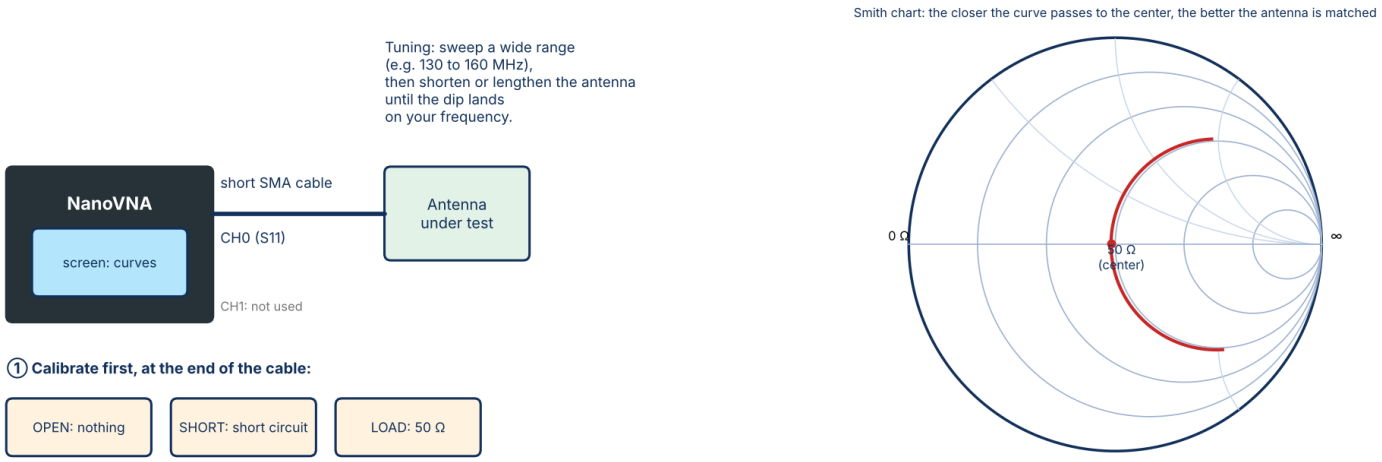
| Antenna | Principle and size | Gain / impedance | Use |
|---|---|---|---|
| **Half-wave dipole** | 2 elements of λ/4, centre feed, horizontal or inverted V | 2.15 dBi, 50 to 73 Ω | Basis of every HF antenna, good for listening on 40 and 80 m |
| **λ/4 vertical** | Vertical element + at least 4 radials of λ/4 | ~0 dBi, 36 Ω | DX, VHF/UHF FM, CB, car antenna |
| **Long wire / random wire** | 10 to 20 m of wire, with a 9:1 unun or a tuner | Variable | Multi-band HF and SW listening, discreet |
| **EFHW (end-fed)** | λ/2 wire on the low band, 49:1 unun, multi-band (harmonics) | ~2 dBi, ~2,500 Ω at the feed point | Portable, 80/40/20/15/10 m |
| **Full-wave loop** | Perimeter of 306 / f (MHz) | ~3 dBi, 100 to 120 Ω | Less sensitive to noise, good for receiving |
| **Yagi** | Driven element + reflector + directors, boom of 0.5 to 3 λ | 3 el.: 7 dBi; 5 el.: 10 dBi; 10 el.: 13 dBi | Directional: DX, satellites, VHF/UHF |
| **Discone / wideband** | Disc + cone, covers ~25 to 1300 MHz | 0 to 2 dBi, vertical | Scanner, aviation, marine, PMR listening |
| **Collinear / 5/8 λ** | Several stacked sections; 5/8 λ = 187.5 / f | 3 to 6 dBi | VHF/UHF repeaters, ADS-B, base stations |
| **Magnetic loop / active** | Small tuned loop, or short probe + amplifier | Low efficiency, narrow band | HF and LF in a small space, listening |
| **Dish** | Reflector + feed at the focus | 30 to 40 dBi | Satellites, microwaves |

## Practical tips

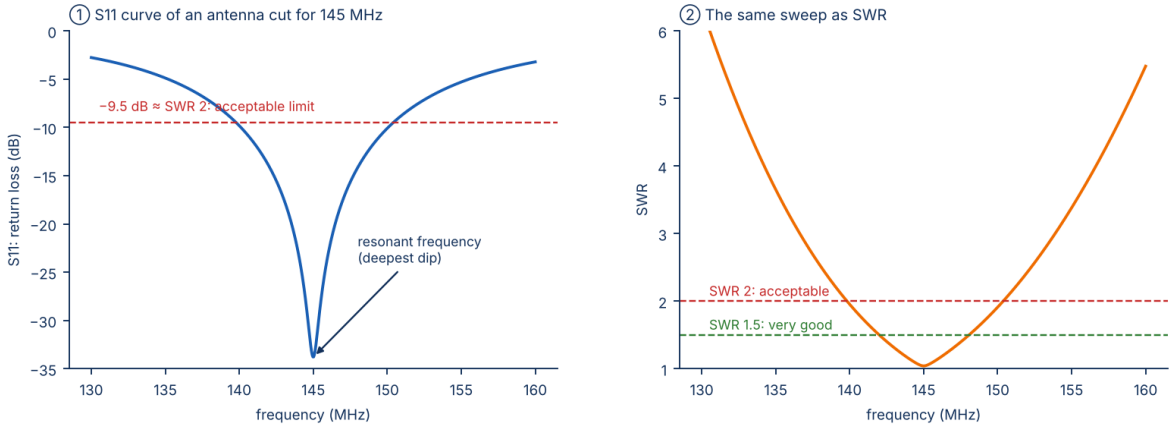
- **Height and clearance** matter more than gain: on VHF/UHF, every extra metre increases the radio horizon. Keep the antenna away from metal walls and cables.
- **Cable**: losses rise with frequency and length (orders of magnitude: RG58 ~15 dB / 100 m at 145 MHz, ~30 dB / 100 m at 435 MHz; RG213 about half). Keep the cable short on UHF.
- **SWR**: below 1.5 is very good, 2 is acceptable. When receiving, a high SWR mostly reveals a fault (contact, length, cable).
- **Safety**: no antenna near power lines; earthing and lightning arrester outdoors.
- **UV-K1**: the original whip is a compromise. A quarter-wave tuned to the band you listen to (table above) on SMA, or an outdoor antenna, clearly improves reception.

## Measuring your antenna: the NanoVNA

**In brief.** The **NanoVNA** is a small, cheap, portable vector network analyser. Connected to the antenna, it plots in a few seconds its **matching curve** over a frequency range: you see where it resonates, what its SWR and impedance are, without transmitting any power. It is the tool of the antenna sheets (sheets A19 to A28) and of the baluns (sheet A20): cut, measure, correct.




*Measurement setup (left) and Smith chart (right): the closer the curve passes to the centre (50 Ω), the better matched the antenna.*




*Illustrative curves, not a real measurement: your readings depend on your antenna and its surroundings.*

- **What it measures**: on channel CH0, the reflection coefficient S11 (return loss in dB, SWR, impedance R + jX, Smith chart); between CH0 and CH1, the transmission S21 (filters, attenuators, cables). The range depends on the model (from a few tens of kHz to 900 MHz on common models, more on the more expensive versions).
- **Calibration**: before each series of measurements, calibrate over the range used with OPEN (nothing), SHORT (short circuit) and LOAD (50 Ω), **at the end of the cable** you use. Change range or cable: recalibrate.
- **Tuning an antenna**: sweep wide, find the S11 dip (lowest SWR), then shorten the elements to go up in frequency or lengthen them to go down. A millimetre matters on VHF and UHF.
- **Reading the result**: an SWR below 2 (S11 below −9.5 dB) is acceptable, below 1.5 very good. A good SWR says nothing about gain or radiation: a 50 Ω dummy load gives a perfect SWR.

- **Precautions**: keep the antenna away from objects and from your body during the measurement; add a choke (sheet A20) if the curve changes when you touch the cable; never connect the NanoVNA to an antenna linked to a transmitter that is on.
- **Software**: NanoVNA-Saver on a computer lets you save the curves, overlay them and export the files.

# Listening settings

Modes, bandwidth, squelch, gain and audio connection



## The settings that make the difference

| Setting | Usual guidelines |
|---|---|
| **Modulation** | **FM**: marine, VHF/UHF amateur radio, PMR, radiosondes (narrow FM). **AM**: civil and military aviation, broadcast, CB. **LSB / USB**: HF amateur radio and utility stations (LSB below 10 MHz, USB above), all digital modes in USB. **CW**: Morse. |
| **Bandwidth** | SSB: 2.4 to 3 kHz. Aviation AM: 8 to 10 kHz. Narrow FM: 12.5 kHz; wide FM (FM radio): about 150 kHz. CW: 300 to 500 Hz. Narrower = less noise, but the signal may be cut. |
| **Frequency step** | 12.5 or 25 kHz on VHF/UHF; 8.33 kHz on some aviation channels (set to the nearest step, 6.25 kHz); 9 kHz (Europe) or 10 kHz for MW; 5 kHz on shortwave. |
| **Squelch** | Cuts the noise when there is no signal. Open to listen to a weak signal or a digital burst; closed for standby monitoring. CTCSS / DCS sub-tones are muted by the squelch: turn it off to hear everything. |
| **Gain and attenuation** | Too much gain overloads (false lines, ghost signals); too little hides weak signals. Attenuate near a strong transmitter. A band-pass or notch filter helps in crowded areas (FM, DAB). |
| **Drift correction (ppm)** | For an SDR without a TCXO, set it on a known signal. Essential for narrow modes (radiosondes, FT8). |

## Which mode for what?

| I want to listen to... | Mode and setting | Sheet |
|---|---|---|
| **Aviation (towers, approach)** | AM, 118 - 137 MHz, 25 kHz step, low squelch | 9 |
| **Marine VHF** | FM, 156 - 162 MHz, channel 16 on standby | 21 |
| **HF amateur radio** | USB / LSB, 2.4 kHz bandwidth | 6 |
| **DATA modes (FT8, RTTY, SSTV...)** | USB, audio sent to a decoder on a PC or smartphone | 23 |
| **Live aircraft, boats, sondes, sensors** | SDR with the matching decoder | 10, 13, 15, 16 |

## From receiver to decoder

- **Audio cable**: headphone or speaker output to the computer's line or microphone input, at a medium volume. A signal that is too loud distorts and makes decoding fail.
- **Through the microphone**: with no cable, the smartphone microphone placed near the speaker is enough for slow modes, in a quiet room.
- **Local noise**: switching power supplies, chargers, screens and LEDs radiate noise. Move the antenna and receiver away, add ferrites on the cables.
- **Antenna**: the biggest improvement almost always comes from the antenna, outside and in the clear (sheet 1).
- **Hardware**: choice of receiver and SDR dongle: sheet 3; the UV-K1, its firmwares and its chips: sheets A01 to A07.

## Quick troubleshooting

| Symptom | Leads |
|---|---|
| **No sound** | Squelch too high, wrong modulation, frequency off by one step, antenna disconnected |
| **Distorted or overloaded sound** | Gain or volume too high, signal too strong: attenuate, move the antenna away from the transmitter |
| **Muffled or distorted voice in SSB** | Wrong sideband (USB / LSB) or frequency offset: fine-tune |
| **Constant noise** | Nearby switching supply, charger, screen: move away, add ferrites, try on battery |
| **Nothing decodes** | Audio level, sideband, dial frequency, clock (for FT8), SDR drift correction. These causes are more frequent than a decoder fault. |
| **Ghost signals** | Receiver overload: reduce gain, add a filter, change location |

# Hardware and firmware

Choosing your receiver: UV-K1, Si4732 module, ATS Mini, SDR dongle



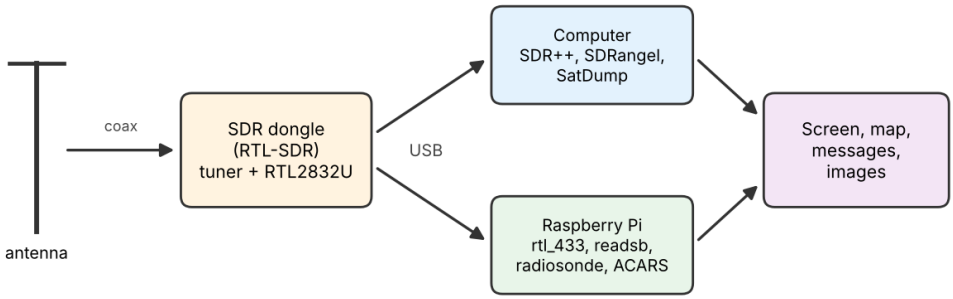
**In brief.** The guide relies on two tools. The **UV-K1** and the **UV-K5 V3** (PY32F071 microcontroller, BK4829 radio chip) listen to VHF/UHF, and to HF with the SI4732 when the hardware and firmware support it. **The UV-K5 V1 and V2 models are not compatible** with the firmwares of this guide. An open source firmware adds applications, a spectrum analyser and tuning tools. The **SDR dongle** takes over for everything that gets decoded: ADS-B, ACARS, AIS, radiosondes, sensors. The following sheets do not repeat these basics: they refer back here.

## Choosing your receiver

| Receiver | What it listens to | Limits |
|---|---|---|
| **UV-K1 (VHF/UHF)** | FM and AM over the range allowed by the firmware: aviation, marine, amateur radio, PMR, beacons | No built-in digital decoding (apart from Labs applications); sensitive to strong signals |
| **UV-K1 modified with SI4732 module (HF)** | 0.15 to 30 MHz in AM, and in LSB/USB with the SSB patch: broadcast, utility stations, amateurs | Modest sensitivity and filters; audio output to send to an external decoder |
| **ATS Mini V4 (Si4732 + ESP32-S3)** **sheet A32** | Ready-to-use pocket receiver: FM 64 to 108 MHz, long, medium and short waves up to 30 MHz, in AM, LSB and USB, with RDS; 1.9-inch colour screen and 800 mAh battery | No VHF/UHF aviation or marine; no transmitting; open source firmware to choose (sheet A32) |
| **RTL-SDR dongle** | About 24 to 1700 MHz: ADS-B (1090 MHz), AIS, radiosondes, 433 / 868 MHz sensors, ACARS, APRS, fax and weather satellites. HF needs "direct sampling" mode or a converter | Instantaneous bandwidth of about 2 MHz; drift correction (ppm) matters, especially around 400 MHz |
| **Wideband SDR (Airspy, SDRplay, HackRF...)** | Wider coverage and larger instantaneous bandwidth: the whole radiosonde band or several channels at once | More expensive, more demanding on the computer |

## The SDR dongle: a receiver driven by a computer

An **SDR dongle** (software-defined radio) is a small USB receiver with no screen or speaker: it digitises the spectrum, and software on a computer, smartphone or Raspberry Pi does the rest (display, demodulation, decoding). The most common dongle is the **RTL-SDR** (RTL2832U chip with an R820T2-type tuner), sold for a few tens of euros. Figures from memory, to verify depending on the model: coverage of about 24 to 1700 MHz, useful instantaneous bandwidth of about 2 MHz (so the software shows one slice of the spectrum at a time), 8-bit sampling.




*Receive chain with an SDR dongle: the antenna matters as much as the dongle; the same software can run on a computer or a Raspberry Pi.*

- **Computer**: SDR++ or SDRangel to listen and see the spectrum, SatDump for weather satellites, plus the decoders of sheets 10 (ADS-B), 10 (ACARS), 12 (AIS), 14 (radiosondes), 15 (rtl_433).
- **Raspberry Pi**: a station that runs on its own, for example an ADS-B receiver or rtl_433 connected to the network; uses a few watts (sheet A33).
- **Antenna**: a dongle supplied with a small telescopic antenna receives poorly; an antenna suited to the band (sheets 1 and A19) changes the result far more than the dongle.
- **HF**: the dongle does not naturally go below 24 MHz; you need "direct sampling" mode (depending on the model) or a converter, with reduced sensitivity.
- **Precautions**: a USB extension cable moves the dongle away from computer noise; an FM broadcast filter avoids overload near a transmitter; the dongle gets hot, let it breathe.

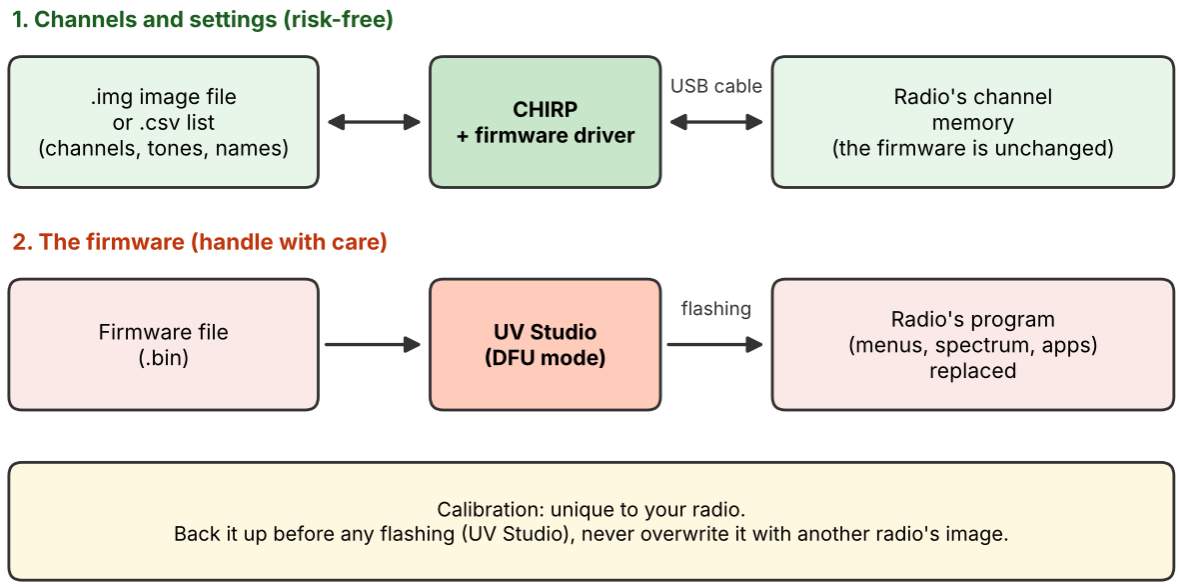
# CHIRP: programming the radio's channels

Channel file, firmware, flashing: do not mix them up; developer mode and Python drivers

1 · Start

**In brief. CHIRP** is free, open source software that writes the **channels** (frequencies, names, tones) and **settings** into the radio. It does **not** replace the firmware: changing the radio's program is a **flash**, done with another tool (UV Studio, sheet A03). A channel file is a list; a firmware is the program that reads it.

## Three things not to mix up




*Two separate paths: channels go through CHIRP, the radio's program through UV Studio. Calibration stays specific to each radio.*

| | Channel file | Firmware | Calibration |
|---|---|---|---|
| **What it is** | The list of programmed stations: frequency, name, mode, tones, power, settings | The microcontroller's program: menus, spectrum, applications, functions | Factory settings of the radio section, specific to each unit |
| **File** | **.img** (radio image) or **.csv** (readable list) | **.bin** (depending on the project) | Backup made with UV Studio |
| **Tool** | CHIRP (this sheet) | UV Studio, DFU mode (sheet A03) | UV Studio |
| **Risk** | Low: you rewrite the channels and can restore everything | Real: a wrong file can make the radio unusable | Another radio's calibration degrades reception and transmission |
| **When** | When you change stations or settings | To change firmware or version | Before any flashing |

**Simple rule.** After flashing, channels from another firmware may be incompatible: **write them with the CHIRP driver of the installed firmware**, not with another one's. A .img image only makes sense with the driver that produced it.

## Programming the channels step by step

- **1. Read the radio first**: Radio → Download from radio (model and port), then **save the image** (.img) before any change. This is your backup.
- **2. Edit**: one row per channel (number, name, frequency, offset, tones, mode, power). Copy and paste from a spreadsheet or import a .csv.
- **3. Write**: Radio → Upload to radio. Keep the radio on and the cable in place until the end.
- **4. Check** on the radio; if the behaviour is odd, restore the saved image.
- **Cable and port**: the UV-K1 connects by USB-C (unless the firmware says otherwise); the UV-K5 V3 uses the programming-plug cable. On Windows, check the COM port; on Linux, your user must belong to the "dialout" group (depending on the distribution).
- **"Wrong model" or "clone" error**: almost always a driver that does not match the installed firmware (see the next section).

# CHIRP: programming the radio's channels

Channel file, firmware, flashing: do not mix them up; developer mode and Python drivers



## A .csv file: a list to share

A channel .csv opens in a spreadsheet. Usual CHIRP columns (from memory):

| Column | Meaning | Example |
|---|---|---|
| **Location** | Channel number in the radio | 1 |
| **Name** | Displayed name (limited length) | ISS |
| **Frequency** | Frequency in MHz | 145.800000 |
| **Duplex, Offset** | Direction and value of the offset (repeater) | -, 0.600000 |
| **Tone, rToneFreq, cToneFreq, DtcsCode** | Sub-tone mode and CTCSS frequency or DCS code | Tone, 88.5 |
| **Mode, Tstep, Skip, Power** | Modulation (FM, AM, NFM), step, skip when scanning, power | FM, 12.5, , Low |

*A .csv file can be shared and read on any computer; a .img opens only with CHIRP and the right driver. Listening is free, transmitting is not: never write into the radio a frequency where you are not allowed to transmit (sheet A00a).*

## Drivers: why a firmware has its own

CHIRP knows the memory layout of each radio through a **driver** (a Python file). A modified firmware, such as Fusion or GOGUFW, **adds or moves settings** in memory: the standard driver does not see them or reads them wrongly. Projects therefore ship **their own driver** with each release (sheets A03, A08 and A12): take the one from the archive of the firmware you flashed, not an old one.

## Developer mode and loading a Python driver

> **Read this first.** A loaded driver is a **Python program run on your computer**. Only load files from the firmware's official repository (sheet A03). An unknown file can do anything on your machine.

- **1. CHIRP version**: take a recent version. Drivers for recent firmwares load in the current CHIRP ("chirp-next", written in Python 3). Check on the firmware's page which one is expected.
- **2. Get the driver**: a .py file in the firmware release archive (for example the F4HWN one, the GOGUFW one, or the "chirp" folder of the Falcon archive), to save on your computer.
- **3. Enable developer mode**: Help menu, "Developer Mode" (from memory); a new menu appears after you confirm, and a CHIRP restart may be requested.
- **4. Load the module**: File → "Load Module", choose the .py file and confirm the security warning.
- **5. Choose the radio**: in the model list, a new model matching the driver appears; select it, then download from the radio. Loading must be repeated at each start (to verify depending on the version).

## What developer mode adds

- **Load a module** (the .py driver): the most useful one for our radios.
- **Analysis tools** for raw memory (hexadecimal view, comparison of two images): handy to understand what a firmware changes, useless for everyday use.
- **Log and more detailed diagnostic messages** when a read error occurs.

## When it does not work

| Symptom | Likely cause | What to do |
|---|---|---|
| **Radio not detected** | Wrong cable or port, radio off | Another cable, another port, radio on; check the COM port or the dialout group |
| **Read error or "clone" error** | Driver does not match the firmware | Load the driver shipped with the flashed version |
| **Odd names or settings** | Image written with another driver | Restore the saved image with the right driver |
| **Your firmware's model is gone** | Module not loaded | Developer mode, then Load Module, to repeat at start-up |
| **Radio no longer starts after flashing** | Interrupted flash or wrong file | Reflash with UV Studio in DFU mode (sheet A03) |

# CHIRP: programming the radio's channels

Channel file, firmware, flashing: do not mix them up; developer mode and Python drivers

1 · Start

## Personal notes (CHIRP version, drivers used, image backups)

# Recognising a signal by ear

From the sound heard to the service and the sheet to consult

1 · Start

## A four-question method

- **On which frequency?** The band already gives the service: 118 - 137 MHz aviation, 156 - 162 MHz marine, 144 MHz amateur radio...
- **In which modulation does the signal become clear?** Clear voice in AM or FM: speech. Whistles or crackles: data.
- **How many tones, and what rhythm?** One chopped tone (CW), two alternating tones (RTTY, NAVTEX, DSC), several tones in a row (Selcall, FT8).
- **When and at what rate?** A single burst after a voice, repetition every second, 15 s cycle, continuous loop...

## What I hear on VHF, UHF and above

| What I hear | What is it? | Where? | Sheet |
|---|---|---|---|
| **Voice in a loop: "Information Alpha, wind..."** | Aeronautical ATIS | VHF 118 - 137 MHz, AM | 9 |
| **Short burst of "bzzz" every few minutes** | ACARS, aircraft-ground text messages | VHF about 131.5 to 131.9 MHz, AM | 11 |
| **A few Morse letters, fixed tone, in a loop** | Identification of an ILS or navigation beacon | VHF 108 - 112 MHz (ILS), AM | 12 |
| **Nothing audible but pulses on the SDR** | ADS-B, aircraft positions | 1090 MHz, SDR | 10 |
| **Data hiss, two channels** | AIS, ship positions | 161.975 and 162.025 MHz, FM | 13 |
| **Short digital burst at the end of a sentence** | Inland waterways ATIS | Marine working channel, FM | 13 |
| **Marine voice then a fast, high "beep-beep"** | DSC, digital selective calling | 156.525 MHz (channel 70), 2187.5 kHz | 14 |
| **Two fast tones, continuous** | NAVTEX, notices to mariners | 518, 490, 4209.5 kHz | 14 |
| **Continuous data, narrow signal, no silence** | Radiosonde | 400 - 406 MHz, narrow FM | 15 |
| **Short burst, no voice, once or twice a minute** | Wireless sensor (weather, probe) | 433.92 and 868 MHz, SDR | 16 |
| **Tone sweeping downward, repeated** | Homing signal of a distress beacon | 121.5 and 243 MHz, AM | 22 |
| **Whistle in packets of 1 to 2 seconds** | APRS, amateur radio positions | 144.800 MHz, FM | 26 |

## On HF and in USB

| What I hear | What is it? | Where? | Sheet |
|---|---|---|---|
| **Fixed tone chopped into dots and dashes** | CW, Morse | Amateur bands, bottom of the band | 34 |
| **Two tones alternating without pause, regular warble** | RTTY | HF, amateur and utility bands | 24 |
| **Sequence of tones in 15 s cycles** | FT8 | Reference frequencies: sheet 6 | 23 |
| **Burst of 4 to 6 short notes in half a second** | CCIR Selcall | VHF / HF, network channels | 25 |
| **Screech that changes pitch, over a minute** | SSTV, slow-scan images | 14.230 MHz, 28.680 MHz... | 28 |
| **Machine noise rhythmed by line, low and high tones** | Weather fax | HF, weather stations | 29 |

**Reference frequencies and emergency frequencies** are grouped only once: distress and emergencies in sheet 21, amateur radio bands in sheet 6. To classify a tone signal automatically, use the SigID application (sheet 32); to find the direction of a transmitter, Sig Finder (sheet 33). For an unknown signal, record a few seconds of audio and look at them in a spectrogram: the shape drawn helps you decide.

*This table gives clues, not certainties: several services use similar sounds, and frequencies vary by country.*

# Amateur radio bands

Allocations (IARU region 1 / France), sub-bands, modes and reference frequencies



| Band | Range | Usual segments | Reference frequencies | Mode / remark |
|---|---|---|---|---|
| **LF, MF and HF** | | | | |
| **2200 m** | 135.7 - 137.8 kHz | CW and slow modes (WSPR, QRSS) | 136.000 kHz (WSPR) | Secondary, limited power |
| **630 m** | 472 - 479 kHz | CW and slow modes | 475.5 kHz (WSPR) | Secondary, limited power |
| **160 m** | 1810 - 2000 kHz | CW 1810-1838 · digital 1838-1843 · voice 1843-2000 | — | LSB. Night band, atmospheric noise |
| **80 m** | 3500 - 3800 kHz | CW 3500-3570 · digital 3570-3600 · voice 3600-3800 | 3580 (RTTY) · 3730 (SSTV) | LSB. Night and regional |
| **60 m** | 5351.5 - 5366.5 kHz | CW, digital and voice within the narrow range | — | USB. Secondary, limited power |
| **40 m** | 7000 - 7200 kHz | CW 7000-7040 · digital 7040-7060 · voice 7060-7200 | 7110 (emergency) · 7171 (SSTV) | LSB. Day and night |
| **30 m** | 10 100 - 10 150 kHz | CW 10 100-10 130 · digital 10 130-10 150 | — | No voice. Reduced power |
| **20 m** | 14 000 - 14 350 kHz | CW 14 000-14 070 · digital 14 070-14 099 · IBP beacons 14 100 · voice 14 101-14 350 | 14 080 (RTTY) · 14 230 (SSTV) · 14 300 (emergency) | USB. The most active for DX |
| **17 m** | 18 068 - 18 168 kHz | CW 18 068-18 095 · digital 18 095-18 109 · voice 18 111-18 168 | — | USB. WARC band |
| **15 m** | 21 000 - 21 450 kHz | CW 21 000-21 070 · digital 21 070-21 149 · voice 21 151-21 450 | 21 340 (SSTV) | USB. Depends on the solar cycle |
| **12 m** | 24 890 - 24 990 kHz | CW 24 890-24 915 · digital 24 915-24 929 · voice 24 931-24 990 | — | USB. WARC band |
| **10 m** | 28.000 - 29.700 MHz | CW 28.000-28.070 · digital 28.070-28.190 · beacons 28.190-28.225 · voice 28.225-29.200 · FM 29.100-29.300 · repeaters 29.520-29.700 | 28.680 (SSTV) · 29.600 (FM calling) | USB; FM above 29.1 MHz |
| **VHF, UHF and microwaves** | | | | |
| **6 m** | 50 - 52 MHz (France: 50.2 - 51.2 MHz, to verify) | CW 50.0-50.1 · SSB 50.1-50.5 · beacons 50.4-50.5 · digital 50.3-50.4 | 50.150 (SSB calling) · 51.510 (FM calling) | Opened by sporadic propagation in summer |
| **2 m** | 144 - 146 MHz | CW 144.000-144.150 · SSB 144.150-144.400 · beacons 144.400-144.491 · FM 145.200-145.5875 · repeaters 145.600-145.7875 · satellites 145.800-146.000 | 144.300 (SSB calling) · 145.500 (FM calling) · 144.800 (APRS) | FM, offset -600 kHz. ISS: 145.800 (voice) and 145.825 (APRS) |
| **70 cm** | 430 - 440 MHz | SSB/CW 432.000-432.100 · beacons 432.400-432.500 · FM 433.400-433.575 · repeaters and digital 430-431.9 and 438-440 · satellites 435-438 | 432.200 (SSB calling) · 433.500 (FM calling) · 433.775 (LoRa APRS) | FM, offset +/-1.6 MHz. Also DMR, D-STAR, Fusion |
| **23 cm** | 1240 - 1300 MHz | Repeaters, ATV, satellites, EME | 1296.200 MHz (SSB calling) | FM, digital, amateur TV |
| **13 cm** | 2300 - 2450 MHz | Links, ATV, EME | 2320.200 MHz (SSB calling) | Shared with 2.4 GHz Wi-Fi |
| **5 cm** | 5650 - 5850 MHz | Links, ATV, satellites | 5760.200 MHz (SSB calling) | Shared with 5 GHz Wi-Fi |
| **3 cm** | 10 - 10.5 GHz | Links, EME, beacons | 10 368.200 MHz (SSB calling) | The most popular microwave band |
| **24 GHz and above** | 24 - 24.25 · 47 - 47.2 · 76 - 81 · 122.25 - 123 GHz | Experimental links | 24 048.200 MHz (SSB calling) | Specialised equipment |

## Reference frequencies for digital modes (kHz, USB unless stated)

| Mode | 160 m | 80 m | 40 m | 30 m | 20 m | 17 m | 15 m | 12 m | 10 m | 6 m |
|---|---|---|---|---|---|---|---|---|---|---|
| **FT8** | 1840 | 3573 | 7074 | 10 136 | 14 074 | 18 100 | 21 074 | 24 915 | 28 074 | 50 313 |
| **FT4** | 1842 | 3575 | 7047.5 | 10 140 | 14 080 | 18 104 | 21 140 | 24 919 | 28 180 | 50 318 |
| **WSPR** | 1836.6 | 3568.6 | 7038.6 | 10 138.7 | 14 095.6 | 18 104.6 | 21 094.6 | 24 924.6 | 28 124.6 | 50 293.0 |
| **PSK31** | 1838 | 3580 | 7040 | 10 142 | 14 070 | 18 100 | 21 070 | 24 920 | 28 120 | 50 305 |

*Frequencies are USB dial settings, usual reference points (not mandatory channels). The PSK31 and FT4 rows are the least certain. WSPR: dial frequency, the signal is offset by 1400 to 1600 Hz. In FT8, most stations occupy 200 to 3000 Hz above the dial. For 144.174 MHz (FT8 on 2 m), the same logic applies.*

# Amateur radio bands

Allocations (IARU region 1 / France), sub-bands, modes and reference frequencies



## Usage, calls and conventions

- **Sideband**: LSB below 10 MHz (160, 80, 40 m), USB from 10 MHz up. Exception: 60 m is USB.
- **Calls**: "CQ" is a general call; "QRZ?" means who is calling me; emergency nets on 3760 and 7110 kHz (region 1) and 14 300 kHz (worldwide).
- **Callsigns, RST reports, CW abbreviations and Q code**: sheet 34. Digital modes: sheet 23.
- **Power**: depends on the licence class and the band, with lower limits on secondary bands (2200 m, 630 m, 60 m). Check the ARCEP decision.
- **Transmitting**: it requires a callsign (ANFR exam). For listening, the UV-K1 covers VHF/UHF in FM and HF with its SI4732 (sheet A06).

# Broadcast bands

Radio and television: frequencies, modulation and listening (region 1 / Europe)



## AM radio (HF and below)

| Band | Range | Modulation / format | Step | Remarks |
|---|---|---|---|---|
| **Long wave (LW)** | 148.5 - 283.5 kHz (Europe: 153 - 279 kHz) | AM, DRM | 9 kHz | Range of several hundred km. Many transmitters have been shut down. |
| **Medium wave (MW)** | 526.5 - 1606.5 kHz (region 1: 531 - 1602 kHz) | AM, DRM, HD Radio (Americas) | 9 kHz (10 kHz in the Americas) | Ground wave by day, sky wave at night. |
| **Short wave (SW)** | 2.3 - 26.1 MHz (14 bands below) | AM, DRM, SSB for utility stations | 5 kHz | International broadcasting, depends on propagation. |

## Short wave: the broadcast bands (kHz)

| Band | Range | Format | Best time | Remarks |
|---|---|---|---|---|
| **120 m** | 2300 - 2495 | AM | Tropical | Mainly tropical regions |
| **90 m** | 3200 - 3400 | AM | Tropical | Mainly tropical regions |
| **75 m** | 3900 - 4000 | AM | Night | Europe, night and regional |
| **60 m** | 4750 - 5060 | AM | Tropical / night | Regional stations |
| **49 m** | 5900 - 6200 | AM, DRM | Night | The most active in the evening |
| **41 m** | 7200 - 7450 | AM, DRM | Day and night | Shared with the 40 m amateur band (7000-7200) |
| **31 m** | 9400 - 9900 | AM, DRM | Day and night | Most used band |
| **25 m** | 11 600 - 12 100 | AM, DRM | Day | Good range in daytime |
| **22 m** | 13 570 - 13 870 | AM, DRM | Day | |
| **19 m** | 15 100 - 15 800 | AM, DRM | Day | Depends on the solar cycle |
| **16 m** | 17 480 - 17 900 | AM, DRM | Day | Depends on the solar cycle |
| **15 m** | 18 900 - 19 020 | AM, DRM | Day | Narrow band |
| **13 m** | 21 450 - 21 850 | AM, DRM | Day | Depends on the solar cycle |
| **11 m** | 25 670 - 26 100 | AM, DRM | Day | Little used |

## FM radio, digital and TV

| Band | Range | Modulation / format | Step / channel | Remarks |
|---|---|---|---|---|
| **FM (band II)** | 87.5 - 108 MHz | Wide FM (WFM), +/-75 kHz | 200 kHz (100 kHz in France) | Stereo: 19 kHz pilot; RDS: 57 kHz subcarrier; 50 µs pre-emphasis. |
| **VOR / ILS (aviation)** | 108 - 118 MHz | AM (VOR: 30 Hz) | 50 kHz | Radionavigation, not public radio (sheet 12). |
| **DAB / DAB+ (band III)** | 174 - 240 MHz (blocks 5A to 13F) | OFDM, COFDM | 1.536 MHz per block | 5A: 174.928 MHz; 13F: 239.200 MHz. |
| **DTT (UHF)** | 470 - 694 MHz (France) | DVB-T / DVB-T2 depending on the multiplex, MPEG-4 (H.264) | 8 MHz per channel | 700 MHz band freed in 2019 for mobile telephony. |
| **Legacy VHF TV** | 47 - 68 · 174 - 230 MHz | Analogue (switched off in France in 2011) | 7 / 8 MHz | The bands have now been reallocated. |
| **Satellite TV (Ku)** | 10.7 - 12.75 GHz | DVB-S2, QPSK / 8PSK | 27.5 to 36 MHz per transponder | Local oscillator 9.75 / 10.6 GHz; Astra 19.2 E, Hotbird 13 E. |
| **Satellite TV (C)** | 3.4 - 4.2 GHz | DVB-S, analogue FM | 36 MHz | Larger dishes, better rain resistance. |

## Time signals (reference)

| Station | Frequency | Modulation / format | Remarks |
|---|---|---|---|
| **WWV / WWVH** | 2.5 · 5 · 10 · 15 · 20 MHz | AM, voice announcements | UTC time, seconds and service tones. |
| **DCF77 (Germany)** | 77.5 kHz | AM, BCD time code | Used by radio-controlled clocks in Europe. |
| **MSF (United Kingdom)** | 60 kHz | AM, time code | Also WWVB (United States) on 60 kHz. |

# Broadcast bands

Radio and television: frequencies, modulation and listening (region 1 / Europe)



## Listening with the UV-K1

- **FM (87.5 - 108 MHz)**: received by the SI4732 as well as by the VHF section, in wide FM. RDS (station name, title) is only readable if the firmware decodes it.
- **LW, MW, SW**: the SI4732 covers 0.15 - 30 MHz in AM (bandwidth 2 to 6 kHz) and in SSB with a patch. SSB is for utility and amateur stations, AM for broadcasting.
- **Tuning step**: on MW, channels are spaced 9 kHz apart in Europe; on SW, 5 kHz. Set the firmware's frequency step accordingly.
- **DRM, DAB and DTT** are digital: the UV-K1 only gets noise from them, you need an SDR or a dedicated receiver.
- **Tips**: listen to MW at night and SW according to the time (49 m in the evening, 25 and 19 m by day). Broadcasters' schedules and frequencies change with the seasons (A and B): check the HFCC schedules.

# Licence-free radio: PMR446, CB, LPD433, LoRa

Licence-free radio in France: frequencies and listening



## Comparison

| Service | Frequencies | Mode | Power | Range |
|---|---|---|---|---|
| **PMR446** | 446.00625 - 446.19375 MHz (16 channels) | Narrow FM, or digital (dPMR / DMR Tier 1) | 0.5 W | 1 to 3 km |
| **CB 27 MHz** | 26.965 - 27.405 MHz (40 channels) | AM, FM, SSB | 4 W (AM/FM), 12 W (SSB) | A few km, more with propagation |
| **LPD433** | 433.075 - 434.775 MHz (69 channels) | FM | 10 mW | Under 1 km |
| **LoRa** | 863 - 870 MHz (and 433 MHz) | Spread spectrum (chirp) | ~25 mW, 1% duty cycle | Several km to tens of km |

PMR446, CB, LPD433 and LoRa are licence-free in France: no licence or fee, with equipment that complies with regulations (approved devices).

## PMR446: the 16 channels (narrow FM, 12.5 kHz step)

| Channel 1 | Channel 2 | Channel 3 | Channel 4 | Channel 5 | Channel 6 | Channel 7 | Channel 8 |
|---|---|---|---|---|---|---|---|
| 446.00625 | 446.01875 | 446.03125 | 446.04375 | 446.05625 | 446.06875 | 446.08125 | 446.09375 |
| Channel 9 | Channel 10 | Channel 11 | Channel 12 | Channel 13 | Channel 14 | Channel 15 | Channel 16 |
| 446.10625 | 446.11875 | 446.13125 | 446.14375 | 446.15625 | 446.16875 | 446.18125 | 446.19375 |

- Users add CTCSS or DCS codes (sub-tones): the squelch hides them, so turn them off to hear everything (sheet 2).
- Digital (dPMR, DMR) is not decoded by the UV-K1: you hear a "buzz" or nothing.

## CB 27 MHz: reference channels

| Channel | Frequency | Usual use |
|---|---|---|
| **1** | 26.965 MHz | Start of band |
| **9** | 27.065 MHz | Emergency channel by convention (sheet 21). Watch not guaranteed. |
| **19** | 27.185 MHz | Truckers and traffic info |
| **40** | 27.405 MHz | End of band |

SSB (USB/LSB) is used on some upper channels of the band, with long-distance contacts when propagation opens (summer, daytime).

## LPD433 and the 433 MHz ISM band

- 69 channels of 25 kHz from 433.075 to 434.775 MHz, FM, 10 mW. Channel 1: 433.075 MHz.
- The 433 MHz band also hosts remote controls, weather stations, doorbells and sensors: 433.920 MHz is the most common frequency. They transmit in OOK or FSK, in short bursts (sheet 16).
- The band overlaps the 70 cm amateur band (secondary use).

## LoRa: usual frequencies

| Use | Frequency | Typical parameters |
|---|---|---|
| **LoRaWAN EU868** | 868.1 / 868.3 / 868.5 MHz | BW 125 kHz, SF7 to SF12 |
| **Meshtastic (Europe 868)** | 869.525 MHz | LongFast: BW 250 kHz, SF11 |
| **LoRaWAN EU433** | 433.175 / 433.375 / 433.575 MHz | BW 125 kHz |
| **LoRa APRS** | 433.775 MHz | BW 125 kHz, SF12 (amateur callsign required to transmit) |

- LoRa modulates the frequency in a fast sweep ("chirp"): SF7 is fast and short range, SF12 is slow and long range.
- **Detection only**: at best the UV-K1 hears a brief hiss. To decode, you need an SDR or a LoRa module (SX127x) set to the same settings (SF, BW).

## Listening with the UV-K1

- **PMR446 and LPD433**: narrow FM, 12.5 kHz step (PMR) or 25 kHz (LPD). Store the channels in memory and scan.
- **CB**: 26.965 - 27.405 MHz in AM or SSB with the HF receiver (SI4732). CB FM is only received if the firmware goes low enough in VHF.
- PMR446 and CB require approved devices to transmit.

# Aviation and aeronautical weather

Listening to the air: bands, traffic, ATIS, VOLMET, METAR, military, balloons and sondes



## Bands to know

| Range | Mode | Use |
|---|---|---|
| **118.000 - 136.975 MHz** | AM | Civil aviation: towers, approaches, centres, ATIS |
| **138 - 144 MHz** | AM / FM | Military in several countries (depends on the state) |
| **225.000 - 399.975 MHz** | AM | Military aviation (NATO) |
| **400.15 - 406.0 MHz** | Narrow FM (GFSK) | Weather radiosondes (sheet 15) |
| **406.0 - 406.1 MHz** | BPSK | Distress beacons (sheet 22) |
| **1,090 MHz** | Pulses | ADS-B: aircraft positions, SDR required (sheet 10) |

**Aviation emergencies**: frequencies 121.500 and 243.000 MHz, search and rescue: sheet 21. The transponder codes **7700** (emergency), **7600** (radio failure) and **7500** (hijacking) are visible in ADS-B (sheet 10).

## Civil traffic (indicative allocation, check in the AIP)

| Range | Typical use |
|---|---|
| **118.0 - 121.4 MHz** | Control towers and approaches |
| **121.6 - 121.975 MHz** | Ground: taxiing, departure clearances |
| **122.0 - 123.65 MHz** | Self-information, gliders, microlights, air-to-air. 123.450 MHz: unofficial air-to-air, often used by crews |
| **124 - 136 MHz** | Approaches and en-route centres (Reims, Brest, Paris, Bordeaux, Marseille): cruising aircraft |
| **131.5 - 131.9 MHz** | ACARS (sheet 11) |
| **136.7 - 136.975 MHz** | VDL Mode 2, data (sheet 11) |

- **Callsigns**: AFR (Air France), RYR (Ryanair), EZY (easyJet), DLH (Lufthansa). Languages: English and French.
- **8.33 kHz spacing**: since 2018, many channels use it. The channel name is not the exact frequency: set the UV-K1 to the closest value (6.25 kHz step), method below.
- **Range**: line of sight only. An aircraft at 10,000 m can be heard up to ~400 km; on the ground, only nearby airports.

## Weather by ear: ATIS, VOLMET and tower

| Service | Content | Where |
|---|---|---|
| **ATIS** | Weather and information for one aerodrome on a loop: runway in use, wind, visibility, clouds, temperature, QNH. Identified by a letter (Alpha, Bravo...). | VHF 118 - 137 MHz, AM, one frequency per aerodrome |
| **VOLMET** | METAR messages (observations) and sometimes TAF (forecasts) for a list of airports, read continuously. | HF in USB (long range) or VHF depending on the country |
| **Tower / approach** | Weather messages given to crews (wind, QNH). | VHF AM, local |

| HF VOLMET (kHz, USB) | Frequencies | Area |
|---|---|---|
| **Shannon VOLMET (Ireland)** | 3413 · 5505 · 8957 · 13264 | Europe, North Atlantic |
| **Gander VOLMET (Canada)** | 3485 · 6604 · 10051 · 13270 | North America, North Atlantic |

*VOLMET frequencies from memory, to verify. The low ones work mostly at night, the high ones by day. For ATIS and VHF VOLMET in France, see the AIP (SIA).*

## Reading a METAR (fictional example)

```
METAR LFPG 081930Z 24012KT 9999 FEW040 BKN090 18/12 Q1015 NOSIG
```

| Element | Meaning |
|---|---|
| **LFPG** | Airport, ICAO code (Paris-Charles de Gaulle) |
| **081930Z** | Day 08, 19:30 UTC (Z = UTC time) |
| **24012KT** | Wind from 240°, 12 knots (G25 = gusts to 25 kt; VRB = variable) |
| **9999** | Visibility 10 km or more (in metres). CAVOK = good visibility, no low cloud |
| **FEW040 BKN090** | Few clouds at 4000 ft, broken at 9000 ft (height in hundreds of feet) |
| **18/12** | Temperature 18 °C, dew point 12 °C |
| **Q1015** | QNH pressure 1015 hPa |
| **NOSIG** | No significant change expected in the next 2 h |

| Clouds | Coverage |
|---|---|
| **FEW** | 1 to 2 oktas |
| **SCT** | 3 to 4 oktas |
| **BKN** | 5 to 7 oktas |
| **OVC** | 8 oktas (overcast) |

| Code | Phenomenon |
|---|---|
| **RA / DZ** | Rain / drizzle |
| **SN** | Snow |
| **FG / BR** | Fog / mist |
| **TS / SH** | Thunderstorm / shower |

Intensity: '-' light, '+' heavy (e.g. +RA = heavy rain). TAF: forecast covering 9 to 30 h, with BECMG (becoming), TEMPO (temporary), PROB30 / PROB40 (probability). Aeronautical Q codes: QNH, QFE, QFU, QDM, QDR (sheet 34).

## Military

- UHF band 225 - 400 MHz in AM: exercises, patrols, air-to-air refuelling, air bases. Frequencies are published in the military AIP and in listener hobbyist databases.
- **Partial reception**: many exchanges use frequency hopping (HAVE QUICK) or encryption: you hear bursts, a hiss or nothing.
- 243.000 MHz remains the military emergency guard frequency: store it in memory with 121.500 MHz.

## Hot-air balloons, balloons and sondes

| Type | Frequency / band | Remark |
|---|---|---|
| **Hot-air balloon (ground crew)** | PMR446: 446.0 - 446.2 MHz (FM); CB 27 MHz | Common, unofficial use: listening to the recovery crew (sheet 8). |
| **Hot-air balloon (in flight)** | Aviation VHF 118 - 137 MHz, AM | When the craft carries a radio, on the local frequency. |
| **Weather radiosondes** | 400.15 - 406.0 MHz | Reception, models and tracking: sheet 15. |
| **Amateur stratospheric balloons** | 434 MHz (RTTY, LoRa); APRS 144.800 MHz; WSPR 14.0956 MHz | Tracking on habhub / SondeHub (sheets 24 and 26). |

## 8.33 kHz spacing: from channel name to tuned frequency

Since 2018, a 25 kHz channel is split into three 8.33 kHz channels. Controllers and charts give the **channel name** (6 digits, like 124.835), which is not the actual transmitted frequency. A radio that only tunes 25, 12.5 or 6.25 kHz steps cannot display this name: you have to convert it.

| Name ending (mod 25 kHz) | Channel type | Actual frequency (carrier) | Examples |
|---|---|---|---|
| **00, 25, 50, 75** | 25 kHz | the name itself | 118.000 → 118.000; 121.500 → 121.500 |
| **05, 30, 55, 80** | 8.33 kHz | 25 kHz block, + 0 kHz | 118.005 → 118.000; 124.830 → 124.825 |
| **10, 35, 60, 85** | 8.33 kHz | 25 kHz block, + 8.333 kHz | 118.010 → 118.0083; 124.835 → 124.8333 |
| **15, 40, 65, 90** | 8.33 kHz | 25 kHz block, + 16.667 kHz | 118.015 → 118.0167; 124.840 → 124.8417 |

**Method**: 1) take the last two digits of the name; 2) subtract the remainder of the division by 25 to get the start of the block (35 gives 25 + 10); 3) add 0, 8.333 or 16.667 kHz depending on the remainder (5, 10 or 15); 4) round to your radio's step.

> **Example: your aerodrome's tower is announced on 124.835** The remainder is 10: block start 124.825, plus 8.333 kHz = **124.8333 MHz** actual. Your radio does not offer this value. With a 6.25 kHz step: you can tune **124.83125** (offset 2.1 kHz, good AM reception). With a 12.5 kHz step: **124.8375** (offset 4.2 kHz, borderline but audible). With a 25 kHz step only: 124.825 (offset 8.3 kHz), the sound is weak or distorted. Preferably program one memory per channel with the name as the label.

| Channel name | Remainder | Actual (MHz) | 6.25 kHz step | 12.5 kHz step |
|---|---|---|---|---|
| **118.005** | 5 | 118.0000 | 118.00000 | 118.0000 |
| **118.010** | 10 | 118.0083 | 118.00625 | 118.0125 |
| **118.015** | 15 | 118.0167 | 118.01875 | 118.0125 |
| **127.440** | 15 | 127.4417 | 127.44375 | 127.4375 |

*Reception only. A stronger adjacent channel can bleed in (wide AM receiver). If your firmware offers an 8.33 kHz step (to verify), enter the actual value.*

# Aviation and aeronautical weather

Listening to the air: bands, traffic, ATIS, VOLMET, METAR, military, balloons and sondes



## UV-K1 settings and tips

- **Aviation**: AM mode (AIR), low squelch, 25 kHz step. Store in memory: 121.500 / 243.000 MHz, your nearby airport (tower, ATIS), an en-route centre. Scan 118 - 137 MHz: note the frequency, time and callsign of each exchange.
- **ATIS and VOLMET**: ATIS in AM on the aerodrome frequency (line-of-sight range). HF VOLMET in USB with the SI4732; the voice is read fast, in English: note the airport, UTC time, wind and QNH.
- **ADS-B and ACARS**: cannot be decoded on the UV-K1, see sheets 10 and 11 (SDR).
- **Never transmit** on aeronautical frequencies: it endangers crews. To prepare a flight, use official sources (Météo-France Aero, aviationweather.gov): radio listening is not reliable for navigation.

# ADS-B

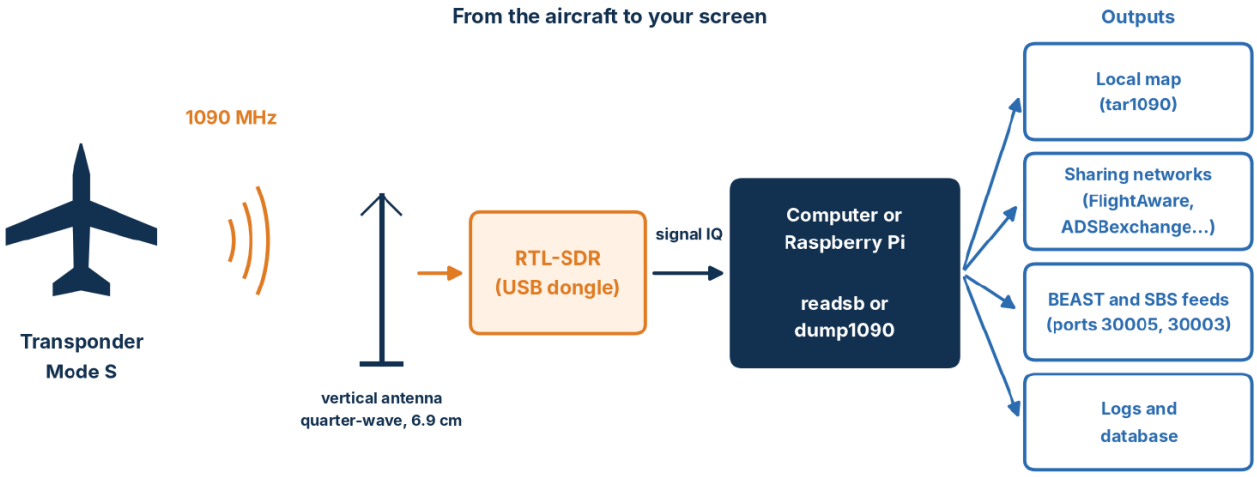
Aircraft positions on 1,090 MHz: pulses, frame, decoding, reception



**In brief.** ADS-B (Automatic Dependent Surveillance-Broadcast) is a system where each equipped aircraft automatically broadcasts, in clear, its identity, GNSS position, altitude and speed, about once a second or more. It mainly uses 1,090 MHz ("1090ES", an extension of the Mode S transponder). In the United States, general aviation flying below 18,000 ft may also use 978 MHz (UAT). The equipment is mandatory in many airspaces: since 1 January 2020 in the United States, and for some aircraft since 2017 in Europe. Messages are neither encrypted nor authenticated: any receiver in range can read them.

| 1,090 MHz | 1 Mbit/s | 112 bits | 24 bits |
|---|---|---|---|
| ADS-B frequency (1090ES) | data rate, pulse position modulation | length of an ADS-B frame | aircraft ICAO address |

## How it flows

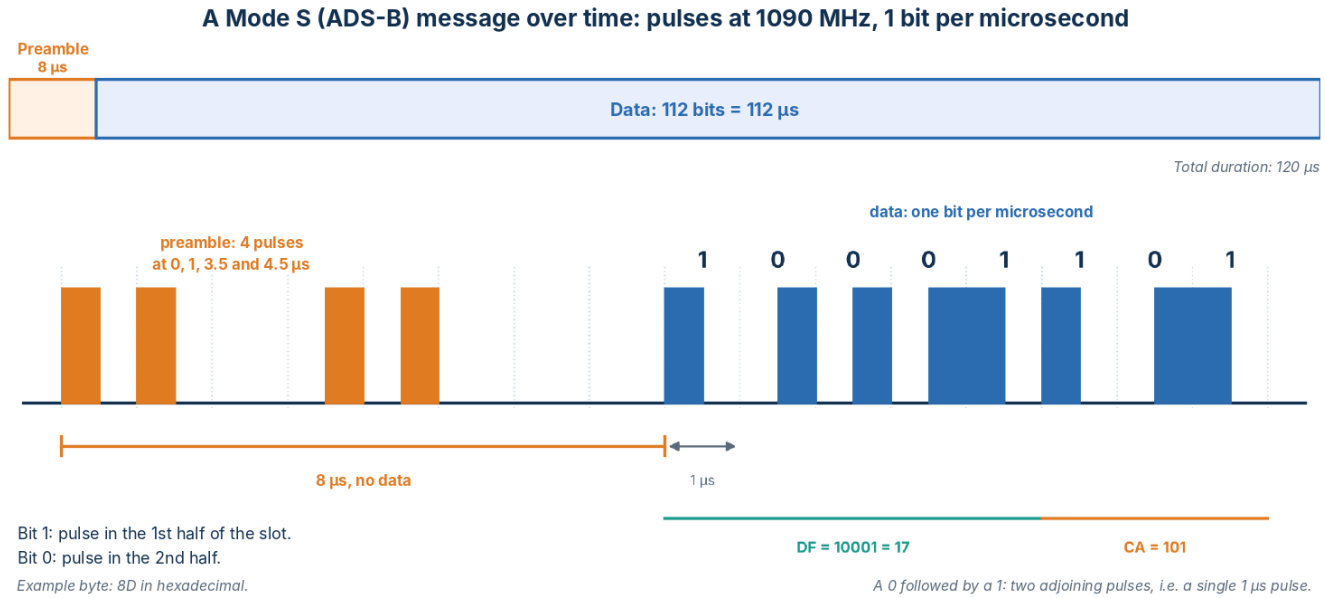



The receiver digitises the band around 1,090 MHz, detects preambles, demodulates the pulses and keeps only frames with a valid CRC. The software then groups messages by aircraft (ICAO address) and publishes them on a map or on the network.

## Setting up and running

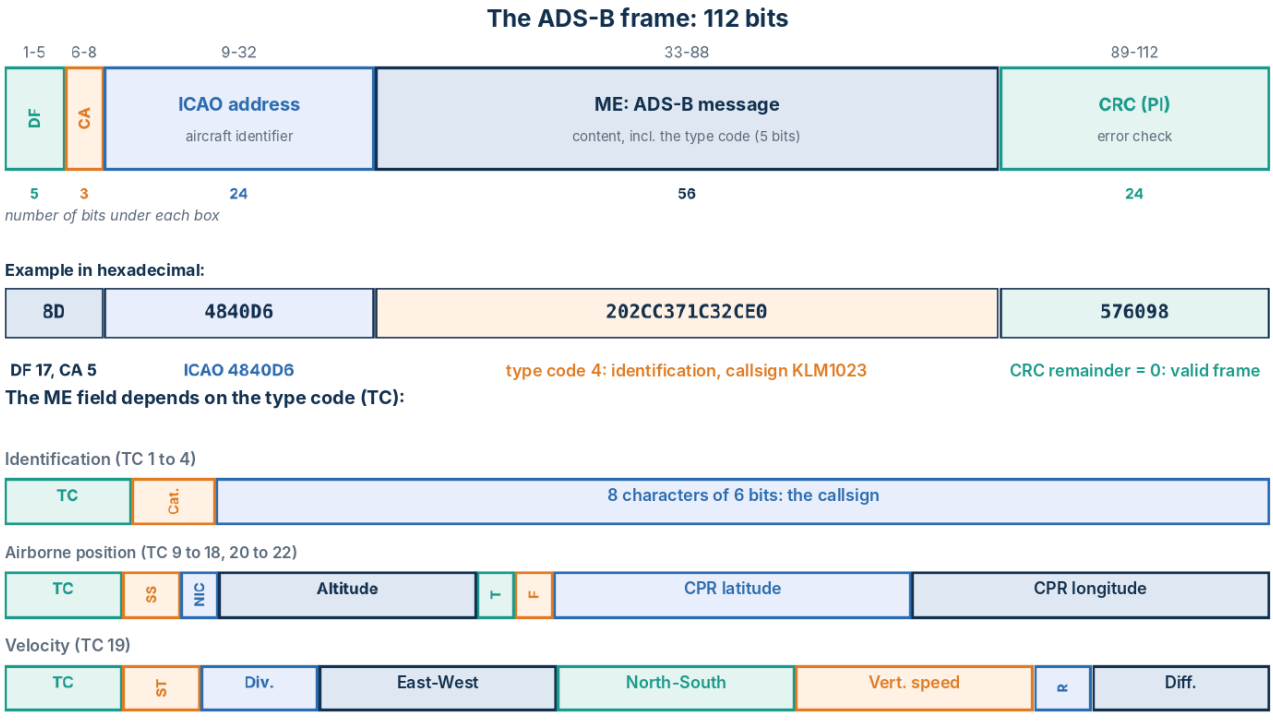
| Item | Detail |
|---|---|
| **Receiver** | RTL-SDR dongle (sheet 3). A 1,090 MHz filter and a low-noise amplifier near the antenna improve range (optional) |
| **Antenna** | Quarter-wave vertical (size: sheet 1, ADS-B line). Place it high, in the clear, with a short cable |
| **Software** | readsb or dump1090 (including the dump1090-fa variant) to decode, tar1090 for the web map |
| **First test** | dump1090 --interactive --net: table of received aircraft. rtl_adsb shows raw frames in hexadecimal |
| **Sharing** | FlightAware, FlightRadar24, ADS-B Exchange, OpenSky or airplanes.live accept feeds from personal receivers |

## The pulses




- The **preamble** lasts 8 microseconds: four 0.5-microsecond pulses, at 0, 1, 3.5 and 4.5 microseconds. It lets the receiver detect a message and synchronise.
- Each **bit** lasts 1 microsecond: a 1 is a pulse in the first half-microsecond, a 0 in the second (pulse position modulation).
- ADS-B frames are long: 112 bits, i.e. 120 microseconds with the preamble. Short Mode S replies are 56 bits.

## The frame




*F: even (0) or odd (1) frame for position decoding. SS: surveillance status. T: time stamp. NIC: integrity indicator.*

For a DF 17 frame, the 24-bit field at the end of the frame is a CRC (generator polynomial 0xFFF409): dividing the 112 bits, the remainder must be zero. A damaged frame is rejected. DF 17 denotes ADS-B sent by a transponder, DF 18 that from equipment without a transponder or relayed by TIS-B.

# ADS-B

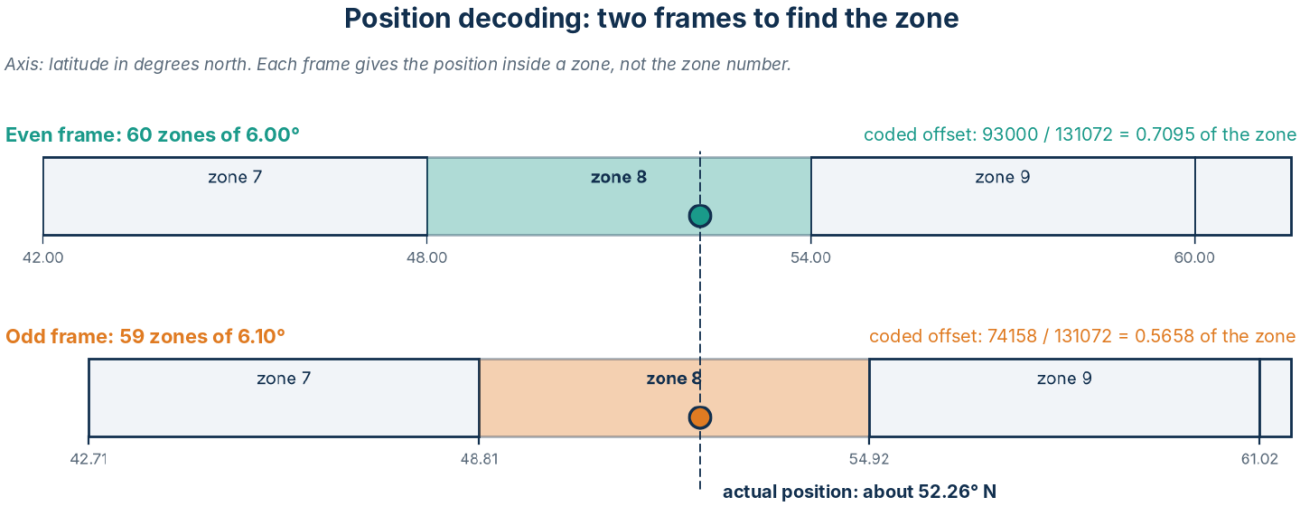
Aircraft positions on 1,090 MHz: pulses, frame, decoding, reception



## Message types (type code)

| Type code | Content | Detail |
|---|---|---|
| **1 to 4** | Identification | Callsign (8 characters) and aircraft category |
| **5 to 8** | Surface position | Position, speed and track on the runway or taxiways |
| **9 to 18** | Airborne position | Barometric altitude, latitude and longitude in CPR |
| **19** | Airborne velocity | Ground speed, track, vertical rate |
| **20 to 22** | Airborne position | GNSS height, latitude and longitude in CPR |
| **28, 29, 31** | Aircraft status, target state and settings, operational status | Emergency, targets, equipment capabilities |

## Position: CPR decoding




*Only one zone number is consistent with both subdivisions: this gives the latitude, here 52.2572° N, and likewise the longitude, here 3.9194° E.*

- Latitude and longitude take 17 bits each: compact, but ambiguous. **CPR** (Compact Position Reporting) encodes the position within a zone.
- The aircraft alternates even and odd frames, which use 60 and 59 latitude zones. By combining one frame of each, the receiver recovers the zone.
- With a nearby reference position (receiver or previous), a single frame may be enough.

## A worked decoding example

| Frame | Decoded content |
|---|---|
| 8D4840D6202CC371C32CE0576098 | Identification: ICAO 4840D6, type code 4, callsign KLM1023, valid CRC |
| 8D40621D58C382D690C8AC2863A7 | Position, even frame: ICAO 40621D, altitude 38,000 ft, CPR latitude 93000, CPR longitude 51372 |
| 8D40621D58C386435CC412692AD6 | Position, odd frame: same aircraft, CPR latitude 74158, CPR longitude 50194 |
| even + odd | Position: 52.2572° N, 3.9194° E (most recent frame is the even one) |
| 8D485020994409940838175B284F | Velocity: ICAO 485020, type code 19: 159 kt, track 182.9°, vertical rate -832 ft/min |

*Altitude: 12-bit code; if the Q bit is 1, altitude = N × 25 - 1000 ft. Velocity (subtype 1): East-West and North-South components, giving ground speed and track. Values checked by a full decoding of these frames.*

# ADS-B

Aircraft positions on 1,090 MHz: pulses, frame, decoding, reception



## Other Mode S messages

| DF | Message | Remark |
|---|---|---|
| **17** | ADS-B (extended squitter) | Broadcast spontaneously by the transponder, 112 bits |
| **18** | Non-transponder ADS-B, TIS-B | Equipment that does not answer interrogations |
| **11** | All-call reply | Announces the aircraft's presence to radars and collision avoidance |
| **4, 5** | Altitude and identity replies | Replies to a radar interrogation, 56 bits |
| **20, 21** | Comm-B replies | Onboard data returned on radar interrogation |
| **0, 16** | Collision avoidance (ACAS) | Dialogue between nearby aircraft |

*Formats other than DF 17 and 18 depend on radar interrogations: a passive receiver mostly picks up the replies, not always their context.*

## Range and limits

- **Line-of-sight range**: about 200 to 250 nautical miles for a cruising aircraft with a good, unobstructed antenna. It drops near hills and buildings.
- **Invisible aircraft**: aircraft without ADS-B Out (older ones, some military) do not appear; some networks locate them by multilateration (MLAT).
- **Congestion**: 1,090 MHz is shared with secondary radars and collision avoidance; in dense areas, messages overlap and are lost.

## Troubleshooting

| Symptom | Things to try |
|---|---|
| **No aircraft** | Antenna outside and vertical, short cable, gain on automatic, check that the dongle is detected |
| **Short range** | Raise the antenna, add a 1,090 MHz filter and a low-noise amplifier near the antenna |
| **Many rejected frames** | Gain too high (overload): reduce it. Move the dongle away from the computer with a USB extension |
| **Wrong or late positions** | Only one position frame received: wait for the pair, or provide a reference position |

## Going further

- **mode-s.org**: online book on Mode S and ADS-B (frame formats, detailed decoding).
- **pyModeS**: Python library to decode Mode S and ADS-B frames.
- **tar1090** and **graphs1090**: web map and reception statistics.

**Good to know.** ADS-B messages are public, but they remain sensitive: privacy programmes exist (LADD in the United States, temporary ICAO address PIA) and some countries restrict or monitor ADS-B receivers. Check your local legislation. **Never transmit on 1,090 MHz**: this frequency is used for air safety (radars, collision avoidance), and broadcasting false positions is illegal and dangerous.

# ACARS

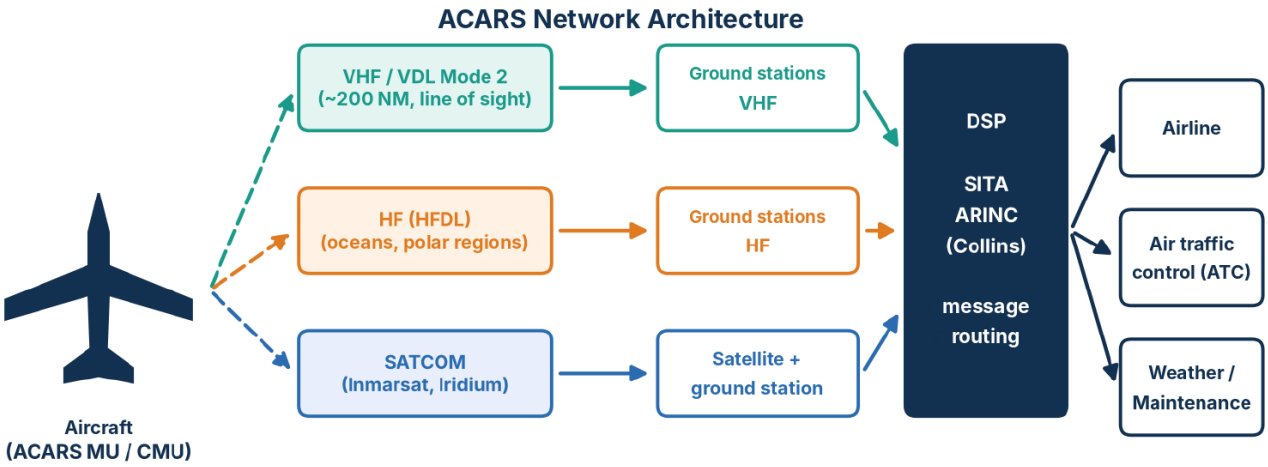
Aircraft-ground text messages on VHF: links, modulation, frame, reception



**In brief.** ACARS is a digital text messaging system between aircraft and the ground, created in 1978 by ARINC. It is used mainly by airlines (operations, maintenance, weather) and, through dedicated applications, by air traffic services. It is not pilot/ATC voice: these are short automatic messages or messages typed by the crew, carried over several types of radio links.

| 1978 | 2400 bit/s | 131.550 MHz | ~220 chars |
|---|---|---|---|
| year of creation | data rate of classic VHF ACARS | worldwide VHF frequency | max. text per block |

## How it flows




*Messages flow in both directions: uplink (ground to aircraft) and downlink (aircraft to ground).*

The onboard equipment (ACARS management unit) picks the available link. Ground stations or the satellite pass the message to the data service provider (DSP, e.g. SITA or ARINC), which routes it to the airline or organisation concerned.

## Links used

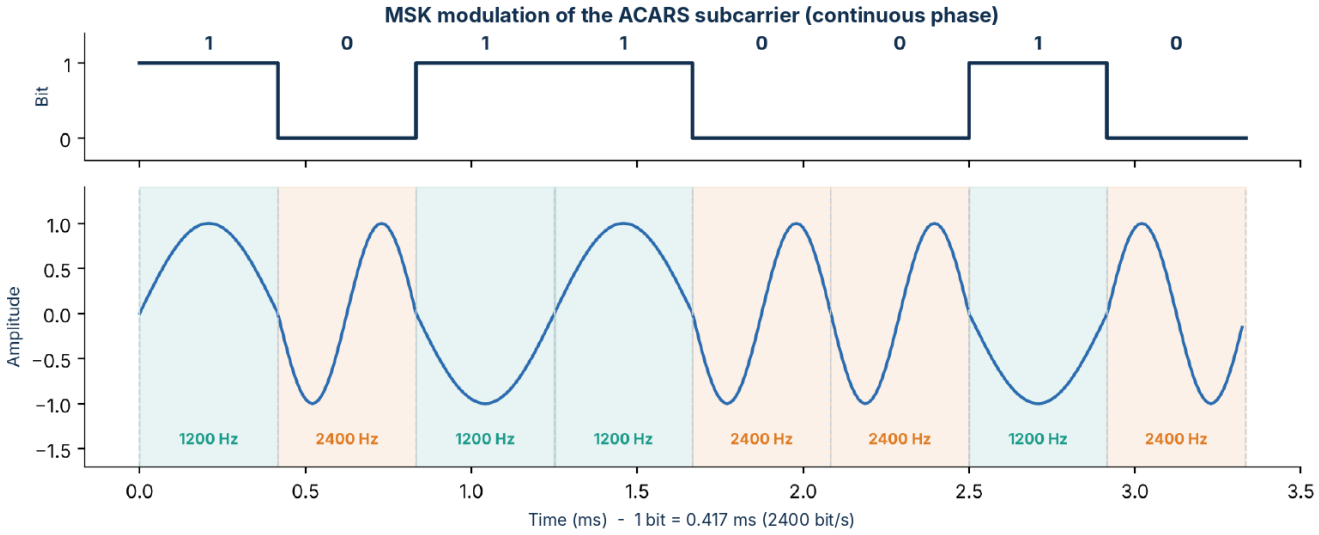
| Link | Band | Data rate | Use |
|---|---|---|---|
| **VHF (classic ACARS)** | 129 - 132 MHz | 2400 bit/s (AM, MSK) | Continental areas, the most widespread |
| **VDL Mode 2** | 136.7 - 136.975 MHz | 31.5 kbit/s (D8PSK) | Gradually replacing classic VHF |
| **HFDL (HF)** | 2 - 22 MHz | 300 to 1800 bit/s | Oceans and polar regions, beyond VHF range |
| **Satellite** | Inmarsat ~1545 MHz; Iridium ~1616 - 1626 MHz | variable | Worldwide coverage |

## Classic VHF frequencies

| Area | Frequencies |
|---|---|
| **Worldwide (common)** | 131.550 MHz |
| **Europe** | 131.725 / 131.525 / 131.825 MHz |
| **United States** | 129.125 / 130.025 / 130.425 / 130.450 MHz |

*25 kHz channels, amplitude modulation (AM). Frequencies vary by region and change over time: check local allocations before programming a receiver.*

## Modulation: MSK on a subcarrier

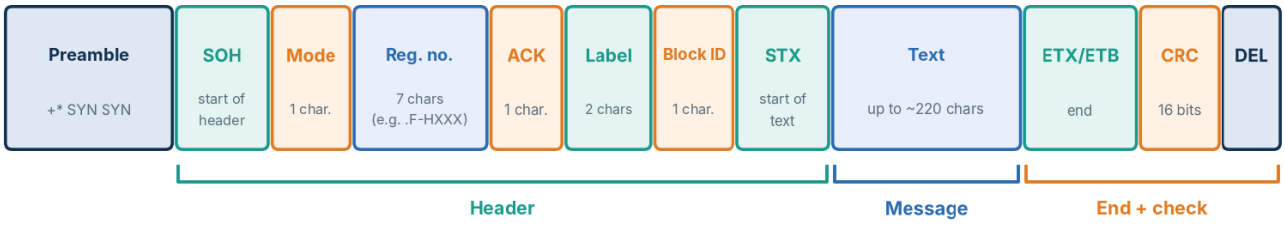



- The audio signal is a subcarrier at 1200 Hz and 2400 Hz, then transmitted in AM on the VHF carrier.
- A 1 bit corresponds to 1200 Hz (half a period per bit), a 0 bit to 2400 Hz (a whole period).
- The phase stays continuous at bit changes: this is what defines MSK and limits spectral spreading.

## The frame

**Structure of an ACARS block (classic VHF)**

*Characters are 7 bits + 1 odd-parity bit. The CRC-16 covers the block content.*




*A block holds at most ~220 characters of text: long messages are split (ETB) then reassembled.*

The preamble is used to synchronise the receiver. The registration identifies the aircraft, the label (2 characters) indicates the message type, and the CRC-16 detects transmission errors.

## What the messages contain

The **OOOI** events punctuate a flight and are sent automatically: OUT (leaving the gate), OFF (takeoff), ON (landing), IN (arriving at the gate).

- Weather (requests and reports), positions and estimated times of arrival (ETA)
- Departure clearances (PDC) and air traffic control messages
- Engine and maintenance data, fault reports
- Free text between the crew and the airline

| Label | Usual meaning |
|---|---|
| **Q0** | Link test |
| **SQ** | Ground station identification (squitter) |
| **H1** | Data messages (reports, ATC applications depending on context) |
| **5Z** | Airline-defined message |

*Non-exhaustive list: usage varies between airlines.*

## Decoding them

| Need | Detail |
|---|---|
| **Receiver** | RTL-SDR dongle (or other SDR covering 130 MHz), sheet 3 |
| **Antenna** | VHF quarter-wave (size: sheet 1, ACARS line) |
| **acarsdec** | Classic VHF ACARS decoder (command line) |
| **dumpvdl2** | VDL Mode 2 decoder |
| **JAERO** | Satellite ACARS decoder (Inmarsat) |
| **acarshub** | Web interface to view and archive received messages |
| **airframes.io** | Community site that aggregates messages received by volunteers |

**Legality.** Simply receiving these signals is generally tolerated, but disclosing the content of messages may be regulated (confidentiality of correspondence). Rules vary by country: check local legislation before publishing or sharing what you decode.

# ILS

Instrument landing system: localizer, glide slope, DDM, markers

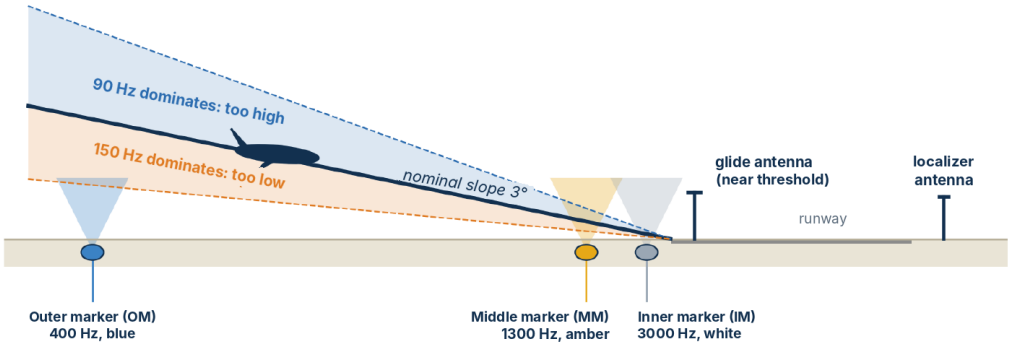


**In brief.** The ILS is an instrument landing system: it guides the aircraft precisely, laterally (localizer) and vertically (glide slope), down to the runway, even without visibility. The principle relies on two radio beams amplitude-modulated by two tones, 90 Hz and 150 Hz. The aircraft measures the difference in depth of modulation (DDM) between these two tones: it is zero on the centreline and changes sign on either side. The pilot, or the autopilot, corrects to bring the needle back to the centre. The system is standardised by ICAO (Annex 10).

| 108 - 112 MHz | 329 - 335 MHz | 90 / 150 Hz | 3° |
|---|---|---|---|
| localizer (VHF) | glide slope (UHF) | the two modulation tones | nominal slope |

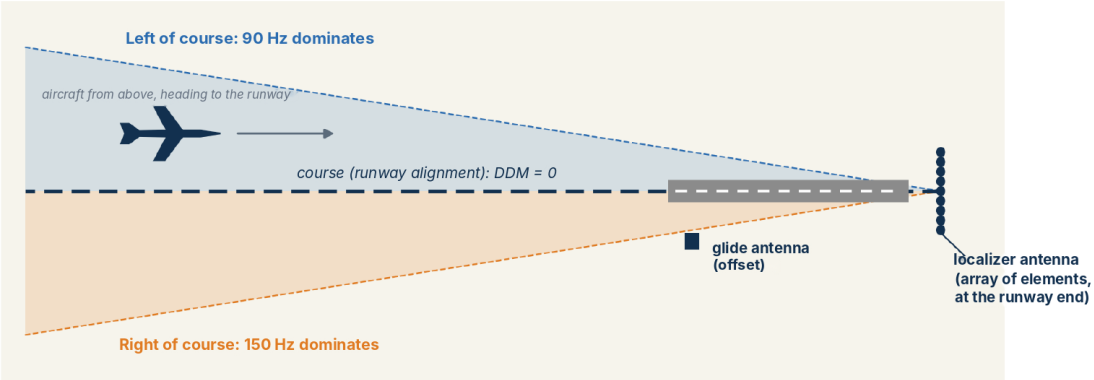
## Overview

**Side view: the glide slope (descent path)**




*Vertical scale greatly exaggerated. Markers are used less and less, replaced by DME.*

**Top view: the localizer (runway centerline)**



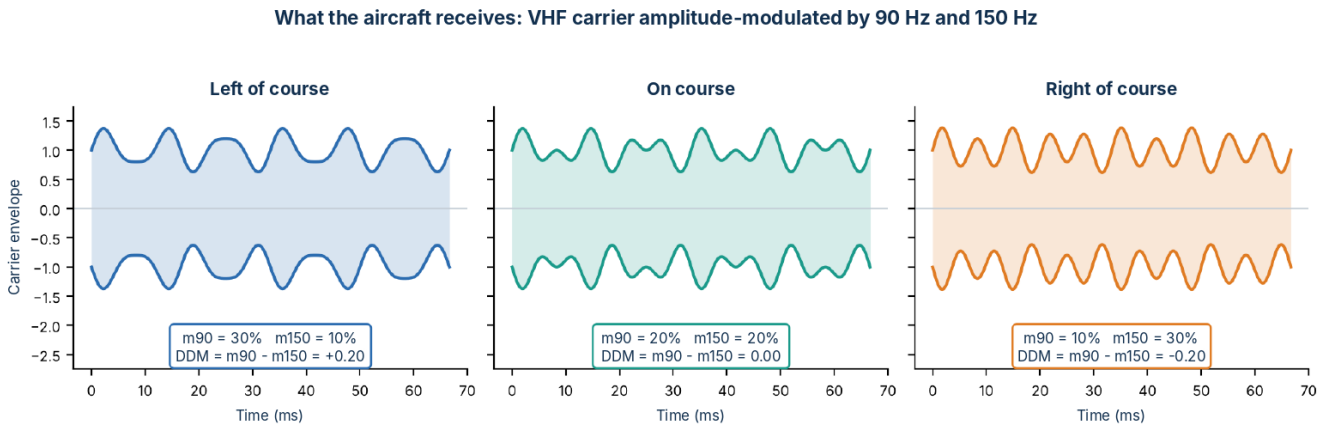

Both beams are radiated from the ground. The localizer marks the runway centreline, the glide slope marks the descent path. Seen from an aircraft on approach, 90 Hz dominates on the left (or above), 150 Hz on the right (or below).

## Components

| Element | Role | Key points |
|---|---|---|
| **Localizer (LOC)** | Lateral guidance, runway centreline | VHF, 108.10 to 111.95 MHz. Antenna at the end of the runway |
| **Glide slope (GS)** | Vertical guidance, slope | UHF, 329.15 to 335.00 MHz. Antenna near the threshold, offset from the centreline |
| **Markers** | Distance marks to the threshold | 75 MHz. Increasingly rarely used |
| **DME** | Distance to the runway | Frequency paired with the LOC, often replaces the markers |
| **Approach lighting** | Handover to visual guidance | Lights aligned on the centreline, before the threshold |
| **Monitors** | Continuous supervision | Switch to the standby transmitter or shut down transmission if the signal goes out of tolerance |

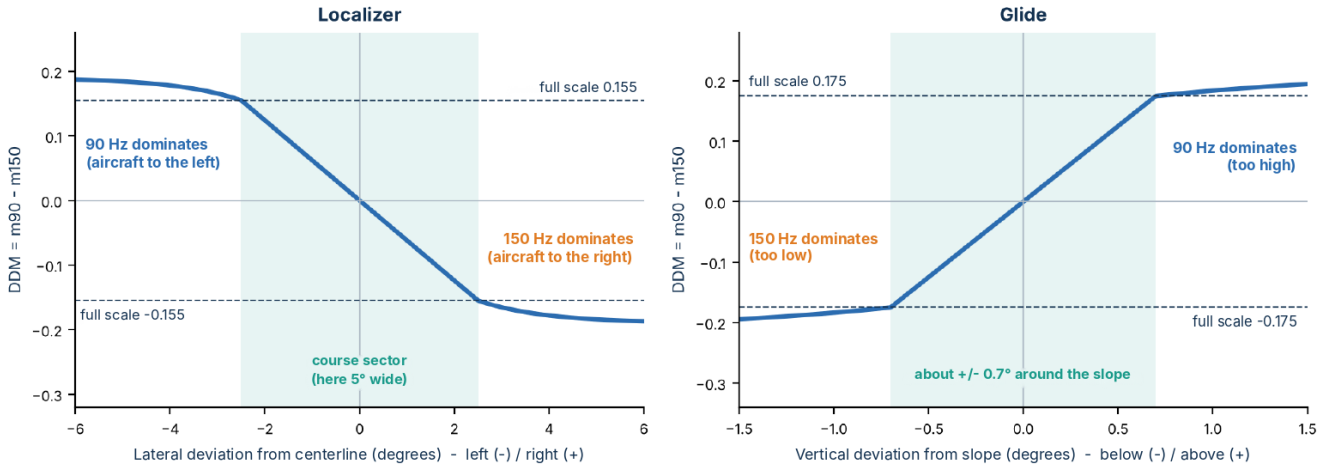
## The principle: two tones

- The depth of modulation of each tone varies with the aircraft's position in the beam.
- On the centreline, both tones have the same depth: the DDM is zero. To the side, one of the tones takes over.
- The receiver separates the two tones by filtering, measures their amplitudes and deduces the deviation.




*What the aircraft receives: VHF carrier amplitude-modulated by 90 Hz and 150 Hz.*

## DDM versus offset




Near the centreline, the DDM grows linearly with the offset: this gives proportional guidance. Beyond full scale, it keeps increasing slowly, which preserves the direction of correction out to large offsets.

## Reading it in the cockpit

| Aircraft position | Dominant tone | Needle | Correction |
|---|---|---|---|
| **Left of centreline** | 90 Hz | Course needle to the right | Correct to the right |
| **Right of centreline** | 150 Hz | Course needle to the left | Correct to the left |
| **Above the slope** | 90 Hz | Glide needle down | Descend |
| **Below the slope** | 150 Hz | Glide needle up | Climb or reduce the descent |

*The needle shows where the path to follow is: it "goes toward the centreline". Flying toward the needle brings the aircraft back onto the beam.*

# ILS

Instrument landing system: localizer, glide slope, DDM, markers



## The radio in numbers

| | Localizer | Glide slope |
|---|---|---|
| **Band** | 108.10 to 111.95 MHz (VHF) | 329.15 to 335.00 MHz (UHF) |
| **Channels** | 40 channels. The 100 kHz digit is odd (108.10, 108.15, 108.30...) | Frequency paired with each localizer frequency |
| **Modulation** | AM, 90 Hz and 150 Hz, 20 % each on the centreline | AM, 90 Hz and 150 Hz, 40 % each on the centreline |
| **Dominant tone** | 90 Hz on the left, 150 Hz on the right (seen from the approach) | 90 Hz above, 150 Hz below |
| **Full scale** | DDM 0.155 (course sector of about 5°, adjustable from 3° to 6°) | DDM 0.175 (about ±0.7° around the slope) |
| **Coverage** | ±10° up to 25 NM, ±35° up to 17 NM | ±8° up to 10 NM, from 0.45 to 1.75 times the slope angle |
| **Identification** | Morse code of 3 to 4 letters starting with I, 1020 Hz tone | No identification of its own |
| **Polarisation** | Horizontal | Horizontal |

*Frequency pairs: 108.10 MHz with 334.70 MHz, 108.15 with 334.55, 110.30 with 335.00, 110.50 with 329.60, 111.90 with 331.10, 111.95 with 330.95.*

## Markers

| Marker | Tone | Light | Typical position |
|---|---|---|---|
| **Outer (OM)** | 400 Hz, dashes | Blue | About 7 km from the threshold |
| **Middle (MM)** | 1300 Hz, alternating dots and dashes | Amber | About 1 km from the threshold |
| **Inner (IM)** | 3000 Hz, dots | White | A few hundred metres from the threshold |

*Positions vary by runway and are published in approach charts.*

## Approach categories

| Category | Decision height | RVR (runway visual range) |
|---|---|---|
| **CAT I** | 60 m (200 ft) or more | 550 m or more (or visibility of 800 m) |
| **CAT II** | 30 to 60 m (100 to 200 ft) | 300 m or more |
| **CAT IIIA** | Less than 30 m, or none | About 200 m or more |
| **CAT IIIB** | Less than 15 m, or none | About 50 to 200 m |
| **CAT IIIC** | None | None (no common operational use) |

*Indicative values (ICAO and EASA); the exact rules depend on the country and the operator. The category actually usable depends on the ground installation, the aircraft equipment and the crew qualification together.*

## Pitfalls and limits

- **False glide slopes**: the glide slope also radiates spurious lobes above the true slope (around 6° and 9°). The glide slope is therefore intercepted from below.
- **Back beam**: the localizer also radiates backwards, with reversed indications if followed without care.
- **Critical areas**: vehicles and aircraft must stay behind the holding points, as they reflect and distort the beams.
- **Terrain and obstacles**: reflections from the ground or buildings create bends in the centreline or the slope.
- **A single path**: the ILS imposes a straight-in approach. GLS (GBAS) or LPV (SBAS) approaches are satellite-based alternatives.

## Receiving and observing them

| Need | Detail |
|---|---|
| **Receiver** | RTL-SDR dongle, in AM mode (sheet 3) |
| **Antenna** | Horizontal dipole (horizontal polarisation): sizes in sheet 1, ILS lines (two equal elements, each half the length given) |
| **Software** | SDR++, SDR# or GQRX to listen, Audacity or a Python script (FFT) to measure |
| **What you hear** | The localizer's Morse identification code (1020 Hz) |
| **What you measure** | In the audio spectrum, two lines at 90 Hz and 150 Hz. Their amplitude ratio varies with the position in the beam |

*Reception is only possible near an airport and especially along the approach axis: the localizer radiates ahead of the runway and little to the sides. The glide slope is even more directional.*

> **Good to know.** The ILS is a safety-critical navigation aid. Never use it to navigate and **never transmit on these frequencies**: interference can endanger lives and is heavily penalised.

# AIS and ATIS

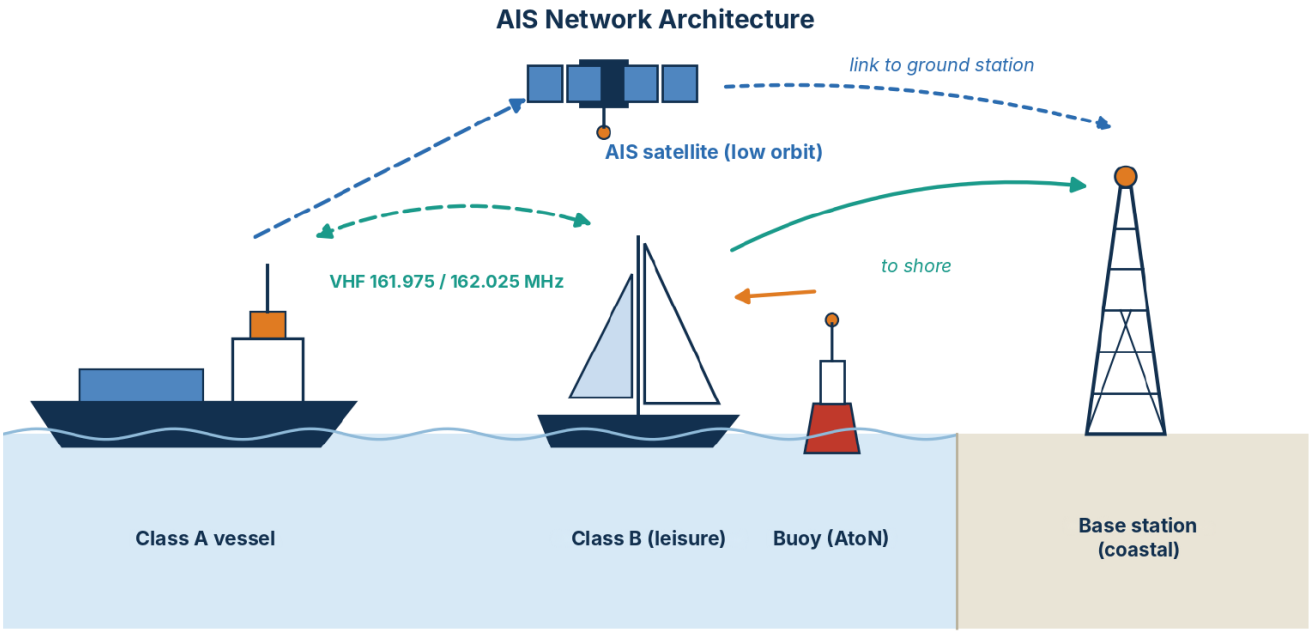
Identifying ships by VHF radio: AIS and ATIS



**En bref.** AIS is a system that identifies and locates ships by VHF radio. Each transmitter automatically broadcasts, in clear, its identity, position, course and speed. The aim: avoid collisions, help the authorities track traffic and make rescue at sea easier. Carriage has been mandatory under the SOLAS convention (IMO) for the merchant ships concerned since 2002, phased in until 2004.

| 2 channels | 9600 bit/s | 2250 | 9 digits |
|---|---|---|---|
| 161.975 and 162.025 MHz | GMSK, 25 kHz channels | slots per minute per channel | MMSI identifier (sheet 14) |

## How it travels




*Each transmitter broadcasts in the clear, with no specific recipient: any receiver in range can decode.*

There is no addressee: all transmitters (ships, base stations, buoys) broadcast and all receivers in range decode. Coastal stations relay the data to traffic centres; low-orbit satellites pick up the same signals over the oceans.

## Equipment classes

| Equipment | Who for | Power | Access | Transmit rate |
|---|---|---|---|---|
| **Classe A** | Ships subject to SOLAS: passenger ships, cargo ships of 300 GT and over on international voyages, 500 GT and over otherwise | 12.5 W | SOTDMA | 2 s to 3 min depending on speed and manoeuvres |
| **Classe B (CS)** | Pleasure craft, fishing, small vessels | 2 W | CSTDMA | about 30 s (3 min when stopped) |
| **Classe B (SO)** | Same, higher-performance version | 5 W | SOTDMA | more frequent than CS |

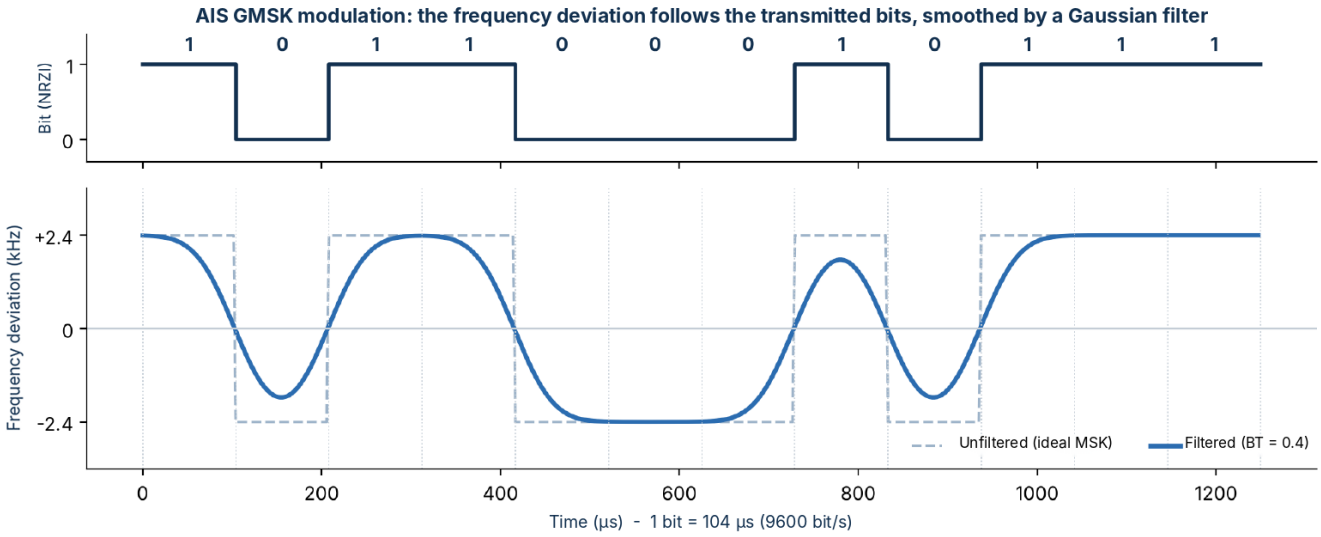
*Class A rate: about 3 min at anchor or moored, 10 s up to 14 knots, 6 s from 14 to 23 knots, 2 s above, and faster still while changing course.*

## The radio layer

| Parameter | Value |
|---|---|
| **Channels** | AIS 1: 161.975 MHz (channel 87B) / AIS 2: 162.025 MHz (channel 88B), the transmitter alternates between the two |
| **Spacing** | 25 kHz |
| **Modulation** | GMSK, BT = 0.4, frequency deviation of ±2.4 kHz |
| **Data rate** | 9600 bit/s |
| **Coding** | HDLC frame, NRZI coding, bit stuffing |
| **Range** | About 20 to 40 nautical miles between ships (line of sight, depending on antenna height) |

*Satellites pick up the same signals (satellite AIS). In very dense areas, transmissions overlap within their field of view, which limits detection. Message 27 was defined for long range.*

## Modulation: GMSK

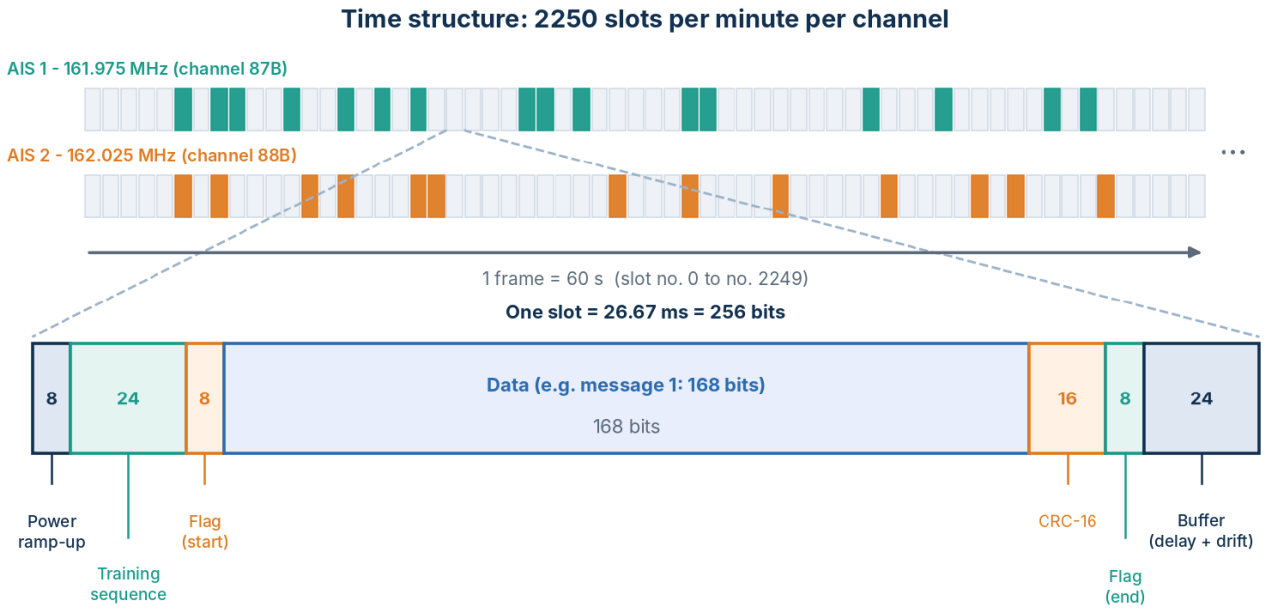



- GMSK is a continuous-phase frequency modulation: the frequency deviation switches from +2.4 to -2.4 kHz depending on the bit.
- A Gaussian filter (BT = 0.4) rounds the transitions: the spectrum stays narrow enough to fit in a 25 kHz channel, at the cost of slight overlap between consecutive bits.
- With NRZI coding, a 0 bit causes a frequency change, while a 1 bit does not.
- Bit stuffing inserts a 0 after five consecutive 1s, so that the frame flag (01111110, i.e. 7E in hexadecimal) never appears by chance in the data.

## Identifying a transmitter: the MMSI

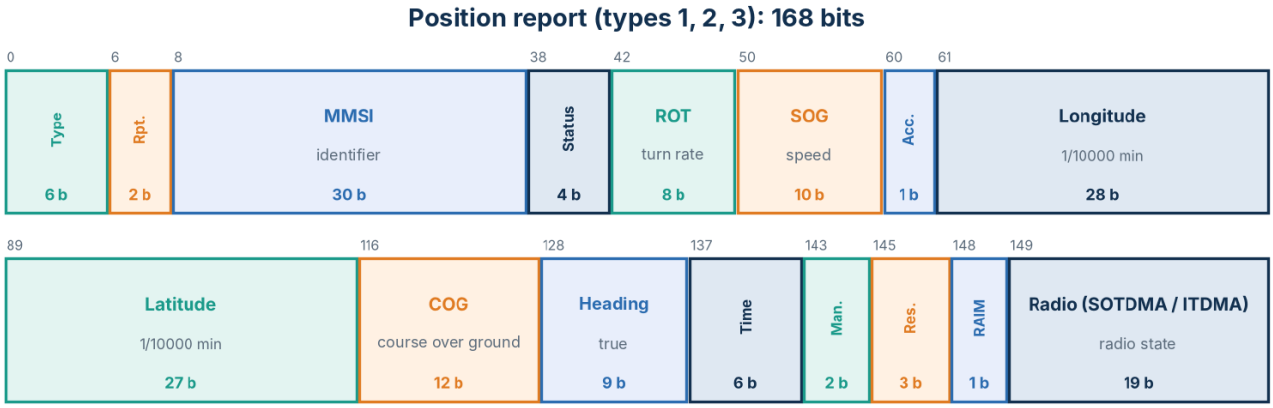
The MMSI has 9 digits; the first three (MID) give the country for a ship. The full formats (ship, coastal station, aid to navigation, AIS-SART beacons) and the MID list are in sheet 14.

## Time sharing: TDMA




- **SOTDMA** (class A): each station picks a free slot and announces its future slots in the radio status field. Stations are synchronised to UTC time, usually via GPS.
- **CSTDMA** (class B CS): the station listens before transmitting and only uses a slot if it is free.
- **ITDMA**: one-off reservation, for example on entering the network or for longer messages.
- **FATDMA**: fixed slots reserved by base stations.

## A position message




*Position in 1/10000 of an arc minute, speed (SOG) in 1/10 of a knot, course (COG) in 1/10 of a degree. Small number above a field: index of the first bit.*

A position message fits in a single slot (168 bits of data). Longer messages, such as the type 5 static data (424 bits), occupy several slots.

## Common messages

| Type | Content |
|---|---|
| **1, 2, 3** | Class A position report |
| **4** | Base station report (UTC time and position) |
| **5** | Static and voyage data: name, callsign, ship type, dimensions, draught, destination, ETA (2 slots) |
| **9** | Position of a search and rescue aircraft |
| **14** | Free-text safety message (also used by AIS-SART beacons) |
| **18, 24** | Class B position and static data |
| **21** | Aid to navigation (AtoN): buoy, beacon, including virtual ones |
| **27** | Long-range position report (satellite detection) |

## Decoded example

```
!AIVDM,1,1,,B,177KQJ5000G?tO`K>RA1wUbN0TKH,0*5C
```

| Element | Meaning |
|---|---|
| !AIVDM,1,1,,B | Received AIS NMEA sentence, fragment 1 of 1, no sequence number, channel B |
| 177KQJ5000G?... | Data in 6-bit ASCII (168 bits) |
| 0*5C | 0 padding bits, checksum (XOR) 5C |
| Decoded | Type 1, MMSI 477553000 (MID 477, Hong Kong), status 5 (moored), speed 0.0 knots, position 47.58283 N and 122.34583 W, course over ground 51.0 degrees, heading 181 degrees, UTC time: second 15 |

## Receiving them

| Need | Detail |
|---|---|
| **Receiver** | RTL-SDR dongle (or another SDR covering 162 MHz), sheet 3 |
| **Antenna** | Quarter-wave vertical (size: sheet 1, AIS row) |
| **AIS-catcher** | Decoder with web interface, for RTL-SDR |
| **rtl_ais** | Command-line decoder, NMEA output |
| **OpenCPN** | Nautical charting software that displays AIS targets |
| **MarineTraffic, VesselFinder, AISHub** | Community aggregators where you can share your received data |

**Good to know.** Transmitting is a different matter from receiving: you need approved equipment and an assigned MMSI, and broadcasting false positions or false beacons is illegal and dangerous. The data is entered by crews: it may be wrong or missing (AIS switched off), so never rely on it alone for navigation.

## Bonus: ATIS on inland waterways

On European rivers and canals, another identification system coexists with AIS: ATIS (Automatic Transmitter Identification System). Instead of transmitting continuously like AIS, the boat sends a short digital burst each time the push-to-talk (PTT) button is released at the end of a VHF voice transmission, on the same channel. It identifies who has just spoken. Not to be confused with aeronautical ATIS, the voice information message of airports (sheet 9).

| | AIS | ATIS |
|---|---|---|
| **Use** | Worldwide, seagoing ships | European inland waterways |
| **Transmission** | Continuous, periodic beacons | One burst after each voice message |
| **Frequency** | 161.975 and 162.025 MHz | The working channel used for voice |
| **Modulation** | GMSK, 9600 bit/s | FSK 1300 / 2100 Hz, 1200 baud, about 285 ms |
| **Content** | Position, course, speed, ship data | Identity only (callsign or coded MMSI) |

*The burst is built on DSC signalling, but on the voice channel. The modulation figures come from Wikipedia; they are not in the application documentation.*

## ATIS RX on UV-K5 / UV-K1 radios (Labs edition)

The ATIS RX application of the experimental Labs firmware decodes this burst directly on the radio. Install the version matching your firmware with UV Studio (sheet A03), then launch it with **F + 7**. First choose an inland waterway working channel, in FM.

| Element | Detail |
|---|---|
| **Decoding** | Ten-digit identity; callsign and country reconstructed when possible. The decoded identity is authoritative, the reconstruction is a convenience |
| **List** | Eight stations in RAM, the most recently received goes to the top. The list disappears when you exit the application |
| **Display** | Callsign or identity, country, signal level, number of receptions, age of the last burst, frequency, recognised marine channel number |
| **Speaker** | Off by default and not needed for decoding; the setting is saved |

| Key | Action |
|---|---|
| **UP / LEFT or DOWN / RIGHT** | Newer or older station (UP / DOWN on UV-K5, LEFT / RIGHT on UV-K1, depending on SetNav) |
| **1** | Turns the speaker sound on or off, opened by the squelch |
| **2** | Clears the station list and the counters |
| **EXIT** | Exits the application |

# DSC, MMSI and NAVTEX

Digital selective calling (ITU-R M.493) and navigational warnings: signal, codes, settings



## DSC (Digital Selective Calling): signals by band

| Band | Frequency(ies) | Modulation | Audio tones | Rate | Note |
|---|---|---|---|---|---|
| **VHF channel 70** | 156.525 MHz | FM, AFSK | 1300 Hz / 2100 Hz | 1200 baud | Distress, urgency, safety, individual or group calls. No voice on this channel. |
| **MF** | 2187.5 kHz (distress); 2177 kHz (ship to ship) | FSK, SSB (J2B) | 1615 Hz / 1785 Hz (170 Hz shift) | 100 baud | Range of a few hundred km. |
| **HF** | 4207.5 · 6312 · 8414.5 · 12,577 · 16,804.5 kHz (distress) | FSK, SSB (J2B) | 1615 Hz / 1785 Hz (170 Hz shift) | 100 baud | Worldwide range. Alerts repeated every 3.5 to 4.5 min until acknowledged. |

## Structure of a distress call (in order)

| Call element | Content | Details |
|---|---|---|
| **Dot pattern** | Sequence of alternating 1s and 0s | Bit synchronisation: 200 bits on MF/HF, 20 bits on VHF |
| **Phasing sequence** | 7 symbols (DX / RX) | Frame synchronisation; the DX and RX channels carry the same message in time diversity |
| **Format specifier (×2)** | 112 = distress | Others: 116 all ships, 120 individual call, 123 semi-automatic individual, 102 geographic area, 114 group |
| **Address (if individual)** | MMSI of the addressee, 9 digits | Distress: "all ships", no specific address |
| **Category** | 112 = distress | Others: 100 routine, 108 safety, 110 urgency |
| **Caller identity** | MMSI, 9 digits | The first three digits (MID) give the country |
| **Nature of distress** | 3-digit code | 100 fire, 101 flooding, 102 collision, 103 grounding, 104 listing, 105 sinking, 106 adrift, 107 unspecified, 108 abandoning ship, 109 piracy, 110 man overboard, 112 EPIRB beacon |
| **Position** | Quadrant + latitude and longitude | 5 symbols, in degrees and minutes (e.g. 48°52' N, 002°20' E) |
| **Time** | 4 digits UTC | HHMM; 8888 = time not available |
| **Subsequent communication** | Mode code | 100 = telephony (156.800 MHz, 2182 kHz); 109 = SSB (J3E); 126 = unspecified |
| **End of sequence (EOS)** | 117, 122 or 127 | 117: acknowledgement requested; 122: acknowledgement; 127: end without acknowledgement |
| **ECC** | Check character | Error check over the whole call |

*A DSC symbol is a 10-bit word: 7 information bits and 3 check bits (number of zeros in the 7 bits). A distress alert is sent 5 times in a row, then repeated until a coast station or a ship acknowledges it. Voice follows on channel 16 or on 2182 kHz.*

## Listening to and decoding DSC

- **VHF (channel 70)**: UV-K1 on 156.525 MHz in FM. You hear a short "bzzz" lasting a few seconds (1200 baud, two fast tones). A decoder on a PC or smartphone, fed with the audio, reads the message.
- **HF**: USB or LSB, tones at 1615 and 1785 Hz (centre around 1700 Hz). The exact dial frequency is set 1700 Hz below the stated carrier, in USB.
- **Decoders**: YADD (Yet Another DSC Decoder) and MultiPSK on Windows; for a home-made decoder, same approach as for NAVTEX: FSK detection, sync on the dot pattern, DX / RX voting, checking the 3 check bits. To verify: availability of this software.
- If you decode a real distress call (format 112, category 112), note the MMSI, position and time, then alert the CROSS (196) or 112 if no acknowledgement is heard (sheet 21).

## MMSI and MID: reading a maritime identifier

| Identifier type | Format | Example / note |
|---|---|---|
| **Ship** | MID + 6 digits | 226 123456: MID 226 = France |
| **Coast station** | 00 + MID + 4 digits | 00 226 1234 (France) |
| **Group of ships** | 0 + MID + 5 digits | Group call |
| **SAR aircraft** | 111 + MID + 3 digits | Aeronautical search and rescue |
| **Aid to navigation (AtoN)** | 99 + MID + 4 digits | Buoy, beacon (sheet 13) |
| **AIS-SART / MOB AIS / EPIRB-AIS** | 970 / 972 / 974 + 6 digits | AIS distress devices (sheet 22) |

# DSC, MMSI and NAVTEX

Digital selective calling (ITU-R M.493) and navigational warnings: signal, codes, settings



| Common MIDs | |
|---|---|
| **France** | 226, 227, 228 |
| **United Kingdom** | 232 to 235 |
| **Germany** | 211, 218 |
| **Spain** | 224, 225 |
| **Italy** | 247 |
| **Netherlands** | 244 to 246 |
| **Belgium** | 205 |
| **Ireland** | 250 |
| **United States** | 338, 366 to 369 |
| **Canada** | 316 |

*The MID is the 3-digit country code.*

## NAVTEX (navigational warnings)

| Parameter | Value |
|---|---|
| **Frequencies** | 518 kHz (English, international); 490 kHz (national language); 4209.5 kHz (HF, supplementary transmissions) |
| **Modulation** | FSK 100 baud, 170 Hz shift, in LSB or USB depending on the decoder (SITOR-B, FEC mode) |
| **Coding** | 7-bit code (CCIR 476): each character contains 4 bits at 1 and 3 bits at 0, which detects errors. Each character is sent twice: direct (DX), then repeated (RX) 4 characters later (time diversity) |
| **Message format** | ZCZC B1 B2 B3 B4 (text) NNNN: B1 = station letter; B2 = type (A navigational warnings, B weather - gale warnings, C ice, D SAR and piracy, E weather forecasts, F pilot service, L additional warnings, Z nothing to broadcast); B3B4 = serial number 00 to 99 (00 = urgent) |
| **Schedule** | Each station transmits in a 10-minute slot, every 4 hours. The slot is offset by 10 minutes per letter (A at 00:00, B at 00:10, C at 00:20...) so that two stations do not interfere. The France area is NAVAREA II |
| **Decoding** | MultiPSK, YAND (Windows), fldigi (NAVTEX / SITOR-B mode in recent versions, to verify); or a home-made 100 baud FSK decoder with phase detection then DX / RX voting |

# Radiosondes

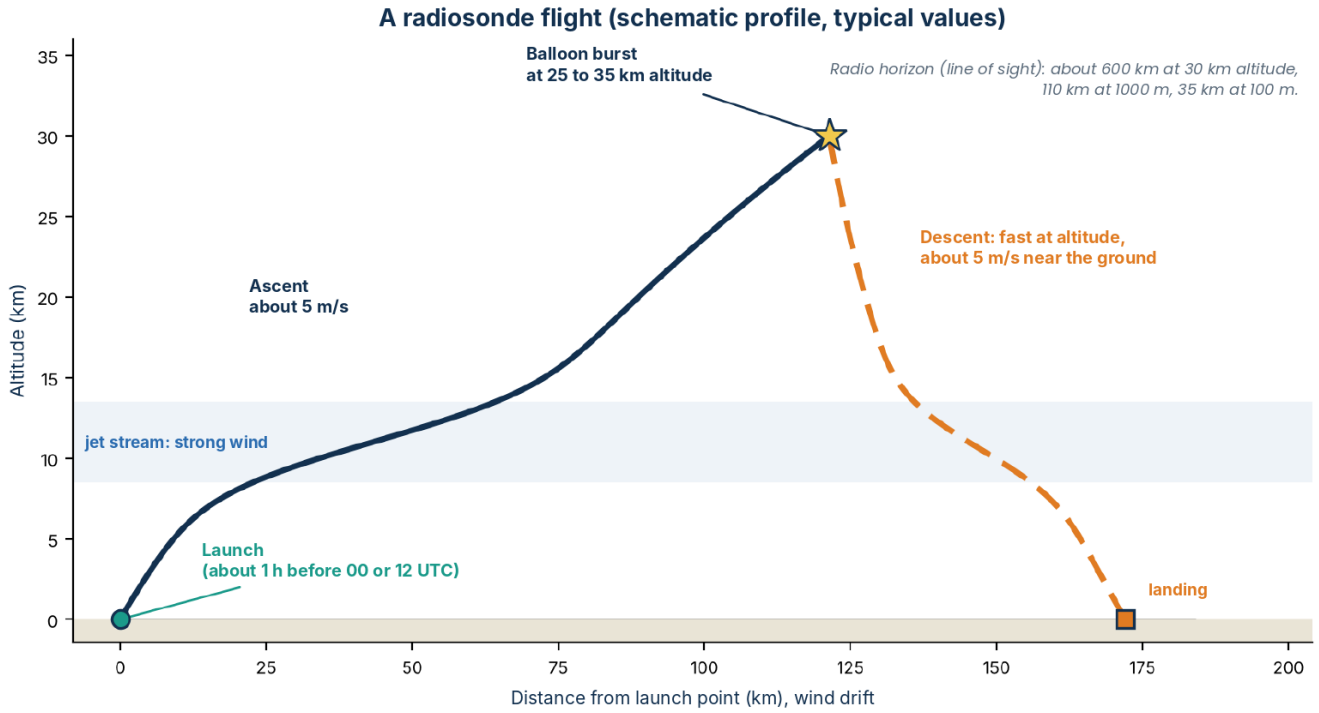
Weather balloons: reception and tracking at 400 - 406 MHz



**In brief.** A radiosonde is a small battery-powered instrument carried by a weather balloon, which measures the atmosphere and sends its measurements by radio. About 1,300 sites worldwide launch balloons, mainly for the 00 and 12 UTC observations (launch about an hour earlier). The sonde measures temperature, humidity and, depending on the model, pressure; its GPS gives the position and, from its movements, the wind. It transmits in UHF around 400 to 406 MHz (some models at 1680 MHz). The free software radiosonde_auto_rx scans the band, identifies sondes, decodes the telemetry and publishes positions on SondeHub.

| 400 - 406 MHz | ~1,300 | 00 and 12 UTC | ~250 g |
|---|---|---|---|
| sonde UHF band | launch sites | observation times | typical mass |

## The flight of a radiosonde




*Typical values: ascent of about 1 h 30 at 5 m/s, burst between 25 and 35 km altitude, then a descent of about forty minutes under parachute. The wind drifts the sonde by several tens to over 100 km.*

## What the sonde measures

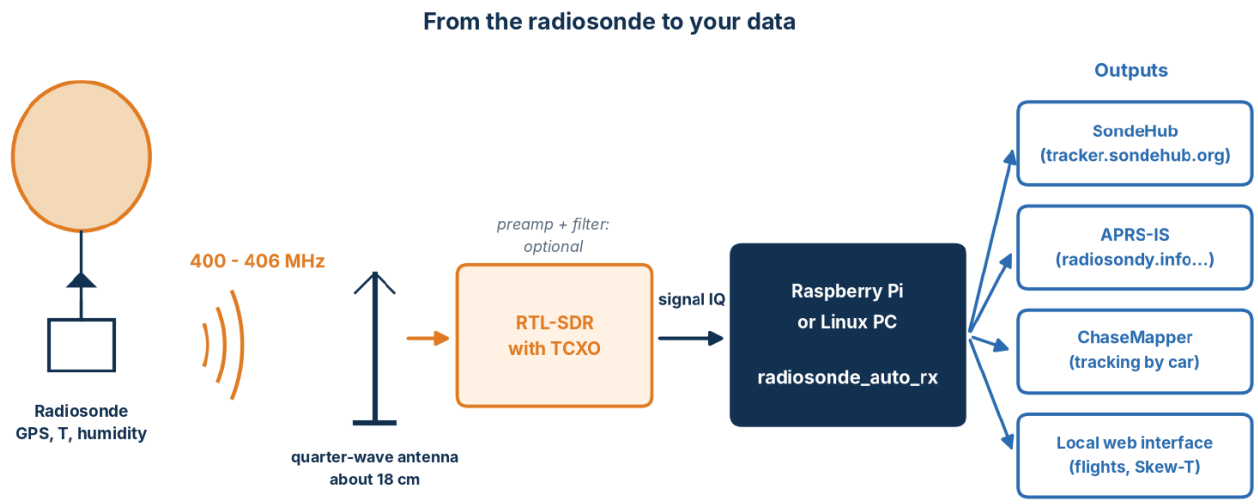
| Quantity | Source |
|---|---|
| **Temperature, relative humidity** | Onboard sensors |
| **Pressure** | Onboard sensor, but some models do not send it |
| **Position** | GPS: latitude, longitude, altitude |
| **Wind** | Derived from GPS movements |

# Radiosondes

Weather balloons: reception and tracking at 400 - 406 MHz



## From the sonde to your data

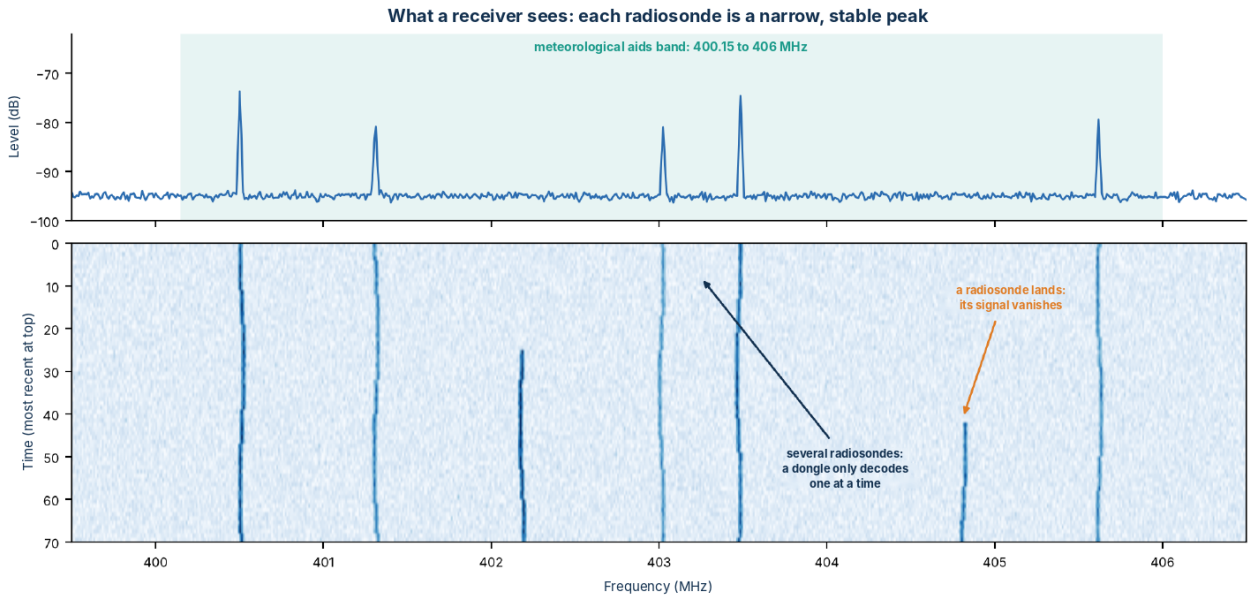



| Item | Detail |
|---|---|
| **Computer** | Raspberry Pi 3 or newer, or any Linux PC |
| **Receiver** | RTL-SDR dongle with a TCXO oscillator, drift correction (ppm) properly set (sheet 3) |
| **Several sondes at once** | One dongle decodes only one; an Airspy (via KA9Q-Radio), a HydraSDR or an SDRPlay covers the whole band |
| **Antenna** | Quarter-wave 400 - 406 MHz (sizes: sheet 1); a 70 cm amateur antenna is probably fine |
| **More range** | Preamplifier and band-pass filter recommended, good results are possible without |

| radiosonde_auto_rx installation | Detail |
|---|---|
| **Docker** | Probably the easiest |
| **Raspberry Pi image** | Preconfigured images, listed in the project wiki |
| **Native** | Raspbian Bullseye or newer, Python 3.6 or newer |
| **Configuration** | station.cfg file: station position, frequency range, upload to SondeHub (enabled by default) |

*Data sent to SondeHub is licensed CC BY-SA 2.0. The local web interface shows the station status, past flights and Skew-T diagrams.*

## The spectrum: one peak per sonde




*A sonde stays on the same frequency throughout the flight, with a slight drift: this is what lets auto_rx follow it. Several sondes are often active together.*

# Radiosondes

Weather balloons: reception and tracking at 400 - 406 MHz

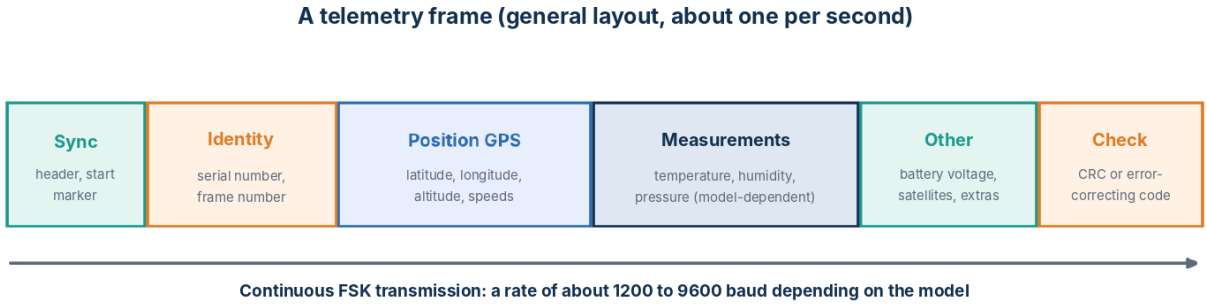


## Which sondes?

| Manufacturer | Supported models |
|---|---|
| **Vaisala** | RS41-SG, RS41-SGP, RS41-SGM; RS92-SGP, RS92-NGP |
| **Graw** | DFM-06, DFM-09, DFM-17 |
| **Meteomodem** | M10, M20 |
| **Intermet Systems** | iMet-4, iMet-54 |
| **Lockheed Martin** | LMS6-400, LMS6-1680 |
| **Meisei** | iMS-100, RS11G |
| **Others** | MRZ-H1 (400 MHz), Meteosis MTS01, Changfeng CF-06, Changwang GTH-6, Meteolabor SRS-C50 |

*The level of support varies by model and by measurement (some do not send pressure).*

## The telemetry




*The decoder locates the header, reads the bits back, verifies the error check, then converts them into physical quantities.*

```
{ "type":"RS41", "serial":"S1234567", "frame":3120, "datetime":"2026-10-15T11:02:14Z",
  "freq":403.5, "lat":48.1234, "lon":2.3456, "alt":15234, "vel_v":5.1, "temp":-55.2, "humidity":4.3 }
```

*Illustrative example, invented values: altitude in metres, vertical speed in m/s, temperature in °C.*

## Tracking and recovering

| Tool | Use |
|---|---|
| **SondeHub Tracker** | Live map of all received sondes: tracker.sondehub.org |
| **APRS-IS** | Positions viewable on radiosondy.info and wettersonde.net |
| **ChaseMapper** | Field tracking, by car, to go and fetch the sonde |
| **auto_rx web interface** | Station status, past flights, Skew-T |

## Troubleshooting

| Symptom | Leads |
|---|---|
| **No signal** | Wait for a nearby launch (00 and 12 UTC), antenna outside, check TCXO and ppm correction |
| **Intermittent decoding** | Preamplifier and filter, clear antenna, dongle away from the computer |
| **Several sondes** | Add dongles or get a wideband SDR |
| **Pressure missing** | Normal for some models |

**Good to know.** Receiving is free, but in the field do not enter private property and avoid power lines, tall trees, water and roads. Sondes belong to weather services and often carry a label saying what to do if you find one. Never transmit on these frequencies.

# Wireless sensors: rtl_433

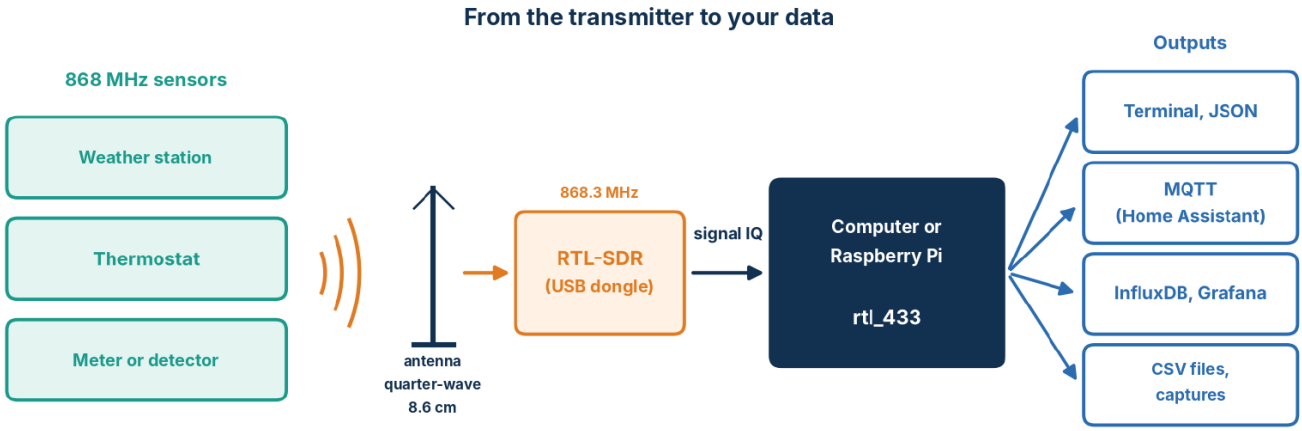
Generic decoder for 433 and 868 MHz sensors, focus on 868 MHz



**In brief.** rtl_433 is a free decoder for wireless sensors in the ISM bands: mainly 433.92 MHz, but also 868 MHz (SRD in Europe), 315, 345 and 915 MHz. It works with an RTL-SDR dongle (or any SDR supported by SoapySDR). Weather stations, thermostats, detectors and meters send short messages in the clear, without pairing, at regular intervals; rtl_433 identifies them among its decoders (numbered up to 386).

| 868.3 MHz | 250 kHz | 433.92 MHz | 386 |
|---|---|---|---|
| very common channel | default sample rate | default frequency | highest decoder number |

## How it flows

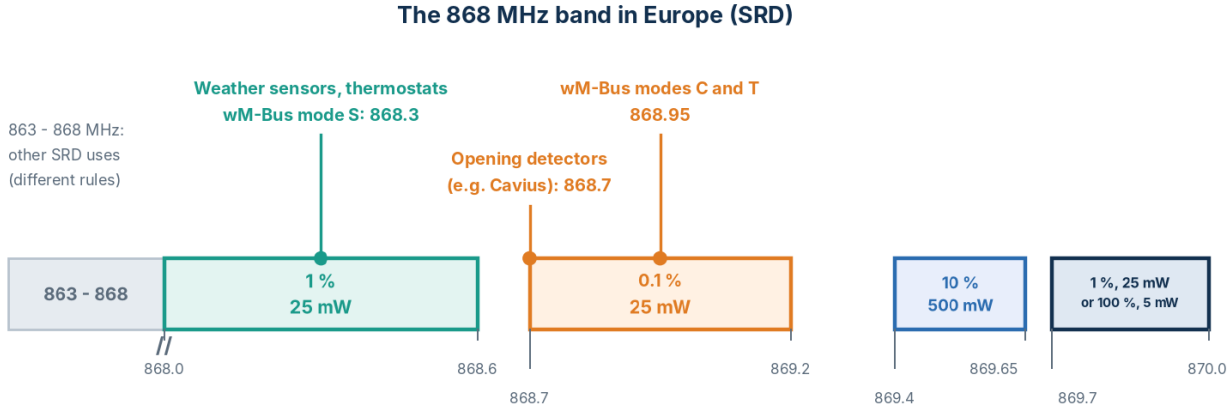



The dongle digitises the band and sends the raw signal (IQ) to rtl_433, which detects pulses, demodulates, splits frames and tests them against each decoder. Most check a checksum before publishing. Quarter-wave antenna: sizes in sheet 1.

## Install and run

| System | Command or method |
|---|---|
| **Debian, Ubuntu** | sudo apt install rtl-433 |
| **macOS** | brew install rtl_433 |
| **FreeBSD** | pkg install rtl-433 |
| **Docker** | Community images (hertzg's rtl_433_docker project) |
| **Sources** | docs/BUILDING.md in the GitHub repository merbanan/rtl_433 |
| **First try** | rtl_433 -f 868.3M: let it run for a few minutes, sensors do not transmit continuously |

## The 868 MHz band




*Maximum duty cycle and effective radiated power per sub-band (indicative values, ERC 70-03 recommendation).*

*LoRa (868.1 / 868.3 / 868.5) and Sigfox (868.13) also use this band, but rtl_433 does not decode them.*

# Wireless sensors: rtl_433

Generic decoder for 433 and 868 MHz sensors, focus on 868 MHz



Each sub-band imposes a maximum duty cycle: at 1%, a transmitter occupies the channel no more than 36 s per hour. Sensors therefore send very short messages, often every 30 to 60 s or more: be patient.

## What you find there

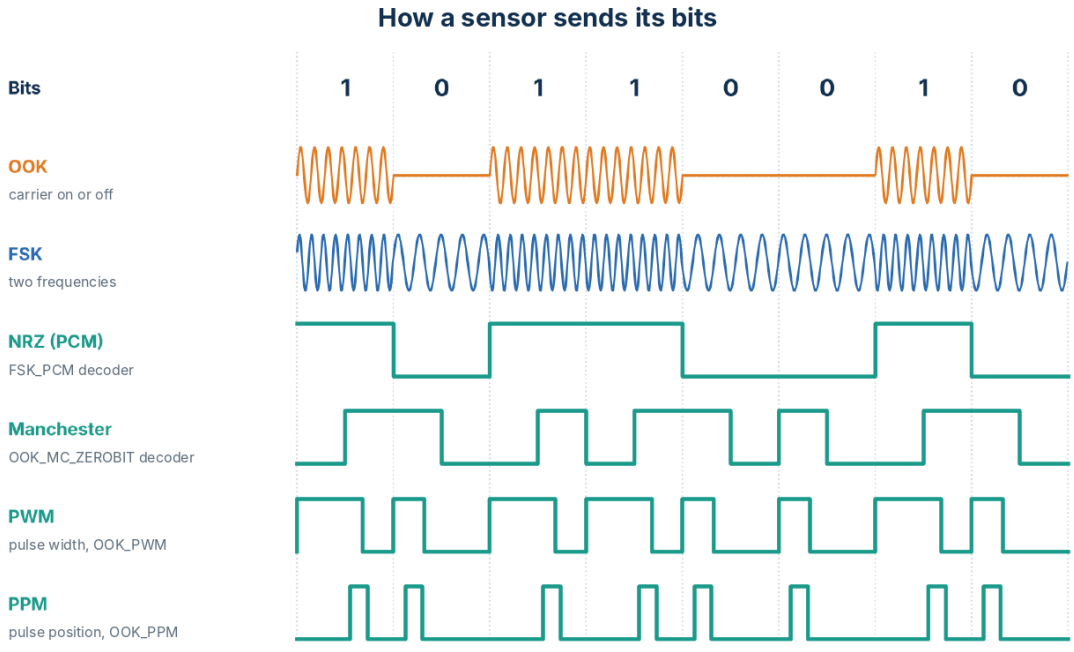
| Family | Examples (decoder no.) | Remark |
|---|---|---|
| **Weather stations** | Bresser 5-in-1, 6-in-1, 7-in-1 (119, 172, 173); Fine Offset WS80 and WS90 (213, 244); Ecowitt, LaCrosse, TFA | 868 MHz version often sold in Europe: check the model |
| **Thermostats** | Eberle Instat 868r1 (358), Watts Vision (366) | Watts Vision: 868.3 MHz |
| **Detectors** | Cavius, door and window (368) | 868.7 MHz |
| **Meters** | Wireless M-Bus modes S, T, C, R, F (104 to 107, 238) | High data rate: sample rate 1000 to 1200 kHz; data sometimes encrypted |
| **Home automation** | Somfy io-homecontrol (189) | Bidirectional, 868 MHz |

## Tuning the reception

| Option | Role |
|---|---|
| **-f 868.3M** | Listening frequency (default 433.92 MHz) |
| **-s 250k** | Sample rate (default 250 kHz); 1000k or 1200k for wM-Bus |
| **-g 0** | Automatic gain; if saturation or silence, try fixed values |
| **-p 20** | Dongle drift correction, in ppm (example value) |
| **-f 433.92M -f 868.3M -H 30** | Alternating reception, switching every 30 s (default 600 s) |

*With frequency hopping, frames from the other band go unnoticed: one dongle per band is more reliable.*

## How sensors encode their bits




*Reading conventions vary between manufacturers (inverted 1 or 0 bit, rising or falling edge).*

- The **modulation** (OOK, FSK) says what varies on the carrier; the **encoding** (NRZ, Manchester, PWM, PPM) says how bits are represented.
- At 868 MHz, FSK is often preferred to OOK, which is more common at 433 MHz: better resistance to noise and distance.
- A frame generally contains preamble, identifier, measurements and a checksum or CRC, which rtl_433 verifies and reports in the mic field.

| No. | 868 MHz decoder | Suggested setting |
|---|---|---|
| **104** | Wireless M-Bus, modes C and T, 100 kbit/s | -f 868.95M -s 1200k |
| **105** | Wireless M-Bus, mode S | -f 868.3M -s 1000k |
| **106, 107** | Wireless M-Bus, modes R and F | Disabled by default |

# Wireless sensors: rtl_433

Generic decoder for 433 and 868 MHz sensors, focus on 868 MHz



| No. | 868 MHz decoder | Suggested setting |
|---|---|---|
| **238** | Wireless M-Bus, mode T | -f 868.3M -s 1000k |
| **358** | Eberle Instat 868r1 | - |
| **366** | Watts Vision (thermostat) | -f 868.3M |
| **368** | Cavius (door and window detector) | -f 868.7M |

```
{ "time":"2026-10-14 07:12:03", "model":"Bresser-6in1", "id":12345678, "channel":0,
  "battery_ok":1, "temperature_C":14.2, "humidity":71, "mic":"CRC" }
```

*Illustrative example, invented values. Fields vary by sensor; -M level adds the signal level.*

## Useful commands

| Command | Effect |
|---|---|
| **rtl_433 -f 868.3M** | Listens on 868.3 MHz and prints decoded messages |
| **rtl_433 -f 868.3M -F json -M level -M protocol** | JSON output with level and protocol number |
| **rtl_433 -f 868.3M -s 1000k -R 105** | Decoder 105 only (wM-Bus mode S) |
| **rtl_433 -f 868.3M -F mqtt://localhost:1883** | Publishes to MQTT (Home Assistant, Node-RED) |
| **rtl_433 -f 868.3M -A** | Pulse analyser for an unknown signal |
| **rtl_433 -f 868.3M -S unknown** | Saves undecoded signals |
| **rtl_433 -r capture.cu8** | Replays a capture without hardware |
| **rtl_433 -f 868.3M -T 300** | Stops after 300 s |

*Other outputs: csv, influx, kv, syslog, http. The generic decoder -X lets you describe a simple protocol yourself.*

## Troubleshooting

| Symptom | Leads |
|---|---|
| **No message** | Wait several minutes, move the antenna closer, check the sensor's actual frequency |
| **Noise without decoding** | Gain too high or too low: -g 0 then fixed values |
| **Frequency offset** | Correct with -p or get a TCXO dongle |
| **Interference** | Move the dongle away from the computer (USB extension cable), take care over the power supply |
| **Unknown signal** | -A and -S unknown, then ask on the GitHub repository with the capture |

**Good to know.** Receiving is generally permitted, but the data from your neighbours' sensors (meters, alarms, thermostats) remains private: do not publish or exploit it. Never transmit to replay a command (shutter, door, alarm): that is intrusion. To go further: merbanan/rtl_433 repository, Home Assistant, Node-RED, InfluxDB and Grafana.

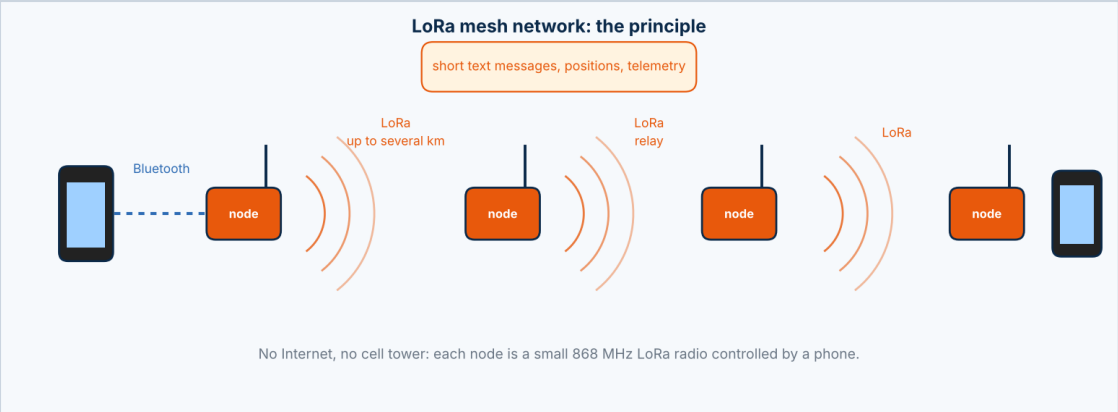
# Meshtastic

LoRa mesh network: principle, roles, EU 868 MHz settings, getting started

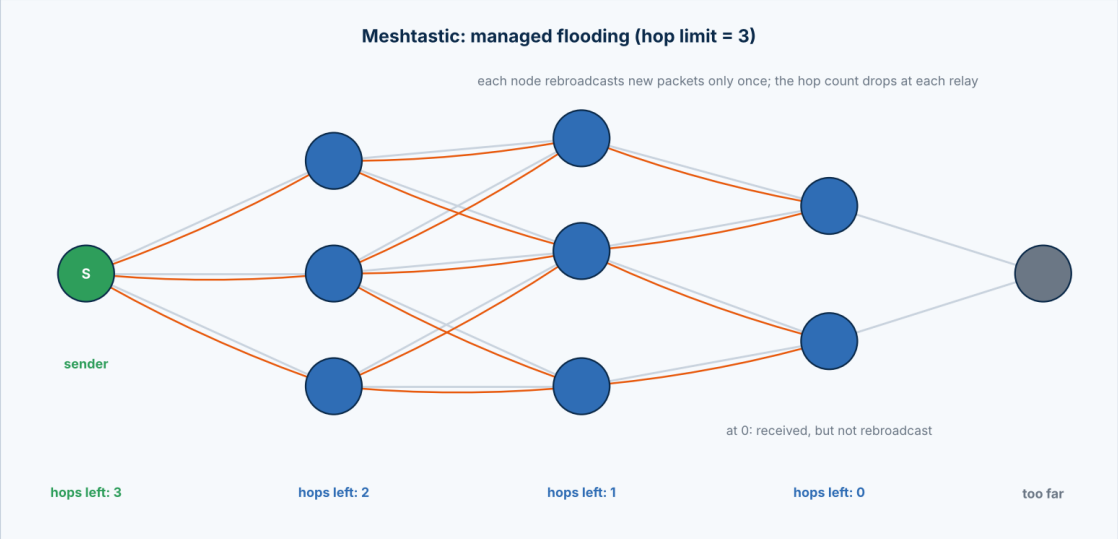
· Radiosonde

**In brief.** Meshtastic is open-source firmware that turns a small LoRa board (Heltec, LilyGO T-Beam or T-Deck, RAK, Seeed XIAO) into a messaging device that needs no Internet or cell tower, controlled from a phone. Each board is a node: it sends its own messages and relays those of others. It does not talk to MeshCore (sheet 18), even though both use the same LoRa modulation. In Europe it lives in the SRD sub-band 869.4 - 869.65 MHz, next to the 868 MHz sensors of sheet 16.

| 869.5 MHz | 10 % | 27 dBm | 3 / 7 |
|---|---|---|---|
| centre of the EU sub-band (869.4 - 869.65) | max. duty cycle | max. power (500 mW) | hops default / maximum |




## How a message travels




- **Managed flooding**: a node that hears a new packet rebroadcasts it once, unless it has already heard another node do so. The farther a node is from the sender (weak signal), the earlier it rebroadcasts, so the packet advances quickly.
- **Hop limit**: 3 by default, 7 at most. Each relay removes one hop; at zero, the packet is received but no longer rebroadcast. Raising the limit loads the network: keep 3 unless you really need more.
- **Direct messages**: since 2.6, the first message to a contact is flooded to learn the route; later ones follow the learned neighbour, which reduces congestion.
- **Encryption**: each channel is encrypted with AES using a key (PSK) shared by all its members; the default channel has a public key known to everyone. Direct messages use public-key encryption (check against your version).

## Node roles (Device Role setting)

| Role | Behaviour | Use for |
|---|---|---|
| **CLIENT** | Relays a packet only if nobody else has | Normal use with the phone (default) |
| **CLIENT_MUTE** | Takes part but does not relay | Nodes that must not load the network |
| **CLIENT_BASE** | Relays the traffic of favourite nodes | Well-placed node (roof, attic) for your other nodes |
| **ROUTER** | Infrastructure relay: rebroadcasts every packet once, visible in the list | Strategic high point, used sparingly |
| **ROUTER_LATE** | Rebroadcasts after all the others | Covering a local dead zone |
| **REPEATER** | Like ROUTER but hidden; deprecated since 2.7.11 | No longer recommended for new installations |
| **TRACKER / SENSOR** | Priority to position / telemetry | Tracking a person or object; low-power measurements |
| **TAK, TAK_TRACKER, LOST_AND_FOUND, CLIENT_HIDDEN** | Special cases: ATAK, finding a device, discretion | Special uses |

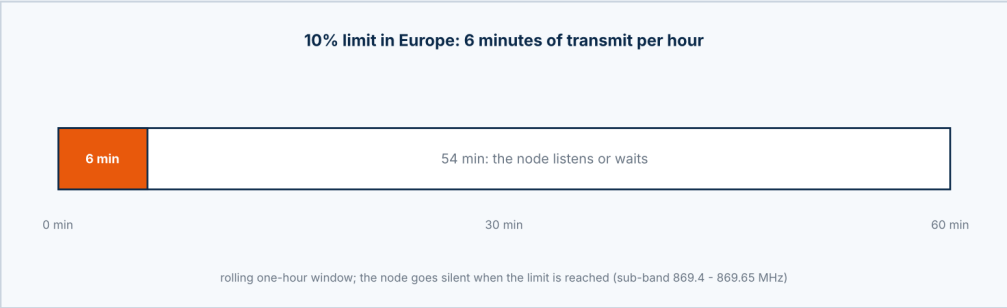
*Too many routers degrade a network: one good high point is worth more than ten routers on the ground.*

## Settings in Europe

| Region | Frequencies | Duty cycle | Max. power |
|---|---|---|---|
| **EU_868** | 869.4 - 869.65 MHz | 10 % | 27 dBm |
| **EU_N_868** | 869.4 - 869.65 MHz (narrow channel) | 10 % | 27 dBm |
| **EU_866** | 865.6 - 867.6 MHz | 2.5 % (10 % for ROUTER) | 27 dBm |
| **EU_433** | 433.0 - 434.0 MHz | 10 % | 10 dBm |

| Setting | Value and remark |
|---|---|
| **Preset (modem preset)** | LONG_FAST by default (250 kHz, SF 11, CR 5). Ranges from SHORT_TURBO (fast, short range) to LONG_MODERATE / LONG_SLOW (slow, long range). The "TURBO" presets (500 kHz) are restricted depending on the region |
| **Narrow presets** | NARROW_FAST / SLOW (62.5 kHz) for EU_N_868; LITE_FAST / SLOW (125 kHz) for EU_866; TINY (15.6 kHz) reserved for the 2 m amateur radio regions and boards with a TCXO |
| **Frequency slot** | If 0, derived from the primary channel name. All nodes must use the same slot |
| **Hop limit / power** | 3 hops; power 0 = the region's legal maximum by default |

*Nodes must share the same region, preset, slot and channel key to talk to each other. The power shown by the firmware is a software ceiling, not the legal limit in your country.*

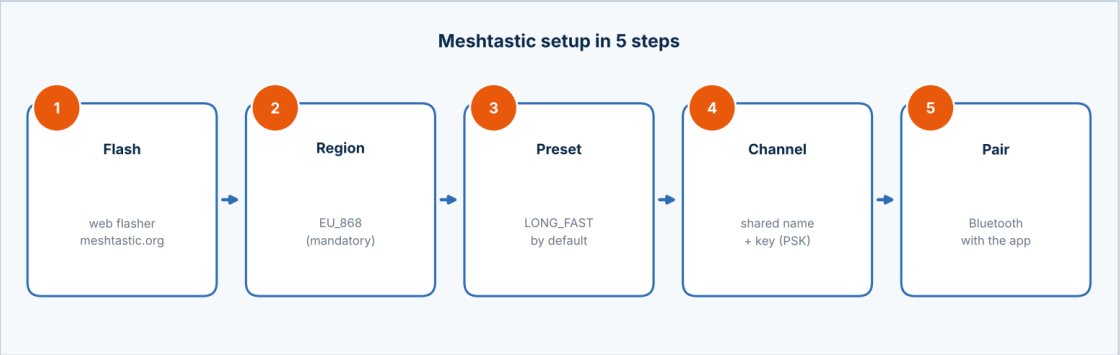


# Meshtastic

LoRa mesh network: principle, roles, EU 868 MHz settings, getting started



## Getting started




- Flash the board with the web tool on meshtastic.org, set the **region** (until it is set, the node does not transmit), then pair the Android or iOS app over Bluetooth.
- Never power a board without its antenna: the transmit stage can be damaged. Quarter-wave antenna for about 869 MHz: see sheet 1.
- Boards with an nRF52 processor (RAK4631) use the least power; ESP32 boards (Heltec V3) are cheaper but drain faster.
- Privacy: position and telemetry broadcasts use airtime. Keep intervals long; in public, avoid showing your precise position.

## What changes with 2.8 (preview version seen online)

- New nodes in the United States: default preset LONG_TURBO instead of LONG_FAST. The two cannot hear each other: to join a LONG_FAST network, select that preset again.
- Node ID derived from the public key (no longer from the MAC address), packet signing, telemetry and position disabled by default, traffic management (anti-duplicates, rate limiting) and new amateur radio regions (70 cm, 1.25 m, 2 m).
- The page consulted listed 2.8.0 as alpha and 2.8.1 as the latest release, with no date: check the stable version before flashing.

## Listening with the UV-K1 and the SDR dongle

- **UV-K1**: useless. LoRa modulation (chirps) cannot be demodulated in FM or AM; the radio only shows activity. GOGUFW Messenger (sheet A12) is the equivalent for handhelds, in FSK.
- **RTL-SDR dongle**: a waterfall centred on 869.5 MHz (SDR++ or GQRX) shows the packets as small slanted lines. They cannot be decoded with rtl_433 (sheet 16); you read them with a Meshtastic node.

**Legality.** These are transmissions: the 869.4 - 869.65 MHz sub-band imposes a 10 % duty cycle and 500 mW e.r.p. (27 dBm); antenna gain and cable losses count. An amateur radio callsign gives no extra rights on this band; encrypted messages are not allowed on the amateur bands. Check your regulator's rules.

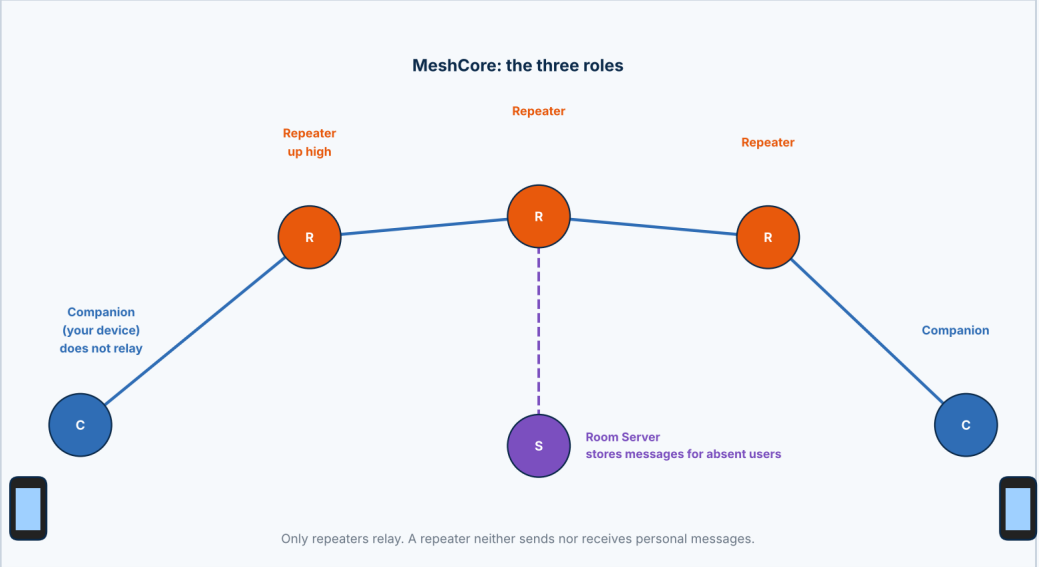
## Personal notes (hardware, channel, settings of your local network)

*Sources: Meshtastic documentation (LoRa Configuration, Device Config), firmware release notes, community guides.*

**In brief.** MeshCore is another open-source LoRa messaging firmware, designed for larger networks. Instead of having every node relay every message, it separates the roles: **personal devices (companions)** send and receive, dedicated **repeaters** relay, and a **room server** keeps messages for those who are away. The first message to a contact is flooded to discover the path; later ones follow it. MeshCore does not talk to Meshtastic (sheet 17): the two systems do not mix on the same network.

| 64 | 3 | 868 MHz | 1 byte |
|---|---|---|---|
| max. hops (direct path) | roles: companion, repeater, room | Europe version of the board | ID of a repeater in the path |

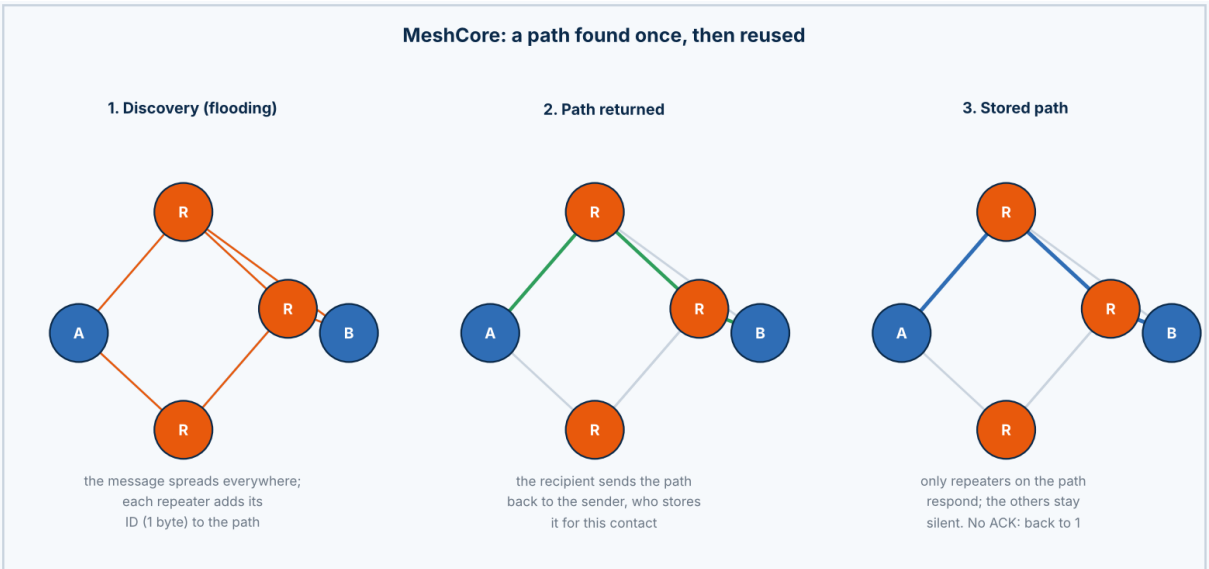
## The three roles




| Role | What it is for | Good to know |
|---|---|---|
| **Companion** | Your personal device, paired to the phone over Bluetooth | Does not relay other people's traffic. Two companions in range talk directly |
| **Repeater** | Infrastructure: relays packets | Neither receives nor sends personal messages; place it high up (roof, window), with power and an antenna |
| **Room Server** | Stores the history and delivers it to those who were offline | Anonymous login is rate-limited |

*The role is set by the firmware: to change it, you must reflash the board. Start with a companion; add infrastructure later.*

## How a message finds its path




- **1. Discovery**: with no known path, the message is flooded. Each repeater that retransmits it adds its one-byte ID.
- **2. Path return**: the recipient sends this path back reversed to the sender, who keeps it for that contact.

- **3. Direct path**: later messages carry the path. Each repeater checks whether the leading byte is its own: if so, it removes it and forwards; otherwise it stays silent.
- **Self-healing**: if no acknowledgement arrives after several tries, the path is erased and the message goes out again as a flood.
- **Filters**: a repeater ignores a packet it has already seen or whose path reaches its *flood_max* limit (adjustable from 0 to 64). The best-heard nodes retransmit first; hearing a duplicate cancels its own retransmission.

## Adverts: making yourself known

| Type | Range | Use |
|---|---|---|
| **Advert Zero Hop** | Direct neighbours only; repeaters do not retransmit it | Announce your presence locally |
| **Advert Flood** | Whole network, retransmitted by repeaters | Get discovered far away, to be used sparingly |

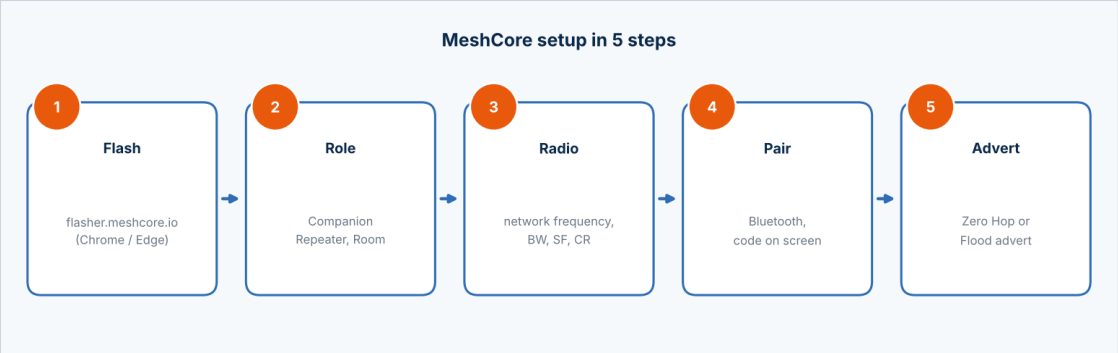
*Passive discovery takes 2 to 5 minutes; the "radio" button in the app forces an advert. Since 1.16.0, the default flood advert interval is 47 h (12 h before), and flood.max.advert limits these adverts to 8 hops by default.*

## Radio settings in Europe

| | Value | Remark |
|---|---|---|
| **Band** | 868 MHz board (Europe); 433 MHz possible | Choose the right board variant when buying |
| **Example (Essex Ham)** | 869.618 MHz, 62.5 kHz, SF 8, CR 8 | Settings vary from one network to another: a Czech group publishes 869.525 MHz, SF 7, CR 5. Ask for those of your local network |
| **Bandwidth** | 62.5 kHz by default | Narrower than Meshtastic LONG_FAST (250 kHz) |
| **Duty cycle** | Adjustable from 1 to 100 %, 50 % by default | In Europe the limit is 10 %: lower it (see the diagram on sheet 17) |
| **Client repeat (EU)** | 869.495 MHz since 1.16.0 | Replaces 869.0 MHz |

*All nodes in the network must have exactly the same frequency, bandwidth, SF and CR.*

## Getting started




- **Flash** from flasher.meshcore.io with Chrome or Edge (WebSerial; Firefox and Safari do not support it). Connect the antenna **before** powering the board, use a data cable, and hold the boot button while plugging in if the board is not detected. Choose the model, the "Companion" variant, then Flash (1 to 2 minutes).
- **Mobile app**: MeshCore on Android and iOS (the Meshtastic app does not work). Bluetooth pairing with the code shown on the screen. Community web clients also exist; the map of active nodes is at map.meshcore.io.
- **Channels**: a public channel exists at the start; for a private channel, share a code or a QR code (Channels, then Add Channel), or join a "hashtag" channel such as #yourlocalnetwork. Name your channels clearly.
- **Common hardware**: Heltec LoRa 32 V3, RAK4631, Seeed Wio Tracker L1, SenseCAP T1000-E, Seeed XIAO ESP32-S3 with SX1262, LilyGO T-Deck. nRF52 boards last several weeks on battery, ESP32 a few days.
- **Repeater**: place it as high as possible, with a good antenna. A badly placed repeater does not help; two well placed ones are often enough.

# MeshCore

LoRa mesh network with repeaters: roles, path routing, settings, getting started



## Meshtastic or MeshCore?

| | Meshtastic (sheet 17) | MeshCore |
|---|---|---|
| **Routing** | Managed flooding; direct messages: route learned since 2.6 | Flood once, then remembered path; back to flooding if the ACK is missing |
| **Hops** | 3 by default, 7 maximum | Up to 64 |
| **Roles** | All nodes relay (adjustable) | Companion does not relay; dedicated repeaters |
| **Background noise** | Periodic position, telemetry and node info | Adverts at long intervals, few automatic broadcasts |
| **Applications** | Android, iOS, web, Python CLI, ATAK, MQTT, Home Assistant | Android, iOS, community web clients, T-Deck, Home Assistant |
| **Better for** | Beginners, small group, lots of documentation, GPS and telemetry | Larger planned network, confirmed delivery, repeaters up high |

*Both use the same boards: switching from one to the other is done by reflashing. MeshCore weak points reported by users: short documentation, Bluetooth pairing sometimes temperamental, unclear channel sharing, and a companion placed between two others relays nothing, which looks like a fault.*

## Versions

- Release 1.16.0 (6 June 2026): variables flood.max.unscoped and flood.max.advert, longer preamble at low SF, ACKs extended to 6 bytes (update the **repeaters first**), power savings, new boards (T-Echo Card, T-Impulse Plus, Station G3, Heltec Mesh Node T1, new XIAO S3 variants), flood advert every 47 h.

**Legality.** The rules of sheet 17 apply here too: 10 % duty cycle, 500 mW e.r.p., no encrypted messages on the amateur bands. Do not disclose the content of other people's traffic that you overhear.

## Personal notes (local repeaters, channels, local settings)

*Sources: MeshCore 1.16.0 release notes, EastMesh wiki (routing), community guides, comparisons published in 2026.*

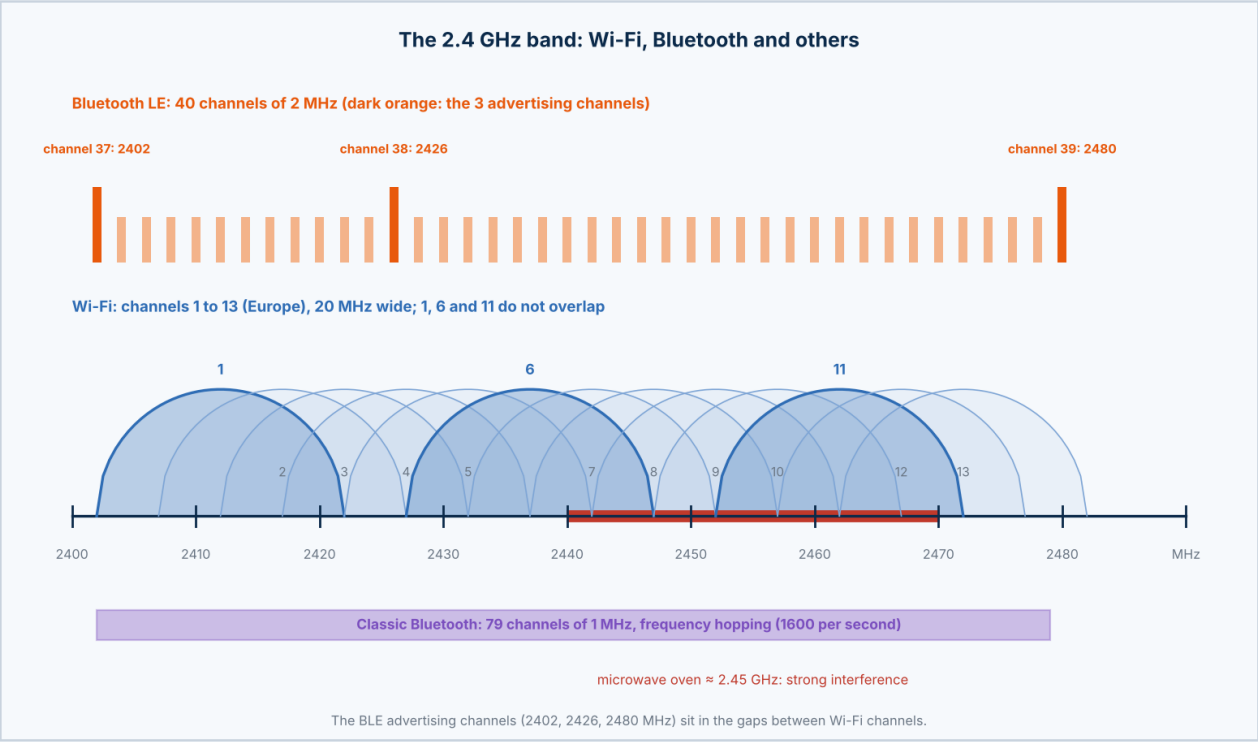
# Bluetooth and Wi-Fi

The 2.4 GHz band and 5 / 6 GHz: channels, powers, what you can observe



**In brief.** Bluetooth and Wi-Fi are the two most widespread licence-free links: they live in the **2.4 GHz** ISM bands (shared with Zigbee, microwave ovens, drones...), then **5 GHz** and **6 GHz** for recent Wi-Fi. Neither the UV-K1 nor a standard RTL-SDR dongle (limited to about 1.7 GHz) reaches these frequencies: to observe them, use a phone, an ESP32 board, a Wi-Fi adapter or a wideband SDR.

| 2.4 GHz | 40 | 13 | 100 mW |
|---|---|---|---|
| ISM band: 2400 - 2483.5 MHz | Bluetooth LE channels (2 MHz) | 2.4 GHz Wi-Fi channels in Europe | max. e.i.r.p. at 2.4 GHz |




## Bluetooth: classic and low energy

| | Classic Bluetooth (BR/EDR) | Bluetooth Low Energy (BLE) |
|---|---|---|
| **Channels** | 79 channels of 1 MHz, frequency hopping (1600 per second) | 40 channels of 2 MHz (2402 to 2480 MHz): 3 advertising (37, 38, 39) and 37 data |
| **Modulation** | GFSK, then PSK for faster data rates | GFSK |
| **Use** | Audio (headsets, speakers), car, controllers | Sensors, watches, beacons, trackers, Meshtastic and MeshCore nodes (sheets 17 and 18) |
| **Power** | Class 1: 100 mW; class 2: 2.5 mW; class 3: 1 mW | Up to 100 mW in Bluetooth 5, often 1 to 10 mW |
| **Range** | About 10 m (class 2) | A few tens of metres; the Bluetooth 5 "long range" mode goes further |

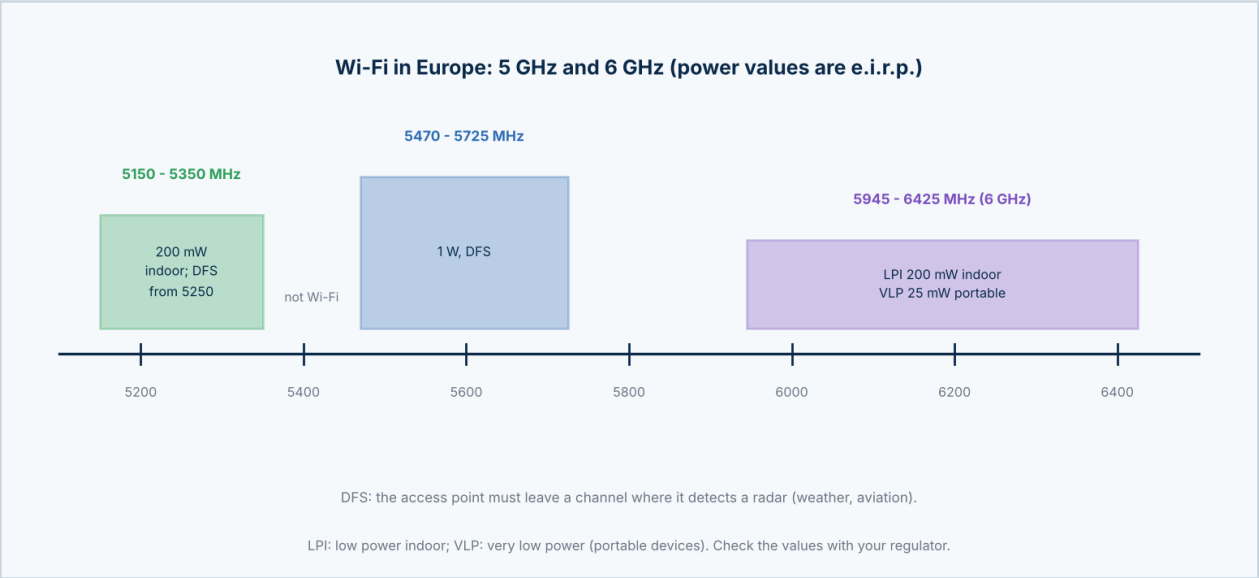
*Adaptive frequency hopping avoids the channels occupied by Wi-Fi. A BLE device announces itself on the three advertising channels; a phone scanner listens to them and shows the name, device type and signal strength (RSSI), which gives a very rough idea of the distance.*

# Bluetooth and Wi-Fi

The 2.4 GHz band and 5 / 6 GHz: channels, powers, what you can observe



## Wi-Fi: bands and channels in Europe




| Band | Channels | Max. power | Good to know |
|---|---|---|---|
| **2.4 GHz** | 1 to 13 (2412 to 2472 MHz), 20 MHz wide, one every 5 MHz; channel 14 (2484 MHz) exists only in Japan | 100 mW e.i.r.p. | Very crowded. Channels 1, 6 and 11 overlap the least |
| **5 GHz (5150 - 5350)** | Channels 36 to 64, 20 to 160 MHz wide | 200 mW e.i.r.p. | Indoor only; DFS from 5250 MHz |
| **5 GHz (5470 - 5725)** | Channels 100 to 140 | 1 W e.i.r.p. | DFS mandatory: the access point leaves a channel where it detects a radar |
| **6 GHz (5945 - 6425)** | 24 channels of 20 MHz (up to 320 MHz bonded) | LPI 200 mW (indoor); VLP 25 mW (portable) | Wi-Fi 6E and 7; no DFS; recent hardware needed |

*5 GHz channel number: centre frequency = 5000 + 5 × number (channel 36 is at 5180 MHz). The values come from European regulation as summarised by technical guides: confirm those of your country with your regulator.*

## Wi-Fi generations

| Name | Standard | Bands |
|---|---|---|
| **Wi-Fi 4** | 802.11n | 2.4 and 5 GHz |
| **Wi-Fi 5** | 802.11ac | 5 GHz |
| **Wi-Fi 6** | 802.11ax | 2.4 and 5 GHz |
| **Wi-Fi 6E** | 802.11ax | 2.4, 5 and 6 GHz |
| **Wi-Fi 7** | 802.11be | 2.4, 5 and 6 GHz, channels up to 320 MHz |

## Observing without breaking anything

| I want to see... | Tool | Remark |
|---|---|---|
| **Bluetooth LE devices around me** | Phone app (nRF Connect, LightBlue), or an ESP32 board that scans | Name, address, RSSI; does not read encrypted content |
| **Wi-Fi networks and their channels** | Wi-Fi analyser app, or analyser on a computer | Pick the least loaded channel |
| **Frame details** | Wi-Fi adapter in monitor mode, Wireshark or Kismet; BLE sniffer (nRF52840 dongle); Ubertooth for classic Bluetooth | Passive only; see the legality note below |
| **The radio signal** | Wideband SDR: HackRF One, PlutoSDR, bladeRF (about 6 GHz). A standard RTL-SDR dongle stops at about 1.7 GHz and needs a converter | Instantaneous bandwidth of about 20 MHz: one channel at a time |

*On a waterfall, 2.4 GHz Wi-Fi forms a block about 20 MHz wide that lights up in bursts; Bluetooth, narrower, hops from one channel to another; the microwave oven is a very wide block around 2.45 GHz.*

## Common interference at 2.4 GHz

- **Microwave oven** (≈ 2.45 GHz), **Zigbee, Thread and Matter** (channels 11 to 26, 2405 to 2480 MHz), wireless cameras and intercoms, drones and video links, wireless keyboards and mice.

# Bluetooth and Wi-Fi

The 2.4 GHz band and 5 / 6 GHz: channels, powers, what you can observe



- **Tips**: prefer 5 GHz for fixed devices close to the access point; on 2.4 GHz, stay on channels 1, 6 or 11; keep the access point away from the oven and from poorly shielded USB 3.0 cables.
- **Antenna**: at 2.44 GHz, a quarter wave is about 2.9 cm (formula in sheet 1, factor 0.95); at 5.5 GHz, about 1.3 cm.

**Legality.** Picking up public announcements (network names, Bluetooth advertising) is allowed, but **intercepting, recording or decrypting other people's exchanges is prohibited** in most countries, as is connecting to a network without permission. Jammers, deauthentication attacks and amplifiers or antennas that push power above the allowed limits are prohibited. Power limits vary by country; check those of your regulator.

## Personal notes (channels used, devices, measurements)

*Sources: Bluetooth LE specifications and datasheets, SigID Wiki, summaries of European Wi-Fi regulation (rfessentials, Qorvo); to confirm with your country's regulator.*

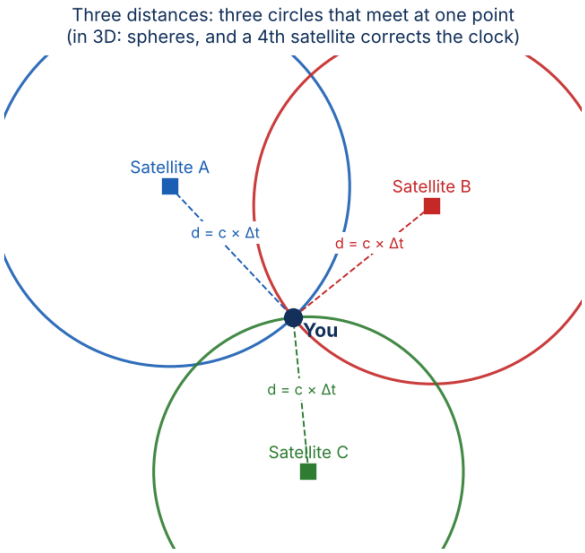
# GPS and GNSS: finding your position by satellite

GPS, GLONASS and Galileo: principle, signals and uses in this guide



**In brief. GNSS** (Global Navigation Satellite System) gives position and exact time anywhere on Earth. Three systems are covered in the guide: the American **GPS**, the Russian **GLONASS** and the European **Galileo** (the Chinese BeiDou exists too). A receiver only **receives**: it transmits nothing. The UV-K1 has no GPS; the position is entered by hand (sheet 26 for APRS) or comes from an external module, a smartphone or a beacon.

## How it works



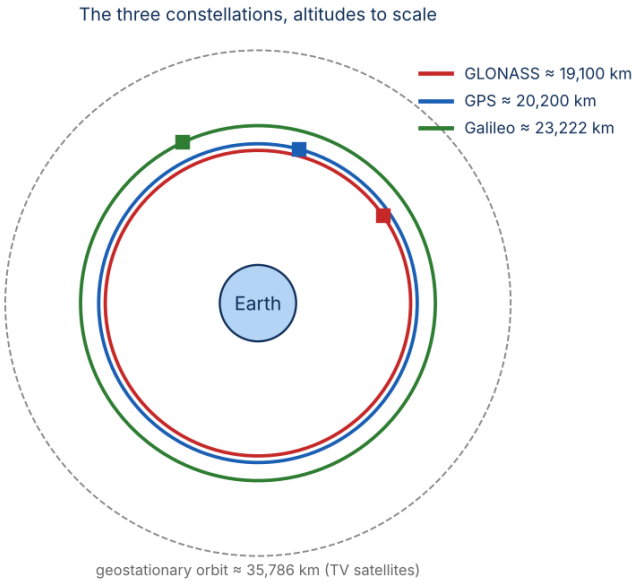

*Each satellite sends the time of its atomic clock and its position. The signal delay gives the distance, and three distances give a point.*

- **Distance**: the signal travels at the speed of light; by multiplying its delay Δt by c, the receiver gets the distance to each satellite (about 70 to 90 ms of travel).
- **Position**: three distances define a point in space; a **fourth satellite** is needed because the receiver's clock, far less accurate than an atomic clock, adds an unknown error that the calculation removes.
- **Very weak signal**: on arrival, the signal is weaker than the noise (on the order of −130 dBm). It is spread in frequency by a code (CDMA for GPS and Galileo) and then recovered by correlation, which is why active antennas are used.
- **Accuracy**: a few metres with a simple receiver, better with several constellations and two frequencies. Time is accurate to a few tens of nanoseconds: it is what synchronises networks, radiosondes and clocks.

## The three systems

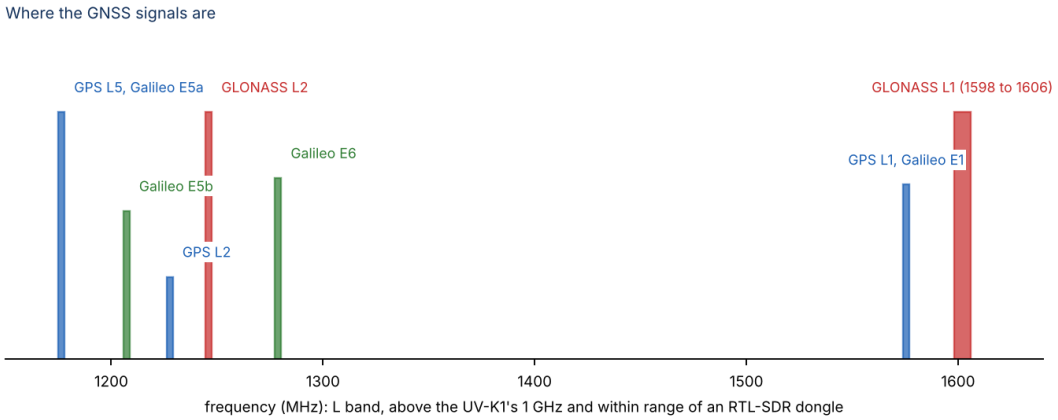
| | GPS (United States) | GLONASS (Russia) | Galileo (Europe) |
|---|---|---|---|
| **Satellites in service** | About 31 | About 24 | About 28 to 30 |
| **Altitude** | ≈ 20,200 km | ≈ 19,100 km | ≈ 23,222 km |
| **Inclination, planes** | 55°, 6 planes | 64.8°, 3 planes | 56°, 3 planes |
| **Civil frequencies** | L1 1575.42 MHz; L2 1227.60 MHz; L5 1176.45 MHz | L1 ≈ 1598 to 1606 MHz; L2 ≈ 1246 MHz | E1 1575.42; E5a 1176.45; E5b 1207.14; E6 1278.75 MHz |
| **Sharing principle** | CDMA: same frequency, different codes | Historically FDMA: one frequency per satellite (CDMA added on recent ones) | CDMA |
| **Special feature** | The oldest and most widespread | Useful at high latitudes; often combined with GPS | Civil; search and rescue service linked to 406 MHz beacons (sheet 22) |

*Rounded orders of magnitude, which change with launches and decommissioning: check the current figures (official status pages of each system).*




*Altitudes to scale: the three systems orbit between 19,000 and 23,000 km, about half as high as the geostationary orbit (sheet 31).*

## Where the signals are




*L band: above the range of the radio, but within that of an RTL-SDR dongle (up to about 1.7 GHz) with a suitable active antenna.*

## A GNSS receiver in practice

- **Modules**: small modules with a u-blox chip (NEO-6M to M10 series) or equivalent give the position over a 9600 baud serial link as **NMEA** sentences. A recent module tracks several constellations at once (GPS, GLONASS, Galileo, BeiDou), which speeds up the position fix.
- **Antenna**: a small active patch antenna (powered with 3 to 5 V by the module) laid flat, with a clear view of the sky. Inside a building or under a dense tree, the position is lost.
- **Start-up**: "cold" (no data in memory) the first fix takes from 30 seconds to a few minutes; "hot" a few seconds.
- **Time**: the module also outputs a one-second pulse (PPS), used to set a clock or a computer.

```
$GNGGA,092750.000,4807.038,N,01131.000,E,1,10,0.9,545.4,M,46.9,M,,*47
$GNRMC,092750.000,A,4807.038,N,01131.000,E,022.4,084.4,230394,003.1,W*6A
(UTC time 09:27:50; latitude 48°07.038 N; longitude 11°31.000 E; 10 satellites; altitude 545.4 m)
```

*Example NMEA sentences: GGA gives the position, altitude and number of satellites; RMC the position, speed and date. The GN prefix indicates several constellations; GP, GL and GA designate GPS, GLONASS and Galileo alone.*

## What GNSS brings to the other sheets

| Use | Role of GNSS | Sheet |
|---|---|---|
| **Radiosondes** | The sonde locates itself by GPS and sends its position: this is what lets you track and recover it. | 15 |
| **Meshtastic, MeshCore** | Nodes share their position on the network map if a GNSS module is present. | 17, 18 |
| **APRS** | A beacon sends its position from a GPS; the UV-K1 APRS Edit app receives it by hand. | 26 |
| **Distress beacons** | Modern 406 MHz beacons add their GNSS position to the message; Galileo sends back the acknowledgement. | 22 |
| **Transmitter hunting** | Position of the receiver and of the measurement point to cross-check a bearing. | 33 |
| **Radiosonde and GeoNav** | The GeoNav app computes azimuth and distance from your "home" position. | A15 |

**Jamming and spoofing.** GNSS signals are tiny: a small jammer is enough to drown them, and a fake signal can fool a receiver (spoofing). Conflict zones, ports and the surroundings of sensitive sites suffer from this regularly. As a receiver user, never trust a single source for safety: compare with a map, a watch, a compass. It is illegal to transmit on these frequencies.

## Personal notes (module used, antenna, position tests)

# Emergency frequencies

Receivable from 1 kHz to 20 GHz: frequency, modulation, coding, listening with the UV-K1



**Reading the table. Direct**: voice can be heard as is (AM, FM, SSB). **External decoding**: you hear data sounds to feed into a decoder (PC, smartphone, SDR). **Detection only**: a burst or a pulse, with no readable content. **Not receivable**: encrypted or out of reach of an ordinary receiver.

| Frequency | Modulation | Coding / format | Use | Listening |
|---|---|---|---|---|
| **MF AND HF (1 kHz to 30 MHz): nothing receivable below 457 kHz** | | | | |
| **457 kHz** | OOK | Short pulse about every second | Avalanche transceivers (victim detectors) | Detection only |
| **500 kHz** | A1A (CW) | Morse | Former distress frequency, abandoned in 1999 | Historical |
| **518 · 490 · 4209.5 kHz** | FSK 100 bd, 170 Hz | SITOR-B, 7 elements | NAVTEX: notices, distress and relayed SAR (sheet 14) | External decoding |
| **2174.5 kHz** | FSK 100 bd | NBDP (radio telex) | MF printer distress | External decoding |
| **2182 kHz** | SSB, USB | Voice | MF distress and calling | Direct |
| **2187.5 kHz** | FSK 100 bd | DSC (ITU-R M.493) | MF digital alert (sheet 14) | External decoding |
| **3023 · 5680 kHz** | SSB, USB | Voice | On-scene SAR, aircraft and ships | Direct |
| **4125 · 6215 · 8291 · 12290 · 16420 kHz** | SSB, USB | Voice | Maritime HF voice distress (4125 also SAR) | Direct |
| **4177.5 · 6268 · 8376.5 · 12520 · 16695 kHz** | FSK 100 bd | NBDP | HF printer distress | External decoding |
| **4207.5 · 6312 · 8414.5 · 12577 · 16804.5 kHz** | FSK 100 bd | DSC | HF DSC alerts, worldwide range (sheet 14) | External decoding |
| **8364 kHz** | SSB, USB | Voice | Lifeboats and life rafts | Direct |
| **3760 · 7110 (LSB) · 14 300 kHz (USB)** | SSB | Voice | Amateur emergency networks, IARU Region 1; 18,160 and 21,360 kHz in USB | Direct |
| **27.065 MHz (CB 9)** | | | | |
| **VHF (30 - 300 MHz)** | | | | |
| **121.500 MHz** | AM | Voice; ELT beacons: siren 1600 → 300 Hz | Aviation distress, final homing of beacons | Direct |
| **123.100 MHz** | AM | Voice | On-scene SAR, aviation | Direct |
| **156.800 MHz (channel 16)** | FM | Voice | Maritime distress, urgency, safety; CROSS on watch | Direct |
| **156.525 MHz (channel 70)** | FM, AFSK 1200 bd | DSC | Maritime digital alert (sheet 14) | External decoding |
| **156.300 MHz (channel 6)** | FM | Voice | Ship to ship; SAR coordination | Direct |
| **161.975 · 162.025 MHz** | GMSK 9600 bd | AIS (sheet 13) | AIS-SART, AIS MOB, AIS EPIRB in distress | External decoding |
| **144.800 MHz** | AFSK 1200 bd | AX.25 (APRS) | Positions and messages used in emergencies | External decoding |
| **145.500 MHz** | FM | Voice | Amateur VHF calling, emergency networks | Direct |
| **162.400 - 162.550 MHz** | FM | SAME: AFSK 520.83 bd | NOAA Weather Radio, alerts (United States, not in France) | External decoding |
| **UHF (300 MHz - 3 GHz)** | | | | |
| **243.000 MHz** | AM | Voice | Military distress | Direct |
| **225 - 400 MHz** | AM; HAVE QUICK | Voice, sometimes encrypted | Military aviation and emergencies | Direct (partial) |
| **380 - 400 MHz** | TETRAPOL / TETRA | Encrypted digital | French emergency services (ANTARES, INPT): SDIS, SAMU, police | Not receivable |
| **406.0 - 406.1 MHz** | BPSK ±1.1 rad, 400 bit/s | Manchester, 112 or 144 bits, BCH | EPIRB, PLB, ELT: Cospas-Sarsat (sheet 22) | Detection only |
| **433.500 MHz** | FM | Voice | Amateur UHF calling, emergency networks | Direct |

# Emergency frequencies

Receivable from 1 kHz to 20 GHz: frequency, modulation, coding, listening with the UV-K1



| Frequency | Modulation | Coding / format | Use | Listening |
|---|---|---|---|---|
| **1090 MHz** | PPM, 1 Mbit/s | Mode S, 112 bits, CRC-24 | ADS-B: emergency (squawk 7700, sheet 10) | External decoding |
| **1537 - 1545 MHz** | BPSK 600 bd | Inmarsat-C, SafetyNET | Maritime alerts by satellite | Dedicated antenna |
| **1544.1 · 1544.5 MHz** | Satellite link | 406 MHz translation | Cospas-Sarsat downlink | Not receivable |
| **SHF (3 - 20 GHz): nothing receivable with an ordinary receiver** | | | | |
| **9.2 - 9.5 GHz** | X-band radar | Reply: 12 dots on the radar screen | Radar SART | Not receivable |

## With the UV-K1

VHF and UHF in AM and FM depending on the firmware range; HF with the SI4732 in AM and SSB. FM can only be received on the VHF/UHF part. Data sounds go through an audio cable or the microphone to a decoder (sheet 3).

## The three levels of voice call (said three times)

| Word | Meaning |
|---|---|
| **MAYDAY** | Distress: life or vessel in grave and imminent danger |
| **PAN-PAN** | Urgency: serious situation without imminent danger |
| **SECURITE** | Important safety notice (weather, obstacle) |

## If you hear a real distress

- Note the time, the position, the vessel's name or MMSI and the nature of the problem.
- Listen for a CROSS or a ship acknowledging.
- No acknowledgement: call the CROSS on 196 (at sea) or 112, by phone.
- Respect the confidentiality of other exchanges: do not disclose their content.

**Reminder.** Never transmit on a distress frequency without real need: false alerts are punished and tie up rescue services.

# Distress beacons

EPIRB, PLB, ELT, AIS-SART, radar SART: signals, alert and listening



## The beacon families

| Type | For whom | Activation | Signal | Frequency |
|---|---|---|---|---|
| **EPIRB** | Ships, boats | Automatic (hydrostatic) or manual | Digital burst + homing | 406 MHz; homing 121.5 MHz |
| **PLB** | People (mountain, hiking, boating) | Manual | Digital burst + homing | 406 MHz; 121.5 MHz |
| **ELT** | Aircraft | Automatic on impact or manual | Digital burst + homing | 406 MHz; 121.5 MHz (older: 243 MHz) |
| **AIS-SART** | Life rafts and survival craft | Manual | AIS message "SART ACTIVE" (MMSI 970xxxxxx) | 161.975 and 162.025 MHz |
| **AIS MOB / AIS EPIRB** | Man overboard, boat beacon | Manual or automatic | AIS message (MMSI 972 / 974xxxxxx) | 161.975 and 162.025 MHz |
| **Radar SART** | Rafts, boats | Manual | Replies to a radar: 12 aligned dots | 9.2 - 9.5 GHz (X band) |

## The 406 MHz signal

| Parameter | Value |
|---|---|
| **Band** | 406.0 - 406.1 MHz (beacon channels around 406.025 - 406.076 MHz) |
| **Modulation** | BPSK, phase ±1.1 rad, 400 bit/s; Manchester; no usable FM |
| **Burst** | About 0.5 s, repeated every ~50 s (interval slightly random) |
| **Frame** | 112 or 144 bits, BCH correction: unique ID, country (MID), beacon type, position if GPS |
| **Position** | With GPS: a few tens of metres. Without GPS: Doppler effect, a few km |
| **Final homing** | 121.5 MHz in AM: descending two-tone sound, continuous, for rescuers with a direction finder |

## The alert path

| Step | Role |
|---|---|
| **1. Beacon** | Sends the burst on 406 MHz |
| **2. Cospas-Sarsat satellites** | Low orbit (Doppler), geostationary (immediate alert), medium orbit (precise position) |
| **3. Ground station and control centre** | Compute the position, identify the owner in the registration database |
| **4. CROSS (in France)** | Contacts the owner to rule out a false alert, then launches the rescue |

## Listening with the UV-K1 and the SDR dongle

- **406 MHz**: detection only. In FM on 406.025 - 406.076 MHz, at best you hear a brief crackle and the S-meter jumps. BPSK can only be decoded with an SDR and a decoder such as dec406. A burst detector (squelch + 50 s timer) fits in a small app (sheet 32).
- **121.5 MHz**: AM mode (AIR); continuous descending two-tone sound.
- **AIS-SART** (161.975 / 162.025 MHz): cannot be decoded with the UV-K1; SDR + AIS decoder (sheet 13).
- Registration is mandatory (in France, with the ANFR); battery of about 5 years; test only with the "self-test" function, never by triggering the beacon.
- Never transmit on 406 MHz. If you pick up a real alert, it is not a test: warn the CROSS (196) or 112.

# HF DATA modes

FT8, FT4, PSK31, Olivia, Feld Hell: signal, settings and decoding

| Mode | Modulation | Speed / duration | Width | Coding | Reference (kHz, USB) | Decoders |
|---|---|---|---|---|---|---|
| **FT8** | 8-GFSK, 8 tones spaced 6.25 Hz apart | 6.25 bd; 15 s slots (12.6 s of transmission) | 50 Hz | 77 bits + CRC 14, LDPC (174,91), 79 symbols including 21 sync | 14,074 (20 m) | WSJT-X, JTDX; Android: FT8CN |
| **FT4** | 4-GFSK, 4 tones spaced 20.83 Hz apart | 20.83 bd; 7.5 s slots (4.5 s) | ~90 Hz | Same coding as FT8, 103 symbols | 14,080 (20 m) | WSJT-X, JTDX |
| **PSK31** | BPSK 31.25 bd (QPSK31 with FEC) | ~50 words/min | ~31 Hz | Varicode, no FEC | 14,070 (20 m) | fldigi, MultiPSK |
| **Olivia** | MFSK 4 to 64 tones with FEC | 8/250, 16/500, 32/1000 (tones/Hz) | 125 to 2000 Hz | Block FEC, Walsh-Hadamard | Digital segments (e.g. 14,070 - 14,099) | fldigi, MultiPSK |
| **Feld Hell** | OOK, ~980 Hz tone | 122.5 bd, ~2.5 char./s | ~250 Hz | None: text drawn as pixels | Digital segments (e.g. 14,063) | fldigi, MultiPSK |

## By ear and on screen

| Mode | What you perceive | Threshold |
|---|---|---|
| **FT8** | Steady chirping, starting at seconds 00 / 15 / 30 / 45; clock within ±1 s (NTP or GPS) | -21 dB (2500 Hz bandwidth) |
| **FT4** | Faster than FT8, designed for contests | -17.5 dB |
| **PSK31** | Soft purring, very thin trace in the waterfall; variants PSK63, PSK125, PSKxxxR (with FEC) | |
| **Olivia** | Burst of gliding tones; very robust (fading, QRM) | ~-14 dB |
| **Feld Hell** | Fast Morse-like rhythm; the text appears as an image (double line) that you read by eye | |

## Settings and UV-K1

- USB, moderate volume without clipping, 2.4 to 3 kHz filter. For FT8 and FT4, the PC or phone clock must be synchronised (NTP or GPS).
- UV-K1: HF with the SI4732 in USB (LSB below 10 MHz), audio to a decoder on a PC or smartphone by cable or microphone. None of these modes can be decoded inside the radio.
- Decoding FT8 or FT4 with a home-made FSK decoder is possible in principle, but needs clock synchronisation and LDPC decoding: much heavier than RTTY.
- Frequencies of all the amateur radio bands: sheet 6. Transmitting as a ham requires a callsign.

# RTTY

Radioteletype: listening and decoding sheet

· DATA mod

**What is it?** Text transmission in FSK: two tones alternate, one for bit 1 (mark), the other for bit 0 (space). It is the oldest digital mode still active, used by amateur radio (mostly in contests), for weather and by commercial or military stations.

| Parameter | Value |
|---|---|
| **Coding** | Baudot (ITA2): 5 bits, two character sets (letters LTRS and figures FIGS) |
| **Frame** | 1 start bit (space) + 5 bits (least significant first) + 1.5 stop bit (mark): 7.5 bits per character |
| **Speed** | 45.45 bd (amateur radio, ~60 words/min); 50 bd (weather, Europe); 75 and 100 bd |
| **Shift** | 170 Hz (amateur radio); 425 / 450 Hz; 850 Hz (commercial, military) |
| **Usual tones** | 2125 Hz (mark) / 2295 Hz (space) for 170 Hz shift |
| **Width** | About 250 to 300 Hz for 170 Hz / 45.45 bd |
| **Sideband** | LSB by convention. In USB, mark and space are swapped: enable "Reverse" in the decoder |

## Where to listen

| Frequency | Parameters | Remark |
|---|---|---|
| **3.580 - 3.600 MHz (80 m)** | 45.45 bd, 170 Hz, LSB | Amateur radio |
| **7.035 - 7.045 MHz (40 m)** | 45.45 bd, 170 Hz, LSB | Amateur radio |
| **14.080 - 14.100 MHz (20 m)** | 45.45 bd, 170 Hz, LSB | The most active, crowded during contests |
| **21.080 - 21.100 / 28.080 - 28.100 MHz** | 45.45 bd, 170 Hz, LSB | 15 m and 10 m depending on propagation |
| **4583 · 7646 · 10,100.8 kHz** | 50 bd, 450 Hz, LSB | DWD weather; some transmissions may have been stopped |
| **518 · 490 kHz** | 100 bd, 170 Hz (SITOR-B) | NAVTEX: sheet 14 |

## Baudot ITA2 alphabet (LTRS set)

| A | B | C | D | E | F | G | H | I | J | K | L | M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 00011 | 11001 | 01110 | 01001 | 00001 | 01101 | 11010 | 10100 | 00110 | 01011 | 01111 | 10010 | 11100 |
| N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| 01100 | 11000 | 10110 | 10111 | 01010 | 00101 | 10000 | 00111 | 11110 | 10011 | 11101 | 10101 | 10001 |

*Commands: LTRS 11111 · FIGS 11011 · space 00100 · CR 01000 · LF 00010. In the FIGS set, the keys Q W E R T Y U I O P give 1 2 3 4 5 6 7 8 9 0. Classic test: RYRYRY.*

## Listening and decoding

- By ear: two notes alternating non-stop, about 6 characters per second at 45.45 bd, with no pause between packets (unlike APRS).
- UV-K1 (SI4732): LSB on 14.080 - 14.100 MHz, audio to fldigi (PC), MMTTY or 2Tone (Windows). See sheet A06.
- Garbled text: try "Reverse", then check the speed (45.45 or 50 bd) and the shift (170 or 450 Hz). Typical CQ: "CQ CQ DE F4ABC F4ABC K".

# CCIR Selcall

Selective calling by sequential tones: listening and decoding sheet



**What is it?** Selcall calls a specific station with a series of audio tones: one digit = one frequency. The receiver only rings if it recognises its number (4 or 5 digits). The CCIR set is used on HF and VHF: commercial networks, NGOs, remote medical services, fleets. Not to be confused with DTMF (two simultaneous tones): here only one tone at a time, several in sequence. The signal goes through the audio channel, in FM (VHF/UHF) or SSB (HF); the frequency depends on the network.

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| 1981 | 1124 | 1197 | 1275 | 1358 | 1446 | 1540 | 1640 |
| 8 | 9 | A | B | C | D | E (repeat) | |
| 1747 | 1860 | 2400 | 930 | 2247 | 991 | 2110 | |

*CCIR tones in Hz. E marks a repeated digit; A, B, C, D are service or group tones depending on the network. From one tone to the next the gap is 70 to 120 Hz: a good signal-to-noise ratio is needed.*

## The signal at a glance

| Parameter | Value |
|---|---|
| **Tone duration** | 100 ms (CCIR-1, the most common) or 70 ms |
| **Call length** | 4 to 6 tones, i.e. ~0.5 to 0.6 s for 5 tones at 100 ms |
| **Repeat** | Two identical consecutive digits: the second becomes E (2110 Hz) |
| **Transmission** | Audio over FM or SSB, before or after the voice message |

| Number | Sequence (Hz) | Remark |
|---|---|---|
| **12345** | 1124 - 1197 - 1275 - 1358 - 1446 | Five different digits |
| **11234** | 1124 - 2110 - 1197 - 1275 - 1358 | The second "1" becomes E |
| **55021** | 1446 - 2110 - 1981 - 1197 - 1124 | Repeat of the 5, then 0, 2, 1 |

| Other set | Tone duration | Reference |
|---|---|---|
| **CCIR** | 100 or 70 ms | Digits 1124 to 1981 Hz, repeat 2110 Hz |
| **ZVEI-1** | 70 ms | Digits 1060 to 2400 Hz, repeat 2600 Hz (Germany, fire services, industry) |
| **EEA** | 40 ms | Same range as CCIR, shorter tones (United Kingdom) |
| **EIA** | 33 ms | Digits from 600 to 1869 Hz (United States) |

*To identify the set: measure the duration of a tone and the frequency of the highest tone.*

## Listening and decoding

- By ear: a burst of 5 short notes of different pitches in just over half a second, like a short melody, often at the start of an exchange.
- UV-K1: FM (or SSB on HF with the SI4732) on the network frequency, low squelch so as not to miss the call. The radio does not decode selcall.
- Decoding: sequential-tone decoder on a PC or smartphone; Goertzel detection on the 15 frequencies, 100 ms window, is enough. Simple output format: "11234".
- Idea for SigID (sheet 32): recognise 4 to 6 tones of 100 ms among these 15 frequencies and display "CCIR Selcall".

# APRS

Automatic Packet Reporting System: listening and decoding sheet



**What is it?** Amateur radio network for positions, messages, weather and telemetry, created by Bob Bruninga (WB4APR). Stations send short packets, relayed by digipeaters and iGates to the Internet (APRS-IS, visible on aprs.fi). Packets are AX.25 UI in 1200 baud AFSK (Bell 202) on a VHF FM carrier. Transmitting requires an amateur radio callsign.

| Area | Frequency | Remark |
|---|---|---|
| **Europe / France** | 144.800 MHz | FM, 1200 bd: the main one to monitor |
| **ISS (digipeater)** | 145.825 MHz | FM, 1200 bd, when active; Doppler ±3.5 kHz |
| **North America** | 144.390 MHz | FM, 1200 bd |
| **Australia / Japan** | 145.175 / 144.640 MHz | FM, 1200 bd |
| **HF (30 m)** | 10.1476 MHz | USB, 300 bd, long distance |

| Parameter | Value |
|---|---|
| **Modulation** | AFSK: 1200 Hz (mark) / 2200 Hz (space), 1200 bd, over FM |
| **Coding** | NRZI, bit stuffing, 0x7E flags, CRC-16 (FCS) |
| **Frame** | AX.25 UI: source, destination, path (digipeaters), data |
| **Timing** | One beacon every 1 to 10 min, faster when moving (smart beaconing) |

## A decoded packet

```
F4ABC-9>APRS,WIDE1-1,WIDE2-1:!4852.00N/00220.00E>Test mobile
```

| Element | Meaning |
|---|---|
| **F4ABC-9** | Source callsign (fictitious), SSID 9 = mobile |
| **APRS** | Generic destination (identifies the software, or encodes the position in Mic-E) |
| **WIDE1-1,WIDE2-1** | Requested path: up to 2 relays |
| **!** | Position without timestamp or messaging |
| **4852.00N / 00220.00E** | 48°52.00' N, 002°20.00' E |
| **/ ... >** | Symbol: primary table + code > = car |
| **Test mobile** | Free comment |

| SSID | Use | Start | Type |
|---|---|---|---|
| **-1** | Digipeater / fixed | ! or = | Position |
| **-5** | Smartphone | / or @ | Position + time |
| **-7** | Handheld | ` or ' | Mic-E position |
| **-9** | Mobile | : | Message |
| **-10** | Internet iGate | ; | Object |
| **-11** | Balloon / aircraft | _ | Weather |
| **-15** | Generic | > | Status |

## Listening with the UV-K1

- 144.800 MHz, narrow FM, squelch open or low: short bursts of "screeching" lasting 0.5 to 1 s.
- The radio does not decode APRS: audio to Direwolf, APRSdroid or Xastir by cable or microphone (TNC link: sheet 27). To see stations received by others: aprs.fi.

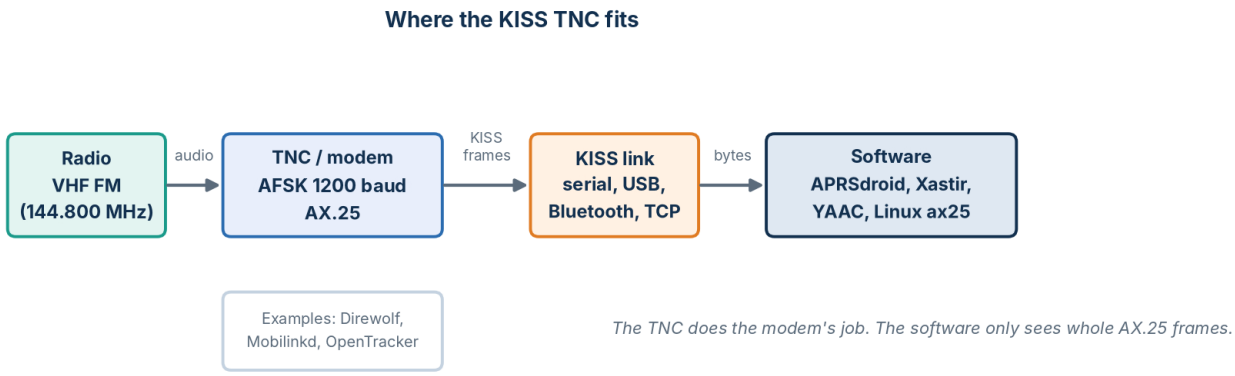
# KISS TNC

KISS protocol between a TNC and software: frame, commands, connection, APRS

**In brief.** KISS (Keep It Simple, Stupid) is a very simple protocol between a TNC (Terminal Node Controller, the packet modem) and a computer or phone. The TNC handles the radio side: 1200 baud AFSK modulation, detection, AX.25 frames. KISS only carries whole frames, byte by byte, over a serial, USB, Bluetooth or TCP link. All the intelligence (APRS, connected packet, BBS) stays in the software. Created by Mike Chepponis and Phil Karn in 1987, it is the most widespread interface between a TNC and APRS software (sheet 26).

| 0xC0 | 1200 bd | 8001 | 1987 |
|---|---|---|---|
| FEND: start and end of frame | AFSK on VHF, APRS | Direwolf KISS TCP port | year of the specification |

## Where the TNC fits

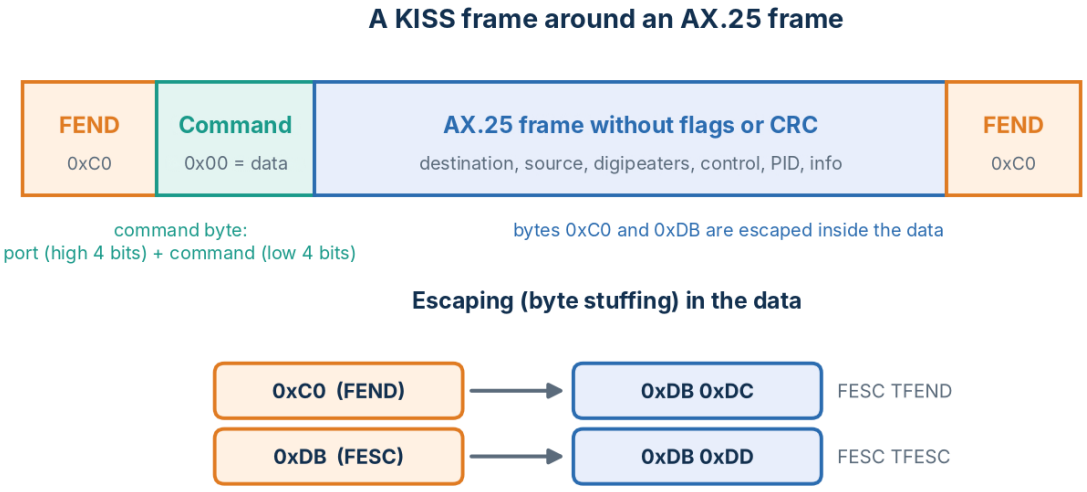



*The TNC does the modem's job. The software only sees whole AX.25 frames.*

A TNC can be a box (Mobilinkd, OpenTracker, Kantronics), a board (TNC-Pi) or software: Direwolf turns a computer's sound card into a TNC and exposes a KISS interface.

| Layer | Role | Who handles it |
|---|---|---|
| **APRS** | Meaning of the data: position, message, weather | Software (APRSdroid, Xastir, YAAC) |
| **AX.25** | Addresses, relays, frame type, CRC | TNC (the software reads the addresses in the frame) |
| **KISS** | Transport of frames between TNC and software | Both ends of the link |
| **AFSK 1200** | 1200 / 2200 Hz tones over FM, channel access | TNC and radio |

## The KISS frame




*Without this escaping, a 0xC0 byte in the data would be taken for the end of the frame.*

- FEND (0xC0) opens and closes each frame. Two consecutive FENDs are ignored: you can send an extra one to resynchronise.
- The command byte carries the port number (high 4 bits, for a TNC with several radios) and the command (low 4 bits).
- The data is the AX.25 frame without flags or CRC: the TNC adds them when transmitting and removes them when receiving.

- In the data, 0xC0 becomes 0xDB 0xDC and 0xDB becomes 0xDB 0xDD (escaping).

## The commands

| Code | Name | Role | Usual value |
|---|---|---|---|
| **0x00** | Data | AX.25 frame to transmit, or received | - |
| **0x01** | TXDELAY | Delay between transmitter key-up and the first bit, in units of 10 ms | 50 (500 ms) |
| **0x02** | P | Persistence: probability of transmitting when the channel is free, (P + 1) / 256 | 63 (25 %) |
| **0x03** | SlotTime | Length of a waiting slot, in units of 10 ms | 10 (100 ms) |
| **0x04** | TXtail | Delay after the end of the frame (obsolete) | 0 |
| **0x05** | FullDuplex | 0 = half duplex, any other value = full duplex | 0 |
| **0x06** | SetHardware | Hardware-specific setting of the TNC | model-dependent |
| **0xFF** | Return | Leaves KISS mode (depending on the TNC) | - |

*Default values from the specification; Direwolf sets these parameters in its configuration file.*

## A decoded example

```
C0 00 82 A0 A4 A6 40 40 60 8C 68 82 84 86 40 72 AE 92 88 8A 62 40 62
AE 92 88 8A 64 40 63 03 F0 21 34 38 35 32 2E 30 30 4E 2F 30 30 32 32
30 2E 30 30 45 3E 54 65 73 74 C0
```

| Bytes | Content |
|---|---|
| **C0 00** | FEND, command 0: data frame on port 0 |
| **82 A0 A4 A6 40 40 60** | Destination APRS, SSID 0: each letter is shifted one bit to the left (0x41 << 1 = 0x82) |
| **8C 68 82 84 86 40 72** | Source F4ABC-9 (fictitious): the SSID 9 is in byte 72 |
| **AE 92 88 8A 62 40 62** | Relay WIDE1-1 |
| **AE 92 88 8A 64 40 63** | Relay WIDE2-1: the least significant bit set to 1 marks the end of the addresses |
| **03 F0** | UI frame (03), protocol "no layer 3" (F0) |
| **21 34 38 … 54 65 73 74** | ASCII text: !4852.00N/00220.00E>Test, APRS position without timestamp |
| **C0** | FEND: end of the frame |

*Illustrative example, invented callsign, calculated by hand. No byte equals 0xC0 or 0xDB: nothing to escape.*

## Connecting and using

| Link | Detail |
|---|---|
| **Serial or USB** | Often 9600 baud, 8N1; some TNCs accept 115200. Same speed on both sides |
| **Bluetooth** | Mobilinkd and others: serial profile. On Android, APRSdroid picks the TNC from the paired list |
| **TCP** | Direwolf listens on port 8001 by default; the software connects to the computer's address |
| **Linux** | kattach or kissattach (ax25-tools package) creates a network interface ax0 from the KISS link |

| Item | Use |
|---|---|
| **Direwolf** | Software TNC: sound card to KISS (TCP, pseudo-terminal, serial), APRS iGate and digipeater |
| **APRSdroid** | Android: KISS TNC over Bluetooth, USB or TCP |
| **Xastir, YAAC** | Computer: APRS maps, via the KISS port |
| **Mobilinkd TNC** | Bluetooth box, on the audio output of a handheld radio |
| **OpenTracker, TNC-Pi** | Boards and boxes, with or without GPS |
| **Linux ax25** | Kernel AX.25 stacks: connected packet, BBS, network applications |

## With the UV-K1

- The UV-K1 has no built-in TNC (apart from Labs applications). For APRS, connect its audio output to a TNC: a box, or Direwolf on a computer (sheet 3).
- Setting: 144.800 MHz in narrow FM, low squelch, medium volume. Too high a volume overloads the TNC input.
- To transmit, the TNC must control the PTT: suitable cable or VOX, and an amateur radio callsign. For listening only: a phone with APRSdroid and a Mobilinkd on the headphone output is enough.

## Troubleshooting

| Symptom | Things to check |
|---|---|
| **No frame received** | Audio level, 144.800 MHz frequency, squelch, identical serial speed |
| **Garbled characters** | Wrong serial speed, or TNC not in KISS mode (text command mode) |
| **Many rejected frames** | Audio level too high or too low, power supply noise, frequency offset |
| **Transmission has no effect** | PTT wired wrongly, TXDELAY too short, channel busy (P and SlotTime) |
| **Port 8001 refused** | Direwolf not running, or firewall between the computer and the phone |

**Good to know.** Default values, ports and speeds are from memory, based on the KISS specification: confirm them with the documentation of your TNC or Direwolf. APRS positions are public, but do not share those of people who did not want it.

# SSTV

Slow-Scan Television: listening and decoding sheet

**What is it?** Still images sent as sound, line by line, in a few tens of seconds to a few minutes. The brightness of a pixel is an audio frequency: 1500 Hz = black, 2300 Hz = white; line syncs at 1200 Hz (same principle as fax, sheet 29). The signal fits in 3 kHz: SSB on HF, FM on VHF. Transmitting requires an amateur radio callsign.

| Frequency | Mode | Remark |
|---|---|---|
| **145.800 MHz** | FM | ISS (ARISS) during SSTV events; ~10 min pass, Doppler ±3.5 kHz |
| **14.230 MHz** | USB | International calling frequency (20 m), the most active |
| **28.680 / 21.340 MHz** | USB | 10 m and 15 m depending on propagation |
| **7.171 MHz** | LSB | 40 m, mostly Europe |
| **3.730 MHz** | LSB | 80 m, regional traffic |

## Common modes

| Mode | Duration | Resolution | VIS | Remark |
|---|---|---|---|---|
| **Robot 36** | 36 s | 320 x 240 | 8 | Fast, very common on VHF |
| **Robot 72** | 72 s | 320 x 240 | 12 | |
| **Martin M1 / M2** | 114 s / 58 s | 320 x 256 | 44 / 40 | Very common in Europe |
| **Scottie S1 / S2** | 110 s / 71 s | 320 x 256 | 60 / 56 | Very common in the Americas |
| **Scottie DX** | 269 s | 320 x 256 | 76 | Long distance, slow |
| **PD50 / PD90** | 50 s / 90 s | 320 x 256 | 93 / 99 | |
| **PD120** | 126 s | 640 x 496 | 95 | Usual ISS mode |
| **PD180** | 187 s | 640 x 496 | 96 | High quality |

## Start of a transmission (VIS header)

| Element | Frequency | Duration |
|---|---|---|
| **Calling tone (leader)** | 1900 Hz | 300 ms |
| **Break** | 1200 Hz | 10 ms |
| **Calling tone** | 1900 Hz | 300 ms |
| **Start bit** | 1200 Hz | 30 ms |
| **7 VIS code bits (least significant first)** | 1100 Hz = 1 / 1300 Hz = 0 | 30 ms each |
| **Parity (even) then stop bit** | 1100 or 1300 Hz, then 1200 Hz | 30 ms each |

## Listening and decoding

- ISS: UV-K1 on 145.800 MHz, narrow FM, moderate volume; correct the Doppler in steps of 1 to 2 kHz during the pass.
- HF: USB above 10 MHz, LSB below; a frequency drift slants the image, tune finely.
- The radio does not decode the image: audio to Robot36 (Android), Black Cat SSTV (iOS), MMSSTV (Windows) or QSSTV (Linux), in a quiet room.

| Fault | Probable cause |
|---|---|
| **Slanted image** | Frequency drift or inaccurate audio clock |
| **Shifted colours, torn image** | Wrong mode, or sync lost (weak signal) |
| **Streaks, noise** | Weak signal, QSB, interference or volume too high |
| **Image cut off at the top** | Start missed (VIS header not received) |

# Weather fax

HF radiofax: weather charts by radio



**What is it?** Official weather stations broadcast charts on HF (analyses, forecasts, waves, ice, satellite images), widely used by offshore sailors. Same principle as SSTV (1500 Hz black, 2300 Hz white), but much slower. Reception is free and open.

| Parameter | Value |
|---|---|
| **Modulation** | FM of an audio subcarrier, transmitted in USB |
| **Audio frequencies** | Centre 1900 Hz; black 1500 Hz; white 2300 Hz (±400 Hz) |
| **Speed** | 120 lines/min (2 per second), the most common; 60, 90 and 240 exist |
| **IOC** | 576 (standard, ~1810 points per line) or 288 |
| **Start** | Tone ~5 s (300 Hz for IOC 576, 675 Hz for IOC 288), then phasing ~30 s |
| **End** | 450 Hz tone ~5 s, then black line |
| **Tuning** | In USB, 1.9 kHz below the published frequency (assigned carrier) |

| Station | Frequencies (carriers, kHz) | Remark |
|---|---|---|
| **DWD (Germany)** | 3855 · 7880 · 13,882.5 | Europe and North Atlantic: the easiest from France |
| **Northwood (United Kingdom)** | 2618.5 · 4610 · 8040 · 11,086.5 | North Atlantic |
| **Boston NMF (United States)** | 4235 · 6340.5 · 9110 · 12,750 | Northwest Atlantic |
| **Point Reyes NMC (United States)** | 4346 · 8682 · 12,730 · 17,151.2 · 22,527 | Pacific |

*Frequencies and schedules from memory, which may change or be discontinued: check the official schedules (DWD, NOAA, UK Met Office). Lower frequencies mostly at night, higher ones by day.*

## What you receive

- Surface analyses (isobars, fronts), forecasts at 24, 48, 72 and 96 h.
- Wind, wave and swell charts; sea temperature; ice; satellite images; broadcast schedule.
- APT from satellites at 137 MHz is a cousin of fax (2400 Hz subcarrier in AM, 2 lines/s): sheet 31.

## Listening and decoding

- By ear: a low, regular "tick-tick", 2 lines per second, with pitch variations depending on light and dark areas.
- UV-K1 (SI4732): USB on the published frequency minus 1.9 kHz; audio to fldigi (WEFAX 576 and 288), MultiPSK, Black Cat HF Weather Fax (iOS) or HF Weather Fax (Android).

| Fault | Probable cause |
|---|---|
| **Slanted image** | Wrong speed (60/90/120/240) or inaccurate audio clock |
| **Shifted image** | Phasing missed: realign or wait for the next chart |
| **Negative image** | Wrong sideband (LSB instead of USB) or "invert" option |
| **Noise, streaks** | Weak signal, fading: change frequency (day / night) |

# Images in the waterfall

Drawing an image in the spectrum: principle, calculations, reception

**Principle.** You build a sound whose spectrogram draws an image. Each line of the image is a piece of sound of duration T; each column is an audio frequency; the pixel brightness becomes the amplitude of the corresponding sine wave. In a waterfall (SDR, WSJT-X, fldigi), the image appears. There is no decoding: no coding, no error correction, no fixed frequency; the image is read by eye, in the SSB audio band (about 300 to 2700 Hz). Feld Hell (sheet 23) already does this for text.

| Calculation | Formula | Example |
|---|---|---|
| **Column width** | df = B / N (B audio band, N columns) | B = 2400 Hz, N = 100: df = 24 Hz |
| **Minimum line duration** | T ≥ 1 / df; take 1.5 to 2 times more for a sharp image | df = 24 Hz: T ≥ 42 ms, so ~80 ms |
| **Image duration** | Timage = number of lines x T | 60 lines x 80 ms = 4.8 s |
| **Column frequency** | fk = f0 + k x df | f0 = 300 Hz, k = 50: 1500 Hz |
| **Signal of one line** | s(t) = sum Ak sin(2π fk t + φk), Ak = brightness | Random phases φk to avoid a power peak |
| **Trade-off** | More columns: more detail but longer lines | 200 columns at 12 Hz: T ≥ 83 ms |

## Reception and good practice

- Reception: USB, moderate volume, waterfall set to 0 to 3 kHz. In LSB the image is reversed (mirrored). Adjust the FFT size: too fine, it smears the image; too coarse, it blurs it.
- Test without transmitting: generate the sound on your computer and view it in Spectrum Lab or Audacity (spectrogram). No radio needed.
- Transmitting: only if your callsign and the band allow it. The signal is wide: stay away from active frequencies, FT8, distress frequencies and beacons.

# Satcom

Weather, amateur radio, Inmarsat, Iridium and military UHF: frequencies, formats and equipment



| Service | Frequency | Format | Content | Equipment / decoding |
|---|---|---|---|---|
| **Civil and amateur radio satellites (downlink)** | | | | |
| **NOAA APT** | 137.100 · 137.620 · 137.9125 MHz | FM, 2400 Hz subcarrier (AM), 2 lines/s | Polar weather image (NOAA 15, 18, 19 shut down or at end of life: check) | V-dipole 52 cm; RTL-SDR; satdump, WXtoImg |
| **Meteor-M2 LRPT** | 137.1 · 137.9 MHz | QPSK 72 kbit/s | High-resolution digital images | SDR, SatDump |
| **Orbcomm** | 137 - 138 MHz | SDPSK 4800 bit/s | Fleet data | SDR, dedicated decoder |
| **GOES HRIT / EMWIN** | 1694.1 MHz | DVB-S2 (QPSK) | Images, charts, notices (Americas) | Dish, SDR, SatDump; not visible from Europe |
| **Meteosat (MSG)** | 1695.15 MHz (HRIT) | BPSK + FEC | Geostationary images | Dish + SDR; also EUMETCast (Ku) |
| **FM amateur radio satellites** | Downlink 435 - 438 MHz; uplink 145 MHz | FM, sometimes CTCSS 67 Hz | Voice (SO-50, AO-91...), beacons | UV-K1 + handheld Yagi; Doppler ±10 kHz |
| **QO-100 (Es'hail-2)** | | | | |
| **Commercial and military satcom (SDR)** | | | | |
| **Inmarsat-C (STD-C)** | 1537 - 1545 MHz | BPSK 600 bit/s, TDM | SafetyNET, EGC: maritime notices, distress | 1.5 GHz patch, SDR, STD-C decoder |
| **Inmarsat AERO** | 1545 - 1555 MHz | BPSK / QPSK 600 to 1200 bit/s | ACARS by satellite | Patch, SDR; partly encrypted |
| **Iridium** | 1616 - 1626.5 MHz | QPSK 25 kbaud, TDMA | Voice, SMS, ACARS, signalling | SDR, gr-iridium, iridium-toolkit |
| **Military UHF** | 243 - 270 MHz | AM, FM, DAMA 25 kHz, encryption | Military voice and data | SDR + 250 MHz Yagi; partial monitoring |

## Reading and limits

- UV-K1: FM amateur radio satellites (435 - 438 MHz, ISS voice on 145.800 MHz) with a suitable antenna. 137 MHz needs a wider bandwidth than narrow FM: SDR. Inmarsat, Iridium, GOES and QO-100 are out of reach of the radio.
- 406 MHz beacons: sheet 22. NOAA images: close to weather fax (sheet 29) through the 2400 Hz subcarrier.
- Listening is free; do not transmit without a callsign; commercial and military satcom are restricted; do not disclose the content of private exchanges.

# SigID

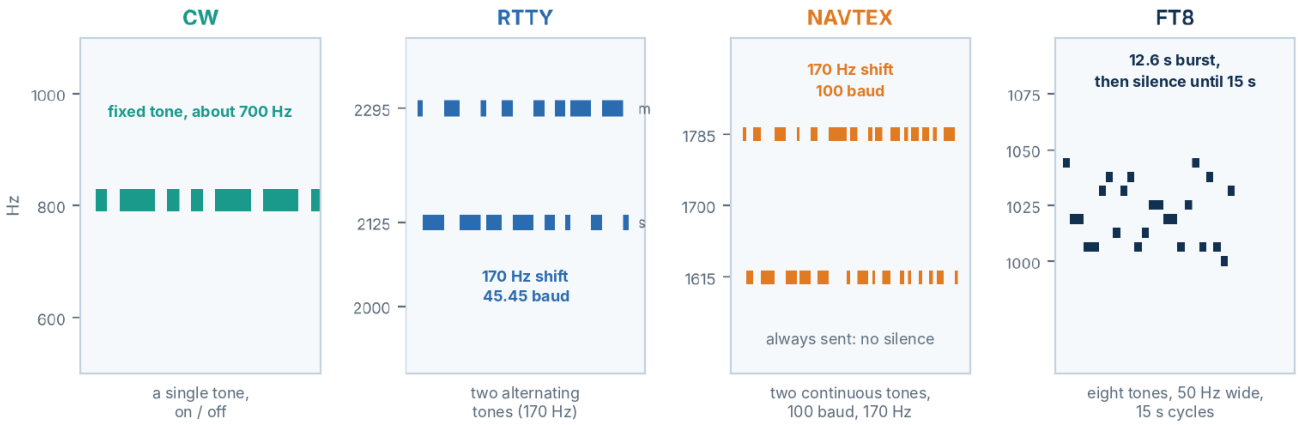
Identifying a radio signal by its tones (UV-K1)

**In brief.** SigID is a small application for the UV-K1 that identifies the type of signal received (CW, RTTY, NAVTEX, FT8) and displays it. It does not decode the content: once the mode is known, decoding is done on a smartphone or PC. It starts from the SelCall CCIR application (Labs overlay F4HWN v6.0.0), which measures the energy of a few tones with the Goertzel algorithm on audio sampled at 8 kHz. Main constraint: 4 KB maximum once compiled.

| 4 KB | 8 kHz | 4 modes | Goertzel |
|---|---|---|---|
| maximum size | audio sampling | CW, RTTY, NAVTEX, FT8 | analysis of a few tones |

## Four signatures to recognise

**Four signatures: audio frequency (vertical) versus time (horizontal)**

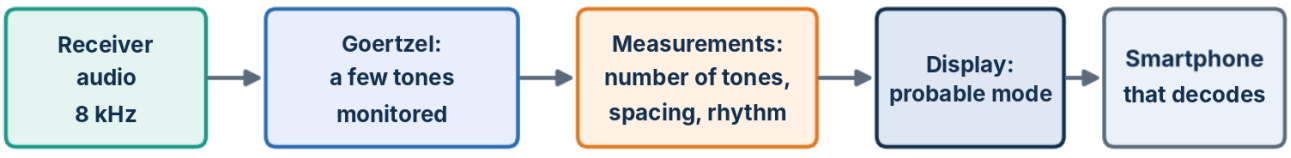



*Illustrative diagram: each mode leaves a different footprint in the audio time / frequency plane.*

| Mode | Tones | Rate and shift | Timing clue | Where to listen |
|---|---|---|---|---|
| **CW** | 1 | Tone on / off, variable speed | A dash lasts about three dots | Amateur radio bands |
| **RTTY** | 2 | 45.45 bd, 170 Hz shift (amateur radio); 50 bd, 425 or 450 Hz (commercial, weather) | Continuous, 5-bit Baudot | Amateur radio bands, HF utility (sheet 24) |
| **NAVTEX** | 2 | 100 bd, 170 Hz shift, CCIR 476 (SITOR-B) | 10 min slot per station, letter A at 00:00 UTC, every 4 h; phasing of at least 10 s then ZCZC | 518 kHz, 490 kHz, 4209.5 kHz (sheet 14) |
| **FT8** | 8 | ~6.25 bd, tones spaced ~6.25 Hz apart | 12.64 s burst every 15 s (00, 15, 30, 45) | 7.074; 14.074; 21.074; 28.074 MHz (sheet 23) |

## The chain inside the radio

**What the app does in the radio, then on the smartphone**



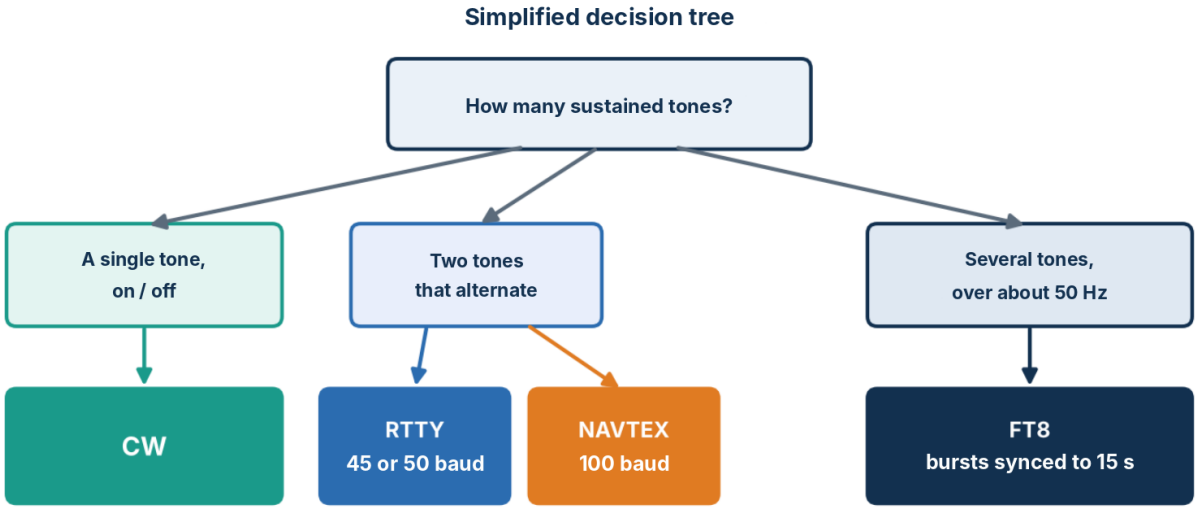

*Full decoding (NAVTEX text, FT8 messages) stays on the smartphone: the radio only identifies the signal.*

## Goertzel: measuring a tone on the cheap

A full FFT needs too much memory for 4 KB. The Goertzel algorithm computes the energy of a single frequency with two state variables and one multiplication per sample; a few are run in parallel, one per monitored tone.

| Concept | Key point |
|---|---|
| **Channel width** | fs / N. At 8 kHz with N = 160: 50 Hz per channel, one measurement every 20 ms |
| **Trade-off** | The larger N is, the more precise but slower. NAVTEX (10 ms bit) requires a short N, FT8 (6.25 Hz tones) a long N |
| **Cost** | One coefficient 2cos(2πk/N) per tone, then s = x + c·s1 - s2 at each sample: possible in integers |
| **Decision** | Compare the energy of the tones with each other and with the noise, using an adaptive threshold |

## Decision tree




*The number of tones sorts first, then the rhythm separates RTTY and NAVTEX, and synchronisation on 15 s confirms FT8.*

## Limits and difficulties

| Difficulty | Approach |
|---|---|
| **RTTY and NAVTEX are close** | Same 170 Hz shift: only the rhythm (45.45 or 50 bd versus 100 bd) tells them apart; measure over several seconds |
| **FT8 is very narrow** | Tones spaced ~6.25 Hz apart: long N, slow analysis; rely on the 15 s structure |
| **Weak signal** | Require several consistent measurements, display "unknown" when in doubt |
| **Tuning offset** | Measure the gap between tones rather than their absolute value |
| **Receiver** | NAVTEX at 518 kHz and HF FT8 require a medium-wave and shortwave receiver in a suitable mode (SI4732) |
| **Size** | Few monitored tones, no big tables, integer calculations |

| Identified mode | Decoding on smartphone or PC |
|---|---|
| **CW** | Morse decoder from the sound |
| **RTTY** | RTTY decoder (shift and rate depending on identification): sheet 24 |
| **NAVTEX** | NAVTEX FEC 100 bd decoder, text between ZCZC and NNNN: sheet 14 |
| **FT8** | FT8 application with a well-synchronised clock: sheet 23 |

**Going further.** Add other modes (SelCall sheet 25, DTMF, SSTV, AIS, ATIS), display the level, duration and measured rate, test on known references (NAVTEX 518 kHz, beacons). This sheet gives the principle, not the code: the values of N, the thresholds and the keys are to be adapted to your version.

# Sig Finder

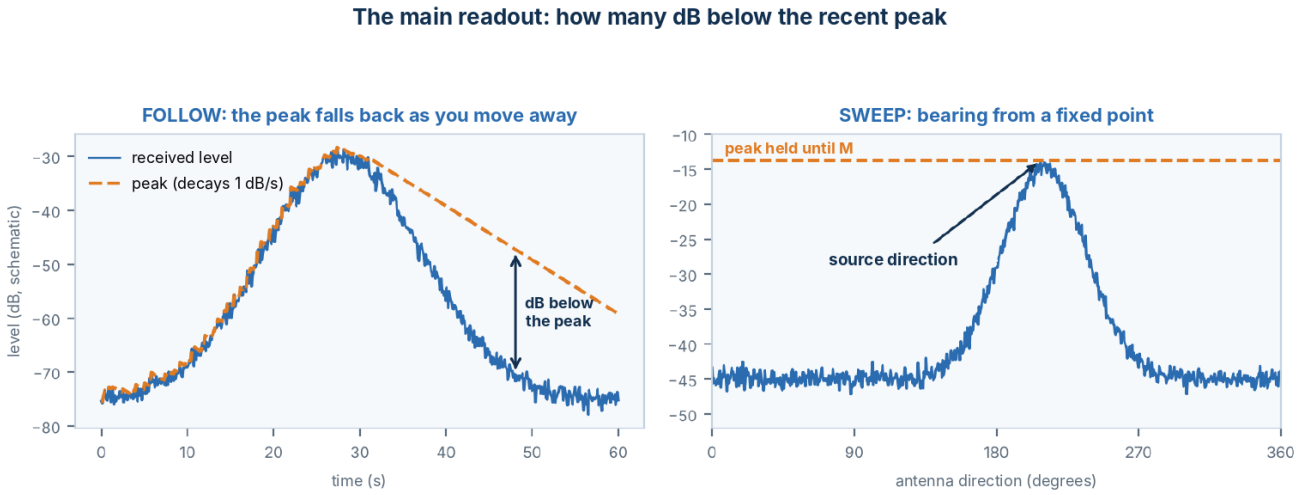
Direction finding on UV-K5 / UV-K1 (Labs edition)

· DATA mod

**In brief.** Sig Finder is a Labs edition application that helps find the direction of a carrier: fox in a fox hunt, jammer, or homing signal at 121.5 / 243 MHz from a distress beacon. Its main indication is not the absolute level but the number of dB below the strongest recent level. It also recognises the downward-swept tone of a homing signal and displays ELT. Receive only.

| dB below peak | 2 modes | 121.5 / 243 | F + 7 |
|---|---|---|---|
| main indication | FOLLOW and SWEEP | MHz, homing signal | launch |

## Principle: getting closer by comparison




*In FOLLOW, the stored peak falls back at a chosen rate: if the dB gap grows, you are moving away. In SWEEP, the peak stays frozen until you press M.*

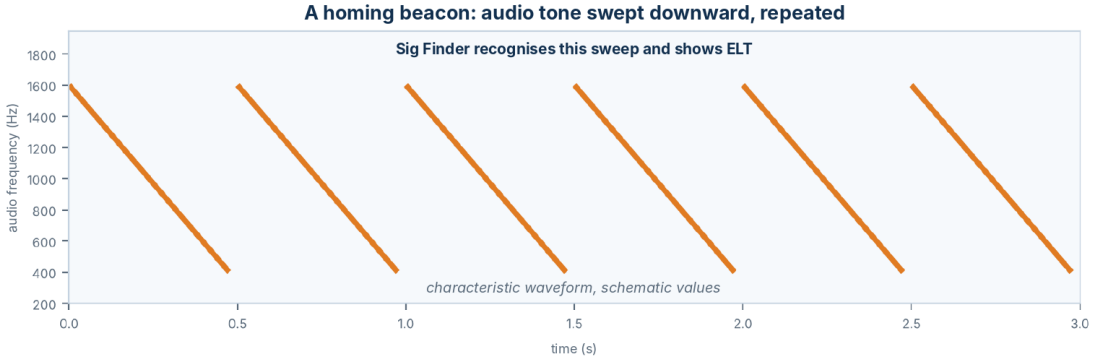
## Preparing and launching

| Step | Detail |
|---|---|
| **Install** | With UV Studio, the version of the application matching the firmware (sheet A03) |
| **Set the VFO** | AM for 121.5 / 243 MHz, otherwise the suitable modulation |
| **Launch** | F + 7 |
| **Navigation** | Depending on SetNav: UP / DOWN on UV-K5, LEFT / RIGHT on UV-K1 (UP/LEFT and DOWN/RIGHT denote the equivalent keys) |

## FOLLOW and SWEEP

| Mode | Behaviour | Use |
|---|---|---|
| **FOLLOW** | The peak decays at an adjustable rate: 0.5; 1; 2 or 4 dB/s | You move towards the source |
| **SWEEP** | The peak is kept; M resets it to zero for a new bearing | Bearing from a fixed point |

**Automatic attenuator.** Close to the source, the signal can saturate the receiver. The attenuator adds attenuation before saturation and removes it when the signal weakens; you can also set it by hand (key 3 or UP/LEFT and DOWN/RIGHT).
**Recognising a homing signal: ELT.** Sig Finder detects the tone that sweeps downward and displays ELT. The muted-audio and listening modes detect it; the proximity beeps mode does not, because it cuts the sound between beeps.



# Sig Finder

Direction finding on UV-K5 / UV-K1 (Labs edition)

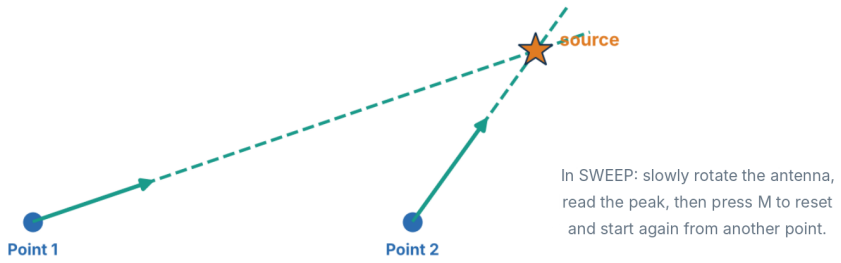
## The keys

| Key | Action |
|---|---|
| **1** | Toggles FOLLOW / SWEEP and resets the peak to zero |
| **2 or F then 2** | Next or previous audio mode: muted, proximity beeps, normal listening, USB listening |
| **3, UP/LEFT, DOWN/RIGHT** | Changes the attenuation step and switches to manual; F then 3: previous step |
| **4** | Turns automatic attenuation on or off |
| **5 or F then 5** | Peak decay rate in FOLLOW (0.5; 1; 2; 4 dB/s) |
| **6** | Backlight kept on, or normal timeout |
| **M** | Resets the peak to zero; in SWEEP, starts a new sweep |
| **Hold F** | Locks or unlocks the application keypad |
| **EXIT** | Quits the application (keypad unlocked) |

*Saved settings: mode, audio, attenuation, auto attenuation, decay rate, backlight.*

## Field tips

**Two bearings from two points: the source is at their intersection**




- SWEEP to take a bearing, then FOLLOW to advance towards the source.
- Reduce the attenuation by hand if the signal is weak; leave it on automatic near the source.
- Several bearings from different points: the crossing gives the position.
- Lock the keypad (hold F) to avoid accidental presses while walking.

**Warning.** Never transmit a test homing signal on 121.5, 243 or 406 MHz: these are monitored distress frequencies. Sig Finder is a receive-only tool.

# Morse, alphabet, Q and CB codes, prefixes

Cheat sheet for hams and listeners



## Alphabet, Morse and spelling

| | Morse | ICAO | | Morse | ICAO |
|---|---|---|---|---|---|
| A | •— | Alfa | N | —• | November |
| B | —••• | Bravo | O | ——— | Oscar |
| C | —•—• | Charlie | P | •——• | Papa |
| D | —•• | Delta | Q | ——•— | Quebec |
| E | • | Echo | R | •—• | Romeo |
| F | ••—• | Foxtrot | S | ••• | Sierra |
| G | ——• | Golf | T | — | Tango |
| H | •••• | Hotel | U | ••— | Uniform |
| I | •• | India | V | •••— | Victor |
| J | •——— | Juliett | W | •—— | Whiskey |
| K | —•— | Kilo | X | —••— | X-ray |
| L | •—•• | Lima | Y | —•—— | Yankee |
| M | —— | Mike | Z | ——•• | Zulu |

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| ————— | •———— | ••——— | •••—— | ••••— | ••••• | —•••• | ——••• | ———•• | ————• |

*Digits, international pronunciation: Zero, Wun, Too, Tree, Fower, Fife, Six, Seven, Ait, Niner.*

## Punctuation and procedural signals

| Sign | Morse | Signal | Meaning | Morse |
|---|---|---|---|---|
| . | •—•—•— | AR | End of message | •—•—• |
| , | ——••—— | SK | End of contact | •••—•— |
| ? | ••——•• | BT | Break (=) | —•••— |
| / | —••—• | KN | Over to you only | —•——• |
| = | —•••— | AS | Wait | •—••• |
| + | •—•—• | CT | Start of message | —•—•— |
| - | —••••— | SOS | Distress | •••———••• |
| ( ) | —•——• / —•——•— | Error | Correction | •••••••• |
| : " ' @ | ———••• / •—••—• / •————• / •——•—• | | | |

*Timing: dot = 1 unit, dash = 3, gap between signs = 1, between letters = 3, between words = 7. Dot duration (ms) = 1200 / speed in words per minute: 60 ms at 20 wpm.*

## RST reports and CW abbreviations

- RST: R = readability (1 to 5), S = strength (1 to 9), T = tone (1 to 9, CW only). 599 = perfect in CW; 59 in phone.
- Typical call: CQ CQ CQ DE F4ABC F4ABC K; reply: F4ABC DE DL1XYZ K.

| Abbr. | Meaning | Abbr. | Meaning | Abbr. | Meaning |
|---|---|---|---|---|---|
| CQ | General call | 73 | Best regards | UR | Your |
| DE | From ("this is") | 88 | Love and kisses | FB | Fine business |
| K | Over to you | GM / GA / GE | Good morning / afternoon / evening | DX | Long distance |
| KN | Over to you only | YL / XYL | Lady operator / wife | ES | And |
| AR | End of message | RIG / ANT | Station / antenna | BK | Break (back to you) |
| SK | End of contact | PWR | Power | R | Received |
| PSE | Please | WX | Weather | OM | Old man (friend) |
| TNX / TKS / TU | Thanks | AGN | Again, repeat | | |

# Morse, alphabet, Q and CB codes, prefixes

Cheat sheet for hams and listeners



## Q code (the most useful)

| Code | Meaning | Code | Meaning |
|---|---|---|---|
| **QRA** | Station name | QRU | I have nothing for you |
| **QRG** | Exact frequency | QRV | I am ready |
| **QRH** | My frequency drifts | QRX | Wait, I will call again |
| **QRI** | Quality of the tone | QRZ | Who is calling me? |
| **QRK** | Readability of the signals | QSA | Signal strength (1 to 5) |
| **QRL** | Frequency busy | QSB | Fading |
| **QRM** | Interference from other stations | QSL | Received; QSL card |
| **QRN** | Atmospheric noise | QSO | Radio contact |
| **QRO** | Increase power | QSP | Relay a message |
| **QRP** | Reduce power / low power | QSY | Change frequency |
| **QRQ** | Send faster | QTC | Messages to send |
| **QRS** | Send slower | QTH | Position, location |
| **QRT** | Stop transmitting, end of traffic | QTR | Exact time |

*A Q code followed by "?" is a question (QRZ? = who is calling me?). QNH, QFE, QFU, QDM and QDR are aeronautical Q codes (sheet 9).*

## CB code (10-codes and slang)

On CB (27 MHz, channels in sheet 8), people use the "10-codes" that came from the United States, often mixed with slang. Usage varies by region and group: they are guidelines, not a standard.

| Code | Meaning | Code | Meaning |
|---|---|---|---|
| **10-1** | Poor reception | 10-20 | Location (your location?) |
| **10-2** | Good reception | 10-21 | Call me by phone |
| **10-3** | Stop transmitting | 10-22 | Meet me in person |
| **10-4** | Received, OK | 10-23 | Stand by |
| **10-5** | Relay the message | 10-24 | Assignment completed |
| **10-6** | Busy, wait | 10-26 | Disregard last information |
| **10-7** | Out of service, leaving the air | 10-27 | I am moving to channel ... |
| **10-8** | In service, listening | 10-28 | Give your identification |
| **10-9** | Repeat | 10-32 | Reception test (radio check) |
| **10-10** | End of transmission, I keep listening | 10-33 | Emergency traffic |
| **10-13** | Road or weather conditions | 10-36 | Exact time |
| **10-16** | Take charge, pick up | 10-42 | Traffic accident |
| **10-18** | Anything for us? | 10-62 | Unreadable, call me by phone |
| **10-19** | Nothing for you, return to base | 10-70 | Fire |

| Slang word | Meaning |
|---|---|
| **Break (break 19)** | I ask to speak (on channel 19, for example) |
| **Over** | End of message, it is your turn to speak; "out": end of contact |
| **Copy, roger** | Received |
| **Affirmative, negative** | Yes, no |
| **Radio check** | Request for a reception test |
| **Mobile, base** | Station in a vehicle, fixed station |

*Values from memory, to verify: several codes differ from one group to another. Channel 9 is used as an emergency channel by convention (sheet 21). CW punctuation and abbreviations (73, 88, QSL...) are above.*

## Amateur radio callsigns

Structure: prefix (country) + digit + suffix. F4ABC = F (France), 4, ABC. Suffixes: /P portable, /M mobile, /MM boat at sea, /AM aircraft.

| Prefix | Territory | Prefix | Territory |
|---|---|---|---|
| **F** | Metropolitan France | TK | Corsica |

# Morse, alphabet, Q and CB codes, prefixes

Cheat sheet for hams and listeners



| Prefix | Territory | Prefix | Territory |
|---|---|---|---|
| **FM** | Martinique | FG | Guadeloupe |
| **FS** | Saint-Martin | FY | French Guiana |
| **FR** | Réunion | FH | Mayotte |
| **FP** | Saint-Pierre-et-Miquelon | FK | New Caledonia |
| **FO** | French Polynesia | FW | Wallis-et-Futuna |
| **FT** | French Southern Lands (TAAF) | TM / TO | French special callsigns |

| Country | Pref. | Country | Pref. | Country | Pref. |
|---|---|---|---|---|---|
| Germany | DL | Czech Rep. | OK | South Africa | ZS |
| United Kingdom | G/M | Hungary | HA | United States | K/W/N |
| Spain | EA | Romania | YO | Canada | VE |
| Italy | I | Croatia | 9A | Mexico | XE |
| Belgium | ON | Slovenia | S5 | Brazil | PY |
| Netherlands | PA | Bulgaria | LZ | Argentina | LU |
| Switzerland | HB9 | Russia | UA | Chile | CE |
| Luxembourg | LX | Ukraine | UR | Japan | JA |
| Portugal | CT | Turkey | TA | China | BY |
| Ireland | EI | Monaco | 3A | India | VU |
| Austria | OE | Andorra | C3 | South Korea | HL |
| Poland | SP | Morocco | CN | Thailand | HS |
| Sweden | SM | Algeria | 7X | Indonesia | YB |
| Norway | LA | Tunisia | 3V | Israel | 4X |
| Denmark | OZ | Senegal | 6W | Australia | VK |
| Finland | OH | Ivory Coast | TU | New Zealand | ZL |
| Greece | SV | Madagascar | 5R | | |

**Little challenge.** Decode this message (letters separated by a space, words by /): **.. -.-. ...-- .-. .. -.-. ....- / .. -.-. . .-. .. -.-. -.-**. Hint: the rest is on GitHub, at **ic3fr**.

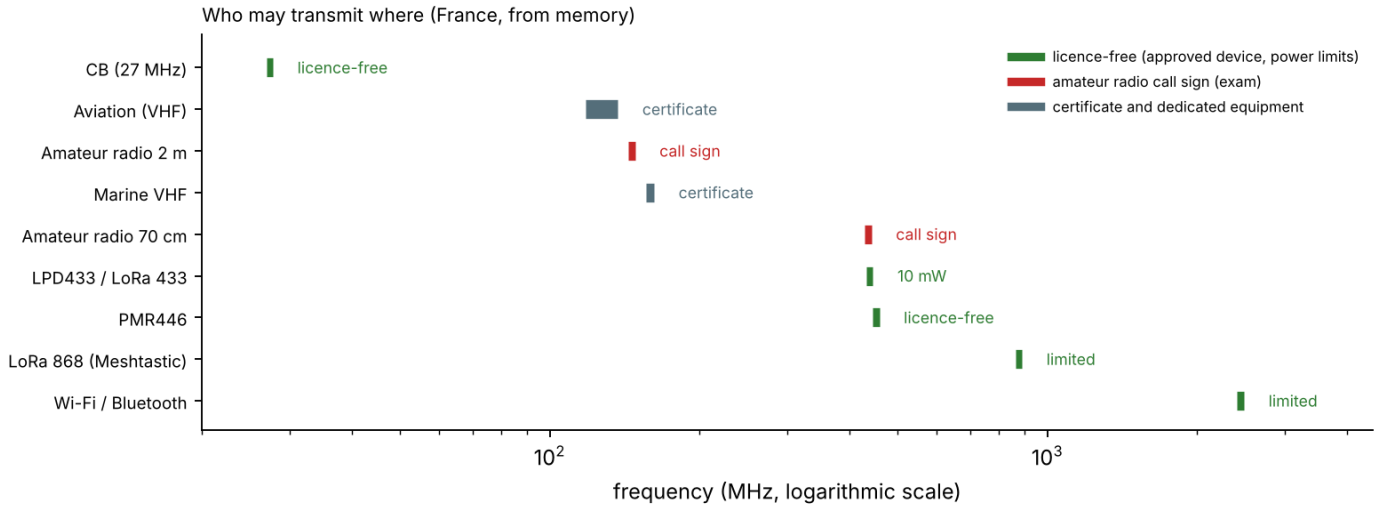
# Transmitting: licences, licence-free bands and procedures

Who can transmit on what, how to make a call and a distress message



**In brief.** Listening is free, transmitting is not. Depending on the service, you need an amateur radio callsign (exam), an approved device or a certificate. In this guide, transmitting is a way to **be heard at the right time with the right means**: a licence-free band at short range, or HF with a callsign. The values in this sheet are orders of magnitude for France: check the rules in force (ANFR, ARCEP) before transmitting.

## Who can transmit where




*Orders of magnitude for each service: in green, approved devices without a licence; in red, a callsign is required; in grey, a certificate and dedicated equipment.*

## Licence-free services

| Service | Frequencies | Usual limits | Key points |
|---|---|---|---|
| **PMR446** | 446.0 to 446.2 MHz, 16 analogue channels (32 digital) | 0.5 W, fixed antenna, no repeaters | Off-the-shelf walkie-talkie, a few hundred metres to a few km (sheet 8) |
| **LPD433** | 433.075 to 434.775 MHz, 69 channels | 10 mW | Very short range; approved devices only (sheet 8) |
| **CB** | 26.965 to 27.405 MHz, 40 channels | 4 W in AM and FM, 12 W in SSB (to verify) | Channel 9: emergency by convention, channel 19: road (sheet 21) |
| **LoRa, Meshtastic, MeshCore** | 433 MHz and 868 MHz (licence-free bands) | Limited power and airtime | Mesh network of short messages (sheets 17 and 18) |
| **Wi-Fi, Bluetooth** | 2.4 and 5 GHz | Limited power, approved devices | Sheet 19 |

**No transmitting with the UV-K1 on these services.** With unlocked firmware the UV-K1 can transmit well beyond the amateur bands, but **it is not an approved device** for PMR446, CB or the marine and aviation bands. With it, transmit only on the amateur bands and with a callsign. For the licence-free bands, use a device designed for them.

## Becoming a radio amateur

- **Exam**: regulation and technical questions taken with the ANFR (or an approved body); it gives you a personal **callsign** and the right to transmit on the amateur bands (sheet 6).
- **What it opens up**: VHF/UHF (2 m and 70 cm) in FM with repeaters and APRS (sheet 26), HF in voice, in CW (sheet 34) and in digital modes (sheet 23), satellites. Power and sub-bands depend on the licence class.
- **Why prepare it now**: it is the only legal way to transmit on HF and on 2 m with a radio like the UV-K1. The amateur emergency networks (3760, 7110 and 14,300 kHz) are already organised (sheet 21).

## Distress, urgency, safety

**Basic rule.** In a real distress (grave and imminent danger), any means is allowed to call for help: this is a principle of the Radio Regulations. Outside a distress, **never** transmit on the distress frequencies (121.5; 156.8; 243; 406 MHz; 2182 kHz), and never simulate an alert (sheets 21 and 22).

| Signal | When | Example opening |
|---|---|---|
| **MAYDAY** | Distress: grave and imminent danger to a life or a vessel | "MAYDAY, MAYDAY, MAYDAY. This is..." |
| **PAN-PAN** | Urgency: serious situation but no imminent danger | "PAN-PAN, PAN-PAN, PAN-PAN. All stations..." |

# Transmitting: licences, licence-free bands and procedures

Who can transmit on what, how to make a call and a distress message



| Signal | When | Example opening |
|---|---|---|
| **SECURITE** | Safety notice (weather, danger to navigation) | "SECURITE, SECURITE, SECURITE. All stations..." |

## Making a voice distress message

- **1.** MAYDAY three times, then "THIS IS" and your name or callsign three times, then MAYDAY and your name once.
- **2. Where**: position (coordinates, or a known landmark: road, kilometre marker, place name) and, if possible, your GPS position (sheet 20).
- **3. What**: nature of the distress (injured, lost, fire, breakdown), help requested, number of people.
- **4.** Useful information (weather, condition, colour or sign to help spot you), then "OVER".
- **5.** Repeat at regular intervals while changing means: a call that gets no answer must move to the next level of the plan (sheet A00b). Radio silence "SEELONCE MAYDAY" is imposed by the station directing the rescue: respect it.

## Good transmitting practice

- **Listen before transmitting**: is the frequency free? In AM and SSB, wait a few seconds.
- **Short and clear**: who I am calling, who I am, what I want. Spell what matters (names, callsigns, coordinates) with the phonetic alphabet (sheet 34).
- **Keywords**: "THIS IS" (I am), "OVER" (I am waiting), "ROGER" (understood), "SAY AGAIN" (repeat), "OUT" (end of exchange).
- **Discretion**: no private names or sensitive places on a frequency open to everyone; radio is public.
- **Minimum power**: just enough to be heard: you save battery (sheet A33) and disturb others less.

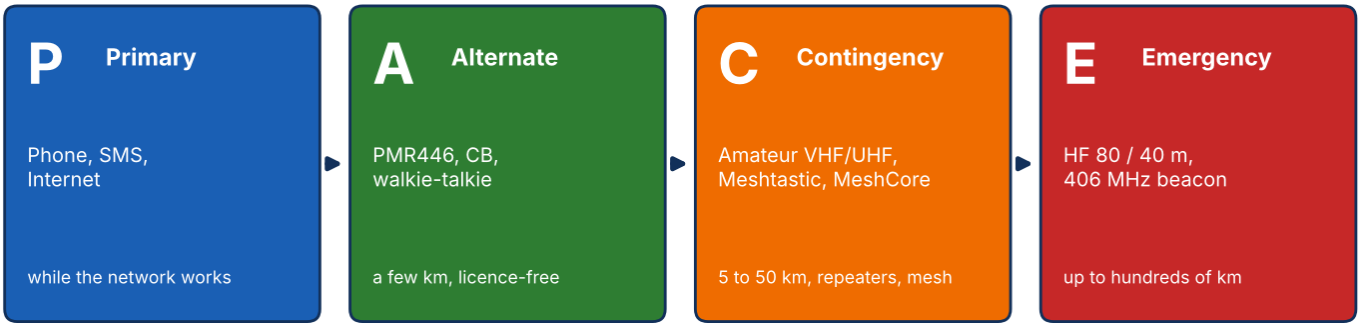
## Personal notes (callsign, licence class, exam date)

# Crisis communication plan: the PACE method

Who calls whom, on what, at what time, and how to move to the next means



**In brief.** A good communication plan fits on one page and is prepared **before** the crisis. The **PACE** method (Primary, Alternate, Contingency, Emergency) sets four ranked means; when one stops answering, you move to the next. The plan says **who**, **what** (the means), **when** (the listening time) and **where** (the frequency or channel).




*The PACE method: you never depend on a single means. Ranges are orders of magnitude.*

## An example of a four-level plan

| Level | Means | Frequency / channel | Range | Sheets |
|---|---|---|---|---|
| **P: Primary** | Phone, SMS, messaging | Mobile network or Wi-Fi | As long as the network works | 18 |
| **A: Alternate** | PMR446 or CB | PMR channel 1 (to agree), CB channel 19 or another agreed channel | A few hundred m to a few km | 7, 20 |
| **C: Contingency** | Amateur VHF/UHF; Meshtastic / MeshCore | 145.500 MHz (FM calling) or 433.500 MHz; agreed Meshtastic channel | 5 to 50 km, more with a repeater or a mesh network | 5, 16, 17, 27 |
| **E: Emergency** | HF (NVIS); 406 MHz distress beacon if life is in danger | 7110 kHz (LSB) by day, 3760 kHz at night, 14 300 kHz (USB) | Up to several hundred km (NVIS) or worldwide (DX) | 1, 5, 21 |

*The frequencies in the "Contingency" and "Emergency" rows come from the amateur emergency networks (sheet 21): they require a callsign. Without a callsign, stay on levels P and A, and on the licence-free means of level C (Meshtastic).*

## Preparing the plan

- **Who**: list the people to reach (family, neighbours, group), each with **a role** ("relay station", "watch keeper") and their radio.
- **Same settings for everyone**: same frequency, same mode, same sub-tone (CTCSS); write them on a laminated card with the UV-K1 and on a paper kept next to the radio (sheet 2).
- **Listening windows**: a simple example, a 10-minute listen at fixed times (for example 8:00, 12:00 and 18:00). Outside the window, save battery (sheet A33). If nobody answers, try again at the next window, then move to the next level.
- **Meeting point**: an agreed fallback place if all radios go silent.
- **Password**: a recognition word or phrase, to be sure you are talking to the right person.
- **Practice**: test each level every month; a plan never tested fails the day you need it.

## A standard four-line message

| Line | Content | Example |
|---|---|---|
| **1. Who** | My name or callsign, and who I am talking to | "This is Alpha One for Base" |
| **2. Where** | My location or position | "At the north fallback point, safe and sound" |
| **3. Status and need** | Health, urgent need | "Two people, one lightly injured, need water" |
| **4. Next** | Next listening time and fallback means | "Next listen at 18:00, PMR channel 1 then 145.500" |

# Crisis communication plan: the PACE method

Who calls whom, on what, at what time, and how to move to the next means



## Plan to fill in

| Level | Means | Frequency / channel | Listening time | Who |
|---|---|---|---|---|
| **P: Primary** | | | | |
| **A: Alternate** | | | | |
| **C: Contingency** | | | | |
| **E: Emergency** | | | | |

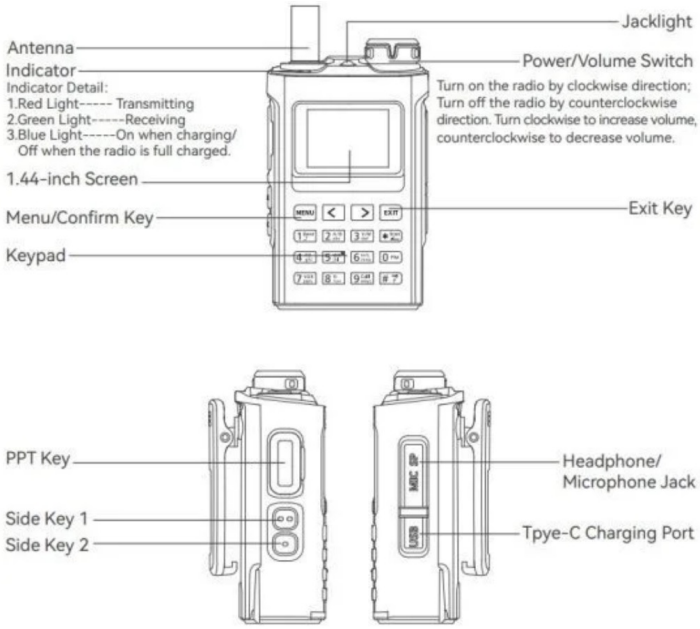
**Net and control station.** For a group of more than three people, a **control station** ("base") runs the exchange: it calls each station in turn, notes what is said, relays important messages and decides when to move to the next level. The others speak only when called, except in an emergency.

## Personal notes (group members, tests, remarks)

# UV-K1 external controls

Front, sides and top: what each marking does






*Front, sides and top of the UV-K1 (manufacturer diagram, labels in English).*

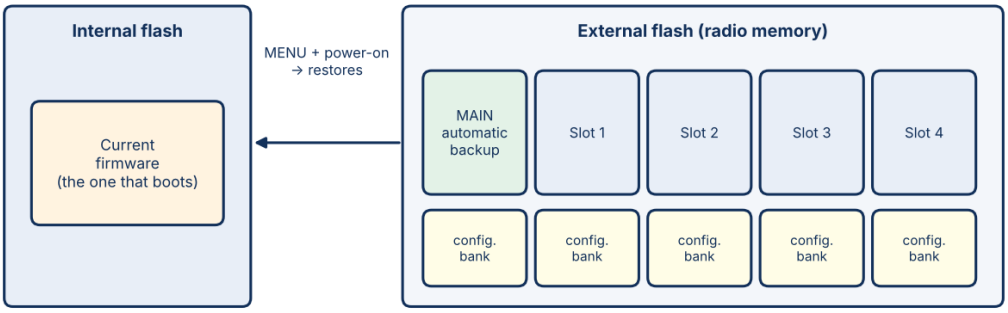
| Marking (English) | Function |
|---|---|
| **Indicator** | LED: red when transmitting, green when receiving, blue while charging and off when the battery is full. |
| **Power / Volume Switch** | Knob on top: turned clockwise it switches the radio on and raises the volume; counter-clockwise it lowers the volume, then switches off. |
| **Jacklight** | Built-in flashlight, above the screen. |
| **MENU / Confirm, arrows, EXIT** | Menu navigation. The left and right arrows act as NAV in overlay applications. EXIT leaves. |
| **Keypad** | Numeric keypad: the secondary functions depend on the firmware (sheet A08). |
| **PPT, Side Key 1, Side Key 2** | Transmit, and two programmable side keys. |
| **MIC / SP jack** | Headset and microphone socket (accessories and decoding adapters). |
| **USB-C** | Charging and link to the computer or smartphone. |

# Multiboot: several firmwares in the same radio

Since F4HWN v6: slots, flash memory, switching firmware



**Since F4HWN version 6.** The external flash memory keeps a "Main" backup (the first firmware installed) and **four slots** for other firmwares or editions, each with its own configuration bank. Each image is checked before it is restored. You switch firmware without a computer: radio off, hold **MENU** while powering on, release when the multiboot screen appears, **UP / DOWN** to choose, **MENU** to restore, **EXIT** to leave without changing anything. **Never cut the power during a restore**: the internal flash would be left half-written. Common layout: Main = a safe Fusion, one Fusion slot for everyday use, Transfer for programming, FieldOps for hunts, Labs for experiments (sheet A08).




*Memory organisation: the external flash keeps Main and 4 slots, each with its configuration bank; the selection copies an image into the internal flash.*

# The firmwares in this guide: choosing, flashing, backing up

Fusion, Labs, FieldOps, Transfer, GOGUFW, Apex, Sonic, Falcon, and how to install them



## The firmwares in this guide

| Firmware | What it is | Key points | Checked |
|---|---|---|---|
| **Fusion (F4HWN)** sheet A08 | Reference edition of the project by armel and muzkr for UV-K1 and UV-K5 V3. | UV Studio (flash, calibration, Live Viewer), CHIRP driver supplied with each version, spectrum, APRS. Hidden menu 76/77 SetNav: set UP/DOWN or LEFT/RIGHT depending on the model. | Online |
| **Labs** sheet A09 | Experimental F4HWN edition with installable "overlay" applications. | Applications installed with UV Studio, launched with F + 7: full list of applications: sheets A14 to A18 (ATIS RX, Sig Finder, APRS, SSTV, sondes...). An application is compiled for a specific firmware version. | Guide |
| **FieldOps** sheet A08 | F4HWN edition based on Fusion for the field. | Adds RescueOps, Fox Hunt (receiver) and Beacon (Morse beacon): fox hunts, SOTA / POTA, emergency communications. Fits in a multiboot slot. | Online |
| **Transfer** sheet A08 | F4HWN edition based on Fusion for cloning radios. | Adds AirCopy (bank transfer over the air), Beam (single-channel transfer), cable cloning and full flash copy. Useful for a club. | Online |
| **GOGUFW** sheet A12 | Fork of F4HWN, stable version 3.3.2 seen online. UV-K1 and UV-K5 V3 only. | Messenger: FSK text messages between GOGUFW radios. Also HEARD, Range Check, CALLTX and memory spectrum. Since 3.0.0: 16 application slots. Dedicated CHIRP module, "multiboot" image. | Online |
| **Apex (NR7Y CW)** sheet A13 | CW (Morse) firmware by NR7Y, based on Fusion (F4HWN). UV-K1 and UV-K5 V3, version 1.4. The project documentation calls it "NR7Y CW Firmware". | Automatic iambic keyer (modes A, B, Ultimatic, Bug), paddle on PTT / Side1 or external, recorded macros, Code Practice mode, RIT / XIT, CW decoding displayed. Long 0: FM, AM, USB, CW; long 5: practice oscillator; long 7: break-in. | Online docs |
| **Sonic (Robby69400)** sheet A11 | Fork of F4HWN with code from NTOIVOLA, egzumer and Dual Tachyon. Sonic 1K and 4K versions. UV-K1 (USB-C) and RS232 versions. Repository: github.com/Robby69400/Sonic. | Spectrum at ~300 channels/s with 50 scan lists and 50 custom bands (CHIRP), ULTRA WATCH ~600 channels/s, history of 200 detections, close call, no hidden menu. No applications; logo and calibration to be handled with F4HWN. | Manual |
| **Falcon** sheet A10 | Current version 3.0. UV-K5 V3 and UV-K1. | Spectrum and Sentinel (LIST, BAND, RANGE), history of 200 detections, dual, triple, quad or multi-list watch, discreet Falcon mode, broadcast FM radio. CHIRP driver supplied in the archive. | Manual |

## Flashing and backing up

- **UV Studio** (armel.github.io/uvstudio), in Chrome, Edge or Chromium on a computer: flash in DFU mode, calibration backup, Live Viewer (the radio's screen on the computer), Apps page for the applications.
- **Back up the calibration right after the first flash.** It is specific to your radio and is restored before going back to the original firmware.
- **CHIRP** (sheet 4): use the driver supplied with the flashed version (GOGUFW driver for GOGUFW). A driver from another version reads the settings wrongly.
- **quansheng_toolkit** (open source, Apache 2.0): reads and writes channels, calibration and DFU flash with a list of compatible firmwares.
- The wrong firmware can make the radio unusable. Each firmware targets specific models: read its sheet before flashing.

# UV-K1 internals and the BK4829 radio chip

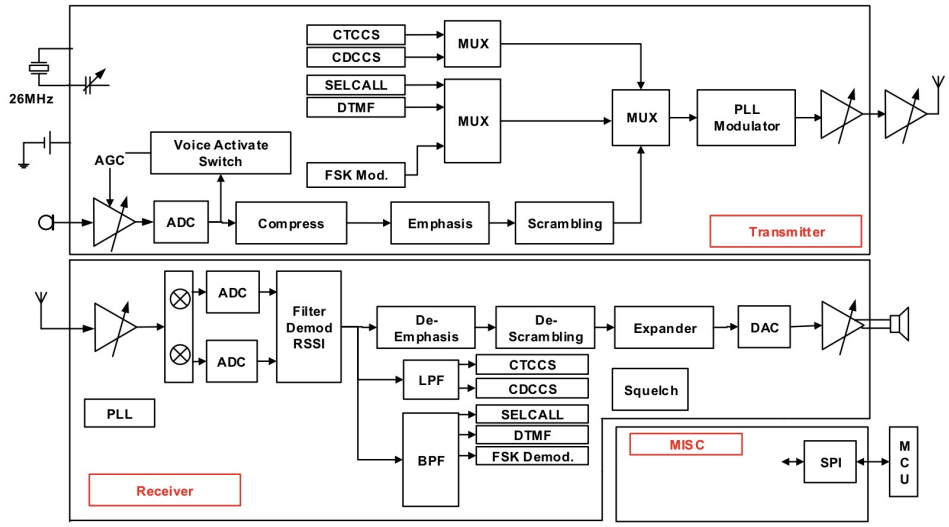
What is inside the radio, then the block diagram of the VHF/UHF chip



## What is inside

| Component | Function |
|---|---|
| **PY32F071 microcontroller (Puya, ARM Cortex-M0+ core)** | Runs the firmware: screen, keypad, memories, dialogue with the radio chips, applications. This is what you reflash (UV Studio, CHIRP for the settings). |
| **BK4829 radio chip (Beken)** | VHF/UHF core: frequency synthesiser, FM and AM demodulation, CTCSS / DCS tones, DTMF, RSSI. The microcontroller drives it over a serial bus; the received range depends on the firmware. |
| **BK1080 FM chip (Beken)** | Broadcast FM receiver (65 to 108 MHz, stereo) controlled by the microcontroller: the firmware's FM radio. Replaced by the Si4732 if you install the module (sheet A06). |
| **Si4732-A10 receiver (Silicon Labs), optional** | Module to add by modifying the radio (fitted as standard on the future UV-K10 Pro). FM, AM, shortwave and longwave receiver with built-in digital signal processing (DSP) and RDS. It provides HF, with the SSB patch in compatible firmwares. |
| **1.44-inch screen** | Monochrome or colour depending on the version; the firmware draws spectrum, waterfall and applications on it. |
| **External flash memory** | Configuration, calibration, logo, logs and overlay applications (see the Flash Map application, sheet A18). |
| **Power** | 1400 mAh lithium battery supplied, a 2500 mAh pack is available as an option. USB-C port: charging, programming, KISS and CW depending on the firmware. |
| **Published measurements** | DARC test (OV 38): about 4.0 to 4.8 W at high power on 145 and 435 MHz, sensitivity of about 0.07 µV for 12 dB SINAD, narrow band 12.5 kHz. |

## The BK4829 radio chip: block diagram



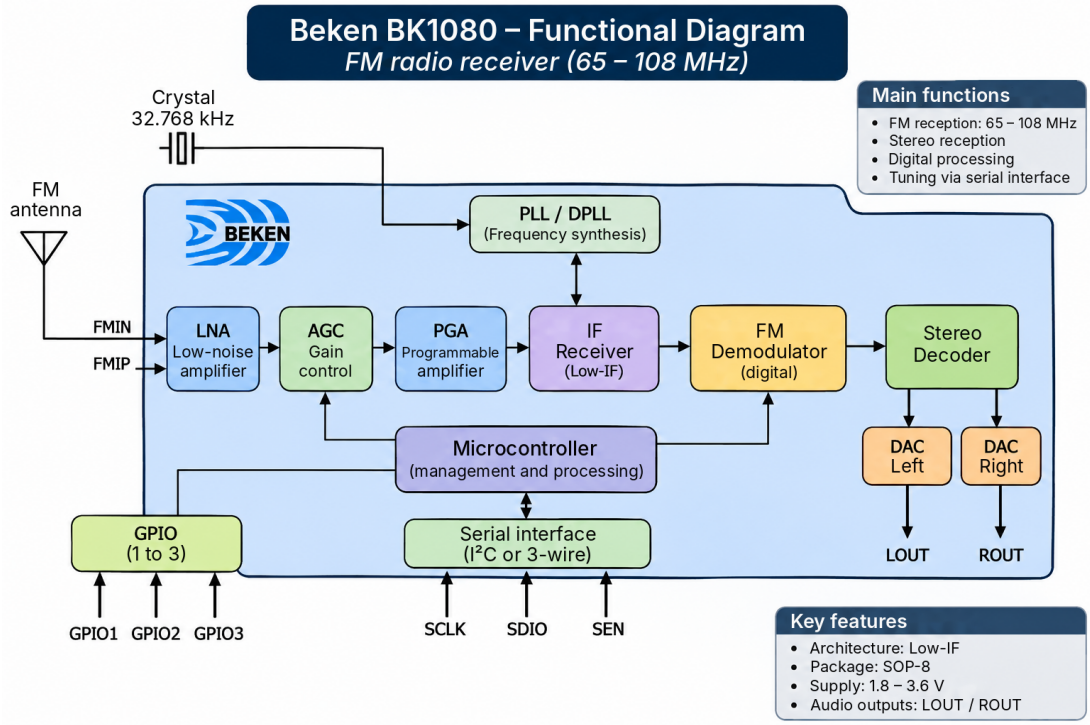

*Block diagram of the BK4829 chip (Beken): transmitter at the top, receiver at the bottom, SPI interface to the microcontroller (labels in English).*

| Block (English) | What it does |
|---|---|
| **Receiver: mixers, 2 ADCs, Filter / Demod / RSSI** | The amplified signal is converted down to two channels (I and Q), digitised, filtered and demodulated (FM, AM); the received level (RSSI) is used for the squelch and the spectrum analyser. |
| **De-Emphasis, De-Scrambling, Expander, DAC** | Audio processing: de-emphasis, descrambling (spectrum inversion), dynamic range expansion, then conversion to analogue sound for the amplifier. |
| **CTCSS, CDCSS, SELCALL, DTMF, FSK Demod.** | Detection of subtones, keypad signalling (DTMF), selective calling and FSK data: this is the basis of selective listening and of several applications. |
| **Squelch** | Mutes the sound as long as no useful signal is received. |
| **Transmitter: AGC, ADC, Compress, Emphasis, Scrambling** | The microphone is amplified, digitised, compressed, pre-emphasised and optionally scrambled before modulation. A voice detector (VOX) can trigger transmission. |
| **MUX, PLL Modulator, amp** | Mixes the voice and the tones (CTCSS, CDCSS, SELCALL, DTMF, FSK) then modulates the carrier produced by the PLL; the power stage follows. |
| **PLL, 26 MHz crystal** | Frequency synthesiser: generates the receive and transmit frequency that the firmware sets. |
| **SPI, MCU** | The microcontroller reads and writes the chip's registers over a serial link. |

# Beken BK1080 FM chip

The broadcast FM receiver: block diagram and role of the blocks

**For the broadcast FM radio.** The firmware's broadcast FM radio (88 to 108 MHz) is received by a **dedicated FM chip**, separate from the BK4829. The diagram below shows it: 65 to 108 MHz stereo FM receiver, Low-IF architecture, controlled over a serial bus. The original UV-K5 uses the BK1080 for this function; check the part number on your UV-K1. When you install the Si4732 module (sheet A06), this is the component you desolder.




*Block diagram of the Beken BK1080 (65 to 108 MHz FM receiver), as supplied.*

| Block | What it does |
|---|---|
| **LNA, AGC, PGA** | Low-noise amplifier, gain control and programmable amplifier: adapt the level of the signal picked up by the FM antenna (inputs FMIN and FMIP). |
| **IF receiver (Low-IF), PLL / DPLL** | Converts the signal to a low intermediate frequency; the digital PLL generates the tuning frequency from a 32.768 kHz crystal. |
| **Digital FM demodulator, stereo decoder** | Recovers the sound and separates the left and right channels. |
| **Left / right DAC, LOUT and ROUT** | Analogue audio outputs to the radio's amplifier. |
| **Internal microcontroller, serial interface, GPIO** | Manages the settings; controlled over I²C or a 3-wire link (SCLK, SDIO, SEN); three GPIO pins. Supply from 1.8 to 3.6 V. |

# Si4732 receiver: a module to add

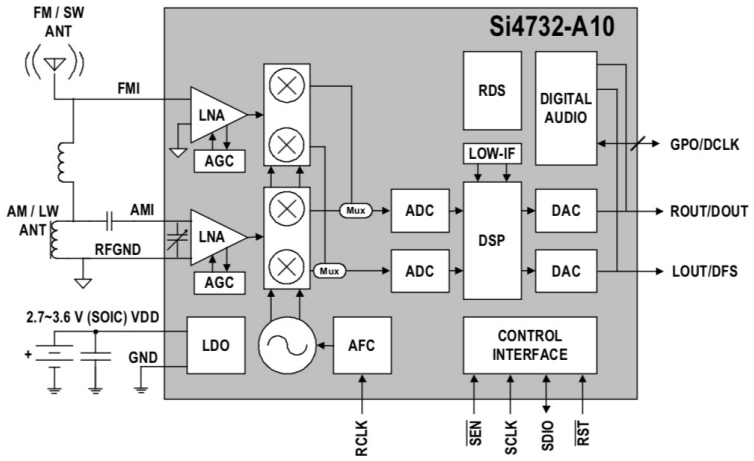
Hardware add-on: modifying the radio, chip diagram, reception ranges



**Replaces the BK1080: it must be desoldered.** The Si4732 module **replaces the BK1080 FM chip** of the UV-K1 (sheet A05): the BK1080 must be **desoldered** from the board before fitting the module. The job is delicate (tiny components, risk of damaging the board) and removes the original FM radio, which the Si4732 then provides along with HF.

**Hardware add-on.** The Si4732-A10 is **not in the original UV-K1**: it is an additional module (hardware add-on) that has to be soldered in, which means **modifying the radio**, with the compatible firmware. The future **UV-K10 Pro** is meant to have it as standard, with no modification. Without it, the UV-K1 only listens to VHF/UHF and what the BK4829 allows.

## The Si4732: chip diagram




*Block diagram of the Si4732-A10 (Silicon Labs): from the antenna to the digital or analogue audio signal.*

| Block | What it does |
|---|---|
| **FM / SW and AM / LW inputs** | Two antenna inputs: one for FM and shortwave, one for medium wave and longwave. |
| **LNA and AGC** | Low-noise amplifier and automatic gain control: prevents overload on strong signals. |
| **Mixers, multiplexer, ADC** | The signal is converted to a low intermediate frequency, then digitised. |
| **DSP** | Filtering, demodulation (FM, AM, SSB with the patch), RDS, bandwidth setting: everything is done by calculation. |
| **DAC, ROUT / LOUT outputs** | Analogue stereo audio output to the radio's amplifier. A digital output (DOUT, DFS) also exists. |
| **AFC, RCLK clock** | Automatic frequency correction and reference clock supplied to the chip. |
| **Control SEN, SCLK, SDIO, RST** | Control bus from the microcontroller and reset. The firmware sends frequency, mode and volume here. |
| **LDO, VDD supply 2.7 to 3.6 V** | Internal regulator: the chip is happy with a 3 V supply. |

*Usual ranges of the Si473x family (datasheet, from memory, to confirm): FM 64 to 108 MHz, longwave 153 to 279 kHz, medium wave 520 to 1710 kHz, shortwave 2.3 to 26.1 MHz. Firmwares and modifications can extend what you can listen to.*

# Which tool for which sheet

Choosing between the UV-K1 and the SDR dongle, and sharing the roles



## Which tool for which sheet

| I want to... | UV-K1 (firmware) | SDR dongle (software) | Sheet |
|---|---|---|---|
| **Listen to aircraft, marine, amateur radio, PMR** | Direct: AM/FM depending on the band, low squelch, memories | Optional: wide spectrum with SDR++ or GQRX | 2, 6, 8, 9 |
| **See aircraft and their messages** | Voice listening only | readsb or dump1090 (ADS-B), acarsdec (ACARS), dumpvdl2 | 10, 11 |
| **Track boats** | ATIS RX (Labs) for inland waterways | AIS-catcher or rtl_ais, OpenCPN | 13 |
| **Track weather balloons** | A crackling carrier | radiosonde_auto_rx, SondeHub | 15 |
| **Read wireless sensors** | Detects a burst only | rtl_433 | 16 |
| **Look for a beacon or a source** | Sig Finder (Labs), AM on 121.5 or 243 MHz | Decoding the 406 MHz frame with dec406 | 22, 33 |
| **Identify a tone signal** | SigID, then decoding on a smartphone (Labs applications: sheets A14 to A18) | SDR + waterfall to see the shape | 5, 32 |
| **Decode FT8, RTTY, SSTV, fax, APRS** | SI4732 in USB, audio to PC or smartphone | Decoder on the computer (fldigi, WSJT-X, Direwolf) | 23 to 29 |
| **Write to another radio without Internet** | GOGUFW Messenger, between GOGUFW radios | Not applicable | this sheet |

## Sharing the roles

- The radio picks up, filters and delivers the signal or the audio. It does not decode heavy digital modes (FT8, AIS, ADS-B).
- The SDR dongle digitises a wide band and decodes, but has no practical squelch and no battery: it stays at home or in the bag with a computer.
- UV Studio's Live Viewer mirrors the radio's screen on the computer: handy for capturing a Sig Finder reading.

**Good to know.** GOGUFW Messenger messages are transmissions: they require a callsign or authorised use on the chosen frequency. Information on Fusion and GOGUFW re-read online; Sonic from its manual; Falcon from its user guide; Labs applications taken from the guide. Apex from the online NR7Y CW documentation.

# Fusion firmware (F4HWN)

The reference firmware: keys, spectrum, UV Studio



## In brief

| Item | Detail |
|---|---|
| **Origin** | Project by armel and muzkr for the UV-K1 and UV-K5 V3. |
| **Repository** | github.com/armel/uv-k1-k5v3-firmware-custom (CHIRP driver: github.com/armel/uv-k5-chirp-driver) |
| **Tools** | UV Studio (flash, calibration, Live Viewer), CHIRP driver shipped with each version. |
| **Strengths** | Spectrum, APRS, hidden menu 76 / 77 (SetNav). It is the base of Labs, GOGUFW, Apex and Falcon. |
| **Editions (since v6.0.0)** | **Fusion**: the reference edition, recommended for beginners and everyday use. **FieldOps**: Fusion plus RescueOps, Fox Hunt and Beacon, for fox hunts (ARDF), SOTA / POTA and emergency communications. **Transfer**: Fusion plus AirCopy and Beam, to clone channels from one radio to another (club, cable, full flash). **Labs**: experimental edition with the overlay applications (sheet A09). Fusion no longer contains AirCopy / Beam, RescueOps, Fox Hunt / Beacon or Breakout: they are in the other editions. |
| **Multiboot (since v6.0.0)** | The external flash keeps a "Main" backup of the first firmware installed and **four firmware slots**, each with its own configuration bank (SetCfg menu). Switch the radio off, hold MENU while powering on, release when the multiboot screen appears: UP / DOWN selects, MENU restores, EXIT leaves without changing anything. **Never cut the power during a restore.** |
| **Version** | The releases page consulted stopped at 6.0.0 (year not shown); check the latest version before flashing. |

**Reading the tables. Long** = long press; **F +** = F key (or F #) then the key. SK1 / SK2 = side keys. UP / DOWN is called LEFT / RIGHT on the UV-K1 depending on the SetNav setting (hidden menu 76 / 77).

## Numeric keypad

| Key | Fusion (F4HWN): long press, then F + |
|---|---|
| **1 BAND** | Long: same as F +. F +: changes band 1 to 7; in memory mode, copies the channel to the VFO |
| **2 A/B** | F +: switches the main VFO (top / bottom) |
| **3 VFO/MR** | F +: toggles frequency / memory |
| **4 FC** | F +: copies frequency and CTCSS; save with M |
| **5 NOAA** | Long: in memory mode, cycles the scan list (OFF, 1 to 24, ALL); in frequency mode, scan range. F +: spectrum analyser |
| **6 H/M/L** | F +: channel power |
| **7 VOX** | Long: VOX on/off. F +: Labs: application launcher |
| **8 R** | Long: reverse (TX on the RX frequency). F +: manual backlight on/off |
| **9 Call** | Long: call channel. F +: end of manual backlight |
| **0 FM** | F +: broadcast FM radio |
| * **SCAN** | Short: DTMF entry. Long: channel or frequency scanner. F +: DCS / CTCSS scanner |

## Control keys

| Key | Fusion |
|---|---|
| **MENU (M)** | Short: menu. Long: "M Long" action; during a scan, excludes the channel. During a scan, keeps the last channel found |
| **EXIT** | Short: exits, deletes a digit. Long: clears the entry, exits DTMF, monitor and ScnRng |
| **UP / DOWN** | Menus and lists. F +: squelch |
| **F #** | Short: activates F +. Long: locks the keypad (SetLck chooses what is locked) |
| **PTT** | Transmits (SetPTT mode: CLASSIC or ONEPUSH). PTT + SK2: 1750 Hz tone. PTT + key: DTMF |
| **SK1 / SK2** | F1Shrt, F1Long, F2Shrt, F2Long programmable. F + SK1 / SK2: changes the step. F + long press: chooses the action |

*SetLck: KEYS locks the keypad; KEYS + ACTIONS also locks the side shortcuts and M Long; KEYS + PTT locks transmission; KEYS + ACTIONS + PTT locks everything.*

# Fusion firmware (F4HWN)

The reference firmware: keys, spectrum, UV Studio



## Personal notes (versions, settings, tips)

*Source: F4HWN wiki (Button functions).*

# Labs firmware (overlay applications)

F + 7 launcher and applications installed with UV Studio



## In brief

| Item | Detail |
|---|---|
| **Origin** | Experimental F4HWN edition (introduced with v6.0.0): applications stored in external flash and loaded into RAM at launch. |
| **Repository** | github.com/armel/uv-k1-k5v3-firmware-custom (same repository as Fusion) |
| **Launch** | F + 7 opens the application launcher. First set the receive VFO (modulation, frequency), then launch the application. |
| **Installation** | Applications are installed with UV Studio (sheet A03, Apps page). They generally need firmware 6.3.0 or later and the matching application version: an application is compiled for a specific firmware version. |
| **Common keys** | **NAV** = UP / DOWN on the UV-K5, LEFT / RIGHT on the UV-K1. **F then X** = F then the key, usually the opposite action. **EXIT** leaves all applications. |
| **Transmission** | Applications marked "transmission" send a signal: an amateur radio callsign or authorised use is required. |

## Where to find each application

| Sheet | Applications |
|---|---|
| **A14** | Sig Finder, FoxHunt, Beacon, Beam, EPIRB 406, QR Map |
| **A15** | ACARS RX, AIS RX, ATIS RX, ATIS DSC RX, Sondes M10 / M20 / RS41, GeoNav |
| **A16** | POCSAG RX, APRS RX / TX / Edit / DIGI, KISS TNC |
| **A17** | ToneCall, CW Decode, CW Keyer, CW USB, SSTV, AF Scope |
| **A18** | AF Spectro, Spectrum3D, SystemInfo, Flash Map, Resistor, Cube3D and Plasma |

*The games are not included. The Labs keys depend on the installed application version.*

## Personal notes (versions, settings, tips)

# Falcon firmware

Spectrum, Sentinel, multi-list watch, FM radio



## In brief

| Item | Detail |
|---|---|
| **Repository** | github.com/Robby69400/Sonic |
| **Version** | **3.0** (current version). UV-K5 V3 and UV-K1. |
| **Installation** | To complete: flashing procedure, CHIRP driver and applications of version 3.0. |
| **Strengths** | Spectrum and Sentinel (LIST, BAND, RANGE), history of 200 detections, dual, triple, quad or multi-list watch, discreet Falcon mode, broadcast FM radio. CHIRP driver included in the archive. |

## For listening

| Action | Keys (VFO) |
|---|---|
| **Spectrum LIST / BAND / RANGE** | F + 7 / F + 8 / F + 9; Sentinel: F + 6 |
| **Scan the channel list / all memories (fast)** | Long press on * / F + * |
| **Scan between two free frequencies (VFO A and B)** | Long press on * when A and B are free |
| **Monitor a frequency** | F + 4 (up to 50 frequencies) |
| **Triple, quad or multi-list watch** | RxMode menu = FULL; priority channels PRICH1 and PRICH2 |
| **Squelch threshold in dBm** | SQLSCN and SQLHYS menus, or F + left / right |
| **Broadcast FM radio** | F + 0; nine favourites, long press 1 to 9 to store |
| **Falcon mode (discreet, backlight off)** | Long press on 7 in SINGLE or DUAL |

## Numeric keypad and controls

| Key | Falcon: long press, then F + |
|---|---|
| **1 BAND** | Long: copies the displayed channel to the VFO at the same frequency. F +: moves the channel to the next list |
| **2 A/B** | Long: switches VFO A/B in DUAL. F + SK1 / SK2: changes the step |
| **3 VFO/MR** | Long: toggles frequency / memory |
| **4 FC** | Long: changes the step. F +: adds or removes the frequency from the watch list (50 max.) |
| **5 NOAA** | Long: wide / narrow bandwidth. F +: applications menu (OA version) |
| **6 H/M/L** | Long: power. F +: Sentinel (LIST / BAND / RANGE) |
| **7 VOX** | Long: Falcon stealth mode. F +: LIST spectrum |
| **8 R** | Long: squelch 1 to 9. F +: BAND spectrum |
| **9 Call** | Long: backlight. F +: RANGE spectrum |
| **0 FM** | Long: modulation FM, AM, USB. F +: broadcast FM radio |
| *** SCAN** | Long: list scan (or range scan between two free VFOs). F +: fast scan of all memories |

| Key | Falcon (in addition to Fusion) |
|---|---|
| **MENU (M)** | Long: torch or configured shortcut. F + MENU: new channel, or EDIT / NEW / DEL |
| **EXIT** | Long: new frequency entry |
| **UP / DOWN** | F + left / right: squelch threshold in dBm |
| **F #** | Same |
| **PTT** | SK1 / SK2: PTT 2 (transmits on the non-selected VFO) |
| **SK1 / SK2** | Same + RXMOD, ScanLst Up, ScanLst Dw |

## In the spectrum and Sentinel

| Key | Action |
|---|---|
| **1 / SK2** | Skips the current reception / blacklists the frequency |
| **2 · 3** | Backlight · normal / fast speed |
| **5 · 6 · 7** | Frequency details · LIST / BAND / RANGE · saves the settings |
| **9 · 0** | History (200 detections) · session modulation |
| *** / F** | Relative squelch up or down |
| **SK1** | NORMAL, FREQ LOCK, MONITOR |
| **MENU · long MENU** | List or band selections · settings page |

# Falcon firmware

Spectrum, Sentinel, multi-list watch, FM radio



*In the spectrum and Sentinel, 9 opens the history, SK1 locks or opens the audio, SK2 blacklists the frequency. Keys taken from the Falcon user guide: to verify on 3.0.*

## Personal notes (versions, settings, tips)

# Sonic firmware (Robby69400)

Fast spectrum, ULTRA WATCH, history and close call



## In brief

| Item | Detail |
|---|---|
| **Origin** | Fork of F4HWN with code from NTOIVOLA, egzumer and Dual Tachyon. Sonic 1K and 4K versions. UV-K1 (USB-C) and RS232 versions. |
| **Repository** | to complete |
| **Strengths** | Spectrum at ~300 channels/s with 50 scan lists and 50 custom bands (CHIRP), ULTRA WATCH ~600 channels/s, history of 200 detections, close call, no hidden menu. |
| **Limits** | No applications; logo and calibration to be handled with F4HWN. |
| **Reset** | Power on holding PTT + SK2 (warning screen, hold for 4 seconds). |
| **LED** | Red when transmitting, green when receiving, flashing yellow during a scan or a close call. |

## Main screen

| Main screen (L = long press) | Action |
|---|---|
| **L0 · L1 · L2** | MR: copies the channel to the VFO · stronger squelch · bandwidth |
| **L3 · L4 · L5** | Modulation AM / FM / USB · power L / M / H · step |
| **L6 · L7** | Toggles MR / VFO · MR: channel list number |
| **L8 · L9** | Torch indication on reception · constant backlight |
| **F2** | MR: deletes the channel; VFO: keeps the frequency |
| **F6 · F7 · F8** | Spectrum by range (RANGE) · by lists · by bands |
| **F0** | Broadcast FM radio |

## In the spectrum / ULTRA WATCH

| In the spectrum / ULTRA WATCH | Action |
|---|---|
| **1 · 2 · 3** | Skips the frequency · backlight · fast or slow scan |
| **4 · 5 · 6** | Scanned lists or bands · options menu (values with 1 and 3) · switches mode lists / bands / range |
| **7 · 8 · 9** | Saves the settings · raw or smoothed spectrum and ULTRA WATCH · modulation |
| **0** | History (200 entries) |
| *** · #** | DYNSQL threshold up or down; long press: DYNSQL off |
| **SK1 · SK2** | Locks the frequency (FL), then monitor · blacklist (# in the history) |
| **MENU** | Low-level registers: LNA, PGA, MIX... (navigate with 2 and 8) |
| **In the history** | 1: scans the history · 2: saves to a free channel · 3: deletes · 5: menu · 6: close call · 8: saves to flash |

## Personal notes (versions, settings, tips)

*Source: Sonic manual.*

# GOGUFW firmware

FSK Messenger between radios, HEARD, Range Check



## In brief

| Item | Detail |
|---|---|
| **Origin** | Fork of F4HWN for the UV-K1 and UV-K5 V3. Stable version read: **3.3.2** (26 October, year not shown). |
| **Repository** | github.com/Gogu-Qs/GOGUFW-UV-K1-Messenger (release v3.3.2) |
| **Messenger** | FSK text messages between GOGUFW radios. |
| **Other functions** | HEARD, Range Check, CALLTX and memory spectrum. |
| **New in 3.3.2** | Text editor with insertion cursor on all input screens (deleting in the middle of the text, immediate space with 0, ABC / abc / 123 indicators); APRS Edit: immediate spaces; FM Memory can use the shared editor; navigation layout kept after power-off (Reset All clears it); spectrum fixed in VFO and memory mode; the "PTT" action becomes "PTT MODE" (toggles SetPTT Classic / OnePush, does not transmit). |
| **Update** | Firmware alone is possible. Applications from 3.3.1 remain compatible. CHIRP module: Gogufw_3.3.2_chirp_module.py. Image of 117,864 / 120,832 bytes. |
| **Applications** | 16 application slots (since 3.0.0), "multiboot" image. |

**To complete.** No key table found online: open the radio's menu or the project manual, then note the keys in the table below.

## Keys and functions (to fill in)

| Key or menu | Action |
|---|---|
| | |
| | |
| | |
| | |

## Personal notes (versions, settings, tips)

# Apex firmware (NR7Y CW)

Morse firmware: iambic keyer, Code Practice, RIT / XIT



## In brief

| Item | Detail |
|---|---|
| **Origin** | CW (Morse) firmware by NR7Y, based on Fusion (F4HWN). UV-K1 and UV-K5 V3. |
| **Name** | The project documentation calls it "NR7Y CW Firmware" and does not mention "Apex". |
| **Repository** | github.com/briand/uv-k1-k5v3-firmware-custom (docs: briand.github.io/cw-firmware-docs) |
| **Version** | Latest release seen on the repository: **v1.3.1** (6 August, year not shown); the documentation's download page announces **1.4** (nr7y.k1-k5v3.v1.4.bin, CHIRP module nr7y.k1-k5v3.chirp.v1.4a.py): to confirm. Repository: "NR7Y CW mod for F4HWN Fusion". |
| **New in 1.3 / 1.3.1** | Fusion 5.7.0 integrated; Code Practice oscillator assignable to an action key; both paddle keys act as a straight key in Port Handkey mode; debounce filter for key and bug; fix for key release (1.3.1). |
| **External paddle** | With a hardware modification, or via a special USB cable with no modification. |
| **Functions** | Automatic iambic keyer (modes A, B, Ultimatic, Bug), paddle on PTT / Side1 or external (with or without hardware modification), macros recorded with the key, Code Practice mode (lamp option) and callsign drills with Farnsworth spacing, CW speed popup, correct roger, timeout if the key stays stuck, RIT / XIT and quick CW actions on the side keys. |
| **Licences** | Firmware Apache 2.0; documentation CC BY-NC-SA 4.0. |

## Flashing and setting up CHIRP

1) Back up the calibration (file name = the radio's serial number). 2) Power on while holding PTT (LED on, black screen). 3) Flash with Armel's UVTools2 or n7six's Multi-firmware tool (K1 / K5 v2 mode); the LED blinks at the start of the update. **CHIRP-next**: enable Help > Developer Mode (once), then File > Load Module with the module of the same version as the firmware. Radio > Download from radio: manufacturer Quansheng, model "UV-K1 & UV-K5 V3 (F4HWN Fusion)", variant NR7Y. The module is loaded for the session only: reload it each time you open CHIRP. The built-in Quansheng models do not handle the CW settings. A recent module refuses to write to a firmware that is too old: flash first.

## CW keys

| Key (long press) | Action |
|---|---|
| **0 FM** | Modulation FM, AM, USB, CW (the squelch opens in USB and CW) |
| **4 FC** | CW filter width (on K1: F then long 4) |
| **5 NOAA** | CW practice oscillator (VFO in CW; on K1: F then long 5) |
| **7 VOX** | CW break-in |
| **8 R** | SSB cross-mode: transmission offset by the tone (CWfreq) |

## Personal notes (versions, settings, tips)

# Overlay applications: Finding a source

Labs, F + 7: Sig Finder, FoxHunt, Beacon, Beam, EPIRB 406, QR Map



## Sig Finder

| | |
|---|---|
| **Function** | Direction-finding aid: relative signal feedback and detection of a distress homing signal (sheet 33) |
| **Keys** | 1: FOLLOW / SWEEP · 2 (F then 2): audio mode · 3 (F then 3) or NAV: attenuation · 4: auto attenuation · 5: decay speed · 6: backlight · M: reset · F held: lock |
| **My notes** | |

## FoxHunt

| | |
|---|---|
| **Function** | Fox hunting aid: signal strength, history, attenuation and audio guidance (sheet 33) |
| **Keys** | 1: bar or history · 2: audio (silence, beeps, sound) · 3: attenuation 0 / 6 / 15 / 27 dB · F then 2 or 3: backwards · M: resets PK, MN and trend · side key: Beacon |
| **My notes** | |

## Beacon (transmit)

| | |
|---|---|
| **Function** | Repeated ARDF-type Morse beacon on the chosen transmit VFO |
| **Keys** | 1: transmit duration 5 to 60 s · 2: silence 5 to 240 s · 3: identifier MOE / MOI / MOS / MOH / MO5 / MO / CALL · M: stops, new silence · side key: FoxHunt |
| **My notes** | |

## Beam (transmit)

| | |
|---|---|
| **Function** | Transfers a channel configuration between compatible radios over the air |
| **Keys** | NAV: transmit or receive · M: sends the channel or waits for a packet · EXIT: stops or quits. Only one channel received per launch |
| **My notes** | |

## EPIRB 406

| | |
|---|---|
| **Function** | Receives and decodes messages from first-generation 406 MHz Cospas-Sarsat beacons (sheet 22) |
| **Keys** | NAV: previous or next message · 4 / 6: -5 / +5 kHz (±50 kHz) · 5: back to VFO · 1: speaker · *: compact view · M: clears · 7 then M: QR Map |
| **My notes** | |

## QR Map

| | |
|---|---|
| **Function** | Shows a position sent by another application as a map QR code to scan with a phone |
| **Keys** | 1 Google · 2 Apple · 3 geo: · 4 OpenStreetMap · 5 text · NAV: next format · EXIT: back to the application · EXIT held: normal screen |
| **My notes** | |

*NAV = UP / DOWN (UV-K5) or LEFT / RIGHT (UV-K1). F then X = F then the key, usually the reverse action. EXIT quits the application. "Transmit" applications require an amateur radio callsign or authorised use.*

# Overlay applications: Aircraft, boats and sondes

Labs, F + 7: ACARS RX, AIS RX, ATIS RX, ATIS DSC RX, Sonde M10, M20, RS41, GeoNav



## ACARS RX

| | |
|---|---|
| **Function** | Receives and browses VHF ACARS messages from aircraft (sheet 11) |
| **Keys** | NAV: scrolls, then newer or older message · *: compact view · 1: audio · F: decoded or raw message · 2: clears (AM, local ACARS frequency) |
| **My notes** | |

## AIS RX

| | |
|---|---|
| **Function** | Receives AIS reports from ships on AIS 1 or AIS 2 (sheet 13) |
| **Keys** | NAV: scrolls, then next ship (8 kept) · *: compact view · 1: raw audio · M: freezes the list · 2: clears (FM, 161.975 or 162.025 MHz) |
| **My notes** | |

## ATIS RX

| | |
|---|---|
| **Function** | Decodes the identity of inland waterway ATIS stations at the end of a transmission (sheet 13) |
| **Keys** | NAV: newer or older station (8 kept) · 1: speaker · 2: clears |
| **My notes** | |

## ATIS DSC RX

| | |
|---|---|
| **Function** | Receives VHF digital selective calling on marine channel 70 (sheet 14) |
| **Keys** | NAV: newer or older call (6 kept) · 1: audio · 2: clears. Distress and urgency: inverted screen, green LED, beep (channel 70, 156.525 MHz) |
| **My notes** | |

## Sonde M10, M20, RS41

| | |
|---|---|
| **Function** | Decode telemetry from Meteomodem M10 and M20 and Vaisala RS41 radiosondes: position, movement, and depending on the model temperature, humidity, battery (sheet 15) |
| **Keys** | 1: speaker · 2 (F then 2): +5 / -5 kHz, ±50 kHz · 3 (F then 3): next / previous attenuation · 4: back to the VFO frequency · 5: decode beep · EXIT: restores the VFO. M10: standard 0x9F frames only; RS41: no temperature, humidity or pressure, and a clean signal is needed (no error correction) |
| **My notes** | |

## GeoNav

| | |
|---|---|
| **Function** | Shows azimuth, distance, altitude and elevation between the APRS "home" position and a decoded radiosonde (sheet 15) |
| **Keys** | M in a sonde application (M10, M20 or RS41) opens GeoNav; EXIT returns to the sonde. "Home" position taken from APRS Edit. Reception is paused during GeoNav |
| **My notes** | |

*NAV = UP / DOWN (UV-K5) or LEFT / RIGHT (UV-K1). F then X = F then the key, usually the reverse action. EXIT quits the application. "Transmit" applications require an amateur radio callsign or authorised use.*

# Overlay applications: POCSAG, APRS and KISS

Labs, F + 7: POCSAG RX, APRS RX, APRS TX, APRS Edit, APRS DIGI, KISS TNC



## POCSAG RX

| | |
|---|---|
| **Function** | Receives and browses POCSAG 512 and 1200 baud messages, with optional tone-only retention |
| **Keys** | NAV: scrolls, then next page · *: compact view · 1: raw audio · 2: clears · 3: details or statistics · 4: keeps tone-only pages (wide FM, 10 pages) |
| **My notes** | |

## APRS RX

| | |
|---|---|
| **Function** | Receives, decodes and browses APRS packets on the receive VFO (sheet 26) |
| **Keys** | NAV: scrolls, then next frame (5 kept) · *: compact view · 1: speaker · 2: clears |
| **My notes** | |

## APRS TX (transmit)

| | |
|---|---|
| **Function** | Sends an APRS position packet from the transmit VFO (sheet 26) |
| **Keys** | PTT or M: sends a frame · 1 (F then 1): tone level · 2 (F then 2): 2200 Hz twist · 3: beacon interval 1 / 5 / 15 / 30 / 60 min (F then 3: off) · 0: APRS Edit · *: view · NAV: scrolls |
| **My notes** | |

## APRS Edit

| | |
|---|---|
| **Function** | Edits callsign, position, path, symbol and comment used by APRS TX (sheet 26) |
| **Keys** | NAV: field · M: opens the editor, confirms · EXIT: cancels the field, or saves and returns to APRS TX · digits: input · *: text mode, hemisphere · F: clears · PTT: preset messages (comment) |
| **My notes** | |

## APRS DIGI (transmit)

| | |
|---|---|
| **Function** | Repeats eligible WIDE1-1 or WIDE2-1 packets, as a fill-in digipeater (sheet 26) |
| **Keys** | 1: W1 or W12 · EXIT: stops. Callsign of 1 to 6 characters in CHIRP (Message Line 1), non-zero squelch |
| **My notes** | |

## KISS TNC

| | |
|---|---|
| **Function** | Links APRSdroid or another KISS client over USB-C (sheet 27) |
| **Keys** | 1: audio · 2: test frame to the host, no radio · 3: receive-only mode · EXIT: stops (sheet 27) |
| **My notes** | |

*NAV = UP / DOWN (UV-K5) or LEFT / RIGHT (UV-K1). F then X = F then the key, usually the reverse action. EXIT quits the application. "Transmit" applications require an amateur radio callsign or authorised use.*

# Overlay applications: Tones, Morse and SSTV

Labs, F + 7: ToneCall, CW Decode, CW Keyer, CW USB, SSTV, AF Scope



## ToneCall (transmit)

| | |
|---|---|
| **Function** | Sends editable ZVEI or CCIR sequential selective calls (sheet 25) |
| **Keys** | List: NAV chooses the preset (1 to 4), M or PTT sends, F edits. Editor: 0 to 9 digits (8 max.), F clears, * standard (ZVEI-1 / 2, CCIR-1 / 2), M keeps |
| **My notes** | |

## CW Decode

| | |
|---|---|
| **Function** | Decodes carrier telegraphy (A1A) with crystal correction and USB listening (sheet 34) |
| **Keys** | NAV: scrolls · 0: audio · *: font · 1 / F then 1: threshold margin · 2 / F then 2: speed · 3: AGC · 4: recalibrate · 5: clears · 6 / F then 6: crystal correction · M: help |
| **My notes** | |

## CW Keyer (transmit)

| | |
|---|---|
| **Function** | Sends Morse by hand with the PTT or the side keys (sheet 34) |
| **Keys** | PTT: carrier · 1: STRAIGHT / PADDLE · 8: AUTO, IAMB A, IAMB B · 2 / 3 / 4 (F then: backwards): speed, tone, delay · 5: swaps dots and dashes · 0: monitoring · 6: crystal · NAV: tuning ±50 kHz · M: help |
| **My notes** | |

## CW USB (transmit)

| | |
|---|---|
| **Function** | Sends Morse with a passive key or paddle on the USB-C (sheet 34) |
| **Keys** | PTT: carrier · SIDE1 / SIDE2 or USB-C: key · 1: STRAIGHT / PADDLE · 2 / 3 / 4 / 6: speed, tone, delay, crystal (F then: decrease) · 5: swap · 7: SIDE / USB port · NAV: tuning 100 Hz · 0: monitoring. Passive adapter only, never on a charger or computer |
| **My notes** | |

## SSTV (transmit)

| | |
|---|---|
| **Function** | Sends the startup logo and receives monochrome SSTV images in 14 modes (sheet 28) |
| **Keys** | M: sends the logo · PTT: ignored · 1 (F then 1): transmit mode · 2: 1 bit / dithered · 3: speaker · 4: image / information · EXIT: interrupts. VFO in FM |
| **My notes** | |

## AF Scope

| | |
|---|---|
| **Function** | Shows the demodulated audio against time, or tunes a stable tone (sheet 32) |
| **Keys** | M: SCOPE, TUNE, AF Spectro · NAV: time base · * held: freezes · 1: STD / RAW · 2: speaker · 3: gain · 4: edge · 5: persistence · 0: clears |
| **My notes** | |

*NAV = UP / DOWN (UV-K5) or LEFT / RIGHT (UV-K1). F then X = F then the key, usually the reverse action. EXIT quits the application. "Transmit" applications require an amateur radio callsign or authorised use.*

# Overlay applications: Measurement, utilities and demos

Labs, F + 7: AF Spectro, Spectrum3D, SystemInfo, Flash Map, Resistor, Cube3D, Plasma



## AF Spectro

| | |
|---|---|
| **Function** | Shows the demodulated audio as a spectrum or a spectrogram (sheet 32) |
| **Keys** | M: SPECTRO, SONO, AF Scope · NAV: range 80 / 60 / 40 dB · * held: freezes · 1: STD / RAW · 2: speaker · 3: sampling 9.6 / 19.2 / 38.4 kHz · 0: clears |
| **My notes** | |

## Spectrum3D

| | |
|---|---|
| **Function** | Live radio spectrum as a scrolling 3D landscape |
| **Keys** | NAV: centre · 1 / 3 (F then: reverse direction): ±6.25 kHz, range · 4 / 6: rotation · 2 / 8: tilt · 5: reset · 0: peak listening · *: speed · M: freezes |
| **My notes** | |

## SystemInfo

| | |
|---|---|
| **Function** | Information on firmware, hardware, battery, memory and channels |
| **Keys** | NAV: scrolls (seven sections: firmware, system, battery, memory, identity, code, wiki) |
| **My notes** | |

## Flash Map

| | |
|---|---|
| **Function** | Layout of the external flash memory and names of firmwares and application slots |
| **Keys** | 1 to 0: regions (config, calibration, logo, log, firmware, apps, Sonic 4K...) · NAV: region (F: first or last) · M: read-only dump · *: HEX / TEXT |
| **My notes** | |

## Resistor

| | |
|---|---|
| **Function** | Calculates resistance, tolerance and temperature coefficient from 3 to 6 colour bands |
| **Keys** | NAV: band · * (F then *): next / previous colour · 0 to 9: colour · M: 3, 4, 5, 6 bands |
| **My notes** | |

## Cube3D, Plasma

| | |
|---|---|
| **Function** | Graphics demos: 3D solid in filled or wireframe rendering; nine animated scenes |
| **Keys** | Cube3D: NAV speed · 1 to 9 shape · * next · F then *: wireframe / filled · M pause. Plasma: NAV speed · 1 to 9 scene · * rendering · 0 invert · F: auto cycle · M pause |
| **My notes** | |

*NAV = UP / DOWN (UV-K5) or LEFT / RIGHT (UV-K1). F then X = F then the key, usually the reverse action. EXIT quits the application. "Transmit" applications require an amateur radio callsign or authorised use.*

# Antennas to build: overview

Choose a plan, lengths to cut, materials and safety



**Before you start.** The lengths come from the formula in sheet 1 (λ = 299.79 / f in m, with f in MHz) and a factor of 0.95 for a conductor in air. They are **theoretical**: cut 3 to 5% too long, then shorten in small steps while watching the SWR on an analyzer (nanoVNA type) or, without an instrument, by listening to a beacon or a known signal. These antennas are for **listening**; to transmit, you must be a licensed ham on your band and hold the corresponding authorisation.

## Which antenna to choose

| Plan | Frequency | For listening to | Difficulty |
|---|---|---|---|
| **1. V dipole** | 137 MHz | Weather satellites (APT), VHF in general | Easy |
| **2. λ/4 vertical** | VHF to UHF | FM, aviation, ADS-B, PMR, radiosondes | Easy |
| **3. J-pole** | 137 to 162 MHz | Marine VHF, AIS, 2 m | Medium |
| **4. Half-wave wire (EFHW)** | HF 3 to 30 MHz | 80 / 40 / 20 m, broadcasting, hams | Medium |
| **5. CJU (loop + rod)** | ~435 MHz | UHF satellites, hand-held | Medium |
| **6. IOio (EA4CYQ)** | 145 and 435 MHz | Amateur FM satellites, hand-held | Hard |
| **7. SATCOM EA1HBX** | 240 to 290 MHz | Military UHF satellites (listening only) | Medium |
| **8. Turnstile I6IBE** | 255.55 MHz | Military UHF satellites, circular polarisation | Hard |

## Lengths to cut (factor 0.95)

| Band / use | f (MHz) | V dipole: each element | Vertical: element; radials | J-pole: stub / radiator / total | J spacing |
|---|---|---|---|---|---|
| **Weather satellite APT** | 137.5 | 51.8 cm | 51.8 cm; 52 cm | 51.8 / 103.6 / 155 cm | 4.0 cm |
| **2 m / APRS** | 145 | 49.1 cm | 49.1 cm; 49 cm | 49.1 / 98.2 / 147 cm | 3.8 cm |
| **Marine VHF** | 156.8 | 45.4 cm | 45.4 cm; 45 cm | 45.4 / 90.8 / 136 cm | 3.5 cm |
| **AIS** | 162 | 44.0 cm | 44.0 cm; 44 cm | 44.0 / 87.9 / 132 cm | 3.4 cm |
| **70 cm / PMR446** | 433 | 16.4 cm | 16.4 cm; 16 cm | 16.4 / 32.9 / 49 cm | 1.3 cm |
| **ADS-B** | 1090 | 6.5 cm | 6.5 cm; 7 cm | not recommended | - |

| EFHW (λ/2 wire) | f (MHz) | Wire length | Counterpoise | Usable bands |
|---|---|---|---|---|
| **80 m** | 3.65 | 39.0 m | 3 to 5 m | 80 m, 40 m (2nd harmonic), 20 m, 15 m |
| **40 m** | 7.1 | 20.1 m | 2 to 3 m | 40 m, 20 m, 10 m |
| **20 m** | 14.2 | 10.0 m | 1.5 to 2 m | 20 m, 10 m |

*Values calculated for elements in air, to be checked with the analyzer: a thick tube or an insulated wire shortens the length by a few percent. These plans are ham classics calculated from memory: check the unun and the J-pole tap point against a reliable diagram before any transmitting use.*

## Materials and safety

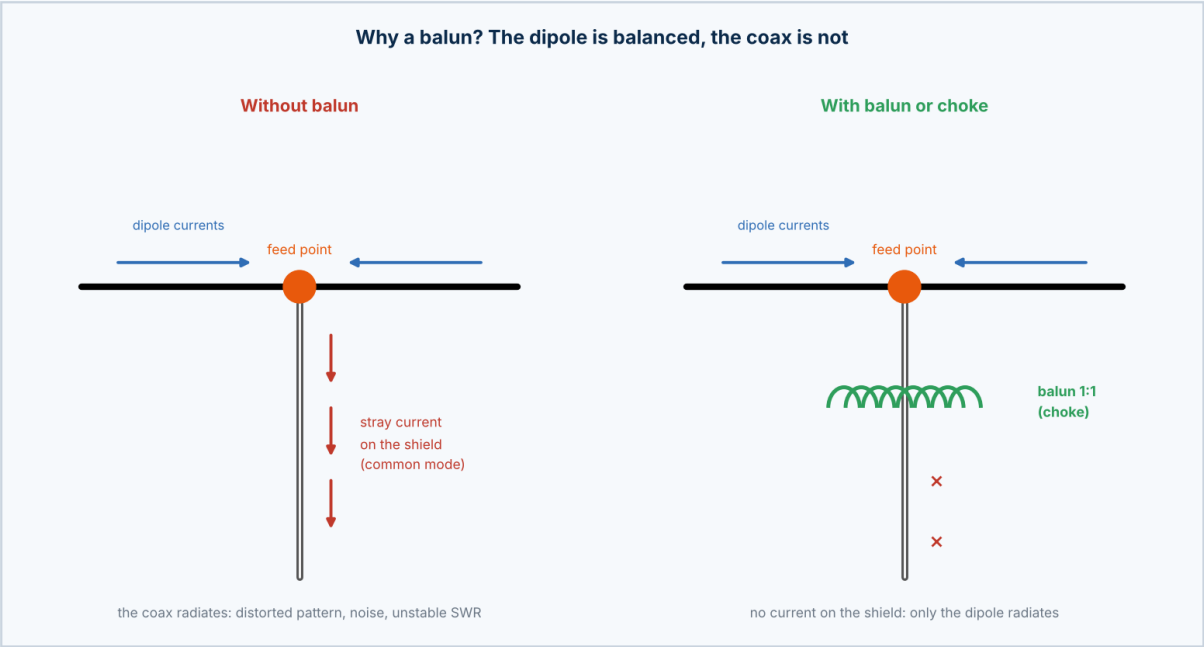
- **Supplies**: 50 Ω coax (RG58, or better RG213 beyond 10 m), SMA / BNC / SO-239 connector base, 2 mm copper wire, heat-shrink, stainless clamps, ferrites, PVC tube for the mast.
- **Soldering**: element on the centre conductor, radials on the connector base ground; then protect against water (varnish, silicone, heat-shrink).
- **Safety**: put them up far from any power line, in calm weather and with two people. A mast or wire that touches a line can be deadly. An outdoor antenna needs earthing and a lightning arrester.

# Baluns and ununs

Matching an antenna to the coax: balanced, unbalanced, impedance ratio



**In brief.** A **balun** (balanced-unbalanced) connects a **balanced** antenna (a dipole: two identical elements) to an **unbalanced** cable (coax: a centre conductor and a shield). An **unun** (unbalanced-unbalanced) only changes the impedance, like the 49:1 transformer of the EFHW antenna (sheet A24). Without them, the coax shield starts to radiate: the pattern is distorted, noise gets into the radio, and the SWR depends on the cable length.




## Which device for which antenna

| Antenna | Impedance at the feed point | Device | Sheet |
|---|---|---|---|
| **Straight or V dipole** | ≈ 50 to 75 Ω (balanced) | 1:1 balun or coax choke | A21 |
| **Folded dipole** | ≈ 288 to 300 Ω | 4:1 balun | - |
| **End-fed half-wave wire (EFHW)** | ≈ 2,500 Ω | 49:1 unun (or 64:1) | A24 |
| **Any wire (long wire, "random wire")** | ≈ 450 Ω, variable | 9:1 unun and counterpoise | - |
| **Quarter-wave vertical (ground plane), J-pole** | ≈ 36 Ω, unbalanced | No balun; a choke is sometimes useful | A22, A23 |

*Usual impedance values, which vary with height, ground and nearby conductors: measure with an analyzer (nanoVNA type) before relying on them.*

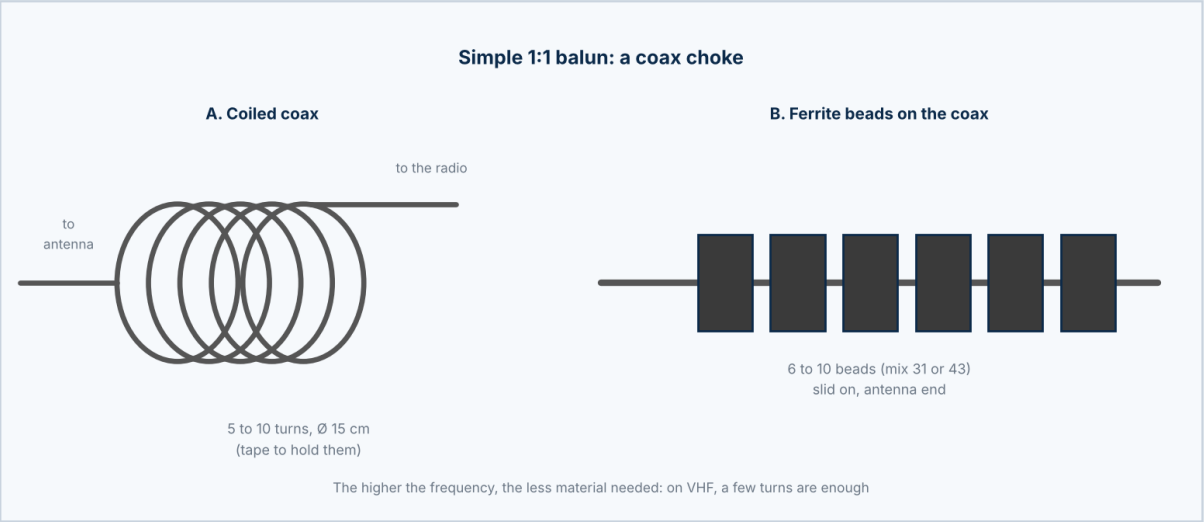
## The three setups to know

- **1:1 balun (choke, or current balun)**: does not change the impedance, it only blocks stray current on the shield. It is the most useful one for a dipole.
- **4:1 balun**: divides the impedance by 4. It is used to go from 200 to 300 Ω (folded dipole, balanced antenna) to 50 to 75 Ω.
- **49:1 unun**: divides the impedance by 49 (≈ 2,500 to 50 Ω) for an end-fed wire. The impedance ratio is the square of the turns ratio: 14 turns against 2 give 7, i.e. $7^2 = 49$.

# Baluns and ununs

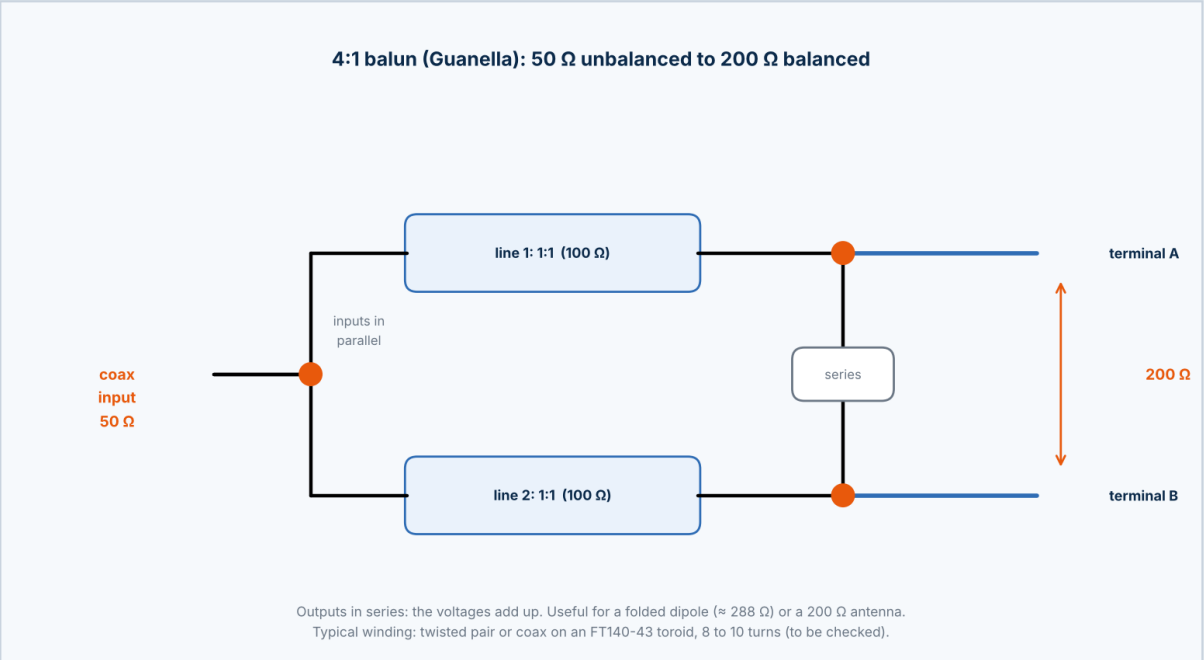
Matching an antenna to the coax: balanced, unbalanced, impedance ratio

## 1:1 balun: the coax choke




- The coax loop is the simplest: 5 to 10 turns of Ø 15 cm at the exit of the feed point (as on the V dipole of sheet A21). It works mainly on VHF and UHF.
- On HF, more impedance is needed: ferrite beads or a toroid (mix 31 for low HF, 43 for high HF and low VHF, 61 for VHF). A few turns of coax on an FT240 or FT140 toroid make a good choke.

## 4:1 balun (Guanella)



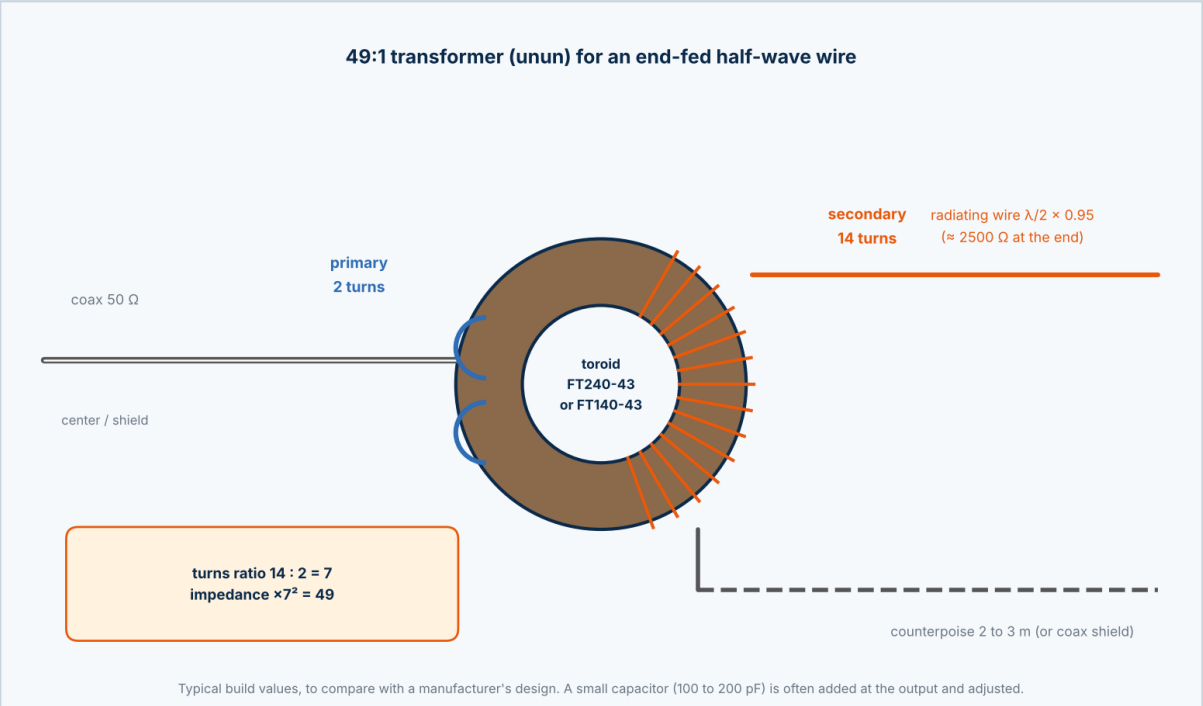

- The Guanella design puts two 1:1 lines in parallel at the input and in series at the output. The voltage doubles, the current is halved: the impedance is multiplied by 4.
- Build it on an FT140-43 toroid with two conductors wound together. The number of turns depends on the band: the lower the band, the more turns are needed.

# Baluns and ununs

Matching an antenna to the coax: balanced, unbalanced, impedance ratio



## 49:1 unun for half-wave wire (EFHW)




- An end-fed λ/2 wire presents a very high impedance, about 2,500 Ω. The toroid brings it down towards 50 Ω with a turns ratio of 14 : 2.
- The counterpoise (2 to 3 m of wire) or the coax shield serves as the return. A few ferrites on the coax at the radio end limit stray current.
- A small capacitor (100 to 200 pF) in parallel on the output is often added to improve the SWR; its value is tuned with the analyzer. Protect the build in a waterproof box.

## Components needed

| Build | Components | Remark |
|---|---|---|
| **Coax choke** | RG58 or RG316 coax (3 to 5 m more than needed); tape or clamps | Ø 15 cm loop, 5 to 10 turns |
| **Ferrite choke** | Beads or toroid (FT240-31, FT140-43 or equivalent); coax | Toroid for HF, beads for VHF |
| **4:1 balun** | FT140-43 toroid; enamelled wire or twisted cable; box; two terminals or SO-239 | Turns to check against a manufacturer diagram |
| **49:1 unun** | FT240-43 or FT140-43 toroid; enamelled wire (≈ 1 mm); waterproof box; SO-239; output terminal; 100 to 200 pF capacitor (optional) | 2 turns and 14 turns |
| **Counterpoise** | 2 to 3 m of insulated wire; lug; insulator | Can replace the ground |

*List drawn up from classic ham builds: check the part numbers and the number of turns before soldering, and check the result with the analyzer.*

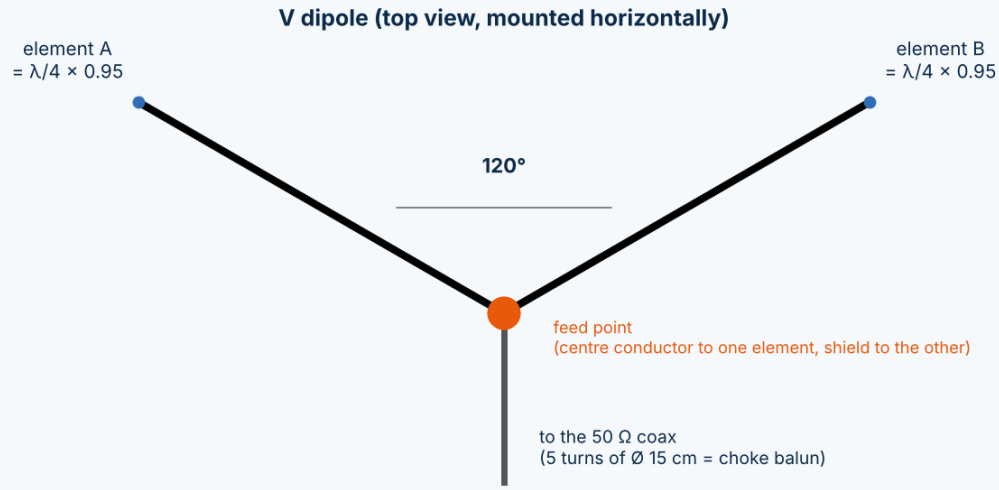
> **Safety.** Ferrite toroids heat up if the build is badly matched: test at low power and touch the toroid. These builds are meant first for **listening**; for transmitting, the power and band depend on your licence.

## Personal notes (toroid used, number of turns, SWR measurements)

# Antenna no. 1: V dipole

137 MHz · easy

A2 · DIY




Example 137.5 MHz (weather satellites): 52 cm elements. Mount on an insulating support (PVC tube), horizontal, clear sky view.

- Two λ/4 × 0.95 elements, spread 120° apart, horizontal, with a clear view of the sky. The feed point is at the centre, on an insulating support.
- The coax comes in from below. Make 5 turns of Ø 15 cm at the exit, to block the currents that climb back up the cable (choke).
- For another frequency, take the element length from the table below.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Radiating elements** | 2 × 51.8 cm (λ/4 × 0.95) | Ø 2 mm copper wire or aluminium rod, spread 120° apart |
| **Central insulating support** | 1 | PVC, treated wood or plastic box; feed point at the centre |
| **Connector base** | 1 | SO-239 or BNC, centre conductor on one element, ground on the other |
| **50 Ω coax** | 5 to 10 m | RG58; 5 turns of Ø 15 cm at the exit (choke) |
| **Mast or tube** | 1 | PVC, 1.5 to 2 m, clear view of the sky |
| **Clamps, heat-shrink, silicone** | as needed | protect the solder joints from water |

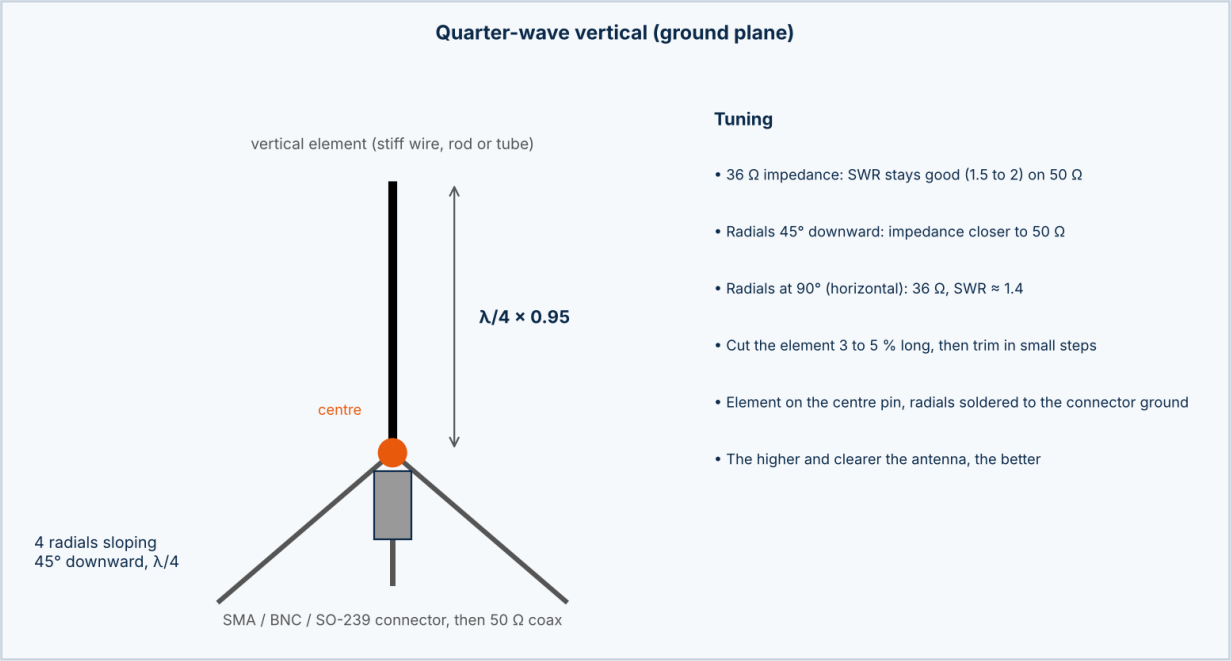
*List drawn up from the plan: check the part numbers and quantities before buying.*

## Personal notes (equipment used, settings, SWR measurements)

# Antenna no. 2: quarter-wave vertical

VHF to UHF · easy

A2 · DIY




- One λ/4 × 0.95 element connected to the centre conductor of a connector base, four λ/4 radials connected to its ground, sloping down at 45°. The antenna is **omnidirectional**, vertically polarised, like the whips of handheld radios and repeaters.
- For the UV-K1, the element alone screws onto the SMA (the radio's case acts as the ground plane); with the radials, put the connector base on a mast or a roof and link it to the radio with a short coax.
- ADS-B variant (1,090 MHz): 6.5 cm element, 7 cm radials, on a piece of PCB or a metal plate.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Radiating element** | 1 × λ/4 × 0.95 | 51.8 cm at 137 MHz; 49.1 cm at 145 MHz; 16.4 cm at 433 MHz |
| **Radials** | 4 × λ/4 | copper wire or rod, sloping down at 45° |
| **Connector base** | 1 | SO-239, BNC or SMA; element on the centre conductor, radials on the ground |
| **50 Ω coax** | short (1 to 5 m) | RG58 |
| **Mast, plate or metal roof** | 1 | supports the connector base; for the UV-K1 alone, the element screws onto the SMA |
| **Heat-shrink, silicone** | as needed | protect the solder joints |
| **ADS-B 1,090 MHz (variant)** | element 6.5 cm, radials 7 cm | on a piece of PCB or a metal plate |

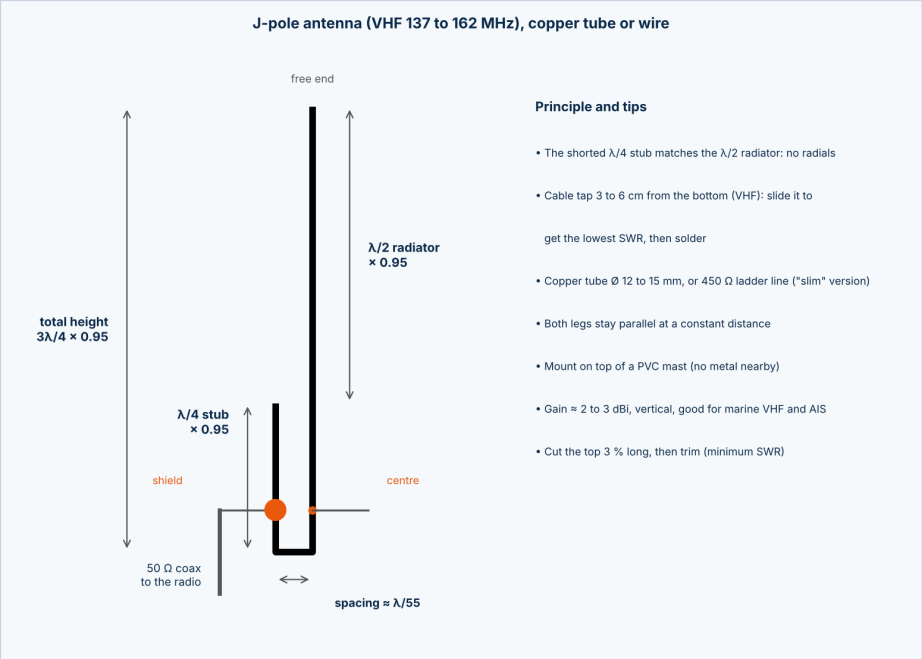
*List drawn up from the plan: check the part numbers and quantities before buying.*

## Personal notes (equipment used, settings, SWR measurements)

# Antenna no. 3: J-pole

137 to 162 MHz · medium

A2 · DIY




- A 3λ/4 tube bent into a J: the short side (λ/4 stub) matches the impedance, the long side (λ/2) radiates. No radials, no balun: the antenna mounts at the top of a mast, away from metal masses.
- The cable tap point is tuned by sliding it (3 to 6 cm from the bottom in VHF), with the analyzer. Solder once the SWR is at its minimum, then protect the joint (heat-shrink, silicone) and plug the top of the tube.
- Gain of 2 to 3 dBi, vertical polarisation: suited to marine, AIS and repeaters.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Copper tube (or rigid wire)** | ≈ 3λ/4 in total | e.g. at 145 MHz: stub 49.1 cm + radiator 98.2 cm (total 147 cm) |
| **Fittings** | 1 × 180° elbow (or 2 × 90°) + 1 tee | copper, to solder; bends into a J: short side λ/4, long side λ/2 |
| **Cap** | 1 | closes the top of the tube |
| **Connector base or coax cable** | 1 | SO-239, or coax soldered directly; tap point 3 to 6 cm from the bottom, to tune |
| **PVC mast and clamps** | 1 | away from metal masses |
| **Heat-shrink, silicone** | as needed | protect the tap point solder joint |
| **Antenna analyzer (nanoVNA)** | 1, to tune | without one: listen to a beacon or a known signal |

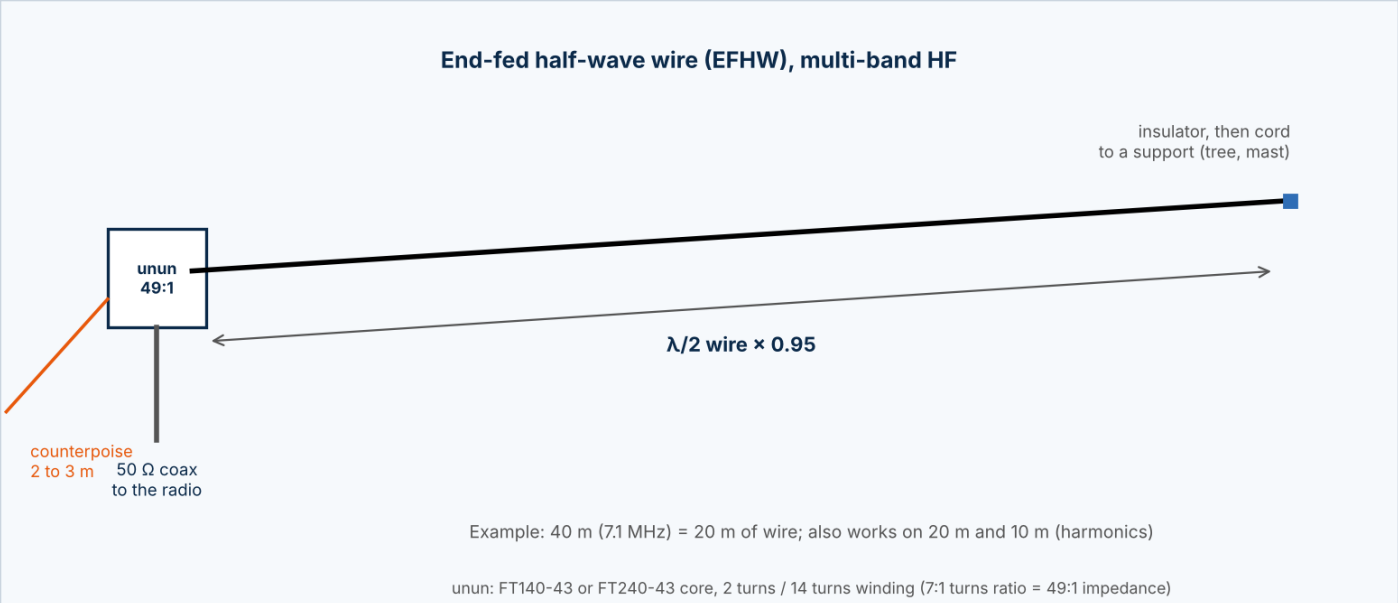
*List drawn up from the plan: check the part numbers and quantities before buying.*

## Personal notes (equipment used, settings, SWR measurements)

# Antenna no. 4: half-wave wire (EFHW)

HF 3 to 30 MHz · medium

A2 · DIY




- A λ/2 × 0.95 wire connected to a 49:1 impedance transformer (unun): the end of a half-wave wire presents about 2,500 Ω, which the unun brings down towards 50 Ω. Cut for 40 m, it also works on 20 m and 10 m (harmonics).
- A 2 to 3 m counterpoise runs from the unun; otherwise, the coax acts as the counterpoise, with a few ferrites near the unun. Stretch the wire 5 m high or more, sloping towards a tree or a mast.
- It is the quickest to set up on HF: one wire, two ropes. The unun winding (2 turns / 14 turns on an FT140-43 or FT240-43 toroid) is given from memory: compare it with a manufacturer diagram.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Radiating wire** | 20.1 m for 40 m | 39.0 m for 80 m; 10.0 m for 20 m; insulated copper wire |
| **Impedance transformer (unun) 49:1** | 1 | FT140-43 or FT240-43 toroid; 2 turns / 14 turns (given from memory: compare with a diagram) |
| **Waterproof box** | 1 | houses the unun |
| **Connector base** | 1 | SO-239 on the box |
| **Counterpoise** | 2 to 3 m | wire connected to the unun ground; or ferrites on the coax |
| **Insulators and rope** | 2 + 2 | to stretch the wire 5 m high or more, sloping |
| **50 Ω coax** | depending on distance | RG58; RG213 beyond 10 m |

*List drawn up from the plan: check the part numbers and quantities before buying.*

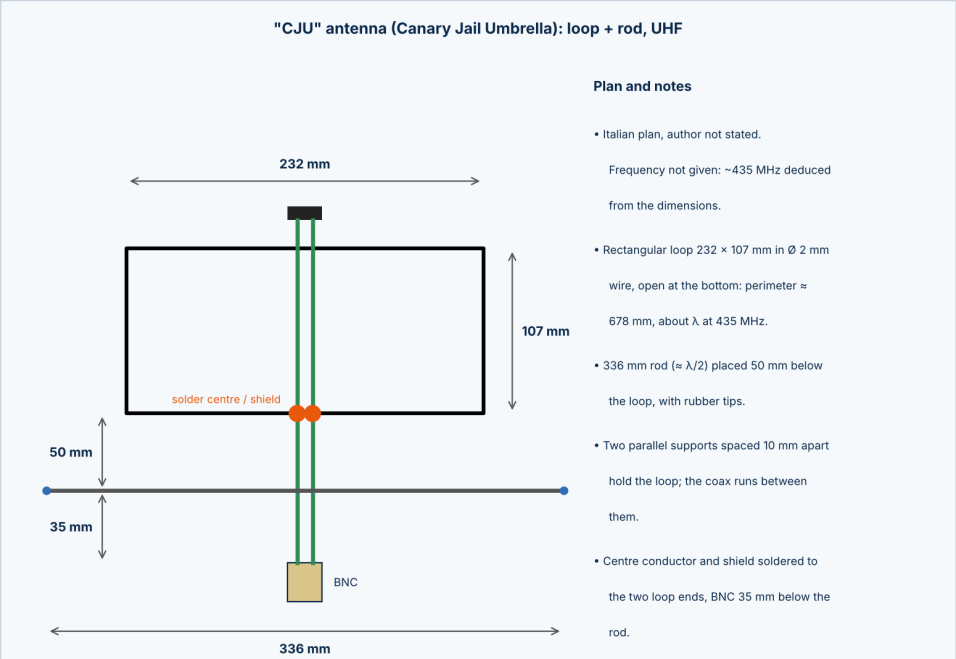
## Personal notes (equipment used, settings, SWR measurements)

# Antenna no. 5: "CJU" (loop and rod)

~435 MHz · medium

A2 · DIY

Satellites pass quickly and change direction: a directional antenna has to be tracked by hand. The antennas in this chapter are therefore **light**, to hold in your hand or place on a tripod, and several have **circular polarisation** (turnstile), which avoids fading caused by the satellite's rotation. The dimensions come from amateur plans that I redrew; I have not built them. For frequencies and what the links carry, see sheet 31. **Listening is free; never transmit** on military satellites.




- A rectangular loop of 232 × 107 mm and a 336 mm rod, 50 mm below it, on two parallel supports: very compact, it fits in a bag.
- Italian plan, author unknown. The frequency is not given: the dimensions correspond to about 435 MHz (loop perimeter ≈ λ, rod ≈ λ/2).
- The original plan gives no tuning: measure the SWR before using the antenna. Ø 2 mm wire, waterproof solder joints.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Ø 2 mm copper wire** | loop of 232 × 107 mm + 336 mm rod | loop perimeter ≈ 678 mm |
| **Parallel insulating supports** | 2 | rod 50 mm below the loop; wood, PVC or rigid plastic |
| **Connector** | 1 | BNC, SMA or SO-239 depending on the radio |
| **50 Ω coax** | short | RG58 or thinner and flexible for the hand |
| **Handle** | 1 | PVC tube or wooden handle |
| **Solder, heat-shrink** | as needed | waterproof solder joints |

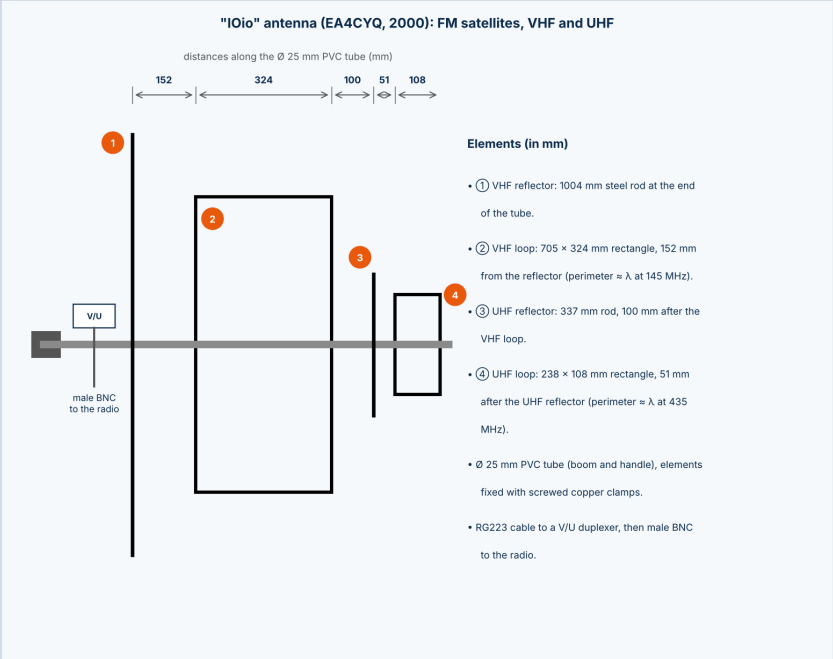
*List drawn up from the plan: check the part numbers and quantities before buying.*

## Personal notes (equipment used, settings, SWR measurements)

# Antenna no. 6: "IOio" (EA4CYQ)

145 and 435 MHz · hard

A2 · DIY




- Two rectangular loops, each with its reflector rod, on a Ø 25 mm PVC tube held in the hand: one for 145 MHz, one for 435 MHz. A V/U duplexer joins the two on a single cable.
- VHF loop 705 × 324 mm (perimeter ≈ λ at 145 MHz), UHF loop 238 × 108 mm (≈ λ at 435 MHz): the frequency is deduced from the dimensions. Plan dated 31-10-2000.
- It is the most complex of the chapter: copper clamps, RG223 cable and a duplexer. Allow half a day.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Ø 25 mm PVC tube** | 1 | handle held in the hand |
| **VHF loop + reflector rod** | loop 705 × 324 mm | copper wire or tube |
| **UHF loop + reflector rod** | loop 238 × 108 mm | copper wire or tube |
| **Copper clamps** | per plan | fixing and contact of the elements on the tube |
| **RG223 cable** | per plan | double shielding; one link per loop |
| **V/U duplexer** | 1 | joins the two bands on a single cable |
| **Connectors** | 2 to 3 | depending on the duplexer and the cable (N, BNC or SMA) |

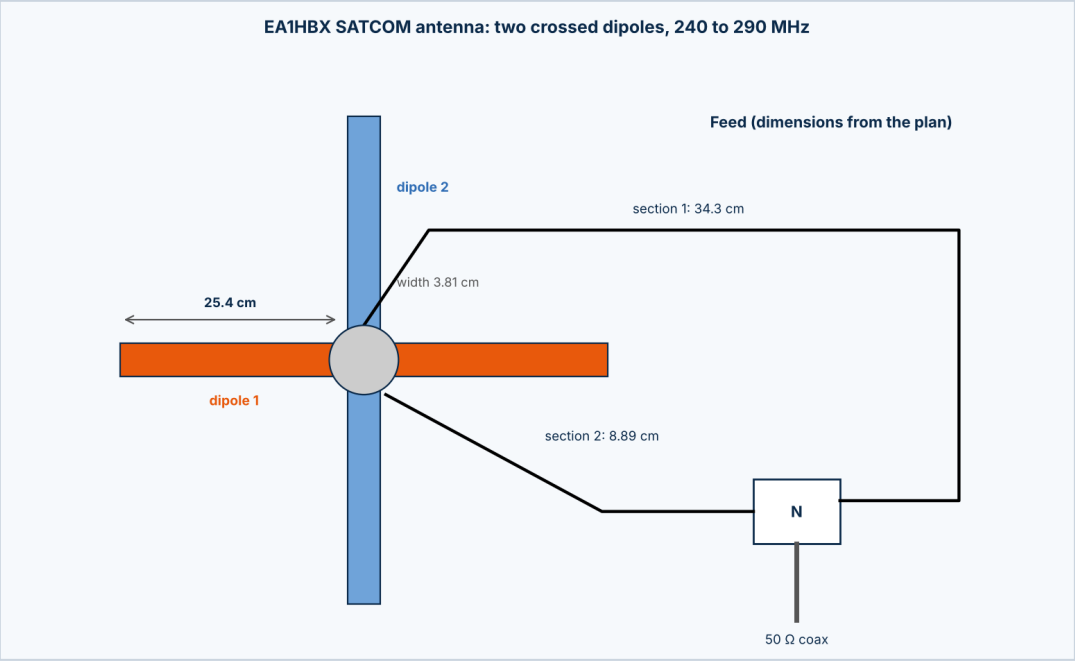
*List drawn up from the plan: check the part numbers and quantities before buying.*

## Personal notes (equipment used, settings, SWR measurements)

# Antenna no. 7: EA1HBX SATCOM

240 to 290 MHz · medium

A2 · DIY




- Four copper or aluminium plates of 25.4 × 3.81 cm form two crossed dipoles. They are fed by two sections of RG59 (75 Ω, vf 0.66) of 34.3 and 8.89 cm, whose centre conductors and shields (braids) are joined on the N connector.
- Stated band: 240 to 290 MHz (military UHF satellites). Dimensions taken from the plan; they are not recalculated, so check the phase difference before relying on it.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Copper or aluminium plates** | 4 × 25.4 × 3.81 cm | form two crossed dipoles |
| **RG59 cable (75 Ω, vf 0.66)** | 2 sections: 34.3 cm and 8.89 cm | centre conductors and shields (braids) joined on the connector |
| **N connector** | 1 | fixed on the support |
| **Cross-shaped insulating support** | 1 | PVC or wood, to hold the four plates |
| **Hardware, solder, heat-shrink** | as needed | dimensions taken from the plan, not recalculated: check the phase |

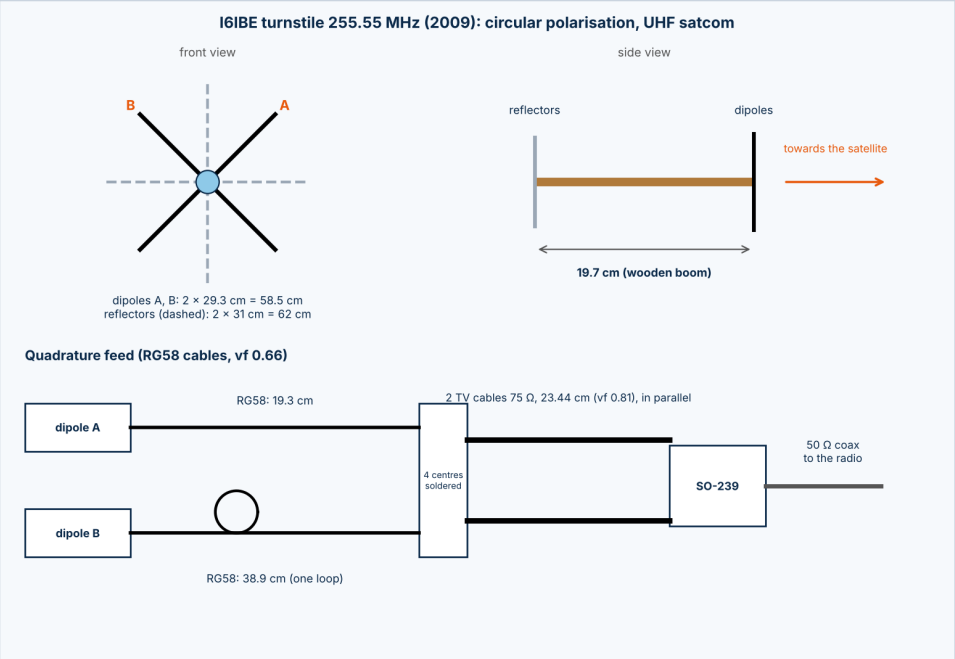
*List drawn up from the plan: check the part numbers and quantities before buying.*

## Personal notes (equipment used, settings, SWR measurements)

# Antenna no. 8: I6IBE turnstile

255.55 MHz · hard






- Two crossed dipoles of 58.5 cm (Ø 2 mm brass on a PCB hub), with two crossed 62 cm reflectors 19.7 cm behind, on a wooden boom. Right-hand circular polarisation.
- Feed: two RG58 cables of 19.3 and 38.9 cm. The 19.6 cm difference makes a quarter wave in the cable and offsets the dipoles by 90°. The four centre conductors are soldered together, then two 75 Ω TV cables of 23.44 cm in parallel lead back to the SO-239 connector.
- Plan by I6IBE (Ivo Brugnera, 2009), designed for receiving and transmitting on military UHF satellites. In this guide, it is for listening only.

## Components needed

| Component | Quantity / dimension | Remark |
|---|---|---|
| **Ø 2 mm brass elements** | 2 dipoles of 58.5 cm | crossed on a PCB hub |
| **Reflectors** | 2 × 62 cm, crossed | 19.7 cm behind the dipoles |
| **Wooden boom** | 1 | carries the dipoles and the reflectors |
| **RG58 cable** | 2 sections: 19.3 cm and 38.9 cm | the 19.6 cm difference = λ/4 in the cable: 90° phase shift |
| **75 Ω TV cable** | 2 × 23.44 cm | in parallel, back to the connector |
| **SO-239 connector** | 1 | output to the 50 Ω coax |
| **PCB hub** | 1 | supports the solder joints of the four centre conductors |

*List drawn up from the plan: check the part numbers and quantities before buying.*

## Personal notes (equipment used, settings, SWR measurements)

# Survival AM radio: diode and transistor

A radio to build from a few salvageable parts, a battery and an earpiece



**In brief.** This radio receives **amplitude-modulated** stations (medium wave, 530 to 1600 kHz) with a coil, a variable capacitor, a diode and a single transistor. It connects to nothing but a wire antenna, a ground and two batteries. It is a classroom circuit: reliable, repairable, and useful for understanding the principle of any radio. AM transmitters are now few in Europe: check what is still on the air near you (shortwave: sheet 7).

## Circuit diagram

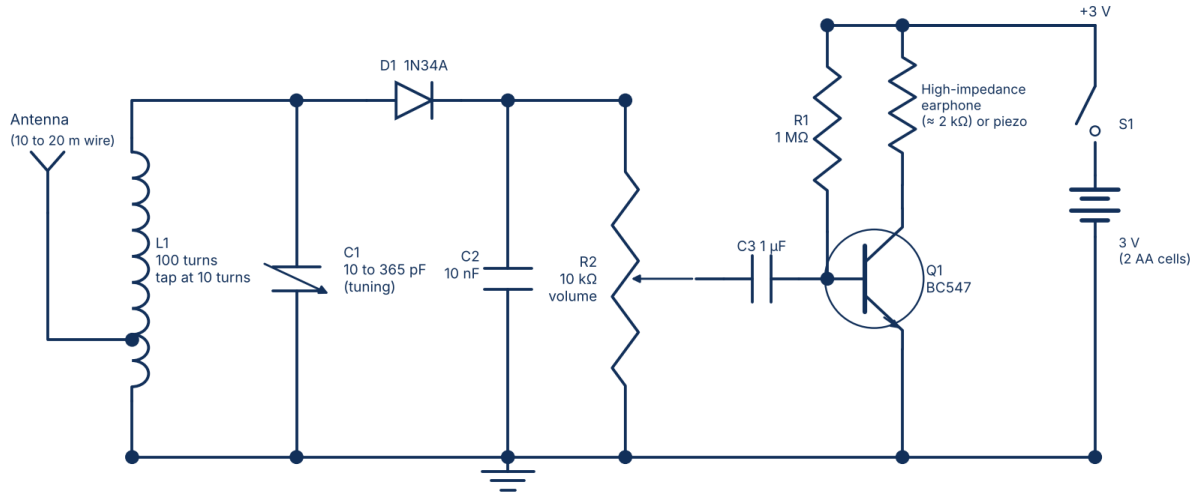



*Diagram redrawn for this guide. The L1 + C1 circuit selects the station, D1 detects, Q1 amplifies.*

**A correction to the original diagram.** The original diagram advertised a radio "with no external power" using a transistor: a transistor needs a battery to amplify. Without a battery, the circuit stops at the **crystal radio** (antenna, L1, C1, D1 and a piezo or high-impedance earpiece): weaker, but it works from the captured signal alone. An 8 to 32 Ω headset is not suitable on this circuit: you need a high-impedance earpiece, or a small matching transformer.

## Parts list

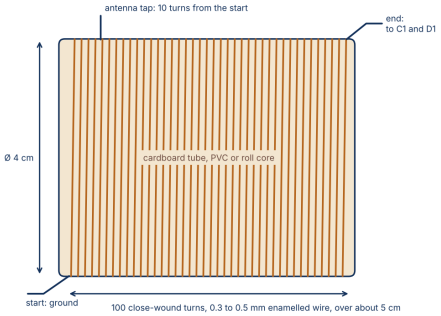
| Ref. | Qty | Value / type | Remark |
|---|---|---|---|
| **Q1** | 1 | BC547 (NPN) or 2N2222 / 2N3904 | Common transistor, very cheap |
| **D1** | 1 | 1N34A or OA91 (germanium); Schottky BAT43 possible | A silicon 1N4148 works less well: it conducts poorly on weak signals |
| **R1** | 1 | 1 MΩ | Biases the base of Q1 |
| **R2** | 1 | 10 kΩ, potentiometer | Volume control |
| **C1** | 1 | 10 to 365 pF, variable | Salvage it from an old AM set or an old tuner |
| **C2** | 1 | 10 nF | Filters the carrier after the diode |
| **C3** | 1 | 1 µF | Coupling between the potentiometer and the base |
| **L1** | 1 | 100-turn coil (≈ 230 µH), tap at 10 turns | Enamelled wire 0.3 to 0.5 mm on a 4 cm diameter tube |
| **Earpiece** | 1 | Piezo or high-impedance (≈ 2 kΩ) | An old crystal-set earpiece works |
| **Power** | 1 | 2 AA batteries (3 V) + switch | A single 1.5 V cell sometimes suffices, with less volume |
| **Antenna, ground** | 1 + 1 | 10 to 20 m of wire; ground stake | The longer and clearer the antenna, the stronger the reception |

# Survival AM radio: diode and transistor

A radio to build from a few salvageable parts, a battery and an earpiece

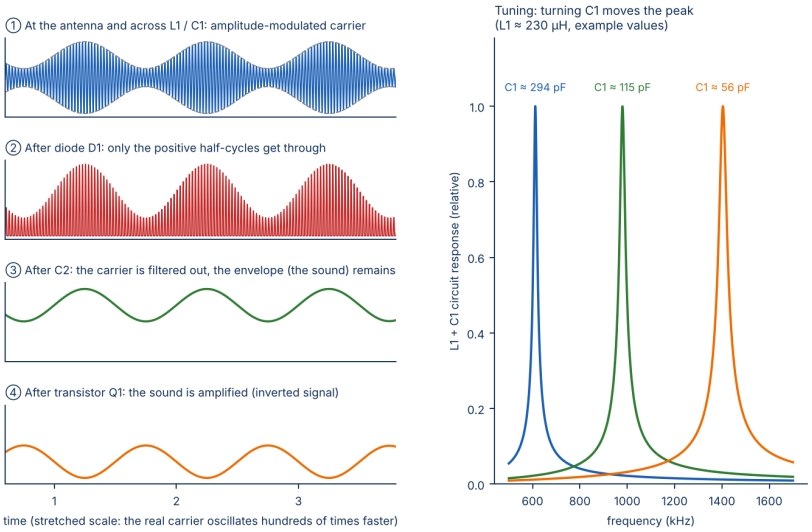


## The L1 coil




*100 tight turns, tap at 10 turns from the ground side (the drawing shows fewer turns to stay readable).*
*Order of magnitude: a 100-turn coil on a 4 cm diameter, 5 cm long tube gives about 230 µH. With C1 from 56 to 294 pF, the circuit covers roughly 610 to 1400 kHz: f = 1 / (2π√(L·C)). If the low end is missing, add a few turns; if the high end is missing, remove some.*

## How it works, with curves




*Illustrative curves (carrier drawn at a very low frequency to stay readable). On the right, turning C1 moves the peak of the tuning curve, and so the station received.*

- **The antenna** picks up all stations at once: tiny signals, in the microvolt range.
- **L1 and C1** form a tuned circuit: it lets the frequency f = 1 / (2π√(L·C)) through and attenuates the others. The 10-turn tap matches the antenna to the circuit without damping it too much.
- **Diode D1** only lets the positive half-cycles through (curve 2), **C2** smooths the carrier: what remains is the envelope, so the sound (curve 3).
- **Q1** amplifies this sound (curve 4) for the earpiece. R1 gives it a quiescent current; C3 blocks DC.

## Assembly and tuning

- First build the crystal part: antenna, ground, L1, C1, D1, C2 and the earpiece directly at the end of D1. You should hear a strong station, faintly.
- Then add R2, C3, Q1, R1 and the battery: the volume increases noticeably. If nothing comes out, check the direction of D1 and of Q1 (collector toward the earpiece).
- Turn C1 slowly: each station appears at a precise point. Change the antenna tap (5 to 20 turns) to find the best compromise between volume and selectivity.

**Safety.** No antenna near power lines. During a thunderstorm, disconnect the antenna and the ground. Take the ground from a stake or a metal water pipe, never from the mains or a gas pipe.

## Personal notes (stations received, settings, variants)

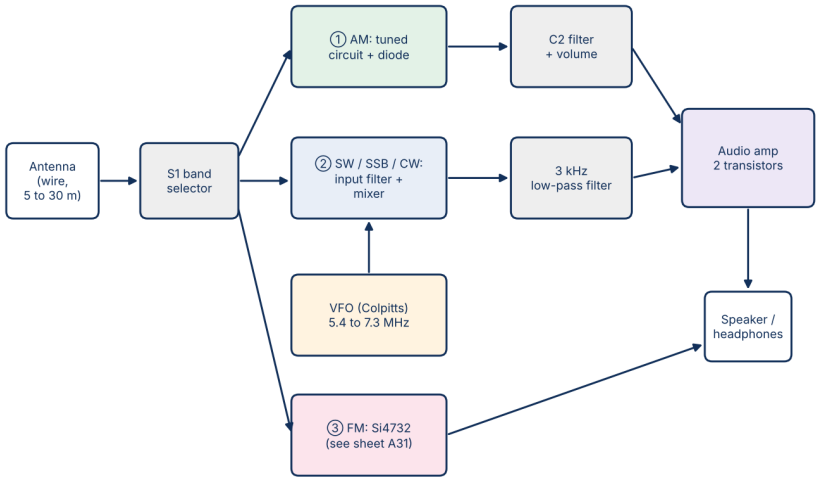
# Multiband radio: AM, shortwave, SSB and CW

A radio without an integrated circuit: principle, oscillator and step-by-step tuning



**In brief.** This sheet describes a radio made of **discrete components** (transistors, diodes, coils) that receives AM, **shortwave** and **SSB** (USB / LSB, SSB) or **CW**. SSB and CW are received by **direct conversion**: a local oscillator (VFO) sits almost on the signal frequency, and mixing gives the sound directly. **FM** is not covered in discrete form (detection and IF filters too delicate for a build without measurements): use the radio of sheet A31. The values are examples to tune with a check receiver and, if possible, a NanoVNA or a frequency counter (sheet 1).

## General architecture

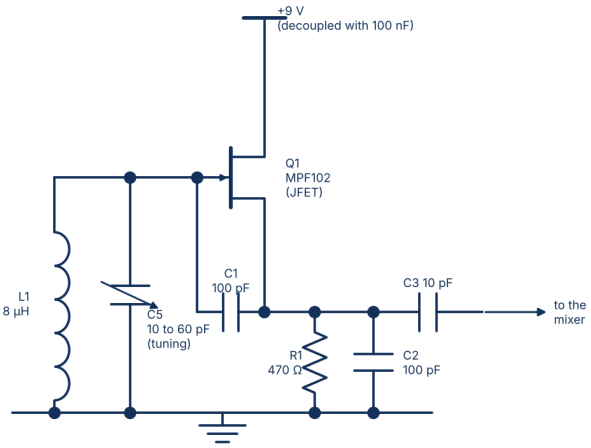



*A single antenna input, three paths: AM, shortwave by direct conversion, FM by a separate module.*

## Path 1: AM

This is the radio of sheet A29, with a coil and a variable capacitor suited to medium wave. It serves as the first check stage: if AM works, the audio amplifier is good.

## Path 2: shortwave, SSB and CW




*JFET Colpitts oscillator: L1 and C5 set the frequency; C1 and C2 provide the feedback.*

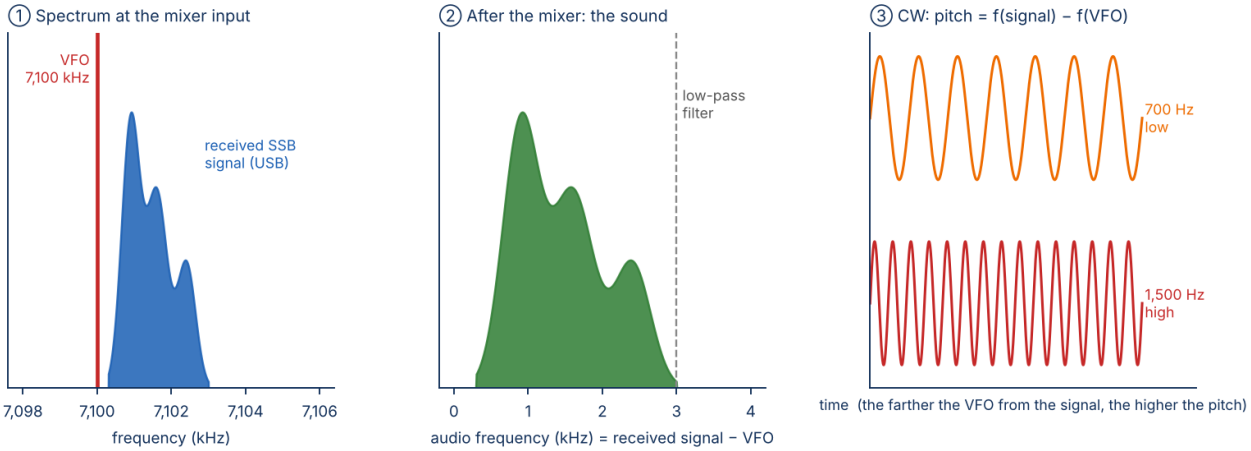
- **VFO frequency**: f = 1 / (2π√(L·C)), with C = C5 + (C1 · C2) / (C1 + C2). With L1 of 8 µH, C1 = C2 = 100 pF (i.e. 50 pF in series) and C5 from 10 to 60 pF, you get about 5.4 to 7.3 MHz. Stray capacitances lower this range a little: compare with a check receiver.
- **Mixer**: a ring of four diodes (1N5711 or BAT43) on two ferrite-toroid transformers, or a transistor / JFET as mixer. The VFO enters one port, the antenna another, and the sound comes out of the third. To be checked against a reference diagram.
- **Low-pass filter** around 3 kHz after the mixer: it cuts the sum frequencies and everything that is not sound.
- **Audio amplifier**: two transistor stages (2N3904 then 2N2222); an LM386 simplifies the build if you accept an integrated circuit.
- **In SSB**, tune the VFO next to the station (USB: VFO slightly below, LSB: slightly above) until the voice sounds natural; **in CW**, the offset sets the pitch of the tone.

# Multiband radio: AM, shortwave, SSB and CW

A radio without an integrated circuit: principle, oscillator and step-by-step tuning



## How it works, with curves




*Illustrative curves: on the left, the SSB signal (in blue) next to the VFO; in the centre, the same signal after the mixer; on the right, the CW tone.*

## List of main components

| Ref. | Value / type | Qty | Role |
|---|---|---|---|
| **Q1** | MPF102 (JFET) | 1 | VFO oscillator |
| **Q2, Q3** | 2N3904, 2N2222 (NPN) | 2 | Audio amplifier |
| **D1 to D4** | 1N5711 or BAT43 (Schottky) | 4 | Mixer ring |
| **L1** | ≈ 8 µH (coil or toroid) | 1 | VFO |
| **C5** | 10 to 60 pF, variable (air-dielectric preferably) | 1 | VFO tuning (stability depends on it) |
| **C1, C2** | 100 pF | 2 | Colpitts feedback |
| **R1** | 470 Ω | 1 | JFET bias |
| **Toroids** | FT37-43 or equivalent | 2 | Mixer transformers |
| **RV1** | 10 kΩ | 1 | Volume |
| **Power** | 9 V battery or 6 × AA, switch | 1 | Standalone power |
| **Antenna, ground** | 5 to 30 m of wire, ground stake | 1 + 1 | See sheet 1 |

## Tuning steps

- **1.** Build and test the audio amplifier alone (a finger on the input should give a hum).
- **2.** Build the VFO and listen to it on a check receiver (or measure it with a frequency counter): set the range with C5 and the number of turns of L1.
- **3.** Fit the mixer, then the low-pass filter: with the VFO left near an SSB or CW station, you should hear a sound that varies as you turn C5.
- **4.** Put everything in a metal case connected to ground and keep the RF and audio stages well separated.
- **5.** Plan for spare parts: transistors and diodes are fragile under the soldering iron's heat.

**Limits.** This radio has modest sensitivity and selectivity, and the VFO drifts with temperature: an integrated-circuit radio does better, and sheet A31 is the example. It is mainly for understanding, practice and emergency repairs.

## Personal notes (windings, measured values, tests)

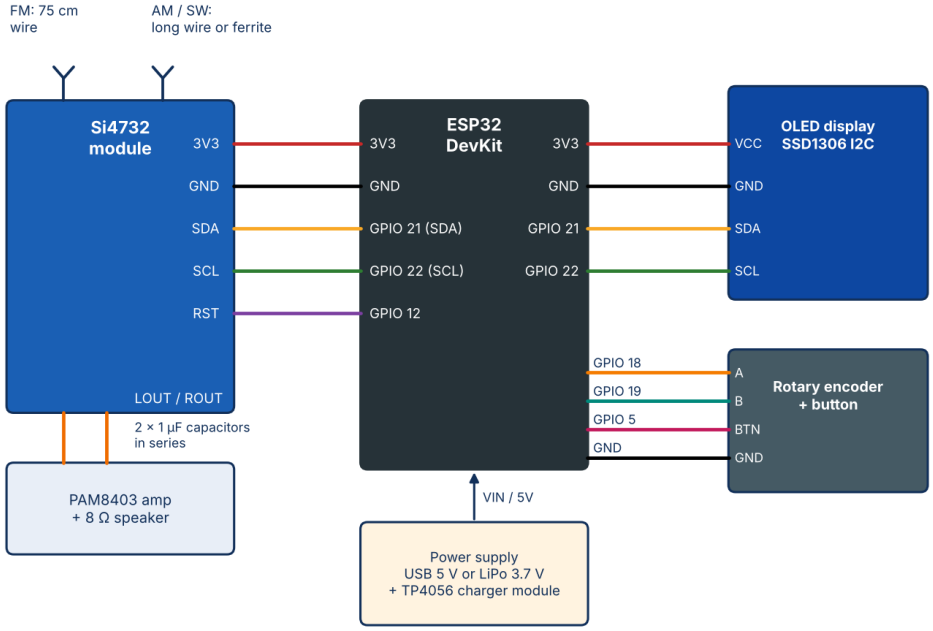
# FM / AM / SW radio with Si4732 and ESP32

A compact receiver with a screen: wiring, library, antennas and settings



**In brief.** The **Si4732** is an almost complete radio receiver in one chip: FM, long, medium and shortwave, with RDS. An **ESP32** (or an Arduino) controls it over the I2C bus and drives a screen and an encoder. It is the simplest route to a reliable FM / AM / SW radio, and it is the same chip as the module added to the UV-K1 (sheet A06). It does not receive aviation VHF signals or digital modes: for those, see sheets 3 and 9.

## Wiring




*Wiring diagram redrawn for this guide. Pin numbers are examples: they can be changed in the program.*

**Two corrections to the original diagram.** The audio does **not** go through the ESP32's converter: the Si4732 itself produces analogue sound on its LOUT and ROUT outputs, which go to the amplifier through two 1 µF capacitors. The module's **RST** pin must also be connected to an ESP32 GPIO: without it, the chip does not start reliably.

## Connections

| Si4732 | ESP32 | Remark |
|---|---|---|
| **VCC** | 3V3 | 3.3 V supply only (the chip accepts 2.7 to 3.6 V) |
| **GND** | GND | Common ground with the screen, encoder and amplifier |
| **SDA** | GPIO 21 | I2C bus, shared with the OLED screen |
| **SCL** | GPIO 22 | I2C bus, shared with the OLED screen |
| **RST** | GPIO 12 | Reset, to be declared in the program |
| **LOUT, ROUT** | Amplifier (PAM8403) or headphone jack | Through two 1 µF capacitors |
| **ANT** | 75 cm wire or ferrite | FM: wire; AM / shortwave: long wire or ferrite rod |

# FM / AM / SW radio with Si4732 and ESP32

A compact receiver with a screen: wiring, library, antennas and settings



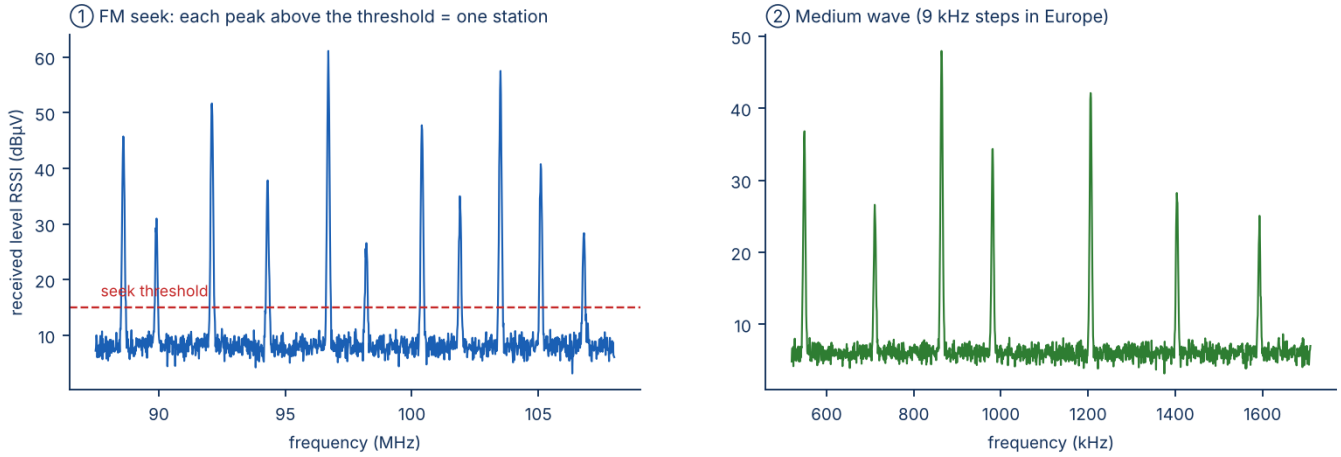
## Parts list

| Component | Quantity | Remark |
|---|---|---|
| **Si4732 module (with 32.768 kHz crystal)** | 1 | Most modules already have it; check that the FM and AM antenna sockets are present |
| **ESP32 DevKit (or Arduino Nano / Uno)** | 1 | With a 5 V Arduino, use a level shifter for I2C |
| **0.96 inch I2C OLED screen (SSD1306)** | 1 | Optional, but useful for frequency and mode |
| **Rotary encoder with button** | 1 | Frequency tuning, mode selection |
| **PAM8403 amplifier + 8 Ω speaker** | 1 | Or simply a headset on the module's output |
| **Capacitors 1 µF, 100 µF, 10 nF** | 2 + 1 + 1 | Audio coupling, decoupling, antenna (option) |
| **3.7 V LiPo battery + TP4056 charger, or USB 5 V** | 1 | The ESP32 itself steps down to 3.3 V for the Si4732 |

## Program

- In the Arduino IDE, install the ESP32 board then the **SI4735** library by Ricardo Caratti (PU2CLR), which handles the whole Si473x family including the Si4732.
- Start from its examples with OLED screen and encoder: adapt the pin numbers (RST, encoder) to your wiring.
- The Si4732's I2C address is 0x11 or 0x63 depending on the state of the module's SEN pin; the library can search for it.
- SSB reception requires loading a **patch** into the chip at each startup: the library's examples do it.

## Useful settings and curves




*Illustrative curves: a real scan's shape depends on the antenna, the location and the time of day.*

*Illustrative scan curves: the program looks for peaks that exceed a received-level threshold.*

- **FM**: automatic station search, RDS name display, 100 kHz step in Europe (50 kHz possible depending on the program).
- **AM, long and medium wave**: 9 kHz step in Europe; shortwave: 5 kHz step.
- **Filter width**: narrow (about 1 to 3 kHz) for crowded stations, wider to hear music better.
- **Antenna**: in FM, the 75 cm wire (quarter wave at 100 MHz) is enough; in AM and shortwave, a ferrite rod or 5 to 10 m of clear wire gains a lot.

> **Listening only.** This build only receives. Private exchanges you hear must not be disclosed.

## Personal notes (actual wiring, program used, stations received)

# ATS Mini (V4): Si4732 pocket receiver and its firmwares

A ready-made ESP32-S3 receiver: what is inside, open firmwares and GitHub addresses



## What is inside

| Item | Detail |
|---|---|
| **Microcontroller** | ESP32-S3-WROOM-1 module: Wi-Fi, Bluetooth, USB, PSRAM memory (OSPI or QSPI depending on the batch) |
| **Receiver** | Si4732-A10: FM, LW, MW, SW, with RDS; SSB goes through a software patch |
| **Screen** | Colour LCD driven by an ST7789 or a GC9307 depending on the batch |
| **Power** | Lithium battery, charging by a TP4056 chip, ME6217 regulator, battery measurement on pin IO4 |
| **Audio** | NS4160 or AD4150B amplifier, speaker and headphone jack |
| **Controls** | Rotary encoder with button, BOOT and RESET buttons (to check: some batches do not have them soldered) |

## The firmwares and their GitHub addresses

| Firmware | Address | What it brings |
|---|---|---|
| **ATS Mini (esp32-si4732)** | github.com/esp32-si4732/ats-mini | Main community firmware, with documentation, Wi-Fi updates and a web interface: esp32-si4732.github.io/ats-mini |
| **ATS_MINI by G8PTN** | github.com/G8PTN/ATS_MINI | The original alternative firmware, from which the previous one derives |
| **ats-mini-pro (sirruf)** | github.com/sirruf/ats-mini-pro | LVGL interface, spectrum and waterfall, control over HTTP and Bluetooth, web interface on the radio |
| **ats-mini PU2QTF (fernandofatech)** | github.com/fernandofatech/ats-mini-pu2qtf | CW introduction, live NCDXF beacons, EiBi tools, USB control tool |
| **tinyradio (lsongdev)** | github.com/lsongdev/tinyradio | AM / FM / SSB receiver in C for ESP32, RDS, aviation band; ATS Mini and GeekNest V5A |
| **13HN2709 fork** | github.com/13HN2709/ats-mini | Variant of the main firmware |

**Before choosing.** On each page, check the mention of the board's V4 revision, the date of the latest release and the licence: these projects evolve quickly and I could not confirm which ones support the V4. The main firmware publishes the files to flash in the "Releases" tab of its repository.

## Flashing safely

- **Choose the right variant**: an OSPI or QSPI version depending on your ESP32-S3's PSRAM (the Settings → About page shows a non-zero PSRAM if the variant is right). The "lilygo-t-embed" variant only suits the Lilygo T-Embed: on any other device, it can damage it.
- **Over Wi-Fi**: set the date and time, then Settings → Update FW, or the Update page of the web interface. An -ota.bin file can also be uploaded by hand, without Internet.
- **Over USB**: from Chrome or a Web Serial-compatible browser (not Firefox or Safari), with the esptool-js tool; or with the Flash Download Tool on Windows (ESP32-S3 chip); or with esptool on the command line.
- **Files**: either four files (bootloader at 0x0, partitions at 0x8000, boot_app0 at 0xe000, firmware at 0x10000), or the merged file at 0x0, which always resets the settings. Do not forget boot_app0.bin if you flash the separate files.
- **Backup**: before flashing, back up the original firmware so you can go back ("backup" and "recovery" pages of the documentation). A failed flash can lock up the board: the BOOT and RESET buttons are used to recover it.

## To remember for this guide

- The ATS Mini is a receiver, not a transmitter: it joins sheet A31 as a DIY build, without soldering.
- For HF listening with the UV-K1, the same Si4732 exists as an add-on module (sheet A06).
- A long wire antenna (5 to 10 m) greatly improves shortwave; a telescopic FM antenna is enough for the FM band.

# ATS Mini (V4): Si4732 pocket receiver and its firmwares

A ready-made ESP32-S3 receiver: what is inside, open firmwares and GitHub addresses



## ATS Mini V4 in brief

| Feature | Value |
|---|---|
| **Radio chip** | Si4732 (Silicon Labs): FM, AM, shortwave, longwave, with digital signal processing |
| **Microcontroller** | ESP32-S3 family: screen, controls, Wi-Fi, firmware update |
| **Advertised coverage** | Long and medium wave 150 to 1710 kHz; shortwave 1711 kHz to 30 MHz; FM 64 to 108 MHz |
| **Modes** | FM (with RDS), AM, LSB and USB (the SSB patch is loaded into the chip) |
| **Screen** | 1.9 inch colour IPS, adjustable brightness |
| **Battery and charging** | 800 mAh rechargeable lithium, USB-C charging |
| **Case** | V4: aluminium alloy with shielded compartments (ABS plastic on the V3) |
| **What's new in V4** | Redesigned circuit board (filtered power, analogue separated from digital), better headphone amplifier, headset cable usable as an FM antenna; built-in CW and RTTY decoding depending on the firmware |
| **Versions V4, V4a, V4b** | The V4 and V4a had a "theremin" effect in SSB, fixed on the V4b, which also limits headset cable noise |

*Source: vu3dxr.in article on the evolution of the ATS Mini from V1 to V4; check the exact specifications with your seller.*

## Other firmwares and related projects

| Firmware | Author | GitHub address | Strengths |
|---|---|---|---|
| **ats-mini (community fork)** | Sunnygold (hardware); Max Arnold R9UCL (organisation) | github.com/esp32-si4732/ats-mini | The most active: Wi-Fi, time via NTP, improved RDS, many features taken from other projects. Releases page and documentation from this address |
| **Original hardware and source** | esp32-si4732 | github.com/esp32-si4732/esp32-si4732-oshwhub | Original schematics and sources of the open hardware |
| **ATS_MINI (G8PTN)** | Dave, G8PTN | github.com/G8PTN/ATS_MINI | More intuitive interface (click and double-click), more logical menus and tuning; files in the FIRMWARE folder |
| **ats20_ats_ex** | Nikita, Goshante | github.com/goshante/ats20_ats_ex | Precise SSB tuning, BFO built into the frequency, clean interface. Originally written for the ATS-20 |
| **SI4735 (ESP32 OLED example)** | Ralph Xavier | github.com/ralphxavier/SI4735 | Code base for driving a Si4732 with an ESP32: a starting point, not a complete firmware |
| **SI4735 library** | Ricardo Caratti, PU2CLR | github.com/pu2clr/SI4735 | The library used by almost all firmwares and by sheet A31 |
| **pocketSI4735DualCoreDecoder** | H.J. Berndt | hjberndt.de/dvb/pocketSI4735DualCoreDecoder.html | Not on GitHub: download page. Station list, signal plot, RDS; RTTY and CW decoders |

*Addresses found online while writing; add the https:// prefix to open them.*

## Updating the firmware

- Get the file of the latest version from the project's releases page, checking that it matches your board (V3 or V4).
- The update is done over USB-C from a computer; the projects describe the procedure in their documentation. Keep the original configuration if the project offers it.
- A firmware is changed the way a radio is tuned: try it, and go back to the previous one if the interface does not suit you.

**Why here, with the radios to build?** It is the same chip as sheet A31, but already assembled, shielded and powered: a good starting point for listening to shortwave and SSB, and for understanding what a Si4732 module can do before modifying a UV-K1 (sheet A06). For SSB and shortwave, a good antenna matters more than the firmware (sheet 1).

# Standalone power: batteries and solar

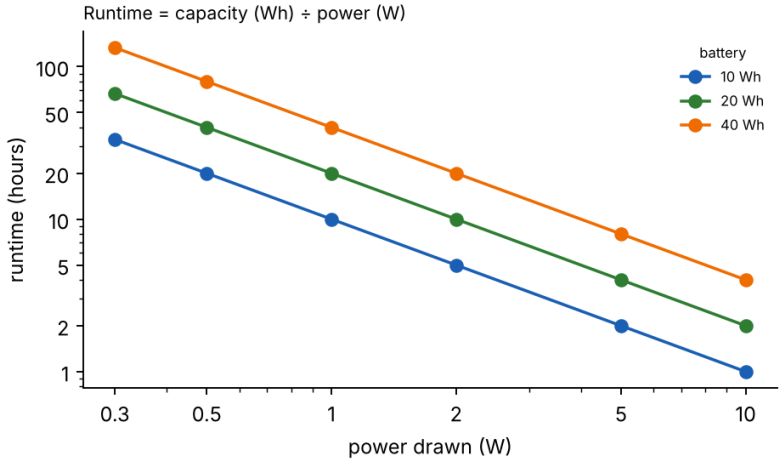
Keeping the radio and its accessories running when the grid goes down

**In brief.** A radio without power is useless. You need to know **how much each device uses**, **how many Wh** you store, and **how you recharge**. The basic rule: runtime (h) = capacity (Wh) ÷ power (W). For capacity in Wh, multiply the battery voltage by its capacity in Ah: a 3.7 V, 3 Ah battery holds 11 Wh.

## How much each device uses

| Device | Typical power | Remark |
|---|---|---|
| **UV-K1 listening** | About 0.3 to 1 W | More with the screen on and the speaker loud; measure on your own radio |
| **UV-K1 transmitting** | Several times more than listening | Transmitting is very power-hungry: keep it short (sheet A00a) |
| **Si4732 radio + ESP32 + screen** | About 0.3 to 0.8 W | Sheets A31 and A32 |
| **RTL-SDR dongle** | About 1 to 1.5 W (USB 5 V) | Not counting the computer or smartphone (sheet 3) |
| **Single-board computer (Raspberry Pi type)** | About 3 to 6 W | Depends on the model and the load |
| **Smartphone charging** | 5 to 15 W | It uses much less on standby |
| **GNSS module (sheet 20)** | About 0.1 W | Very low |

*These values are orders of magnitude: a cheap USB power meter gives the real consumption of each device.*




*Example: a 20 Wh battery powers 1 W for 20 hours, and 5 W for 4 hours.*

## Which battery to choose

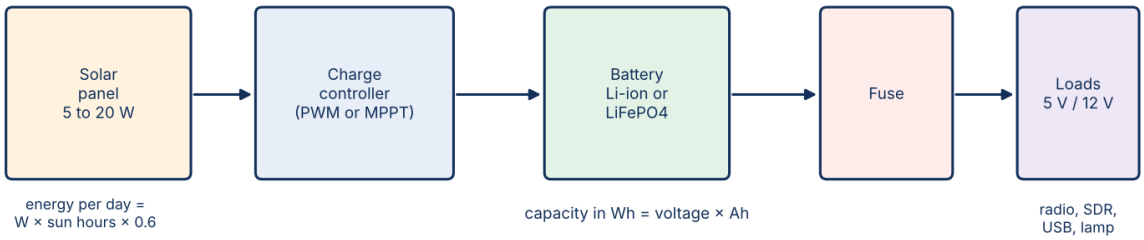
| Type | Voltage | Strengths | Weaknesses |
|---|---|---|---|
| **Lithium-ion (18650, radio battery pack)** | 3.7 V per cell | Light, high energy, very common; TP4056 charging module for one cell (sheet A31) | Dislikes short circuits, overcharging and cold; needs protection |
| **LiFePO4** | 3.2 V per cell | Very durable (thousands of cycles), safer | Slightly heavier, lower voltage |
| **NiMH (rechargeable AA cells)** | 1.2 V per cell | Easy to find, no electronic protection | Self-discharges: choose "low self-discharge" |
| **Lead-acid 12 V (car or AGM battery)** | 12 V | Robust, cheap, large reserve | Heavy; tolerates deep discharge badly (stay above 50% charge) |

## Recharging with the sun

# Standalone power: batteries and solar

Keeping the radio and its accessories running when the grid goes down






*Solar power chain: the fuse protects the battery and wires from a short circuit.*

- **Panel**: a 5 to 20 W panel is enough for a listening station. Point it south, tilted, with no shade, even partial (shade on one cell makes output drop).
- **Energy per day** ≈ panel power × equivalent full-sun hours × 0.6 (losses). In winter at our latitudes, count only a few equivalent hours, and fewer when overcast.
- **Charge controller**: essential. An **MPPT** type recovers more energy than a PWM type, especially with a large panel; for 5 to 20 W, PWM is enough. Use one that matches your battery (lead, lithium).
- **5 V output**: a USB converter (step-down module or 12 V car-to-USB charger) powers the radio, smartphone or SDR dongle.
- **Hand-crank dynamo, generator**: a last-resort stopgap, noisy or inefficient.

## Safety

- **Fuse** on the positive wire, close to the battery: rating a little above the circuit's maximum current.
- **Polarity**: mark red and black, solder or clamp firmly; a reversed wire can destroy the radio.
- **Lithium**: never short-circuit, never charge unattended or inside a bag; use protected cells or a protection module, and keep them away from heat. A swollen or hot cell must be removed and isolated.
- **Listening comes first**: save battery by lowering the backlight and volume, switching off what you don't need, and monitoring only at the times in your plan (sheet A00b).

To build a 12 V battery from 18650 cells (motorcycle terminals, USB, display), see sheet A34.

## Personal notes (batteries owned, panels, measured consumption)

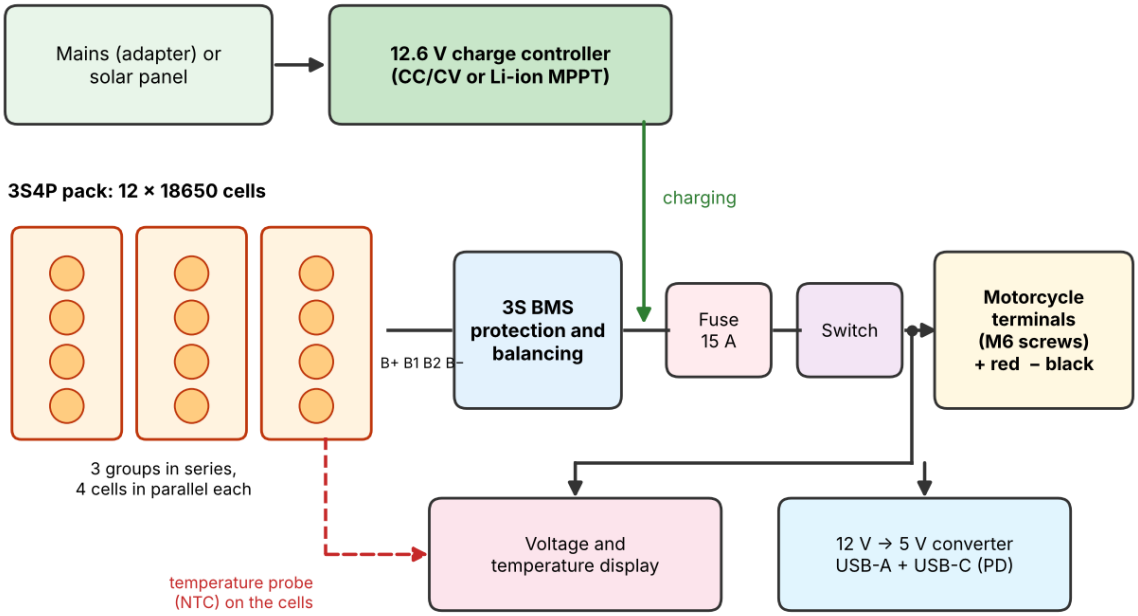
# 12 V battery with 18650 cells

A 3S4P pack with protection, motorcycle terminals, USB-A and USB-C outputs, charge controller and display



**In brief.** Twelve 18650 cells arranged as **3 series groups of 4 parallel cells (3S4P)** make a **12 V** battery (11.1 V nominal, 12.6 V full) of about **130 Wh** with 3 Ah cells (12 Ah × 11.1 V). A **BMS** protects it, **motorcycle terminals** make it easy to connect, a module converts to **USB-A and USB-C**, a **charge controller** recharges it from mains or sun, and a **display** shows voltage and temperature. This build involves lithium cells: read the "Safety" section before starting (principles: sheet A33).

## Overall diagram




*Block diagram: everything goes through the BMS; the charge controller and the outputs connect to the same bus (+ and −) as the terminals.*

## How many cells: choosing the pack

| Pack | Cells | Capacity (3 Ah cells) | Energy (≈ 11.1 V) | Runtime at 1 W | Recommended continuous current |
|---|---|---|---|---|---|
| **3S2P** | 6 | 6 Ah | ≈ 67 Wh | ≈ 67 h | up to 4 A |
| **3S4P** | 12 | 12 Ah | ≈ 133 Wh | ≈ 133 h | up to 8 A |
| **3S6P** | 18 | 18 Ah | ≈ 200 Wh | ≈ 200 h | up to 12 A |

*Calculation: energy (Wh) = voltage × capacity (Ah); runtime (h) = energy ÷ power drawn (sheet A33). Recommended current: about 2 A per cell for salvaged cells, to be adapted to the datasheet if you have it. Real runtimes are shorter (converter losses, cell age).*

**A "12 V" battery in 3S.** A 3S Li-ion pack varies from 9 V (empty) to 12.6 V (full): most 12 V devices accept it, but not all. For a truly stable 12 V range, use 4 LiFePO4 cells in series (4S, 12.8 V nominal, 14.6 V full) with a BMS and a LiFePO4 charger; the build stays the same, only the charge voltage and the BMS change.

# 12 V battery with 18650 cells

A 3S4P pack with protection, motorcycle terminals, USB-A and USB-C outputs, charge controller and display



## Parts

| Item | Recommended choice | Remark |
|---|---|---|
| **18650 cells** | 12 cells of the same model, same capacity, same age | Test each cell with a charger-tester; discard swollen, dented or low-capacity ones (below 60% of the rated value) |
| **Holders or nickel strip** | Spring holders (no soldering) or pure nickel strip spot-welded | Never solder directly onto a cell with an iron: the heat damages it. Spot welding is done with a dedicated welder |
| **3S BMS** | 3S Li-ion BMS with balancing, 20 to 30 A | Choose a model with overcharge, deep-discharge, overcurrent and short-circuit protection |
| **Charge controller** | 12.6 V mains adapter with current limiting (CC/CV), or MPPT solar controller set for a 3S lithium battery | A 14.4 V lead-acid charger is not suitable without adjustment |
| **Fuse** | Blade or glass fuse holder, 15 A (to adapt) | On the positive wire, as close to the BMS as possible |
| **Switch** | 20 A toggle switch, or cut-off switch | Disconnects the battery during transport |
| **Motorcycle terminals** | Two M6 screw terminals (red and black), or ring-lug plates | Placed on top of the case, with caps |
| **USB converter** | 12 V to USB-A and USB-C (PD) module, 3 A or more | Ready-made "cigarette lighter" modules are the safest; check the input range (9 to 30 V) |
| **Voltage + temperature display** | LCD voltmeter with NTC probe (two-wire model, 4 to 30 V) or small microcontroller with screen | Probe taped between two groups of cells (see below) |
| **Cables** | 1.5 to 2.5 mm² cross-section depending on current, crimped lugs | The higher the current, the larger the cross-section |
| **Case** | Plastic or wooden case, with insulating paper between the cells | Ventilation and protected cable passages |

## Step-by-step build

- **1. Sort and test**: charge all the cells, measure their resting voltage (they must be within 0.05 V of each other before paralleling them) and their capacity if your charger allows it.
- **2. Form the 3 groups**: 4 cells **facing the same way** in parallel (all + together, all − together) for each group. One group gives 3.7 V.
- **3. Put them in series**: the − of one group to the + of the next, with nickel strip or thick wire. The pack then gives 11.1 V.
- **4. Wire the BMS**: balance leads to the pack − (B−), between the groups (B1, B2) and to the + (B+), **in the order given in the manual**; the BMS output (P− and the pack +) forms the bus. Follow the diagram for your model.
- **5. Fuse, then switch** on the positive wire, then the **motorcycle terminals**.
- **6. Connect the charge controller** to the bus (+ and −) or to a dedicated connector: same wire as the output, charging goes through the BMS.
- **7. Add the outputs**: the USB converter and the display on the bus, after the switch, with a small fuse if possible.
- **8. Temperature probe**: tape it onto a group of cells, in contact, with heat-resistant adhesive tape.
- **9. First tests**: measure the no-load voltage (around 11 to 12.6 V depending on charge), then connect a small load (an indicator lamp, a low-power device) before the radio.

## Voltage and temperature display

- **Voltage**: a two-wire LCD voltmeter (4 to 30 V) connects directly to the terminals; it shows the approximate state of charge. Landmarks for a 3S Li-ion pack: 12.6 V full, about 11.1 V at half charge, 9.6 to 10 V almost empty (the BMS cuts off lower).
- **Temperature**: a 10 kΩ NTC probe taped to the cells, read by a small display or a microcontroller (like the ESP32 in sheet A31). It is used to spot abnormal heating. Landmarks: while charging, stop above 45 °C; while discharging, stop above 60 °C; no charging below 0 °C (usual values, to verify in the cell datasheet).
- **A more complete option**: an ESP32 with an OLED screen reads the voltage through a divider (two resistors) and the NTC probe, and shows both; it can also cut the outputs if the temperature gets too high. Reserve it for those who can program.

## Charge controller

- **Mains**: a 12.6 V (CC/CV) adapter with current limiting; do not charge above 0.5 C (about 6 A for a 12 Ah pack), 2 to 4 A is gentler on the cells.
- **Solar**: an MPPT solar controller set for a 3S lithium battery (end of charge at 12.6 V); a 20 to 50 W panel is fine for slow recharging (sheet A33).
- **The BMS does not replace the controller**: it cuts off on overcharge, but it is the controller that must stop charging at the right level.

# 12 V battery with 18650 cells

A 3S4P pack with protection, motorcycle terminals, USB-A and USB-C outputs, charge controller and display



## Safety

**Lithium cells can catch fire.** A short circuit, overcharge or damaged cell can cause overheating and fire. Work on a non-flammable surface, without metal jewellery, and keep a bucket of sand or a suitable extinguisher within reach.

- **Matched cells**: same model, same capacity, same state of charge. Never mix new and old ones.
- **No damaged cell**: torn wrapper, dent, leak or abnormal heating: discard it and isolate it.
- **Insulation**: insulating paper between the cells and the case, covered lugs, no bare wire near the terminals.
- **Fuse and BMS**: always, as close to the pack as possible. Never bypass the BMS.
- **Supervised charging**: first charge and first cycles under supervision, never in a bag or on fabric.
- **Current**: do not draw more than the recommended current from the pack. A radio or SDR dongle uses very little; a heater or a 230 V inverter does not.
- **Storage**: around 3.7 V per cell (50% charge) in a cool place, and check the voltage from time to time.

## Personal notes (cells tested, measured capacity, BMS used, consumption measurements)

## Change list

This page lists the changes to the book, from the most recent to the oldest: the **date**, the **version**, the **user** (nickname) of the person who made the change, and what changed. Anyone who modifies the book adds a line at the top of the list with their nickname.

| Date | Version | User | Change |
|---|---|---|---|
| 01/11/2026 | V3 | ic3ric4 | First English edition (EN): all text, diagrams, images and covers translated. |
| 01/11/2026 | V3 | ic3ric4 | Added this change list before the final page. New file name rule: language, version, month and year (example: ENOpenBookRadio-V3-1126.PDF). |
| 31/10/2026 | V3 | ic3ric4 | Starting point of the list: version V3 beta, 70 sheets (chapters 1 to 8, appendices A00, A1 and A2, from A00a to A34). |

## File name

The book file is always named **(Lg)OpenBookRadio-(Version)-MMYY.PDF**, whoever modifies it:

- **(Lg)**: the two-letter code of the book's language (FR for French, EN, ES, DE...).
- **(Version)**: the version number on two characters (V3, V4...).
- **MMYY**: the month and two-digit year of the change (November 2026: 1126).
- Example for this version: **ENOpenBookRadio-V3-1126.PDF**.


TOGETHER,
WE GO FURTHER
This book is not an end, but the beginning of an adventure that continues thanks to you.
Radio, technology, curiosity and the spirit of mutual help bring us together. Thanks to open source, everyone can learn, share, contribute and innovate.
Whether you are a radio enthusiast, a developer, a tinkerer or simply curious, you are part of this great community.
THANK YOU TO ALL CONTRIBUTORS, PAST, PRESENT AND FUTURE.
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INNOVATION • COMMUNITY
Knowledge is stronger when it is shared.